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Variational Analysis of Confinement and Hypocoercive Stability of Equilibria in Vlasov–Poisson–Fokker–Planck Systems

Master's Thesis

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Abstract

The ability to confine plasma in stable, spatially localised configurations is a central challenge in plasma physics, motivating the mathematical study of mechanisms that generate and stabilise stationary states. This thesis studies the confinement problem and the dynamical stability of stationary states for the Vlasov–Poisson–Fokker–Planck system. Its central object is the nonlinear mapping from external potentials to stationary states, which provides a unified framework for the characterisation of equilibria, the analysis of their regularity, the design of confinement strategies, and the investigation of their dynamical stability. The analysis begins with a variational characterisation of stationary states as minimisers of the free energy functional, establishing their existence and uniqueness and identifying them with weak stationary solutions under suitable regularity assumptions. Building on this foundation, a functional-analytic framework is developed to analyse the dependence of stationary states on the confining potential and to establish regularity properties of the potential-to-equilibrium map. This framework is subsequently applied to an inverse confinement problem, in which external potentials are constructed so that the associated stationary spatial densities approximate prescribed target profiles in the Kullback–Leibler metric. It also provides the basis for the study of the linearised dynamics around equilibrium: using hypocoercivity methods, exponential convergence towards stationary states is established under suitable regularity assumptions on the external potentials. Numerical investigations reveal slowly converging regimes, motivating the development of control strategies to accelerate convergence towards equilibrium. Taken together, these results establish the potential-to-equilibrium map as a common analytical framework linking the existence, regularity, design, and stability of stationary states for the Vlasov–Poisson–Fokker–Planck system.

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Chapter 1

Introduction

A recurring and physically central question in kinetic plasma physics is that of *confinement*: whether, and by what mechanism, a body of plasma can be maintained in a stationary, spatially localised configuration rather than dispersing under its own transport and collision dynamics. Beyond spatial localisation, confinement is essential for maintaining stable macroscopic states over physically relevant timescales. In the absence of an effective confining mechanism, transport processes drive plasma particles away from a prescribed region, preventing the formation of stationary equilibria and limiting the ability to control the evolution of the plasma. Understanding how confinement can be achieved, characterised, and optimised is therefore central both to plasma applications and to the mathematical study of kinetic equations.

The Vlasov–Poisson–Fokker–Planck (VPFP) system with external potential provides a kinetic framework for studying the confinement problem. This system arises as a fundamental model in the physics of collisional plasmas [8], where collisions redistribute particle velocities and drive the velocity marginal towards a Maxwellian equilibrium. At the microscopic level, particles evolve under deterministic transport driven by their velocities, the external potential and the self-consistent field (Vlasov–Poisson dynamics), while simultaneously undergoing velocity-space diffusion and friction (Fokker–Planck effects). These latter two terms admit a probabilistic interpretation: they are the Fokker–Planck counterpart of Langevin-type stochastic dynamics on individual particles, in which each particle's velocity is subject to a deterministic drift together with a Brownian random forcing [46]. Since the external potential provides a mechanism for influencing the spatial density of particles, in this thesis we treat it not as a fixed datum but as a design variable to be chosen so as to yield stationary states with prescribed confinement properties.

The VPFP system is analysed on the d-dimensional flat torus with 2π-periodicity:

$$\mathbb{T}^d = \mathbb{R}^d/(2\pi\mathbb{Z})^d \cong \left(\mathbb{R}/2\pi\mathbb{Z}\right)^d.$$

Throughout this manuscript, we shall refer to $\mathbb{T}^d$ simply as the torus. After nondimensionalisation, all physical parameters are taken to be equal to unity, except for the particle mass and the induced background charge density. This yields the nondimensionalised VPFP system, governed by the following equations:

$$\frac{\partial f}{\partial t} + v \cdot \nabla_x f - (\nabla_x u + \nabla_x \varphi_f) \cdot \nabla_v f = \Delta_v f + \nabla_v \cdot (vf)\,, \tag{VPFPa}$$

$$-\Delta_x \varphi_f = \rho_f - \rho_0 := \int_{\mathbb{R}^d} f \mathrm{d}v - \rho_0\,, \tag{VPFPb}$$

where

- $f : [0, +\infty) \times \mathbb{T}^d \times \mathbb{R}^d \to [0, +\infty)$ is the phase-space density, where $f(t, x, v)$ represents the density of particles at position x with velocity v at time t,
- $\varphi_f : [0, +\infty) \times \mathbb{T}^d \to \mathbb{R}$ is the self-consistent electrostatic potential generated by the particle density,
- $u : \mathbb{T}^d \to \mathbb{R}$ is an external potential,

- $\rho_f : [0, +\infty) \times \mathbb{T}^d \to [0, +\infty)$ is the macroscopic particle density obtained by integrating the phase-space density over velocities, and $\rho_0 > 0$ denotes the constant background charge, which is given by (2.2).

In the kinetic equation (VPFPa), the term $v \cdot \nabla_x f$ governs spatial transport, representing the advection of particles in physical space according to their velocity v. The term $-(\nabla_x u + \nabla_x \varphi_f) \cdot \nabla_v f$ accounts for acceleration in velocity space induced by the self-consistent field and the external field. The second-order term $\Delta_v f$ models velocity diffusion, describing random fluctuations in particle velocities caused by thermal noise, and $\nabla_v \cdot (vf)$ represents friction in velocity space, driving particles toward equilibrium.

The Poisson equation (VPFPb) determines the self-consistent potential φ_f generated by the particle density. The Laplacian $-\Delta_x \varphi_f$ relates the potential to the local charge of the particle density, and the resulting force $-\nabla_x \varphi_f$ feeds back into the kinetic equation, coupling the density function f and the self-generated field. The resulting VPFP system is nonlinear, because the self-consistent potential of the electric field and the corresponding force depend on the instantaneous particle density.

The VPFP system raises a first, foundational question: does it admit stationary states and, if so, in what sense can they be characterised? Rather than working directly with the stationary VPFP equations, we adopt a variational viewpoint, in which stationary solutions are recovered as minimisers of a free energy functional combining an entropy term with the electrostatic energy of the self-consistent field, the kinetic energy of the state and the energy associated with the external potential. This perspective, classical in the kinetic theory literature, allows existence and uniqueness to be obtained by direct methods for a given external potential u, and yields a notion of solution – the *free energy solution* – whose relationship to weak solutions of the stationary system must itself be established.

Having fixed, for each admissible u, a well-defined stationary state $f_\star$ with corresponding spatial density $\rho_\star$ and self-consistent potential $\varphi_\star$, the natural second question is how this state depends on u. This question is not merely a refinement of the existence theory; it is precisely what is required to treat u not as a fixed datum but as a design variable. Viewing the mappings $u \mapsto (\rho_\star, \varphi_\star)$ as nonlinear solution operators between suitable Banach spaces, we ask how regularity and topological properties of u propagate to the corresponding stationary spatial density and self-consistent potential. Only once continuity and sensitivity of these solution operators are established does it become meaningful to ask whether a given reference density can be achieved by choosing u appropriately.

This is the third question, and it inverts the logic of the first two: rather than computing the stationary state produced by a prescribed external potential, we ask which external potential produces a prescribed stationary state. We formulate this confinement problem variationally, as the minimisation of an entropy-based cost functional over the space of admissible external potentials, and study its well-posedness – a study that relies directly on the continuity properties established above – before illustrating it numerically in a one-dimensional setting.

The confinement problem yields, for each reference density, a specific equilibrium, but does not address its dynamical properties. The fourth and final question is therefore whether the equilibria produced by the inverse problem are dynamically stable, in the sense that nearby solutions of the time-dependent VPFP system converge toward them, and at what rate. We address this at the linearised level, analysing exponential stability through hypocoercivity methods. Numerical experiments then show that the resulting decay rate depends sensitively on the choice of equilibrium and, where convergence is slow, motivate control strategies to accelerate convergence.

The central object underlying the developments of this thesis is the nonlinear map $u \mapsto (\rho_\star, \varphi_\star)$. Chapter 2 constructs and characterises this map through the variational formulation of stationary states; Chapter 3 studies its fundamental regularity properties; Chapter 4 exploits these properties to optimise over the space of external potentials; and Chapter 5 investigates the dynamics around the equilibria contained in its range. In this sense, the thesis studies the construction, analysis, approximate inversion, and dynamical consequences of the nonlinear correspondence between external potentials and confined equilibria. These four aspects are presented in the order in which each naturally builds on the preceding one. Addressing them draws on tools from the calculus of variations, elliptic regularity theory, and hypocoercivity, developed

within separate subfields of the kinetic theory literature. We review the relevant background for each below.

1.1 Literature Review

The VPFP system sits within a family of kinetic and mean-field models that differ primarily in how they represent particle interactions and collisional effects. At one extreme, the pure Vlasov equation describes collisionless, deterministic transport of a phase-space density under a self-consistent or external field, with no velocity-space dissipation [31, 41]. This is the model of choice for weakly collisional or rarefied plasmas, such as solar winds, and collisionless gravitational systems, such as galaxies. At the other extreme, the pure Fokker–Planck equation retains only the velocity-space diffusion and friction terms, with no spatial transport [47]. It describes the evolution of the phase-space density of particles due to the cumulative effect of many small, random collisions, such as in the classical Ornstein–Uhlenbeck process [46]. In particular, the model combines diffusion, which represents the random scattering of particle velocities, and friction, that is, the systematic loss of momentum through interactions with the surrounding medium. The Vlasov–Fokker–Planck equation combines the two models, resulting in the description of the evolution of the phase-space density of particles that undergo spatial transport, velocity-space diffusion and friction, and acceleration under an external electrostatic field. This makes the model particularly suitable for moderately collisional plasmas, including processes such as heating, thermalisation, and transport of energetic electrons through the plasma [52]. When the electrostatic field is closed self-consistently through the Poisson equation (VPFPb), the resulting model is the VPFP system studied in this thesis. This Coulombic coupling is itself a special case of a broader class: replacing the Poisson-derived interaction with a general, possibly non-local, interaction potential yields the McKean–Vlasov–Fokker–Planck equation [35].

The problem of confinement is a central topic that motivates this hierarchy of models. Without some confining mechanism (external, self-consistent, or both) holding the ensemble of particles within a bounded region of phase space, no stationary configuration exists for the kinetic equation to converge to, and the long-time behaviour of the system is governed by dispersal rather than convergence. This phenomenon underlies an extensive body of literature. For instance, a central engineering problem is that of magnetic confinement in fusion reactors [36], namely the design and computation of an external magnetic field so that the resulting equilibrium remains bounded away from the reactor wall. In inertial confinement fusion, where collisional effects cannot be neglected, the same confinement question is instead posed for the Vlasov–Fokker–Planck model [52]. The confinement problem addressed in this thesis is the electrostatic, collisional analogue of these magnetic and inertial formulations: rather than a magnetic field steering the plasma via the Lorentz force, an external electrostatic potential u is considered.

Addressing this confinement problem requires a systematic study of the stationary states, including their existence, uniqueness, and characterisation. This is the subject of an extensive prior literature, with early approaches focusing primarily on the direct analysis of the stationary equations rather than on the variational formulation adopted in Chapter 2. [26] established existence and uniqueness of steady states for the Vlasov–Fokker–Planck equation with interaction and external potentials. Under suitable regularity assumptions on the potentials, the stationary solution is shown to factorise on the phase space $\mathbb{R}^d \times \mathbb{R}^d$ into a spatial density and a Maxwellian velocity marginal. The spatial component is characterised through a nonlinear exponential integral equation obtained by explicitly integrating the stationary equation in the velocity variable. This approach differs from the variational framework adopted in this thesis, where stationary states are characterised directly as minimisers of a free energy functional. This variational perspective was developed systematically for the VPFP system in [23], where minimisers of the free energy are shown to satisfy the stationary system under suitable assumptions on the external confining potential. As in [26], stationary states are shown to factorise into a Maxwellian velocity marginal and a spatial density, while the latter is coupled to the self-consistent potential through the Poisson–Boltzmann–Emden equation. The passage from a free energy minimiser to a weak solution of the stationary VPFP system, however, is not immediate. It requires additional arguments to justify the weak formulation and handle the compactness difficulties arising from the nonlinear coupling. Such arguments may be obtained through

renormalisation methods in the spirit of the Ambrosio–DiPerna–Lions theory (see, e.g., [29]; see also [19, 22] for the broader renormalisation framework), or through approximation procedures of the type developed in [17]. Alternative approaches to stationary solutions of the VPFP system have also been considered. [34] studied steady states under assumptions on the background density different from those of [23], using an existence argument closer in spirit to a fixed-point construction than to free energy minimisation. More generally, the existence of stationary states depends strongly on the nature of the self-consistent interaction and the confining mechanism. In the repulsive Coulombic case, [13] showed that solutions of the VPFP system converge, as $t \to \infty$, towards the unique stationary state with prescribed mass, implying that confinement always occurs in this regime. By contrast, in the attractive Newtonian case, stationary states may fail to exist: their existence depends on the total mass and on the growth properties of the confining potential at infinity, a distinction reflected in the associated Poisson–Boltzmann equation. The setting considered in this thesis – namely, the torus $\mathbb{T}^d$ together with a uniform neutralising background charge ρ_0 – avoids these difficulties entirely. The compactness of the spatial domain and the neutrality condition $\int_{\mathbb{T}^d}(\rho_f - \rho_0)\,\mathrm{d}x = 0$ place the problem within a structurally confined regime in which existence of stationary states is not the main obstruction. Instead, the central question that remains is how the resulting stationary density and self-consistent potential depend on the external potential u, to which we turn next.

In most of the kinetic theory literature discussed above, the external potential is treated as fixed data rather than as a variable to be chosen. The regularity theory needed to treat u as a control variable therefore draws on classical tools rather than a dedicated prior literature: elliptic bootstrapping for equations of Poisson–Boltzmann–Emden type follows the standard scheme of [33], and continuity of nonlinear solution operators between Banach spaces is a classical theme in functional analysis and the calculus of variations. A partial result in this direction is given by [11], who prove the Fréchet differentiability of the map $u \mapsto \varphi_\star$ on convex bounded Lipschitz domains satisfying a bound on their Poincaré constants. However, the specific questions addressed here – whether, and in what topology, the map $u \mapsto (\rho_\star, \varphi_\star)$ is continuous or regularising – do not appear to have been treated systematically in the VPFP literature, and are the object of Chapter 3.

The regularity properties established for the solution operators $u \mapsto (\rho_\star, \varphi_\star)$ provide the analytical foundation for the inverse problem of recovering a control parameter from a prescribed state of the system. This topic has a substantial history in the literature, typically formulated as PDE-constrained optimisation problems in which a cost functional is minimised subject to the state equation. For elliptic and parabolic equations, this theory is classical [38, 42]; for kinetic and mean-field models, comparatively few results address the recovery of an external or confining potential from a reference density. A formulation of this type is given by [11], who pose the confinement problem variationally with an L^2 tracking cost functional and remark that their approach extends to entropy-based functionals such as the Kullback–Leibler divergence. This choice is natural in the present setting: [6] show that Kullback–Leibler divergence is the functional intrinsically tied to the dissipative structure of Fokker–Planck-type equations, controlling the rate of convergence to equilibrium through convex Sobolev inequalities. Moreover, adopting the Kullback–Leibler divergence as tracking cost keeps the inverse problem within the same entropic framework as the free energy formulation of stationary states developed in Chapter 2. This adaptation, and the well-posedness questions it raises, are addressed in Chapter 4.

The entropic formulation of the confinement problem, however, leaves open whether the resulting equilibrium is dynamically stable. For kinetic equations of Fokker–Planck type, this question of stability has been addressed largely through the framework of hypocoercivity, developed to account for the degeneracy of dissipation, which acts only in the velocity variable while spatial transport remains conservative. The general L^2 hypocoercivity framework of [53] constructs a modified energy functional – combining the natural entropy or L^2 norm with auxiliary cross terms coupling position and velocity – whose dissipation controls the full phase-space norm, yielding exponential convergence to equilibrium despite the degeneracy of the diffusion. [25] reformulate this construction in an abstract L^2 setting for linear kinetic equations conserving mass, and [2] build on this analysis to establish exponential decay of trajectories of the linearised

VPFP system under suitable assumptions on the external confining potential. The latter is the framework we adopt in Chapter 5.

The numerical experiments of Chapter 5 show that the exponential decay rate of the linearised VPFP dynamics depends sensitively on the stationary state $f_\star$ around which the system is linearised, and hence, through the results of Chapter 3, on the underlying external potential u. Motivated by this observation, we investigate several control strategies for accelerating convergence to equilibrium, connecting our work to the broader literature on controlling relaxation rates in kinetic equations. [14] formulate an optimal control problem for the kinetic Fokker–Planck equation in which the confinement potential is perturbed, without altering the invariant measure, so as to maximise the rate of convergence to equilibrium. A related, PDE-constrained approach is pursued by [28], who optimise an external electric field for the Vlasov–Poisson system, using the kinetic equation itself as a constraint and deriving the corresponding adjoint equation to compute the functional derivative of the objective with respect to the field. Unlike [14], whose control problem targets the linear rate of convergence to equilibrium of a dissipative kinetic equation, [28] address the collisionless Vlasov–Poisson system, where the relevant notion of stability is not spectral decay but suppression of the nonlinear instabilities intrinsic to the hyperbolic transport structure. A methodologically distinct alternative is the instantaneous control approach of [3], who couple the Vlasov equation with the Maxwell equations to obtain, at each timestep, a feedback electromagnetic field that instantaneously minimises a confinement cost functional, rather than solving a single optimal control problem over the full time horizon via an adjoint equation. This closed-loop strategy is computationally cheaper than the adjoint approach and is shown numerically to steer the plasma away from the domain boundary. In [40], a finite-dimensional Riccati-based feedback law was derived for the McKean–Vlasov equation to achieve convergence towards equilibrium at a prescribed rate. To the best of our knowledge, however, no such control strategies appear to have been considered for the linearised VPFP system, a gap Section 5.4 addresses.

Across these four threads, the literature review reveals a common pattern: existence theory for the VPFP system on the torus is well established, but the questions of most direct relevance to treating the external potential as a control variable – continuity of the solution operators $u \mapsto (\rho_\star, \varphi_\star)$, the entropic inverse confinement problem, and control of the linearised dynamics around a general equilibrium – remain largely unaddressed. The remainder of this thesis develops each of these in turn. We summarise the resulting contributions below.

1.2 Contributions and Outline

A central contribution of this thesis is the functional-analytic framework within which the four questions raised above are formulated and shown to be coherently related. Considered separately, existence of stationary states, regularity of the associated solution operators, well-posedness of an inverse confinement problem, and stability of the resulting equilibria constitute four distinct technical problems, each addressed by methods drawn from a different area of the kinetic theory literature. The construction of this framework – namely, the choice of Banach space topologies under which the results of each chapter constitute a starting point for the next – underlies the contributions summarised below.

Chapter 2 is devoted to the variational characterisation of stationary states. Although the existence and structure of such equilibria are already established in the literature, we provide a self-contained treatment adapted to the functional setting required for the subsequent chapters. The first result concerns the existence and uniqueness of stationary states. For each admissible external potential u, we associate to the VPFP system a free energy functional, combining an entropy term with the kinetic energy of the state, the electrostatic energy of the self-consistent field, and the potential energy induced by u. The functional is bounded from below, weakly lower semicontinuous, and strictly convex on the set of admissible phase-space densities with prescribed mass (Lemma 2.2.14). Moreover, this admissible set is weakly compact (Lemma 2.2.13). Therefore, the direct method of the calculus of variations yields existence and uniqueness of the minimiser $f_\star$, together with its associated spatial density $\rho_\star$ and self-consistent potential $\varphi_\star$; this is the content of Proposition 2.2.16. Computing the first-order optimality conditions, we then derive the

Euler–Lagrange equation satisfied by $f_\star$ in Theorem 2.2.18: as in [23] and [26], the minimiser is shown to factorise into a Maxwellian velocity marginal and a spatial density $\rho_\star$, while the self-consistent potential $\varphi_\star$ is given implicitly as the solution of a nonlinear elliptic equation of Poisson–Boltzmann–Emden type. Theorem 2.2.23 establishes a converse: the convexity of the free energy functional implies that each pair $(f_\star, \varphi_\star)$ satisfying the previous Euler–Lagrange equation and the Poisson–Boltzmann–Emden equation is a minimiser of the free energy. Finally, we show that, under suitable integrability assumptions on $\varphi_\star$, this free energy minimiser is a weak solution of the stationary VPFP system (Theorem 2.2.30).

The second contribution of the thesis, developed in Chapter 3, concerns the regularity and sensitivity of the solution operators $u \mapsto \rho_\star$ and $u \mapsto \varphi_\star$ associated with this stationary problem. The starting point is an elliptic bootstrapping argument for the Poisson–Boltzmann–Emden equation satisfied by $\varphi_\star$, which must be initiated differently according to dimension: for $d \leq 2$, the H^1 regularity of $\varphi_\star$ already available from Chapter 2 suffices to start the bootstrap; for $d \geq 3$, this regularity is no longer sufficient, and an additional integrability assumption on $\varphi_\star$ is required to recover enough regularity of $\rho_\star$ to proceed. Once this preliminary step is resolved, however, the propagation of further regularity from u to $(\rho_\star, \varphi_\star)$ follows a uniform scheme in all dimensions (Proposition 3.1.5). Building on this pointwise regularity, we then view $u \mapsto \rho_\star$ and $u \mapsto \varphi_\star$ as operators into the Banach spaces $L^p(\mathbb{T}^d)$ and $W^{2,p}(\mathbb{T}^d)$ respectively, for a range of exponents p. Within this operator-theoretic framework, Theorem 3.3.1 establishes weak sequential continuity: along sequences of external potentials converging weakly, for which $\rho_\star$ and $\varphi_\star$ remain bounded in the appropriate topology, a chain of compactness arguments identifies the resulting stationary densities and potentials, in the limit, with those associated to the limiting potential. This identification relies on the sufficiency of the Euler–Lagrange and Poisson–Boltzmann–Emden equations for optimality, established in Theorem 2.2.23. In dimensions $d = 1, 2$, we further establish that these solution operators are continuously Fréchet differentiable (Theorem 3.4.7). This result is obtained through an application of the implicit function theorem to the nonlinear Poisson–Boltzmann–Emden equation characterising the stationary state. Beyond providing a differential characterisation of the sensitivity of equilibria with respect to perturbations of the external potential, this differentiability result provides the analytical foundation for optimise-then-discretise approaches to the inverse confinement problem in low dimensions, in the spirit of [11], although such approaches are not pursued in this thesis.

Chapter 4 contains the third contribution: the entropic formulation of the confinement problem, its well-posedness analysis, and a numerical study in the one-dimensional setting. We first show that exact confinement is, in general, unattainable within the free energy framework. More precisely, no external potential $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ can produce a stationary density vanishing on a set of positive measure (Corollary 4.1.2), so that reference densities with localised support lie outside the range of the solution operator $u \mapsto \rho_\star$. This motivates a variational approximation problem, in which confinement is recast as the minimisation of a cost functional combining the Kullback–Leibler divergence between $\rho_\star$ and a reference density ρ_{ref} with a cost on the external potential, subject to additional constraints encoding a variety of confinement properties. We then identify several degeneracies of this preliminary formulation, including the non-smoothness of the $H^1 \cap L^\infty$ penalisation, insufficient compactness to pass to the limit in the constraints, and the lack of regularity required to identify the weak limit of $\varphi_\star$ using the continuity theory developed in Chapter 3. Each of these issues is addressed through an appropriate regularisation, while restricting the admissible external potentials to the zero-mean subspace removes the artificial minimisation over additive constants. The resulting regularised functional is shown to admit a minimiser over the admissible set of potentials by exploiting the continuity properties of $u \mapsto (\rho_\star, \varphi_\star)$ established in Chapter 3; this well-posedness result is stated in Proposition 4.3.2. We conclude the chapter with a numerical study in the one-dimensional setting, illustrating the qualitative behaviour of the resulting optimal potentials and the sensitivity of the recovered stationary density to the choice of regularisation parameters for increasingly irregular reference densities ρ_{ref}.

The fourth and final contribution, developed in Chapter 5, concerns the dynamical stability and control of the equilibria generated by the confinement problem. Working at the linearised level, we adapt the hypocoercivity framework of [2] from the spatial domain $\mathbb{R}^d$ to the periodic setting $\mathbb{T}^d$, establishing

exponential decay estimates for the linearised VPFP system around a free energy stationary solution under suitable assumptions on the confining potential u. Numerical experiments then examine how the resulting decay rate depends on the choice of equilibrium, confirming the predicted exponential convergence while revealing that equilibria associated with weaker dissipative mechanisms converge markedly more slowly, with the bottleneck traced to low-frequency spatial modes. Motivated by this observation, we investigate feedback control strategies for accelerating convergence: a full-frequency feedback controller acting uniformly across the spectrum, a low-frequency spectral feedback controller acting only on the dynamically dominant modes, and a hybrid controller combining reduced-order LQR control of the least stable modes with spectral damping of the remainder modes. The hybrid strategy is shown numerically to consistently outperform purely spectral feedback, both deterministically and under persistent stochastic perturbations, while remaining computationally cheaper than a full-dimensional design. Together, these results indicate that the equilibria constructed through the confinement problem are not only well-defined as static objects but also dynamically stable, and that their convergence can be accelerated through control strategies informed by the slow modes identified analytically.

The remainder of this thesis is organised as follows. Chapter 2 develops the variational characterisation of the stationary states of the VPFP system. Chapter 3 establishes the regularity of the associated solution operators, first pointwise, through elliptic bootstrapping, and then in the operator-theoretic sense, through weak sequential continuity. Chapter 4 builds on this continuity theory to formulate and solve the inverse confinement problem, together with a numerical illustration in one spatial dimension. Chapter 5 closes the thesis by studying the dynamical stability of the resulting equilibria and by proposing control strategies to accelerate convergence towards them. The conclusion in Chapter 6 reviews the scope and limitations of the results, discusses open problems, and highlights future directions for extending the confinement framework developed in this thesis. Appendix A collects auxiliary results on Sobolev spaces on the torus used throughout the manuscript, and Appendix B presents the abstract hypocoercivity method for linear kinetic equations conserving mass underlying the analysis of Chapter 5.

1.3 Notation

Unless otherwise specified, all function spaces are defined over the domain indicated in the notation, equipped with norms induced by the Lebesgue measure, and consist of real-valued functions. In particular, the distinction between spaces defined on $\mathbb{T}^d$ and those defined on $\mathbb{T}^d \times \mathbb{R}^d$ is always made explicit. When a function space is equipped with a measure μ other than the Lebesgue measure, this is indicated explicitly, e.g. $L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)$. We adopt the standard Sobolev convention $W^{0,p}(\Omega) = L^p(\Omega)$ for all $1 \leq p \leq +\infty$. The suffix zm always denotes the zero-mean subspace of the corresponding function space; for instance,

$$H^1_{\mathrm{zm}}(\mathbb{T}^d) := \left\{ u \in H^1(\mathbb{T}^d) : \int_{\mathbb{T}^d} u \mathrm{d}x = 0 \right\}.$$

For an operator T, we denote its domain by $\mathcal{D}(T)$ and its range by $\mathcal{R}(T)$. For two Banach spaces X and Y, $\mathcal{L}(X, Y)$ denotes the space of bounded linear operators from X to Y, while $\mathcal{K}(X, Y)$ denotes the space of compact operators from X to Y. If $Y = X$, we simplify the notation to $\mathcal{L}(X)$ and $\mathcal{K}(X)$, respectively. We denote by $\mathcal{C}^k(X, Y)$ the space of maps from X to Y that are k-times continuously Fréchet differentiable; if $Y = \mathbb{R}$, we simply write $\mathcal{C}^k(X)$.

The space of distributions on an open set $\Omega \subset \mathbb{R}^n$ is denoted by $\mathcal{D}'(\Omega)$, where $\mathcal{D}(\Omega) = C_c^\infty(\Omega)$ is the space of test functions. The action of a distribution $\phi \in \mathcal{D}'(\Omega)$ on a test function $\xi \in \mathcal{D}(\Omega)$ is denoted by $\langle \phi, \xi \rangle$. We define the distribution spaces $\mathcal{D}'(\mathbb{T}^d)$ and $\mathcal{D}'(\mathbb{T}^d \times \mathbb{R}^d)$ analogously.

For a Banach space X, X^* denotes its continuous dual, $\| \cdot \|_X$ its norm, and $\langle \cdot, \cdot \rangle_{X^*, X}$ the duality pairing between X^* and X. Whenever no ambiguity arises, we abbreviate this pairing to $\langle \cdot, \cdot \rangle$. If X is a Hilbert space, we identify X with its dual via the Riesz representation theorem and denote the resulting pairing by $\langle \cdot, \cdot \rangle_X$; for instance, this convention applies when $X = L^2(\mathbb{T}^d)$.

For an operator T, T^* denotes its dual operator (Banach adjoint). When T acts on a Hilbert space,

this notation agrees with the usual Hilbert space adjoint under the Riesz representation theorem. For a Hilbert space $\mathcal{H}$, $(\cdot,\cdot)_{\mathcal{H}}$ denotes its inner product, and $\|\cdot\|_{\mathcal{H}}$ denotes the norm induced by the inner product.

The notation $X \hookrightarrow Y$ denotes an embedding of the space X into the space Y. Unless otherwise stated, the type of embedding (continuous or compact) will be specified explicitly. Weak convergence is denoted by $\rightharpoonup$, while strong convergence is denoted by $\to$. Almost-everywhere convergence is denoted by $\to$ a.e. Unless otherwise stated, the topology in which convergence takes place is specified explicitly.

For $x \in \mathbb{R}^n$, $|x|$ denotes its Euclidean norm. For a Lebesgue measurable set $\Omega \subset \mathbb{R}^n$, $|\Omega|$ denotes its Lebesgue measure. In particular, $|\mathbb{T}^d| = (2\pi)^d$. For a multi-index $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d$, we write

$$|\alpha| = \sum_{i=1}^{d} \alpha_i, \qquad \partial^\alpha = \frac{\partial^{|\alpha|}}{\partial x^\alpha} = \frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}} \cdots \frac{\partial^{\alpha_d}}{\partial x_d^{\alpha_d}}.$$

For a Banach space X and $x_0 \in X$, we let $B^X(x_0, r)$ denote the open ball of centre x_0 and radius $r > 0$; in particular, $B^X(0,1)$ denotes the open unit ball of X.

Symbol	**Meaning**
d	Dimension of the spatial domain and velocity space
$\mathbb{T}^d$	d-dimensional flat torus, $\mathbb{R}^d/(2\pi\mathbb{Z})^d$ (identified with $[0,2\pi]^d$ with periodic boundary conditions)
∇_x, Δ_x	Gradient and Laplacian with respect to the spatial variable x
$\nabla_x \cdot (\cdot)$	Divergence with respect to the spatial variable x
∇_v, Δ_v	Gradient and Laplacian with respect to the velocity variable v
$\nabla_v \cdot (\cdot)$	Divergence with respect to the velocity variable v
∇, Δ	Spatial gradient and Laplacian, used in place of ∇_x, Δ_x when only spatial variables are considered and no ambiguity arises (e.g. in the Poisson equation)
$f(t,x,v)$	Phase-space density of the VPFP system
$\rho_f = \int_{\mathbb{R}^d} f \,\mathrm{d}v$	Spatial density associated with the phase-space density f
φ_f	Self-consistent electrostatic potential generated by f
ρ_0	Uniform neutralising background charge density
u	External potential
$f_\star, \rho_\star, \varphi_\star$	Free energy stationary solution and its associated spatial density and self-consistent potential for a given external potential u; when the dependence on u is relevant, the notation $f_\star(u), \rho_\star(u), \varphi_\star(u)$ is used explicitly
ρ_{ref}	Prescribed reference density in the confinement problem

Chapter 2

Variational Analysis of Stationary Solutions to the VPFP System

This chapter establishes existence and uniqueness of free energy solutions of the stationary VPFP system with an external potential. The approach is based on classical variational methods; see, for instance, [23, 24]. For completeness, we present full proofs adapted to the present setting. This ensures a self-contained exposition and provides the functional framework used throughout the manuscript.

The chapter is organised as follows. In Section 2.1, we study the Poisson equation on the torus, deriving identities and estimates for the Dirichlet energy of the associated solutions. Building on this functional framework, Section 2.2 establishes the existence and uniqueness of free energy solutions to the stationary VPFP system and characterises them variationally through the corresponding Euler–Lagrange and Poisson–Boltzmann–Emden equations.

2.1 Preliminary Results on Elliptic Regularity

This section collects several classical results concerning the Poisson equation on the torus $\mathbb{T}^d$ (see, e.g., [30, 51] for an introduction) and establishes a functional-analytic framework for the analysis carried out in subsequent chapters. We address the weak formulation of the problem for H^{-1} forcing terms, the necessary and sufficient solvability condition, the associated existence and uniqueness theory, and the functional properties of the solution operator. We conclude by stating fundamental properties of the Dirichlet energy of solutions. Full proofs are included for completeness and future reference.

Consider the Poisson equation on the torus with periodic boundary conditions:

$$-\Delta\varphi = g, \tag{2.1}$$

where $g \in H^{-1}(\mathbb{T}^d) := (H^1(\mathbb{T}^d))^*$.

Definition 2.1.1 (Weak Solution to the Poisson Equation).
Let $g \in H^{-1}(\mathbb{T}^d)$ have zero mean. The map $\varphi \in H^1(\mathbb{T}^d)$ is a weak solution to (2.1) *if*

$$\int_{\mathbb{T}^d} \nabla\varphi(x) \cdot \nabla\psi(x)\,\mathrm{d}x = \langle g, \psi\rangle_{H^{-1}(\mathbb{T}^d),H^1(\mathbb{T}^d)}$$

for all test functions $\psi \in H^1(\mathbb{T}^d)$.

The zero-mean assumption on the forcing term is necessary for the well-posedness of the Poisson equation (2.1). Indeed, every $g \in H^{-1}(\mathbb{T}^d)$ admits the Fourier expansion (see Lemma A.2.1)

$$g(x) = \sum_{k\in\mathbb{Z}^d} \hat{g}_k\, e^{ik\cdot x}, \qquad \hat{g}_k := \frac{1}{\sqrt{|\mathbb{T}^d|}}\langle g, e^{-ik\cdot x}\rangle \quad \text{for all } k \in \mathbb{Z}^d.$$

Suppose now that a weak solution $\varphi : \mathbb{T}^d \to \mathbb{R}$ to (2.1) exists. Then, by the Fourier differentiation rule (see Lemma A.2.5), φ admits a Fourier series which satisfies mode by mode

$$|k|^2 \hat{\varphi}_k = \hat{g}_k \qquad \forall\, k \in \mathbb{Z}^d.$$

Since the equality holds for all modes, it follows that

$$0 = \hat{g}_0 = \frac{1}{\sqrt{|\mathbb{T}^d|}} \langle g, 1 \rangle,$$

where $1 \in H^1(\mathbb{T}^d)$ denotes the constant function equal to 1. Therefore, g must have zero mean.

In the following discussion, we let $H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$ denote the zero-mean subspace of $H^{-1}(\mathbb{T}^d)$, that is,

$$H^{-1}_{\mathrm{zm}}(\mathbb{T}^d) := \left\{ f \in H^{-1}(\mathbb{T}^d) : \langle f, 1 \rangle = 0 \right\}.$$

We define similarly $H^s_{\mathrm{zm}}(\mathbb{T}^d)$ for all $s \in \mathbb{R}$, with the understanding that $H^0_{\mathrm{zm}}(\mathbb{T}^d) = L^2_{\mathrm{zm}}(\mathbb{T}^d)$.

Remark 2.1.2. *The zero mean assumption on the forcing g of the Poisson equation forces, in the context of the VPFP system, the identity*

$$\int_{\mathbb{T}^d} (\rho_f(x) - \rho_0) \mathrm{d}x = 0 \qquad \Longrightarrow \qquad \rho_0 = \frac{1}{|\mathbb{T}^d|} \int_{\mathbb{T}^d} \rho_f(x) \mathrm{d}x = \frac{\boldsymbol{m}}{|\mathbb{T}^d|}. \tag{2.2}$$

Hence, ρ_0 acts as a neutralising background charge to ensure that the total mass variation is zero.

The zero-mean condition is also sufficient for well-posedness, as established in the following result.

Lemma 2.1.3. *Let $g \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$. Then the Poisson equation* (2.1) *has a unique weak solution $\varphi \in H^1(\mathbb{T}^d)$ up to an additive constant.*

Proof. Since $g \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$, there holds (see Lemma A.2.3)

$$\sum_{k \in \mathbb{Z}^d \setminus \{0\}} \frac{|\hat{g}_k|^2}{|k|^2} \leq \|g\|^2_{H^{-1}(\mathbb{T}^d)} < +\infty.$$

For $x \in \mathbb{T}^d$, set

$$\varphi(x) := \sum_{k \in \mathbb{Z}^d \setminus \{0\}} \frac{\hat{g}_k}{|k|^2} e^{ik \cdot x}. \tag{2.3}$$

Then, $\varphi \in H^1(\mathbb{T}^d)$. Indeed, for any multi-index $\alpha \in \mathbb{N}^d$ with $|\alpha| \leq 1$, the Fourier differentiation rule yields

$$\left\| \frac{\partial^{|\alpha|} \varphi}{\partial x^\alpha} \right\|^2_{L^2(\mathbb{T}^d)} = \sum_{k \in \mathbb{Z}^d \setminus \{0\}} \frac{|\hat{g}_k|^2}{|k|^4} |(ik)^\alpha|^2 \leq \sum_{k \in \mathbb{Z}^d \setminus \{0\}} \frac{|\hat{g}_k|^2}{|k|^2} \leq \|g\|^2_{H^{-1}(\mathbb{T}^d)}.$$

In particular, in $\mathcal{D}'(\mathbb{T}^d)$ there holds

$$-\Delta \varphi = \sum_{k \in \mathbb{Z}^d} \hat{g}_k \, e^{ik \cdot x} = g.$$

Therefore, φ is a weak solution to (2.1). We next establish uniqueness up to additive constants. To this end, let φ_1, $\varphi_2 \in H^1(\mathbb{T}^d)$ be two weak solutions to (2.1). Then, for all $\psi \in H^1(\mathbb{T}^d)$ there holds

$$\int_{\mathbb{T}^d} \nabla(\varphi_1 - \varphi_2) \cdot \nabla \psi \mathrm{d}x = \langle g, \psi \rangle - \langle g, \psi \rangle = 0.$$

Testing this identity with $\psi = \varphi_1 - \varphi_2$ yields

$$\int_{\mathbb{T}^d} |\nabla(\varphi_1 - \varphi_2)|^2 \mathrm{d}x = 0.$$

We therefore conclude that $\nabla(\varphi_1 - \varphi_2) = 0$ a.e., that is,

$$\varphi_1 - \varphi_2 = c$$

for some $c \in \mathbb{R}$. □

The previous discussion suggests viewing the Poisson problem (2.1) as an operator equation. Specifically, we define the Laplacian operator as

$$-\Delta : H^1(\mathbb{T}^d) \to H^{-1}_{\mathrm{zm}}(\mathbb{T}^d),$$

where $-\Delta$ is understood in the distributional sense:

$$\langle -\Delta\varphi, \psi \rangle = \int_{\mathbb{T}^d} \nabla\varphi \cdot \nabla\psi \mathrm{d}x \quad \text{for all } \varphi, \psi \in H^1(\mathbb{T}^d).$$

The weak solutions to the Poisson equation with forcing term $g \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$ are precisely the preimages of g under the Laplacian operator. For any such g, a preimage exists by Lemma 2.1.3, hence the Laplacian is surjective. Furthermore, the kernel of $-\Delta$ consists exactly of the constant functions. Therefore, the restriction of the Laplacian to the zero-mean subspace of $H^1(\mathbb{T}^d)$, that is,

$$-\Delta : H^1_{\mathrm{zm}}(\mathbb{T}^d) \to H^{-1}_{\mathrm{zm}}(\mathbb{T}^d),$$

is injective. Since the Laplacian is also linear and continuous (see Proposition 2.1.4 below), it is therefore a linear isomorphism of Banach spaces on zero-mean subspaces.

This operator approach to the Poisson problem naturally extends to $H^s(\mathbb{T}^d)$ forcing terms, where $s \in \mathbb{R}$ is arbitrary, provided that the Laplacian is understood distributionally. The following result formalises this discussion.

Proposition 2.1.4. *Let $s \in \mathbb{R}$. The operator $-\Delta : H^s_{\mathrm{zm}}(\mathbb{T}^d) \to H^{s-2}_{\mathrm{zm}}(\mathbb{T}^d)$ is a linear isomorphism of Banach spaces.*

Proof. The operator is well-defined. Indeed, let $\varphi \in H^s_{\mathrm{zm}}(\mathbb{T}^d)$. Then, by Lemma A.2.1 we have

$$\varphi(x) = \sum_{k \in \mathbb{Z}^d \setminus \{0\}} \hat{\varphi}_k e^{ik\cdot x},$$

where the zero mode vanishes due to the mean-zero condition. For all $k \in \mathbb{Z}^d \setminus \{0\}$, the Fourier differentiation rule gives

$$\widehat{-\Delta\varphi}_k = |k|^2 \hat{\varphi}_k.$$

Hence, using the trivial estimate $|k|^2 \leq 1 + |k|^2$ for all $k \in \mathbb{Z}^d$ yields

$$\begin{aligned}
\| -\Delta\varphi \|^2_{H^{s-2}(\mathbb{T}^d)} &= \sum_{k \in \mathbb{Z}^d \setminus \{0\}} (1 + |k|^2)^{s-2} |\widehat{-\Delta\varphi}_k|^2 \\
&= \sum_{k \in \mathbb{Z}^d \setminus \{0\}} (1 + |k|^2)^{s-2} |k|^4 |\hat{\varphi}_k|^2 \\
&\leq \sum_{k \in \mathbb{Z}^d \setminus \{0\}} (1 + |k|^2)^{s-2} (1 + |k|^2)^2 |\hat{\varphi}_k|^2 \qquad (2.4) \\
&= \sum_{k \in \mathbb{Z}^d \setminus \{0\}} (1 + |k|^2)^{s} |\hat{\varphi}_k|^2 \\
&= \| \varphi \|^2_{H^s(\mathbb{T}^d)}.
\end{aligned}$$

Therefore $-\Delta : H^s_{\mathrm{zm}}(\mathbb{T}^d) \to H^{s-2}_{\mathrm{zm}}(\mathbb{T}^d)$ is well-defined and bounded. Linearity follows immediately from linearity of the Fourier coefficients, and thus $-\Delta$ is continuous.

We next show injectivity. Let $\varphi \in H^s_{\mathrm{zm}}(\mathbb{T}^d)$ be such that $-\Delta\varphi = 0$ in $H^{s-2}_{\mathrm{zm}}(\mathbb{T}^d)$. Then, by the Fourier representation,

$$\widehat{-\Delta\varphi}_k = |k|^2\hat{\varphi}_k = 0 \quad \text{for all } k \in \mathbb{Z}^d \setminus \{0\}.$$

Since $|k|^2 > 0$ for $k \neq 0$, it follows that $\hat{\varphi}_k = 0$ for all such k, hence $\varphi = 0$ in $H^{s-2}_{\mathrm{zm}}(\mathbb{T}^d)$.

We finally prove surjectivity. Let $g \in H^{s-2}_{\mathrm{zm}}(\mathbb{T}^d)$ be arbitrary. Define

$$\hat{\varphi}_k := \frac{\hat{g}_k}{|k|^2}, \quad k \in \mathbb{Z}^d \setminus \{0\}.$$

This is well-defined since $\hat{g}_0 = 0$. We now show that $\varphi \in H^s(\mathbb{T}^d)$. Using that $2|k|^4 \geq (1+|k|^2)^2$ for $k \neq 0$, we have

$$\begin{aligned}
\|\varphi\|^2_{H^s(\mathbb{T}^d)} &= \sum_{k\in\mathbb{Z}^d\setminus\{0\}} (1+|k|^2)^s \frac{|\hat{g}_k|^2}{|k|^4} \\
&\leq \frac{1}{2} \sum_{k\in\mathbb{Z}^d\setminus\{0\}} (1+|k|^2)^{s-2}|\hat{g}_k|^2 \\
&= \frac{1}{2}\|g\|^2_{H^{s-2}(\mathbb{T}^d)}.
\end{aligned}$$

Thus $\varphi \in H^s_{\mathrm{zm}}(\mathbb{T}^d)$. Furthermore, by the Fourier differentiation rule follows that $-\Delta\varphi = g$ in $H^{s-2}_{\mathrm{zm}}(\mathbb{T}^d)$.

We conclude that $-\Delta$ is a bijective bounded linear operator between Banach spaces. By the open mapping theorem (see, e.g., Theorem 2.6 in [15]), its inverse is also bounded, hence $-\Delta$ is a linear isomorphism of Banach spaces. □

As a consequence of Proposition 2.1.4, the solution operator

$$(-\Delta)^{-1} : H^s_{\mathrm{zm}}(\mathbb{T}^d) \to H^{s+2}_{\mathrm{zm}}(\mathbb{T}^d)$$

that maps zero-mean forcing terms $g \in H^s_{\mathrm{zm}}(\mathbb{T}^d)$ to the corresponding unique zero-mean solution of the Poisson equation (2.1) is well-defined, linear and continuous. The isomorphism property of the Laplacian extends to general Sobolev spaces $W^{k,p}_{\mathrm{zm}}(\mathbb{T}^d)$. The following proposition is a classical result; see, e.g., [1, 33, 51] for a detailed treatment of elliptic regularity theory.

Proposition 2.1.5. *Let $k \in \mathbb{N}$ and $1 < p < +\infty$. Then, the operator $-\Delta : W^{k+2,p}_{\mathrm{zm}}(\mathbb{T}^d) \to W^{k,p}_{\mathrm{zm}}(\mathbb{T}^d)$ is a linear isomorphism of Banach spaces.*

2.1.1 Dirichlet Energy of Solutions

The *Dirichlet energy* functional $E^D : H^1(\mathbb{T}^d) \to \mathbb{R}$ is defined as

$$E^D(\varphi) := \frac{1}{2}\int_{\mathbb{T}^d} |\nabla\varphi|^2 \mathrm{d}x. \tag{2.5}$$

Let $g \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$, and let $\varphi := (-\Delta)^{-1}g \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ be the unique zero-mean solution of (2.1). Testing the weak formulation with φ itself yields, by definition of weak solution,

$$E^D(\varphi) = \frac{1}{2}\langle g, \varphi\rangle_{H^{-1}(\mathbb{T}^d), H^1(\mathbb{T}^d)}. \tag{2.6}$$

The next result establishes the coercivity of the Dirichlet energy of solutions to the Poisson equation (2.1) with forcing term in $H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$, as well as a fundamental lower bound.

Lemma 2.1.6. *Let $g \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$, and let $\varphi := (-\Delta)^{-1} g \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ be the unique zero-mean solution of* (2.1). *Then*

$$(i) \qquad E^D(\varphi) \geq \frac{1}{4}\|\varphi\|^2_{H^1(\mathbb{T}^d)}.$$

$$(ii) \qquad E^D(\varphi) \geq \frac{1}{4}\|g\|^2_{H^{-1}(\mathbb{T}^d)}.$$

Proof. Since φ has zero mean, (i) follows from the Poincaré inequality (see Lemma A.2.6):

$$E^D(\varphi) = \frac{1}{2}\|\nabla\varphi\|^2_{L^2(\mathbb{T}^d)^d} \geq \frac{1}{4}\|\varphi\|^2_{H^1(\mathbb{T}^d)}.$$

Combining this with (2.4) yields (ii). □

2.2 Stationary Solutions

This section investigates stationary solutions to the VPFP system with an external potential. We first introduce the notion of weak stationary solutions. However, the weak formulation does not close under the available a priori estimates, and a complete existence and uniqueness theory would require delicate arguments to overcome the lack of compactness. Since this level of generality is beyond the scope of this work, we restrict ourselves to a subclass of weak solutions characterised variationally as minimisers of the free energy functional. We refer to minimisers of this functional as *free energy solutions*. The main result of this section is the existence and uniqueness of free energy solutions. Furthermore, they can be identified with weak stationary solutions of the VPFP system.

This section is organised as follows. In Subsection 2.2.1, we introduce the notion of weak stationary solutions and reformulate the problem as the minimisation of the free energy functional over the set of admissible states. In Subsection 2.2.2, we analyse the entropy functional on the admissible set and establish its fundamental properties. These results enable us to study the admissible set and the free energy functional in Subsection 2.2.3. The direct method of the calculus of variations is then employed in Subsection 2.2.4 to establish the existence and uniqueness of free energy solutions to the stationary VPFP system. In Subsection 2.2.5, we derive the Euler–Lagrange equation satisfied by the unique minimiser and show that it is both necessary and sufficient for global optimality. Finally, in Subsection 2.2.6, we identify the unique minimiser of the free energy functional with a weak stationary solution of the VPFP system.

2.2.1 Variational Framework

Consider the stationary VPFP system with external potential u on the phase-space $\mathbb{T}^d \times \mathbb{R}^d$:

$$v \cdot \nabla_x f - (\nabla_x u + \nabla_x \varphi_f) \cdot \nabla_v f = \Delta_v f + \nabla_v \cdot (vf), \tag{2.7a}$$

$$-\Delta_x \varphi_f = \rho_f - \rho_0 = \int_{\mathbb{R}^d} f \mathrm{d}v - \rho_0, \tag{2.7b}$$

where $\rho_0 = \frac{\boldsymbol{m}}{|\mathbb{T}^d|}$ (see Remark 2.1.2). We consider the following notion of weak solution to (2.7).

Definition 2.2.1 (**Weak Solution to the Stationary VPFP System**).
Let $\boldsymbol{m} > 0$ and $u \in H^1(\mathbb{T}^d)$. A pair of functions $(f_\star, \varphi_\star) \in L^1(\mathbb{T}^d \times \mathbb{R}^d) \times H^1_{\mathrm{zm}}(\mathbb{T}^d)$ is a weak solution to the stationary VPFP *system* (2.7) (*equivalently, a weak stationary solution of the* VPFP *system*) *if:*

(*i*) $f_\star \geq 0$ *a.e.,* $\int_{\mathbb{T}^d \times \mathbb{R}^d} f_\star \mathrm{d}x\mathrm{d}v = \boldsymbol{m}$ *and* $\rho_\star(\nabla_x u + \nabla_x \varphi_\star) \in L^1(\mathbb{T}^d)^d$, *where* $\rho_\star := \int_{\mathbb{R}^d} f_\star \mathrm{d}v$.

(*ii*) *For every test function* $\psi \in \mathcal{C}^\infty_c(\mathbb{T}^d \times \mathbb{R}^d)$

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} f_\star \Big(v \cdot \nabla_x \psi - (\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v \psi - \Delta_v \psi + v \cdot \nabla_v \psi \Big) \mathrm{d}x\mathrm{d}v = 0, \tag{2.8a}$$

and for every test function $\zeta \in \mathcal{C}^\infty(\mathbb{T}^d)$

$$\int_{\mathbb{T}^d} \nabla_x \varphi_\star \cdot \nabla_x \zeta \mathrm{d}x = \int_{\mathbb{T}^d} (\rho_\star - \rho_0)\zeta \mathrm{d}x. \tag{2.8b}$$

Remark 2.2.2. *The condition* $\rho_\star(\nabla_x u + \nabla_x \varphi_\star) \in L^1(\mathbb{T}^d)^d$ *ensures that the distribution*

$$\nabla_v f_\star \cdot (\nabla_x u + \nabla_x \varphi_\star) : \mathcal{C}_c^\infty(\mathbb{T}^d \times \mathbb{R}^d) \to \mathbb{R},\ \psi \mapsto \int_{\mathbb{T}^d \times \mathbb{R}^d} f_\star(\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v \psi \mathrm{d}x\mathrm{d}v$$

is well-defined (see Lemma 2.2.3). Although sufficient, this condition is not necessary to guarantee the well-posedness of Definition 2.2.1. Nevertheless, we adopt this stronger notion of weak stationary solution to simplify the subsequent analysis.

The following lemma establishes the well-posedness of Definition 2.2.1.

Lemma 2.2.3. *Let* $\boldsymbol{m} > 0$ *and* $u \in H^1(\mathbb{T}^d)$*. Moreover, let* $(f_\star, \varphi_\star) \in L^1(\mathbb{T}^d \times \mathbb{R}^d) \times H^1_{\mathrm{zm}}(\mathbb{T}^d)$*, and assume that* $f_\star$ *satisfies condition* (i) *in Definition* 2.2.1*. Then, the weak formulation* (2.8) *is well-defined.*

Proof. The mass constraint $\int_{\mathbb{T}^d \times \mathbb{R}^d} f_\star \mathrm{d}x\mathrm{d}v$ enforces the zero-mean condition

$$\int_{\mathbb{T}^d} (\rho_\star - \rho_0)\mathrm{d}x = \int_{\mathbb{T}^d} \rho_\star \mathrm{d}x - \boldsymbol{m} = 0.$$

Hence, the weak formulation (2.8b) is well-posed by Definition 2.1.1 of weak solution to the Poisson equation. We next consider the weak formulation (2.8a) and show that each distribution is well-defined. To this end, let $\psi \in \mathcal{C}_c^\infty(\mathbb{T}^d \times \mathbb{R}^d)$ be arbitrary, and set $K := \mathrm{supp}(\psi)$. Since $f_\star \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$, we have $|v| f_\star \in L^1_{\mathrm{loc}}(\mathbb{T}^d \times \mathbb{R}^d)$. Hence, using that K is compact,

$$\left| \int_{\mathbb{T}^d \times \mathbb{R}^d} f_\star v \cdot \nabla_x \psi \mathrm{d}x\mathrm{d}v \right| \leq \|\nabla_x \psi\|_{L^\infty(\mathbb{T}^d \times \mathbb{R}^d)^d} \int_K |v| \cdot f_\star \mathrm{d}x\mathrm{d}v < +\infty.$$

Similarly, we have

$$\left| \int_{\mathbb{T}^d \times \mathbb{R}^d} f_\star v \cdot \nabla_v \psi \mathrm{d}x\mathrm{d}v \right| \leq \|\nabla_v \psi\|_{L^\infty(\mathbb{T}^d \times \mathbb{R}^d)^d} \int_K |v| \cdot f_\star \mathrm{d}x\mathrm{d}v < +\infty,$$

$$\left| \int_{\mathbb{T}^d \times \mathbb{R}^d} f_\star \Delta_v \psi \mathrm{d}x\mathrm{d}v \right| \leq \|\Delta_v \psi\|_{L^\infty(\mathbb{T}^d \times \mathbb{R}^d)} \int_K f_\star \mathrm{d}x\mathrm{d}v < +\infty.$$

For $\nabla_v f_\star \cdot (\nabla_x u + \nabla_x \varphi_\star)$, we use Tonelli's theorem (see, e.g., Theorem 3.4.5 in [10]) and the condition $\rho_\star(\nabla_x u + \nabla_x \varphi_\star) \in L^1(\mathbb{T}^d)^d$:

$$\begin{aligned}
\left| \int_{\mathbb{T}^d \times \mathbb{R}^d} f_\star(\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v \psi \mathrm{d}x\mathrm{d}v \right| &\leq \|\nabla_v \psi\|_{L^\infty(\mathbb{T}^d \times \mathbb{R}^d)^d} \int_{\mathbb{T}^d \times \mathbb{R}^d} f_\star |\nabla_x u + \nabla_x \varphi_\star| \mathrm{d}x\mathrm{d}v \\
&= \|\nabla_v \psi\|_{L^\infty(\mathbb{T}^d \times \mathbb{R}^d)^d} \int_{\mathbb{T}^d} |\nabla_x u + \nabla_x \varphi_\star| \int_{\mathbb{R}^d} f_\star \mathrm{d}v\mathrm{d}x \\
&= \|\nabla_v \psi\|_{L^\infty(\mathbb{T}^d \times \mathbb{R}^d)^d} \int_{\mathbb{T}^d} \rho_\star |\nabla_x u + \nabla_x \varphi_\star| \mathrm{d}x < +\infty.
\end{aligned}$$

Therefore, the weak formulation (2.8a) is well-defined. □

Although Definition 2.2.1 is natural from the PDE viewpoint, establishing existence of such solutions appears to require a renormalisation argument in the spirit of the Ambrosio–DiPerna–Lions theory (see, e.g., [29]; see also [19, 22] for the renormalisation theory) or approximation schemes of the type developed

in [17] to overcome compactness issues. In the present non-reflexive setting, this would involve delicate a priori estimates and compactness arguments that lie beyond the scope of this work. For this reason, we restrict our analysis to solutions obtained through the variational formulation presented below.

Let $\boldsymbol{m} > 0$ be the mass of the system, and define the admissible class of states

$$\begin{aligned}\tilde{\mathcal{A}}(\boldsymbol{m}) := \Big\{ f \in L^1(\mathbb{T}^d \times \mathbb{R}^d) \ : \ f \geq 0 \ \text{ a.e.}, \int_{\mathbb{T}^d \times \mathbb{R}^d} f \mathrm{d}x\mathrm{d}v = \boldsymbol{m}, \\ \rho_f \in H^{-1}(\mathbb{T}^d), \int_{\mathbb{T}^d \times \mathbb{R}^d} f \log f \mathrm{d}x\mathrm{d}v < +\infty, \int_{\mathbb{T}^d \times \mathbb{R}^d} |v|^2 f \mathrm{d}x\mathrm{d}v < +\infty \Big\},\end{aligned} \tag{2.9}$$

where $\tilde{\mathcal{A}}(\boldsymbol{m})$ is endowed with the $L^1(\mathbb{T}^d \times \mathbb{R}^d)$ topology, and the spatial density is given by

$$\rho_f(x) = \int_{\mathbb{R}^d} f(x,v) \mathrm{d}v.$$

Remark 2.2.4. *Let $f \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$. Then, by Fubini's theorem (see, e.g., Theorem 3.4.4 in [10]) $\rho_f \in L^1(\mathbb{T}^d)$. Hence, for every $\zeta \in \mathcal{C}^\infty(\mathbb{T}^d)$, the quantity*

$$\int_{\mathbb{T}^d} \rho_f(x)\zeta(x) \mathrm{d}x$$

is well-defined as a Lebesgue integral. Assume moreover that $\rho_f \in H^{-1}(\mathbb{T}^d)$. Then, the map

$$\zeta \mapsto \int_{\mathbb{T}^d} \rho_f(x)\zeta(x) \mathrm{d}x, \quad \textit{initially defined on } \mathcal{C}^\infty(\mathbb{T}^d),$$

extends uniquely to a continuous linear functional on $H^1(\mathbb{T}^d)$ by Friedrichs' lemma (see Corollary A.2.2). In particular, this extension coincides with the canonical duality pairing between $H^{-1}(\mathbb{T}^d)$ and $H^1(\mathbb{T}^d)$. Henceforth, we shall not distinguish between the Lebesgue integral formulation and the duality pairing, and we write

$$\int_{\mathbb{T}^d} \rho_f(x) w(x) \mathrm{d}x$$

for all $w \in H^1(\mathbb{T}^d)$, with the understanding that this denotes the continuous extension of the Lebesgue pairing from smooth functions.

Together with Remark 2.2.4, we will repeatedly use the following elementary fact.

Claim 2.2.5. *Let $\rho \in H^{-1}(\mathbb{T}^d)$ and $h \in H^1(\mathbb{T}^d)$ be arbitrary. Then, there holds $|h| \in H^1(\mathbb{T}^d)$ with $\||h|\|_{H^1(\mathbb{T}^d)} \leq \|h\|_{H^1(\mathbb{T}^d)}$ and*

$$|\langle \rho, |h| \rangle_{H^{-1}(\mathbb{T}^d), H^1(\mathbb{T}^d)}| \leq \|\rho\|_{H^{-1}(\mathbb{T}^d)} \|h\|_{H^1(\mathbb{T}^d)}.$$

Proof. We first show that $|h| \in H^1(\mathbb{T}^d)$. The map $\phi : \mathbb{R} \to \mathbb{R}$, $t \mapsto |t|$ is globally Lipschitz on $\mathbb{R}$. Moreover, there holds

$$\|\phi \circ h\|^2_{L^2(\mathbb{T}^d)} = \int_{\mathbb{T}^d} |\phi(h)|^2 \mathrm{d}x = \int_{\mathbb{T}^d} |h|^2 \mathrm{d}x = \|h\|^2_{L^2(\mathbb{T}^d)}.$$

Hence, $\phi \circ h \in L^2(\mathbb{T}^d)$. Since $h \in H^1(\mathbb{T}^d)$, the Sobolev chain rule (see, e.g., Theorem 2.1.11 in [54]) yields $\phi \circ h = |h| \in H^1(\mathbb{T}^d)$, and the weak gradient satisfies

$$\nabla(\phi \circ h) = \phi'(h) \nabla h = \operatorname{sign}(h) \nabla h \quad \text{a.e.},$$

where $\operatorname{sign}(\cdot)$ denotes the sign function, with the convention that $\operatorname{sign}(0) = 0$. Since $|\operatorname{sign}(t)| \leq 1$ for all $t \in \mathbb{R}$, it follows that

$$|\nabla(|h|)| \leq |\nabla h| \quad \text{a.e.}$$

Combining the above estimates yields

$$\||h|\|^2_{H^1(\mathbb{T}^d)} = \||h|\|^2_{L^2(\mathbb{T}^d)} + \|\nabla(|h|)\|^2_{L^2(\mathbb{T}^d)^d} \leq \|h\|^2_{L^2(\mathbb{T}^d)} + \|\nabla h\|^2_{L^2(\mathbb{T}^d)^d} = \|h\|^2_{H^1(\mathbb{T}^d)}.$$

Finally, since $|h| \in H^1(\mathbb{T}^d)$ and $\rho \in H^{-1}(\mathbb{T}^d)$, the duality pairing $\langle \rho, |h| \rangle$ is well-defined, and there holds

$$|\langle \rho, |h| \rangle_{H^{-1}(\mathbb{T}^d), H^1(\mathbb{T}^d)}| \leq \|\rho\|_{H^{-1}(\mathbb{T}^d)} \||h|\|_{H^1(\mathbb{T}^d)} \leq \|\rho\|_{H^{-1}(\mathbb{T}^d)} \|h\|_{H^1(\mathbb{T}^d)}. \qquad \square$$

For $u \in H^1 \cap L^\infty(\mathbb{T}^d)$, we then define the free energy functional $\mathcal{F}^u : \tilde{\mathcal{A}}(\boldsymbol{m}) \to \mathbb{R}$ by

$$\begin{aligned}\mathcal{F}^u(f) &:= \mathcal{E}(f) + E^K(f) + E^u(f) + E^D(\varphi_f) \\ &= \int_{\mathbb{T}^d \times \mathbb{R}^d} f \log f \mathrm{d}x\mathrm{d}v + \frac{1}{2} \int_{\mathbb{T}^d \times \mathbb{R}^d} |v|^2 f \mathrm{d}x\mathrm{d}v + \int_{\mathbb{T}^d \times \mathbb{R}^d} u f \mathrm{d}x\mathrm{d}v + \frac{1}{2} \int_{\mathbb{T}^d} |\nabla \varphi_f|^2 \mathrm{d}x,\end{aligned}$$

where

$$\varphi_f = (-\Delta)^{-1}(\rho_f - \rho_0) \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$$

is the unique zero-mean solution to the Poisson equation (2.1) with forcing term $\rho_f - \rho_0 \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$. Here, E^D is the previously introduced Dirichlet energy (2.5),

$$\mathcal{E} : \tilde{\mathcal{A}}(\boldsymbol{m}) \to \mathbb{R}, \quad f \mapsto \int_{\mathbb{T}^d \times \mathbb{R}^d} f \log f \mathrm{d}x\mathrm{d}v \tag{2.10}$$

is the Boltzmann–Shannon entropy [48] (up to sign convention),

$$E^K : \tilde{\mathcal{A}}(\boldsymbol{m}) \to \mathbb{R}, \quad f \mapsto \frac{1}{2} \int_{\mathbb{T}^d \times \mathbb{R}^d} |v|^2 f \mathrm{d}x\mathrm{d}v \tag{2.11}$$

is the kinetic energy, and

$$E^u : \tilde{\mathcal{A}}(\boldsymbol{m}) \to \mathbb{R}, \quad f \mapsto \int_{\mathbb{T}^d \times \mathbb{R}^d} u f \mathrm{d}x\mathrm{d}v \tag{2.12}$$

is the potential energy associated with the external potential $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. Formally, along solutions of the time-dependent VPFP system the functional $\mathcal{F}^u$ decreases monotonically, with dissipation rate given by the Fisher information of f relative to the local Maxwellian in v; this is the classical free energy inequality for the Vlasov–Fokker–Planck equation (see, e.g., [13]). Stationary solutions are therefore naturally associated with the points where this dissipation vanishes, i.e., the critical points of $\mathcal{F}^u$ over the admissible class $\tilde{\mathcal{A}}(\boldsymbol{m})$. This motivates characterising stationary states variationally, as minimisers of $\mathcal{F}^u$, rather than directly through the weak formulation above, whose existence theory is considerably more delicate.

Remark 2.2.6. *The assumption $u \in L^\infty(\mathbb{T}^d)$ is not essential for the subsequent analysis. Nevertheless, imposing this condition simplifies the exposition and excludes singularities in the external potential, which lie outside the class of regular external fields typically considered in applications.*

We next define the object of interest of the following discussion.

Definition 2.2.7 (**Free Energy Solutions to the Stationary VPFP System**).
Let $\boldsymbol{m} > 0$ and $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. Then $(f_\star, \varphi_\star) \in \tilde{\mathcal{A}}(\boldsymbol{m}) \times H^1_{\mathrm{zm}}(\mathbb{T}^d)$ is a free energy solution to the stationary VPFP *system* (2.7) (*equivalently, a free energy stationary solution of the* VPFP *system*) *if*

$$\mathcal{F}^u(f_\star) = \inf_{f \in \tilde{\mathcal{A}}(\boldsymbol{m})} \mathcal{F}^u(f)$$

and

$$\varphi_\star = (-\Delta)^{-1}(\rho_\star - \rho_0),$$

where $\rho_\star := \int_{\mathbb{R}^d} f_\star \mathrm{d}v$.

In the sequel, we prove that for any $u \in H^1 \cap L^\infty(\mathbb{T}^d)$, the stationary VPFP system admits a unique free energy solution, and the latter is a weak stationary solution to (2.7) in the sense of Definition 2.2.1.

Assume that $\mathcal{F}^u$ is well-defined on $\tilde{\mathcal{A}}(\boldsymbol{m})$, and that

$$\inf_{f \in \tilde{\mathcal{A}}(\boldsymbol{m})} \mathcal{F}^u(f) > -\infty,$$

that is, the infimum is well-defined. Let $\{f_n\}_{n\in\mathbb{N}}$ be an arbitrary minimising sequence. By construction, the sequences

$$\{\mathcal{E}(f_n)\}_{n\in\mathbb{N}}, \qquad \left\{E^K(f_n)\right\}_{n\in\mathbb{N}}, \quad \text{and} \quad \left\{E^D(\varphi_{f_n})\right\}_{n\in\mathbb{N}}$$

are bounded. By Lemma 2.1.6, the boundedness of the Dirichlet energy implies the boundedness of ρ_{f_n} in $H^{-1}(\mathbb{T}^d)$. Since this holds for all minimising sequences, we may restrict the analysis to

$$\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m}) := \Big\{ f \in L^1(\mathbb{T}^d \times \mathbb{R}^d) \ : \ f \geq 0 \text{ a.e.}, \int_{\mathbb{T}^d\times\mathbb{R}^d} f \,\mathrm{d}x\mathrm{d}v = \boldsymbol{m}, \\ \|\rho_f\|_{H^{-1}(\mathbb{T}^d)} \leq C_\rho, \int_{\mathbb{T}^d\times\mathbb{R}^d} f\log f \mathrm{d}x\mathrm{d}v \leq C_\mathcal{E}, \int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 f \mathrm{d}x\mathrm{d}v \leq C_K \Big\}$$

for some constants $C_\rho, C_\mathcal{E}, C_K$ large enough.

Remark 2.2.8. *The constants $C_\rho, C_\mathcal{E}, C_K$ in the definition of $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ are introduced heuristically to enforce uniform bounds, along minimising sequences, on the $H^{-1}(\mathbb{T}^d)$ norm of the spatial densities, the entropy and the kinetic energy, respectively. In Proposition 2.2.16, we show the existence of sufficiently large $C_\rho, C_\mathcal{E}, C_K$ such that*

$$\inf_{f \in \tilde{\mathcal{A}}(\boldsymbol{m})} \mathcal{F}^u(f) = \inf_{f \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})} \mathcal{F}^u(f),$$

justifying the analysis on the restricted space without loss of generality.

In the sequel, we regard the set $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ as depending on the constants $C_\rho, C_\mathcal{E}, C_K$, chosen sufficiently large so that $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ is nonempty. Such constants exist for any given mass $\boldsymbol{m} > 0$. Indeed, consider the function

$$\gamma_\star(x, v) := \frac{\boldsymbol{m}}{|\mathbb{T}^d|}\mathrm{M}(v) = \frac{\boldsymbol{m}}{|\mathbb{T}^d|}\frac{e^{-\frac{|v|^2}{2}}}{(2\pi)^{\frac{d}{2}}}, \tag{2.13}$$

where $\mathrm{M}(v)$ is the normalised d-dimensional Maxwellian distribution. Then, $\gamma_\star \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ for any $C_\rho, C_\mathcal{E}, C_K$ satisfying

$$C_\rho \geq \frac{\boldsymbol{m}}{\sqrt{|\mathbb{T}^d|}}, \qquad C_\mathcal{E} \geq \int_{\mathbb{T}^d\times\mathbb{R}^d} \gamma_\star \log \gamma_\star \mathrm{d}x\mathrm{d}v, \qquad C_K \geq \int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 \gamma_\star \mathrm{d}x\mathrm{d}v. \tag{2.14}$$

For notational convenience, we shall write $\mathcal{A}$ in place of $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$, omitting the explicit dependence on the constants except where it is required to compare arguments over $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ and $\tilde{\mathcal{A}}(\boldsymbol{m})$.

To establish the existence and uniqueness of free energy solutions to the stationary VPFP system, we employ the direct method of the calculus of variations in Subsection 2.2.4. The key ingredients are the weak compactness of the restricted admissible class and the weak lower semicontinuity and strict convexity of the free energy functional, both of which rely on the fundamental properties of the entropy functional established beforehand.

2.2.2 Entropy Functional

We first establish two preliminary results, and then provide an initial characterisation of the entropy functional (2.10). Throughout, we adopt the standard convention $0 \log 0 = 0$.

Lemma 2.2.9. *Let $f, \gamma \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$ be such that $f \geq 0$ and $\gamma > 0$ a.e. Then,*

$$f \log\left(\frac{f}{\gamma}\right) \geq f - \gamma \quad a.e..$$

Proof. Consider the standard estimate $\log y \leq y - 1$ for all $y > 0$, which is equivalent to

$$\log\left(\frac{1}{y}\right) \geq 1 - y \quad \text{for all } y > 0.$$

Applying it to $\frac{\gamma}{f}$ yields

$$\log\left(\frac{f}{\gamma}\right) \geq 1 - \frac{\gamma}{f} \qquad \text{for all } x \in \{f > 0\}.$$

Since $f \geq 0$, multiplying by f both sides gives the desired inequality for a.e. $(x, v) \in \mathbb{T}^d \times \mathbb{R}^d$. □

Lemma 2.2.10. *Let $\boldsymbol{m} > 0$, and let $C_\rho, C_\mathcal{E}, C_K$ satisfy (2.14). Then,*

$$f \log f \in L^1(\mathbb{T}^d \times \mathbb{R}^d) \quad \text{for all } f \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m}).$$

In particular, there exists $\tilde{c}_\mathcal{E} = \tilde{c}_\mathcal{E}(C_\rho, C_\mathcal{E}, C_K, d) > 0$ such that

$$\|f \log f\|_{L^1(\mathbb{T}^d \times \mathbb{R}^d)} \leq \tilde{c}_\mathcal{E} \quad \text{for all } f \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m}).$$

Proof. Let $f \in \mathcal{A} = \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ be arbitrary. We compute

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} |f \log f| \mathrm{d}x\mathrm{d}v = \int_{\{f \geq 1\}} f \log f \mathrm{d}x\mathrm{d}v - \int_{\{f < 1\}} f \log f \mathrm{d}x\mathrm{d}v.$$

Since

$$\mathcal{E}(f) = \int_{\mathbb{T}^d \times \mathbb{R}^d} f \log f \mathrm{d}x\mathrm{d}v = \int_{\{f \geq 1\}} f \log f \mathrm{d}x\mathrm{d}v + \int_{\{f < 1\}} f \log f \mathrm{d}x\mathrm{d}v,$$

rearranging the terms and substituting in the previous equality yields

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} |f \log f| \mathrm{d}x\mathrm{d}v = \mathcal{E}(f) - 2 \int_{\{f < 1\}} f \log f \mathrm{d}x\mathrm{d}v. \tag{2.15}$$

Since the entropy is bounded by definition of $\mathcal{A}$, it suffices to bound the second term. To this end, let $R > 0$ be arbitrary and split the integral as follows:

$$\int_{\{f<1\}} -f \log f \mathrm{d}x\mathrm{d}v = \int_{\{f<1\} \cap \{|v| \leq R\}} -f \log f \mathrm{d}x\mathrm{d}v + \int_{\{f<1\} \cap \{|v| > R\}} -f \log f \mathrm{d}x\mathrm{d}v.$$

Using the standard estimate $0 \leq -t \log t \leq \frac{1}{e}$ for all $t \in [0, +\infty)$ yields

$$\int_{\{f<1\} \cap \{|v| \leq R\}} -f \log f \mathrm{d}x\mathrm{d}v \leq \frac{|\mathbb{T}^d| R^d}{e}.$$

Next, let $\varepsilon \in \left(0, \frac{2}{2+d}\right) \subset (0, 1)$ be arbitrary, where $d \geq 1$ is the dimension of the system. Then, on the set $\{f < 1\}$ there exists $C_\varepsilon > 0$ such that

$$-f \log f = f |\log f| \leq C_\varepsilon f^{1-\varepsilon}.$$

To see this, consider the function

$$h(t) := \frac{-t \log t}{t^{1-\varepsilon}} = -t^\varepsilon \log t \qquad t \in (0, 1),$$

so that the inequality is equivalent to $h(t) \leq C_\varepsilon$ for all $t \in (0,1)$. Performing the change of variables $x = -\log t$ (i.e., $t = e^{-x}$), we have

$$\tilde{h}(x) := h(t = e^{-x}) = xe^{-\varepsilon x}, \qquad x \in (0, +\infty).$$

The function $\tilde{h}$ attains its maximum at $x = \frac{1}{\varepsilon}$. Consequently, it suffices to let $C_\varepsilon = \frac{1}{e\varepsilon}$.

Since $f \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$ and $\varepsilon \in (0,1)$, we have $f^{1-\varepsilon} \in L^{\frac{1}{1-\varepsilon}}(\mathbb{T}^d \times \mathbb{R}^d)$, where $\frac{1}{1-\varepsilon} > 1$ and $\frac{1}{\varepsilon} > 1$ are conjugate exponents. Therefore, the previous estimate, together with Hölder's inequality (see, e.g., Theorem 4.6 in [15]), yields

$$\begin{aligned}
\int_{\{f<1\}\cap\{|v|>R\}} -f\log f \mathrm{d}x\mathrm{d}v &\leq \int_{|v|>R} C_\varepsilon \frac{|v|^{2(1-\varepsilon)}}{|v|^{2(1-\varepsilon)}} f^{1-\varepsilon} \mathrm{d}x\mathrm{d}v \\
&\leq \left(\int_{|v|>R} C_\varepsilon |v|^2 f \mathrm{d}x\mathrm{d}v\right)^{1-\varepsilon} \left(\int_{|v|>R} \frac{1}{|v|^{\frac{2(1-\varepsilon)}{\varepsilon}}} \mathrm{d}x\mathrm{d}v\right)^{\varepsilon} \\
&\leq (C_\varepsilon C_K)^{1-\varepsilon} \left(|\mathbb{T}^d| \cdot |\mathbb{S}^{d-1}| \int_R^\infty r^{d-1-\frac{2(1-\varepsilon)}{\varepsilon}} \mathrm{d}r\right)^{\varepsilon},
\end{aligned}$$

where the last inequality follows by definition of $\mathcal{A}$ and passing to spherical coordinates in the second integral; here, $|\mathbb{S}^{d-1}|$ denotes the surface area of the unit d-dimensional sphere. Since $\varepsilon \in \left(0, \frac{2}{2+d}\right)$, it follows that

$$d - 1 - \frac{2(1-\varepsilon)}{\varepsilon} < -1,$$

and thus the previous integral is upper bounded by a constant that only depends on the dimension of the system and $\mathcal{A}$. Substituting the above estimates in (2.15) and using the definition of $\mathcal{A}$ yields the upper bound

$$\|f\log f\|_{L^1(\mathbb{T}^d\times\mathbb{R}^d)} \leq C_{\mathcal{E}} + 2\frac{|\mathbb{T}^d|R^d}{e} + 2(C_\varepsilon C_K)^{1-\varepsilon}\left(|\mathbb{T}^d| \cdot |\mathbb{S}^{d-1}| \frac{R^{d-\frac{2-2\varepsilon}{\varepsilon}}}{\frac{2-2\varepsilon}{\varepsilon} - d}\right)^{\varepsilon} =: c_{\mathcal{E}}.$$

The result follows by arbitrariness of $f \in \mathcal{A}$. □

Lemma 2.2.11. *Let $\boldsymbol{m} > 0$, and let $C_\rho, C_{\mathcal{E}}, C_K$ satisfy* (2.14)*. Then,*

(*i*) *$\mathcal{E}$ is well-defined on $\tilde{\mathcal{A}}(\boldsymbol{m})$;*

(*ii*) *$\mathcal{E}$ is strongly lower semicontinuous on $\mathcal{A}(C_\rho, C_{\mathcal{E}}, C_K; \boldsymbol{m})$.*

Proof of (*i*). By definition of $\tilde{\mathcal{A}}(\boldsymbol{m})$, $\mathcal{E}(f) < +\infty$ for all $f \in \tilde{\mathcal{A}}(\boldsymbol{m})$. Therefore, it remains to establish that $\mathcal{E}(f) > -\infty$ for all $f \in \tilde{\mathcal{A}}(\boldsymbol{m})$. To this end, let $f \in \tilde{\mathcal{A}}(\boldsymbol{m})$ be arbitrary and γ as in (2.13). Then, by Lemma 2.2.9 we have

$$f\log f = f\log\left(\frac{f}{\gamma}\right) + f\log\gamma \geq f - \gamma + f\log\gamma \qquad \text{a.e.}$$

Integrating the above inequality over $\mathbb{T}^d \times \mathbb{R}^d$ and using the identity

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} f \mathrm{d}x\mathrm{d}v = \int_{\mathbb{T}^d\times\mathbb{R}^d} \gamma \mathrm{d}x\mathrm{d}v = \boldsymbol{m},$$

we obtain

$$\begin{aligned}
\int_{\mathbb{T}^d\times\mathbb{R}^d} f\log f \mathrm{d}x\mathrm{d}v &\geq \int_{\mathbb{T}^d\times\mathbb{R}^d} f\mathrm{d}x\mathrm{d}v - \int_{\mathbb{T}^d\times\mathbb{R}^d} \gamma\mathrm{d}x\mathrm{d}v + \int_{\mathbb{T}^d\times\mathbb{R}^d} f\log\gamma \mathrm{d}x\mathrm{d}v \\
&= \int_{\mathbb{T}^d\times\mathbb{R}^d} f\log\gamma \mathrm{d}x\mathrm{d}v \\
&= \int_{\mathbb{T}^d\times\mathbb{R}^d} f\log\left(\frac{\boldsymbol{m}}{|\mathbb{T}^d|(2\pi)^{\frac{d}{2}}}\right)\mathrm{d}x\mathrm{d}v - \frac{1}{2}\int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 f \mathrm{d}x\mathrm{d}v \\
&= \boldsymbol{m}\log\left(\frac{\boldsymbol{m}}{|\mathbb{T}^d|(2\pi)^{\frac{d}{2}}}\right) - \frac{1}{2}\int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 f \mathrm{d}x\mathrm{d}v.
\end{aligned} \tag{2.16}$$

Since the kinetic energy (2.11) is bounded on $\tilde{\mathcal{A}}(\boldsymbol{m})$, it follows that $\mathcal{E}(f) > -\infty$. □

Proof of (ii). Let $\{f_n\}_{n\in\mathbb{N}} \subset \mathcal{A} = \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ be such that $f_n \to f$ strongly in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$. It suffices to prove that

$$\mathcal{E}(f) \leq \liminf_{n\to\infty} \mathcal{E}(f_n). \tag{2.17}$$

By strong convergence, there exists a subsequence $\{f_{n_k}\}_{k\in\mathbb{N}}$ such that $f_{n_k} \to f$ a.e. (see, e.g., Theorem 4.9 in [15]). This implies in particular that $f \geq 0$ a.e., and thus $f \log f$ is well-defined a.e. We next define

$$\gamma(x,v) := \frac{\boldsymbol{m}}{|\mathbb{T}^d|} \frac{e^{-|v|}}{\int_{\mathbb{R}^d} e^{-|w|} \mathrm{d}w} = \frac{\boldsymbol{m}}{|\mathbb{T}^d|} \frac{e^{-|v|}}{|\mathbb{S}^{d-1}|(d-1)!}, \tag{2.18}$$

where $|\mathbb{S}^{d-1}|$ denotes the surface area of the d-dimensional unit sphere, and we write

$$f \log f = f \log\left(\frac{f}{\gamma}\right) + f \log \gamma, \qquad f_{n_k} \log f_{n_k} = f_{n_k} \log\left(\frac{f_{n_k}}{\gamma}\right) + f_{n_k} \log \gamma \quad \text{for all } k \in \mathbb{N}. \tag{2.19}$$

We aim to apply Fatou's Lemma (see, e.g., Lemma 4.1 in [15]) to $f_{n_k} \log f_{n_k}$ using the decomposition introduced above in order to establish (2.17) along the subsequence $\{f_{n_k}\}$. This then allows to conclude (2.17) for the whole sequence.

Step 1. Kinetic Bounds on f. Since $f \geq 0$ a.e. and $f_{n_k} \to f$ a.e., it follows that $|v|^2 f \geq 0$ a.e. and $|v|^2 f_{n_k} \to |v|^2 f$ a.e. Moreover, by definition of $\mathcal{A}$, there holds $|v|^2 f_{n_k} \geq 0$ a.e. for all $k \in \mathbb{N}$ and

$$\sup_{k\in\mathbb{N}} \int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 f_{n_k} \mathrm{d}x\mathrm{d}v \leq C_K.$$

Hence, Fatou's lemma applies, and we obtain

$$E^K(f) = \int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 f \mathrm{d}x\mathrm{d}v \leq \liminf_{k\to\infty} \int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 f_{n_k} \mathrm{d}x\mathrm{d}v \leq C_K. \tag{2.20}$$

Similarly, since $|f_{n_k} \log f_{n_k}| \to |f \log f|$ a.e., from Lemma 2.2.10 and Fatou's lemma we deduce that

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} |f \log f| \mathrm{d}x\mathrm{d}v \leq \liminf_{k\to\infty} \int_{\mathbb{T}^d\times\mathbb{R}^d} |f_{n_k} \log f_{n_k}| \mathrm{d}x\mathrm{d}v \leq \tilde{c}_\mathcal{E}(C_\rho, C_\mathcal{E}, C_K, d) < +\infty. \tag{2.21}$$

This implies in particular that $f \log f \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$.

Step 2. Uniform integrability of entropy terms. By the strong convergence $f_n \to f$ in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$, we have

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} f \mathrm{d}x\mathrm{d}v = \|f\|_{L^1(\mathbb{T}^d\times\mathbb{R}^d)} = \lim_{n\to\infty} \|f_n\|_{L^1(\mathbb{T}^d\times\mathbb{R}^d)} = \boldsymbol{m}, \tag{2.22}$$

where the last equality follows from the definition of $\mathcal{A}$. Thus, recalling the definition of γ in (2.18), we compute

$$\begin{aligned}
\int_{\mathbb{T}^d\times\mathbb{R}^d} |f \log \gamma| \mathrm{d}x\mathrm{d}v &\leq \left| \log\left(\frac{\boldsymbol{m}}{|\mathbb{T}^d| \cdot |\mathbb{S}^{d-1}|(d-1)!}\right)\right| \int_{\mathbb{T}^d\times\mathbb{R}^d} f \mathrm{d}x\mathrm{d}v + \int_{\mathbb{T}^d\times\mathbb{R}^d} |v| f \mathrm{d}x\mathrm{d}v \\
&= \boldsymbol{m} \left| \log\left(\frac{\boldsymbol{m}}{|\mathbb{T}^d| \cdot |\mathbb{S}^{d-1}|(d-1)!}\right)\right| + \int_{\mathbb{T}^d\times\mathbb{R}^d} |v| f \mathrm{d}x\mathrm{d}v.
\end{aligned}$$

Furthermore, we have

$$\begin{aligned}
\int_{\mathbb{T}^d\times\mathbb{R}^d} |v| f \mathrm{d}x\mathrm{d}v &= \int_{\{|v|\leq 1\}} |v| f \mathrm{d}x\mathrm{d}v + \int_{\{|v|>1\}} |v| f \mathrm{d}x\mathrm{d}v \\
&\leq \int_{\{|v|\leq 1\}} f \mathrm{d}x\mathrm{d}v + \int_{\{|v|>1\}} |v|^2 f \mathrm{d}x\mathrm{d}v \\
&\leq \int_{\mathbb{T}^d\times\mathbb{R}^d} f \mathrm{d}x\mathrm{d}v + \int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 f \mathrm{d}x\mathrm{d}v \leq \boldsymbol{m} + C_K,
\end{aligned}$$

where the last inequality uses (2.22) and (2.20). From the previous computation, we therefore conclude that

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} |f\log\gamma|\mathrm{d}x\mathrm{d}v \le \boldsymbol{m}\left|\log\left(\frac{\boldsymbol{m}}{|\mathbb{T}^d|\cdot|\mathbb{S}^{d-1}|(d-1)!}\right)\right| + \boldsymbol{m} + C_K,$$

which implies that $f\log\gamma \in L^1(\mathbb{T}^d\times\mathbb{R}^d)$. Using the same argument for f_{n_k}, $k\in\mathbb{N}$, we have

$$\sup_{k\in\mathbb{N}}\int_{\mathbb{T}^d\times\mathbb{R}^d} |f_{n_k}\log\gamma|\mathrm{d}x\mathrm{d}v \le \boldsymbol{m}\left|\log\left(\frac{\boldsymbol{m}}{|\mathbb{T}^d|\cdot|\mathbb{S}^{d-1}|(d-1)!}\right)\right| + \boldsymbol{m} + C_K.$$

Combining the above bound with (2.21), the decomposition (2.19) yields

$$\begin{aligned}\int_{\mathbb{T}^d\times\mathbb{R}^d}\left|f\log\left(\frac{f}{\gamma}\right)\right|\mathrm{d}x\mathrm{d}v &\le \int_{\mathbb{T}^d\times\mathbb{R}^d}|f\log f|\,\mathrm{d}x\mathrm{d}v + \int_{\mathbb{T}^d\times\mathbb{R}^d}|f\log\gamma|\,\mathrm{d}x\mathrm{d}v\\ &\le \tilde{c}_{\mathcal{E}}(C_\rho, C_{\mathcal{E}}, C_K; \boldsymbol{m}) + \boldsymbol{m}\left|\log\left(\frac{\boldsymbol{m}}{|\mathbb{T}^d|\cdot|\mathbb{S}^{d-1}|(d-1)!}\right)\right| + \boldsymbol{m} + C_K,\end{aligned}$$

which implies that $f\log\left(\frac{f}{\gamma}\right)\in L^1(\mathbb{T}^d\times\mathbb{R}^d)$. Using the same argument with the sequence f_{n_k}, $k\in\mathbb{N}$, we conclude that

$$\sup_{k\in\mathbb{N}}\int_{\mathbb{T}^d\times\mathbb{R}^d}\left|f_{n_k}\log\left(\frac{f_{n_k}}{\gamma}\right)\right|\mathrm{d}x\mathrm{d}v \le \tilde{c}_{\mathcal{E}}(C_\rho, C_{\mathcal{E}}, C_K; \boldsymbol{m}) + \boldsymbol{m}\left|\log\left(\frac{\boldsymbol{m}}{|\mathbb{T}^d|\cdot|\mathbb{S}^{d-1}|(d-1)!}\right)\right| + \boldsymbol{m} + C_K.$$

Step 3. Convergence of $\int_{\mathbb{T}^d\times\mathbb{R}^d}|v|f_{n_k}\mathrm{d}x\mathrm{d}v$. We prove that

$$\lim_{k\to\infty}\int_{\mathbb{T}^d\times\mathbb{R}^d}|v|f_{n_k}\mathrm{d}x\mathrm{d}v = \int_{\mathbb{T}^d\times\mathbb{R}^d}|v|f\mathrm{d}x\mathrm{d}v. \tag{2.23}$$

Fix an arbitrary $\varepsilon>0$. Then, there exists $R>0$ such that

$$\int_{\{|v|>R\}}|v|f\mathrm{d}x\mathrm{d}v < \frac{\varepsilon}{4}.$$

Indeed, letting $R>\frac{4C_K}{\varepsilon}$, we estimate

$$\int_{\{|v|>R\}}|v|f\mathrm{d}x\mathrm{d}v = \int_{\{|v|>R\}}\frac{|v|^2}{|v|}f\mathrm{d}x\mathrm{d}v < \frac{1}{R}\int_{\mathbb{T}^d\times\mathbb{R}^d}|v|^2 f\mathrm{d}x\mathrm{d}v \le \frac{C_K}{R} < \frac{\varepsilon}{4},$$

where the second inequality uses (2.20). Similarly, since $E^K(g)\le C_K$ for all $g\in\mathcal{A}$, we have

$$\int_{\{|v|>R\}}|v|f_{n_k}\mathrm{d}x\mathrm{d}v < \frac{\varepsilon}{4}\qquad\text{for all } k\in\mathbb{N}.$$

By convergence of the whole sequence $\{f_n\}$ to f, the subsequence $\{f_{n_k}\}$ also converges strongly to f in $L^1(\mathbb{T}^d\times\mathbb{R}^d)$. Hence, there exists $\tilde{k}\in\mathbb{N}$ such that

$$\|f_{n_k}-f\|_{L^1(\mathbb{T}^d\times\mathbb{R}^d)} < \frac{\varepsilon}{2R}\qquad\text{for all } k\ge\tilde{k}.$$

Combining the above estimates, for all $k\ge\tilde{k}$ we have

$$\begin{aligned}\left|\int_{\mathbb{T}^d\times\mathbb{R}^d}|v|f_{n_k}\mathrm{d}x\mathrm{d}v - \int_{\mathbb{T}^d\times\mathbb{R}^d}|v|f\mathrm{d}x\mathrm{d}v\right| &= \left|\int_{\{|v|\le R\}}|v|(f_{n_k}-f)\mathrm{d}x\mathrm{d}v + \int_{\{|v|>R\}}|v|(f_{n_k}-f)\mathrm{d}x\mathrm{d}v\right|\\ &\le R\|f_{n_k}-f\|_{L^1(\mathbb{T}^d\times\mathbb{R}^d)} + \int_{\{|v|>R\}}|v|f_{n_k}\mathrm{d}x\mathrm{d}v + \int_{\{|v|>R\}}|v|f\mathrm{d}x\mathrm{d}v\\ &< \frac{\varepsilon}{2}+\frac{\varepsilon}{4}+\frac{\varepsilon}{4} = \varepsilon.\end{aligned}$$

By arbitrariness of $\varepsilon>0$, we conclude (2.23).

Step 4. Fatou's lemma for $\int_{\mathbb{T}^d\times\mathbb{R}^d} f_{n_k}\log\left(\frac{f_{n_k}}{\gamma}\right)\mathrm{d}x\mathrm{d}v$. From Lemma 2.2.9, we have

$$f\log\left(\frac{f}{\gamma}\right)-(f-\gamma)\geq 0 \qquad \text{a.e.},$$

and similarly for f_{n_k}, $k\in\mathbb{N}$. Furthermore, using

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} f_{n_k}\mathrm{d}x\mathrm{d}v=\int_{\mathbb{T}^d\times\mathbb{R}^d}\gamma\mathrm{d}x\mathrm{d}v=\boldsymbol{m}, \tag{2.24}$$

we have

$$\sup_{k\in\mathbb{N}}\int_{\mathbb{T}^d\times\mathbb{R}^d}\left(f_{n_k}\log\left(\frac{f_{n_k}}{\gamma}\right)-(f_{n_k}-\gamma)\right)\mathrm{d}x\mathrm{d}v\leq\sup_{k\in\mathbb{N}}\int_{\mathbb{T}^d\times\mathbb{R}^d}\left|f_{n_k}\log\left(\frac{f_{n_k}}{\gamma}\right)\right|+2\boldsymbol{m},$$

where the right-hand side is finite by Step 2. Since $f_{n_k}\to f$ a.e., it follows that

$$f_{n_k}\log\left(\frac{f_{n_k}}{\gamma}\right)-(f_{n_k}-\gamma)\to f\log\left(\frac{f}{\gamma}\right)-(f-\gamma)\quad\text{a.e..}$$

Hence, Fatou's lemma applies, and we obtain

$$\int_{\mathbb{T}^d\times\mathbb{R}^d}\left(f\log\left(\frac{f}{\gamma}\right)-(f-\gamma)\right)\mathrm{d}x\mathrm{d}v\leq\liminf_{k\to\infty}\int_{\mathbb{T}^d\times\mathbb{R}^d}\left(f_{n_k}\log\left(\frac{f_{n_k}}{\gamma}\right)-(f_{n_k}-\gamma)\right)\mathrm{d}x\mathrm{d}v.$$

Using (2.22) and (2.24), we conclude that

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} f\log\left(\frac{f}{\gamma}\right)\mathrm{d}x\mathrm{d}v\leq\liminf_{k\to\infty}\int_{\mathbb{T}^d\times\mathbb{R}^d} f_{n_k}\log\left(\frac{f_{n_k}}{\gamma}\right)\mathrm{d}x\mathrm{d}v.$$

Step 5. Lower Semicontinuity along $\{f_{n_k}\}$. From Step 2, there holds $f\log\left(\frac{f}{\gamma}\right)$, $f\log\gamma\in L^1(\mathbb{T}^d\times\mathbb{R}^d)$. Since additionally $f\log f\in L^1(\mathbb{T}^d)$ – see (2.21) – integrating (2.19) yields

$$\begin{aligned}\mathcal{E}(f)&=\int_{\mathbb{T}^d\times\mathbb{R}^d} f\log\left(\frac{f}{\gamma}\right)\mathrm{d}x\mathrm{d}v+\int_{\mathbb{T}^d\times\mathbb{R}^d} f\log\gamma\mathrm{d}x\mathrm{d}v\\&=\int_{\mathbb{T}^d\times\mathbb{R}^d} f\log\left(\frac{f}{\gamma}\right)\mathrm{d}x\mathrm{d}v+\log\left(\frac{\boldsymbol{m}}{|\mathbb{T}^d|\cdot|\mathbb{S}^{d-1}|(d-1)!}\right)\int_{\mathbb{T}^d\times\mathbb{R}^d} f\mathrm{d}x\mathrm{d}v-\int_{\mathbb{T}^d\times\mathbb{R}^d}|v|f\mathrm{d}x\mathrm{d}v\\&=\int_{\mathbb{T}^d\times\mathbb{R}^d} f\log\left(\frac{f}{\gamma}\right)\mathrm{d}x\mathrm{d}v+\boldsymbol{m}\log\left(\frac{\boldsymbol{m}}{|\mathbb{T}^d|\cdot|\mathbb{S}^{d-1}|(d-1)!}\right)-\int_{\mathbb{T}^d\times\mathbb{R}^d}|v|f\mathrm{d}x\mathrm{d}v,\end{aligned}$$

where the last equality uses (2.22). Similarly, since $g\log g\in L^1(\mathbb{T}^d\times\mathbb{R}^d)$ for all $g\in\mathcal{A}$ (see Lemma 2.2.10), from Step 1 we deduce that, for all $k\in\mathbb{N}$,

$$\mathcal{E}(f_{n_k})=\int_{\mathbb{T}^d\times\mathbb{R}^d} f_{n_k}\log\left(\frac{f_{n_k}}{\gamma}\right)\mathrm{d}x\mathrm{d}v+\boldsymbol{m}\log\left(\frac{\boldsymbol{m}}{|\mathbb{T}^d|\cdot|\mathbb{S}^{d-1}|(d-1)!}\right)-\int_{\mathbb{T}^d\times\mathbb{R}^d}|v|f_{n_k}\mathrm{d}x\mathrm{d}v.$$

Combining this with the convergence established in Steps 3 and 4, we obtain

$$\begin{aligned}\mathcal{E}(f)&\leq\liminf_{k\to\infty}\left(\int_{\mathbb{T}^d\times\mathbb{R}^d} f_{n_k}\log\left(\frac{f_{n_k}}{\gamma}\right)\mathrm{d}x\mathrm{d}v+\boldsymbol{m}\log\left(\frac{\boldsymbol{m}}{|\mathbb{T}^d|\cdot|\mathbb{S}^{d-1}|(d-1)!}\right)-\int_{\mathbb{T}^d\times\mathbb{R}^d}|v|f_{n_k}\mathrm{d}x\mathrm{d}v\right)\\&=\liminf_{k\to\infty}\mathcal{E}(f_{n_k}).\end{aligned}$$

By arbitrariness of $\{f_n\}$, the previous argument shows that, for any strongly convergent sequence in $\mathcal{A}$, there exists a subsequence along which (2.17) holds.

Step 6. Conclusion. We establish (2.17). Let $\{f_n\}_{n\in\mathbb{N}} \subset \mathcal{A}$ be such that $f_n \to f$ strongly in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$. By Lemma 2.2.10, we have

$$a := \liminf_{n\to\infty} \mathcal{E}(f) \geq -\tilde{c}_{\mathcal{E}}(C_\rho, C_{\mathcal{E}}, C_K, d) > -\infty.$$

Furthermore, since $\mathcal{E}(g) \leq C_{\mathcal{E}}$ for all $g \in \mathcal{A}$, there holds

$$a \leq C_{\mathcal{E}}.$$

Hence, a is thus finite. By definition of limit inferior, for any $j \in \mathbb{N}$, $j > 0$, there exists $N_j \in \mathbb{N}$ such that

$$\inf_{n\geq N_j} \mathcal{E}(f_n) \leq a + \frac{1}{2j}.$$

By definition of infimum, there exists $n_j \geq N_j$ such that

$$\mathcal{E}(f_{n_j}) \leq a + \frac{1}{2j} + \frac{1}{2j} = a + \frac{1}{j}.$$

Since the limit inferior of a sequence is bounded above by that of any subsequence, we obtain

$$\limsup_{j\to\infty} \mathcal{E}(f_{n_j}) \leq a \leq \liminf_{j\to\infty} \mathcal{E}(f_{n_j}).$$

Hence, the subsequence $\{\mathcal{E}(f_{n_j})\}_{j\in\mathbb{N}}$ converges to a. Moreover, since the whole sequence converges strongly to f in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$, the subsequence $\{f_{n_j}\}_{j\in\mathbb{N}}$ does so too. By Step 5, there exists a further subsequence $\{f_{n_{j_k}}\}_{k\in\mathbb{N}}$ such that

$$\mathcal{E}(f) \leq \liminf_{k\to\infty} \mathcal{E}(f_{n_{j_k}}) = \lim_{k\to\infty} \mathcal{E}(f_{n_{j_k}}) = a = \liminf_{n\to\infty} \mathcal{E}(f_n),$$

where we used that every subsequence of a converging sequence converges to the same limit. By arbitrariness of $\{f_n\}$, we conclude that $\mathcal{E}$ is strongly lower semicontinuous. □

2.2.3 Restricted Admissible Class and Free Energy Functional

We establish that $\mathcal{A}(C_\rho, C_{\mathcal{E}}, C_K; \boldsymbol{m})$ is weakly compact and that $\mathcal{F}^u$ is weakly lower semicontinuous. These facts rely on a preliminary result, which we prove first. We recall that $\mathcal{A}(C_\rho, C_{\mathcal{E}}, C_K; \boldsymbol{m})$ is equipped with the $L^1(\mathbb{T}^d \times \mathbb{R}^d)$ topology.

Lemma 2.2.12. *Let $\boldsymbol{m} > 0$, and let $C_\rho, C_{\mathcal{E}}, C_K$ satisfy (2.14). Then, the map*

$$\rho : \mathcal{A}(C_\rho, C_{\mathcal{E}}, C_K; \boldsymbol{m}) \to H^{-1}(\mathbb{T}^d),\ f \mapsto \rho_f = \int_{\mathbb{R}^d} f \mathrm{d}v$$

is weakly sequentially continuous.

Proof. Let $\{f_n\}_{n\in\mathbb{N}} \subset \mathcal{A} = \mathcal{A}(C_\rho, C_{\mathcal{E}}, C_K; \boldsymbol{m})$ be such that $f_n \rightharpoonup f$ weakly in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$. We aim to show that

$$\rho_{f_n} \rightharpoonup \rho_f \quad \text{weakly in } H^{-1}(\mathbb{T}^d).$$

The linear map

$$T : L^1(\mathbb{T}^d \times \mathbb{R}^d) \to L^1(\mathbb{T}^d),\ g \mapsto \int_{\mathbb{R}^d} g \mathrm{d}v$$

is bounded. Indeed, by Fubini's theorem,

$$\|Tg\|_{L^1(\mathbb{T}^d)} = \int_{\mathbb{T}^d} |Tg| \mathrm{d}x \leq \int_{\mathbb{T}^d} \int_{\mathbb{R}^d} |g| \mathrm{d}v \mathrm{d}x = \int_{\mathbb{T}^d \times \mathbb{R}^d} |g| \mathrm{d}x \mathrm{d}v = \|g\|_{L^1(\mathbb{T}^d \times \mathbb{R}^d)} \quad \text{for all } g \in L^1(\mathbb{T}^d \times \mathbb{R}^d).$$

Since T is a bounded linear operator between Banach spaces, it is also weakly continuous (see, e.g., Theorem 3.10 in [15]). Hence, the weak convergence $f_n \rightharpoonup f$ in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$ yields

$$\rho_{f_n} = Tf_n \rightharpoonup Tf = \rho_f \quad \text{weakly in } L^1(\mathbb{T}^d).$$

This implies that

$$\langle \rho_{f_n}, \psi \rangle \to \langle \rho_f, \psi \rangle \quad \text{for all } \psi \in \mathcal{C}^\infty(\mathbb{T}^d) \subset L^\infty(\mathbb{T}^d).$$

Therefore, $\rho_{f_n} \to \rho_f$ in $\mathcal{D}'(\mathbb{T}^d)$. Next, by definition of $\mathcal{A}$, the sequence $\{\rho_{f_n}\}_{n\in\mathbb{N}}$ is bounded in $H^{-1}(\mathbb{T}^d)$. Since $H^{-1}(\mathbb{T}^d) = (H^1(\mathbb{T}^d))^*$ and $H^1(\mathbb{T}^d)$ is reflexive, $H^{-1}(\mathbb{T}^d)$ is also reflexive (see, e.g., Corollary 3.21 in [15]). Therefore, by standard results (see, e.g., Theorem 3.18 in [15]), there exists a subsequence $\rho_{f_{n_k}}$ such that

$$\rho_{f_{n_k}} \rightharpoonup g \quad \text{weakly in } H^{-1}(\mathbb{T}^d).$$

This implies that $\rho_{f_{n_k}} \to g$ in $\mathcal{D}'(\mathbb{T}^d)$. Combining this with $\rho_{f_{n_k}} \to \rho_f$ in $\mathcal{D}'(\mathbb{T}^d)$, the uniqueness of distributional limits yields

$$\rho_f = g \quad \text{in } \mathcal{D}'(\mathbb{T}^d).$$

Since $g \in H^{-1}(\mathbb{T}^d)$, we conclude that

$$\rho_{f_{n_k}} \rightharpoonup \rho_f \quad \text{weakly in } H^{-1}(\mathbb{T}^d).$$

Since every subsequence of $\{f_n\}$ admits a further subsequence along which the desired weak continuity holds, we conclude that

$$\rho_{f_n} \rightharpoonup \rho_f \quad \text{weakly in } H^{-1}(\mathbb{T}^d).$$

By arbitrariness of $\{f_n\}$, we conclude that ρ is weakly sequentially continuous into $H^{-1}(\mathbb{T}^d)$. □

Lemma 2.2.13. *Let $\boldsymbol{m} > 0$, and let $C_\rho, C_\mathcal{E}, C_K$ satisfy (2.14). Then, $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ is*

(i) nonempty, convex, and bounded;

(ii) strongly closed in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$;

(iii) weakly compact in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$.

Proof of (i). Nonemptiness is a direct consequence of (2.14). Boundedness follows from the mass constraint

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} f \mathrm{d}x\mathrm{d}v = \boldsymbol{m},$$

together with the nonnegativity condition $f \geq 0$ a.e. We now address convexity. The inequality constraint $f \geq 0$ a.e. is convex. Moreover, the maps

$$f \mapsto \int_{\mathbb{T}^d \times \mathbb{R}^d} f \mathrm{d}x\mathrm{d}v \qquad \text{and} \qquad f \mapsto \int_{\mathbb{T}^d \times \mathbb{R}^d} |v|^2 f \mathrm{d}x\mathrm{d}v$$

are linear. Therefore, both the mass constraint and the kinetic energy bound define convex constraints. Furthermore, the map

$$f \mapsto \rho_f = \int_{\mathbb{R}^d} f \mathrm{d}v$$

is also linear. Hence, for all $f_1, f_2 \in \mathcal{A} = \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ and $\theta \in [0,1]$, the triangle inequality yields

$$\|\rho_{(1-\theta)f_1 + \theta f_2}\|_{H^{-1}(\mathbb{T}^d)} = \|(1-\theta)\rho_{f_1} + \theta\rho_{f_2}\|_{H^{-1}(\mathbb{T}^d)} \leq (1-\theta)\|\rho_{f_1}\|_{H^{-1}(\mathbb{T}^d)} + \theta\|\rho_{f_2}\|_{H^{-1}(\mathbb{T}^d)} \leq C_\rho,$$

showing that the constraint $\|\rho_f\|_{H^{-1}(\mathbb{T}^d)} \leq C_\rho$ is convex. Finally, the map $g : t \mapsto t \log t$ is strictly convex on $\{t > 0\}$, since $g''(t) = \frac{1}{t} > 0$ for all $t > 0$. Hence, for all $f_1, f_2 \in \mathcal{A}$ and for all $\theta \in [0,1]$, the strict convexity of g yields

$$(\theta f_1 + (1-\theta) f_2) \log (\theta f_1 + (1-\theta) f_2) \leq \theta f_1 \log f_1 + (1-\theta) f_2 \log f_2 \quad \text{a.e.}$$

Integrating the above inequality yields the convexity of the entropy constraint. □

Proof of (ii). Since $L^1(\mathbb{T}^d \times \mathbb{R}^d)$ is second-countable, it suffices to show that $\mathcal{A}$ is strongly sequentially closed. Let $\{f_n\}_{n\in\mathbb{N}} \subset \mathcal{A}$ be a convergent sequence with limit $f \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$. We prove that $f \in \mathcal{A}$. To this end, up to a subsequence, there holds $f_n \to f$ a.e. [15, Thm 4.9], and thus $f \geq 0$ a.e. Consequently, $f \log f$ is well-defined. Moreover, the lower semicontinuity of the entropy functional on $\mathcal{A}$ (see Lemma 2.2.11) yields

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} f \log f \mathrm{d}x\mathrm{d}v \leq \liminf_{n\to\infty} \int_{\mathbb{T}^d\times\mathbb{R}^d} f_n \log f_n \mathrm{d}x\mathrm{d}v \leq C_{\mathcal{E}}.$$

We next address the kinetic energy. By definition of $\mathcal{A}$, we have

$$\sup_{n\in\mathbb{N}} \int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 f_n \mathrm{d}x\mathrm{d}v \leq C_K.$$

Since $f_n \to f$ a.e. (up to a subsequence) and $|v|^2 f_n$, $|v|^2 f \geq 0$ a.e. for all n, Fatou's lemma yields

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 f \mathrm{d}x\mathrm{d}v \leq \liminf_{n\to\infty} \int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 f_n \mathrm{d}x\mathrm{d}v \leq C_K.$$

Furthermore, the strong convergence ensures that the mass constraint is preserved in the limit:

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} f \mathrm{d}x\mathrm{d}v = \|f\|_{L^1(\mathbb{T}^d\times\mathbb{R}^d)} = \lim_{n\to\infty} \|f_n\|_{L^1(\mathbb{T}^d\times\mathbb{R}^d)} = \boldsymbol{m}.$$

It remains to consider the constraint $\|\rho_f\|_{H^{-1}(\mathbb{T}^d)} \leq C_\rho$. Since $f_n \to f$ strongly in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$, it follows that $f_n \rightharpoonup f$ weakly in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$ [15, Prop 3.5]. Therefore, by Lemma 2.2.12 we have

$$\rho_{f_n} \rightharpoonup \rho_f \quad \text{weakly in } H^{-1}(\mathbb{T}^d).$$

Hence, by weak lower semicontinuity of the Sobolev norm, we conclude that

$$\|\rho_f\|_{H^{-1}(\mathbb{T}^d)} \leq \liminf_{n\to\infty} \|\rho_{f_n}\|_{H^{-1}(\mathbb{T}^d)} \leq C_\rho,$$

where the last inequality follows by definition of $\mathcal{A}$. The previous reasoning shows that $f \in \mathcal{A}$. By arbitrariness of $\{f_n\}$, we conclude that $\mathcal{A}$ is strongly sequentially closed (hence closed) in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$. □

Proof of (iii). We aim to apply the Dunford–Pettis theorem (see, e.g., Theorem 4.30 in [15]). To this end, we first show that $\mathcal{A}$ is equi-integrable, that is, for all $\varepsilon > 0$ there exists $\delta > 0$ such that, for all Lebesgue measurable subsets $X \subset \mathbb{T}^d \times \mathbb{R}^d$ with $|X| < \delta$, there holds

$$\sup_{f\in\mathcal{A}} \int_X f \mathrm{d}x\mathrm{d}v < \varepsilon.$$

Fix an arbitrary $\varepsilon > 0$, and let $X \subset \mathbb{T}^d \times \mathbb{R}^d$ be Lebesgue-measurable with $|X| < \delta$. Then, for all $R > 1$ and for all $f \in \mathcal{A}$, we estimate

$$\int_X f \mathrm{d}x\mathrm{d}v = \int_{X\cap\{f\leq R\}} f \mathrm{d}x\mathrm{d}v + \int_{X\cap\{f>R\}} f \mathrm{d}x\mathrm{d}v \leq R|X| + \int_{X\cap\{f>R\}} f \mathrm{d}x\mathrm{d}v.$$

On the set $\{f > R\}$, since $R > 1$, the monotonicity of the logarithm yields

$$f \leq \frac{1}{\log R} f \log f.$$

Hence, by definition of $\mathcal{A}$, we have

$$\int_{X\cap\{f>R\}} f \mathrm{d}x\mathrm{d}v \leq \frac{1}{\log R} \int_{X\cap\{f>R\}} f \log f \mathrm{d}x\mathrm{d}v \leq \frac{C_{\mathcal{E}}}{\log R}.$$

Therefore, we obtain the uniform bound

$$\int_X f \mathrm{d}x\mathrm{d}v \le R|X| + \frac{C_{\mathcal{E}}}{\log R}.$$

Letting $R > 1$ be large enough so that $\frac{C_{\mathcal{E}}}{\log R} < \frac{\varepsilon}{4}$ and then choosing $\delta < \frac{\varepsilon}{4R}$ yields the desired property.

We next prove that $\mathcal{A}$ is tight, that is, for all $\varepsilon > 0$ there exists $X \subset \mathbb{T}^d \times \mathbb{R}^d$ Lebesgue-measurable such that $|X| < +\infty$ and

$$\sup_{f \in \mathcal{A}} \int_{(\mathbb{T}^d \times \mathbb{R}^d) \setminus X} f \mathrm{d}x\mathrm{d}v < \varepsilon.$$

Let $\varepsilon > 0$ be arbitrary, and choose $R > \sqrt{\frac{2C_K}{\varepsilon}}$. Define

$$X := \{(x, v) \in \mathbb{T}^d \times \mathbb{R}^d : |v| \le R\},$$

where $|X| \le |\mathbb{T}^d| R^d < +\infty$. For all $f \in \mathcal{A}$, we estimate

$$\int_{(\mathbb{T}^d \times \mathbb{R}^d) \setminus X} f \mathrm{d}x\mathrm{d}v = \int_{\mathbb{T}^d \times \{|v| > R\}} \frac{|v|^2}{|v|^2} f \mathrm{d}x\mathrm{d}v \le \frac{1}{R^2} \| |v|^2 f \|_{L^1(\mathbb{T}^d \times \mathbb{R}^d)} \le \frac{C_K}{R^2} < \frac{\varepsilon}{2}.$$

Taking the supremum over $\mathcal{A}$ yields the desired inequality.

Since $\mathcal{A}$ is bounded, equi-integrable and tight, by the Dunford–Pettis theorem $\mathcal{A}$ has compact closure in the weak topology of $L^1(\mathbb{T}^d \times \mathbb{R}^d)$. Since additionally $\mathcal{A}$ is strongly closed and convex, it is also weakly closed (see, e.g., Theorem 3.7 in [15]), hence it is weakly compact in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$. □

Lemma 2.2.14. *Let $\boldsymbol{m} > 0$ and $u \in H^1 \cap L^\infty(\mathbb{T}^d)$.*

(i) $\mathcal{F}^u$ is well-defined and bounded from below on $\tilde{\mathcal{A}}(\boldsymbol{m})$.

(ii) For any $C_\rho, C_{\mathcal{E}}, C_K$ satisfying (2.14), $\mathcal{F}^u$ is strictly convex and lower semicontinuous on $\mathcal{A}(C_\rho, C_{\mathcal{E}}, C_K; \boldsymbol{m})$ with respect to both the strong and weak topologies.

Proof of (i). We first show that $\mathcal{F}^u$ is well-defined on $\tilde{\mathcal{A}} = \tilde{\mathcal{A}}(\boldsymbol{m})$. The entropy functional $\mathcal{E}$ is well-defined on $\tilde{\mathcal{A}}$ by Lemma 2.2.11. Moreover, the kinetic energy (2.11) is finite, due to the nonnegativity of the integrand and the definition of $\tilde{\mathcal{A}}$. For the potential energy (2.12), we estimate

$$\left| \int_{\mathbb{T}^d \times \mathbb{R}^d} u f \mathrm{d}x\mathrm{d}v \right| \le \|u\|_{L^\infty(\mathbb{T}^d)} \boldsymbol{m} < +\infty \qquad \text{for all } f \in \tilde{\mathcal{A}}(\boldsymbol{m}). \tag{2.25}$$

It remains to consider the Dirichlet energy. Let $f \in \tilde{\mathcal{A}}(\boldsymbol{m})$ be arbitrary. Then, from the definition of $\tilde{\mathcal{A}}(\boldsymbol{m})$, it follows that $\rho_f - \rho_0 \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$. By Lemma 2.1.3, we thus deduce that there exists a unique zero-mean solution

$$\varphi_f = (-\Delta)^{-1}(\rho_f - \rho_0) \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$$

to the Poisson equation with forcing term $\rho_f - \rho_0$. Therefore, we have

$$0 \le E^D(\varphi_f) = \frac{1}{2} \int_{\mathbb{T}^d} |\nabla \varphi_f|^2 \mathrm{d}x \le \frac{1}{2} \|\varphi_f\|^2_{H^1(\mathbb{T}^d)} < +\infty.$$

We next prove that $\mathcal{F}^u$ is bounded from below on $\tilde{\mathcal{A}}$. Let $f \in \tilde{\mathcal{A}}$ be arbitrary. Arguing as in Lemma 2.2.11 *(i)* (see (2.16)), we have

$$\mathcal{E}(f) \ge \boldsymbol{m} \log \left(\frac{\boldsymbol{m}}{|\mathbb{T}^d|(2\pi)^{\frac{d}{2}}} \right) - \frac{1}{2} \int_{\mathbb{T}^d \times \mathbb{R}^d} |v|^2 f \mathrm{d}x\mathrm{d}v.$$

Since the Dirichlet energy is nonnegative and using (2.25), we conclude that

$$\mathcal{F}^u(f) \geq \boldsymbol{m} \log\left(\frac{\boldsymbol{m}}{|\mathbb{T}^d|(2\pi)^{\frac{d}{2}}}\right) - \boldsymbol{m}\|u\|_{L^\infty(\mathbb{T}^d)}.$$

Therefore, $\mathcal{F}^u$ is bounded from below on $\tilde{\mathcal{A}}$ by a constant depending only on the geometry, the mass of the system and the external potential. □

Proof of (ii). We prove strict convexity and lower semicontinuity separately.

Strict Convexity. We treat each term separately. The map $g : t \mapsto t\log t$ is strictly convex on $t > 0$, since $g''(t) = \frac{1}{t} > 0$ for all $t > 0$. Hence, for any $f_1, f_2 \in \mathcal{A} = \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ such that $f_1 \neq f_2$ and for any $\theta \in (0,1)$ we have

$$(\theta f_1 + (1-\theta)f_2)\log\left(\theta f_1 + (1-\theta)f_2\right) \leq \theta f_1 \log f_1 + (1-\theta) f_2 \log f_2 \quad \text{a.e.},$$

where $\theta f_1 + (1-\theta)f_2 \in \mathcal{A}$ by convexity of $\mathcal{A}$ (see Lemma 2.2.13). By the strict convexity of g, equality holds a.e. if and only if $f_1 = f_2$ a.e. However, since $f_1 \neq f_2$ by assumption, we have strict inequality on a set of positive measure. Therefore, integrating the previous inequality yields the strict convexity of $\mathcal{E}$:

$$\mathcal{E}(\theta f_1 + (1-\theta)f_2) < \theta\mathcal{E}(f_1) + (1-\theta)\mathcal{E}(f_2).$$

Furthermore, the kinetic energy (2.11) and the potential energy (2.12) define linear functionals, hence they are convex on $\mathcal{A}$. It remains to consider the Dirichlet energy. Let $f_1, f_2 \in \mathcal{A}$ and $\theta \in [0,1]$ be arbitrary. Then,

$$\rho_{\theta f_1 + (1-\theta)f_2} = \int_{\mathbb{R}^d} (\theta f_1 + (1-\theta)f_2)\mathrm{d}v = \theta\rho_{f_1} + (1-\theta)\rho_{f_2}.$$

Since $\theta f_1 + (1-\theta)f_2 \in \mathcal{A}$ by convexity of $\mathcal{A}$, the mass constraint implies that

$$\rho_{\theta f_1 + (1-\theta)f_2} - \rho_0 \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d).$$

Hence, the linearity of the inverse Laplacian $-\Delta^{-1} : H^{-1}_{\mathrm{zm}}(\mathbb{T}^d) \to H^1_{\mathrm{zm}}(\mathbb{T}^d)$ (see Proposition 2.1.4) yields

$$\begin{aligned}
\varphi_{\theta f_1 + (1-\theta)f_2} &= (-\Delta)^{-1}\left((\theta\rho_{f_1} + (1-\theta)\rho_{f_2}) - \rho_0\right) \\
&= \theta(-\Delta)^{-1}(\rho_{f_1} - \rho_0) + (1-\theta)(-\Delta)^{-1}(\rho_{f_2} - \rho_0) \\
&= \theta\varphi_{f_1} + (1-\theta)\varphi_{f_2}.
\end{aligned}$$

Using this identity and the convexity of $\mathbb{R}^d \to \mathbb{R},\ x \mapsto |x|^2$ yields

$$\int_{\mathbb{T}^d} |\nabla\varphi_{\theta f_1 + (1-\theta)f_2}|^2 \mathrm{d}x = \int_{\mathbb{T}^d} |\theta\nabla\varphi_{f_1} + (1-\theta)\nabla\varphi_{f_2}|^2\mathrm{d}x \leq \theta\int_{\mathbb{T}^d} |\nabla\varphi_{f_1}|^2\mathrm{d}x + (1-\theta)\int_{\mathbb{T}^d} |\nabla\varphi_{f_2}|^2\mathrm{d}x.$$

Therefore, the Dirichlet energy is convex. Since $\mathcal{F}^u$ is the sum of convex functionals, and the entropy functional is strictly convex, we conclude that $\mathcal{F}^u$ is strictly convex.

Strong Lower Semicontinuity. It suffices to treat each term separately. The entropy functional is strongly lower semicontinuous on $\mathcal{A}$ by Lemma 2.2.11. For the kinetic energy, let $\{f_n\}_{n\in\mathbb{N}} \subset \mathcal{A}$ be an arbitrary sequence such that $f_n \to f$ strongly in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$. Then, up to a subsequence, $f_n \to f$ a.e. [15, Thm 4.9]. Moreover, by definition of $\mathcal{A}$, there holds

$$\sup_{n\in\mathbb{N}} \int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 f_n \mathrm{d}x\mathrm{d}v < +\infty.$$

Since $|v|^2 f,\ |v|^2 f_n \geq 0$ a.e., by Fatou's lemma we have

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 f \mathrm{d}x\mathrm{d}v \leq \liminf_{n\to\infty} \int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 f_n \mathrm{d}x\mathrm{d}v.$$

By arbitrariness of $\{f_n\}$, the kinetic energy is strongly lower semicontinuous on $\mathcal{A}$. For the potential energy, let $\{f_n\}_{n\in\mathbb{N}} \subset \mathcal{A}$ be an arbitrary sequence such that $f_n \to f$ strongly in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$. Then, $f_n \rightharpoonup f$ weakly in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$ [15, Prop 3.5]. Since $u \in L^\infty(\mathbb{T}^d)$, we have

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} u f_n \mathrm{d}x\mathrm{d}v = \langle u, f_n\rangle_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d),L^1(\mathbb{T}^d\times\mathbb{R}^d)} \to \langle u, f\rangle_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d),L^1(\mathbb{T}^d\times\mathbb{R}^d)} = \int_{\mathbb{T}^d\times\mathbb{R}^d} u f \mathrm{d}x\mathrm{d}v.$$

By arbitrariness of $\{f_n\}$, the potential energy functional is strongly continuous, and thus strongly lower semicontinuous on $\mathcal{A}$. Finally, we address the Dirichlet energy. Let $\{f_n\}_{n\in\mathbb{N}} \subset \mathcal{A}$ be an arbitrary sequence such that $f_n \to f$ strongly in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$. It suffices to prove that

$$\int_{\mathbb{T}^d} |\nabla\varphi_f|^2 \mathrm{d}x \le \liminf_{n\to\infty} \int_{\mathbb{T}^d} |\nabla\varphi_{f_n}|^2 \mathrm{d}x. \tag{2.26}$$

Since $\mathcal{A}$ is strongly closed in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$ (see Lemma 2.2.13), it is also strongly sequentially closed, hence $f \in \mathcal{A}$. Therefore, by definition of $\mathcal{A}$, there holds $\rho_f - \rho_0 \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$ and $\rho_{f_n} - \rho_0 \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$. Hence, by Lemma 2.1.3 the unique zero-mean solutions to the Poisson equation are well-defined:

$$\varphi_f = (-\Delta)^{-1}(\rho_f - \rho_0),\ \varphi_{f_n} = (-\Delta)^{-1}(\rho_{f_n} - \rho_0) \in H^1_{\mathrm{zm}}(\mathbb{T}^d) \quad \text{for all } n \in \mathbb{N}.$$

Since $f_n \to f$ strongly (hence weakly) in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$ and $\{f_n\} \subset \mathcal{A}$, by Lemma 2.2.12 we have

$$\rho_{f_n} \rightharpoonup \rho_f \quad \text{weakly in } H^{-1}(\mathbb{T}^d).$$

In particular, it follows that

$$\rho_{f_n} - \rho_0 \rightharpoonup \rho_f - \rho_0 \quad \text{weakly in } H^{-1}_{\mathrm{zm}}(\mathbb{T}^d).$$

By Proposition 2.1.4, the inverse Laplacian $-\Delta : H^{-1}_{\mathrm{zm}}(\mathbb{T}^d) \to H^1_{\mathrm{zm}}(\mathbb{T}^d)$ is linear and strongly continuous, hence weakly continuous [15, Thm 3.10]. This implies that

$$\varphi_{f_n} \rightharpoonup \varphi_f \quad \text{weakly in } H^1_{\mathrm{zm}}(\mathbb{T}^d).$$

Hence, $\nabla\varphi_{f_n} \rightharpoonup \nabla\varphi_f$ weakly in $L^2(\mathbb{T}^d)^d$. By weak lower semicontinuity of the Sobolev norm, we conclude (2.26). By arbitrariness of $\{f_n\}$, the Dirichlet energy is strongly lower semicontinuous on $\mathcal{A}$. Since $\mathcal{F}^u$ is the sum of strongly lower semicontinuous functionals on $\mathcal{A}$, it is strongly lower semicontinuous on $\mathcal{A}$.

Weak Lower Semicontinuity. We use standard arguments (cf. Corollary 3.9 in [15]). Let $\lambda \in \mathbb{R}$ be arbitrary, and define

$$X_\lambda := \{f \in \mathcal{A} : \mathcal{F}^u(f) \le \lambda\}.$$

Since $\mathcal{F}^u$ is lower semicontinuous in the strong topology, and $\mathcal{A}$ is strongly closed (see Lemma 2.2.13), X_λ is closed in the strong topology of $\mathcal{A}$. Indeed, since $L^1(\mathbb{T}^d \times \mathbb{R}^d)$ is second-countable, it suffices to prove that X_λ is strongly sequentially closed. To this end, let $\{f_n\}_{n\in\mathbb{N}} \subset X_\lambda$ be such that $f_n \to f$ strongly in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$. Since $\mathcal{A}$ is strongly closed, we have $f \in \mathcal{A}$. Furthermore, by the lower semicontinuity of $\mathcal{F}^u$, it follows that

$$\mathcal{F}^u(f) \le \liminf_{n\to\infty} \mathcal{F}^u(f_n) \le \lambda,$$

which implies that $f \in X_\lambda$. By arbitrariness of the sequence $\{f_n\}$, X_λ is strongly sequentially closed. Furthermore, it is convex by the convexity of $\mathcal{A}$ (see lemma 2.2.13) and that of $\mathcal{F}^u$. Indeed, for any $f_1, f_2 \in \mathcal{A}$ and for any $\theta \in [0,1]$, we have

$$\theta f_1 + (1-\theta) f_2 \in \mathcal{A} \qquad \text{and} \qquad \mathcal{F}^u(\theta f_1 + (1-\theta) f_2) \le \theta \mathcal{F}^u(f_1) + (1-\theta)\mathcal{F}^u(f_2) \le \lambda.$$

Hence, $\theta f_1 + (1-\theta) f_2 \in X_\lambda$. By arbitrariness of f_1, f_2 and θ, X_λ is convex. Since X_λ is strongly closed and convex, it is weakly closed in the subspace topology of $\mathcal{A}$ [15, Thm 3.7]. By arbitrariness of $\lambda \in \mathbb{R}$, $\mathcal{F}^u$ is weakly lower semicontinuous on $\mathcal{A}$. □

2.2.3.1 Weak Sequential Lower Semicontinuity of the Free Energy Functional

Although not required for the proof of the main existence theorem, we establish that the free energy functional is weakly sequentially lower semicontinuous via a Mazur-type argument. This property is relevant to the variational framework and is included for possible future extensions.

Lemma 2.2.15. *Let $\boldsymbol{m} > 0$ and $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. For any $C_\rho, C_\mathcal{E}, C_K$ satisfying (2.14), $\mathcal{F}^u$ is weakly sequentially lower semicontinuous on $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$.*

Proof. Let $\{f_n\}_{n\in\mathbb{N}} \subset \mathcal{A} = \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ be an arbitrary sequence such that $f_n \rightharpoonup f$ weakly in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$. We aim to show that

$$\mathcal{F}^u(f) \le \liminf_{n\to\infty} \mathcal{F}^u(f_n).$$

Since $\mathcal{F}^u$ is bounded from below on $\tilde{\mathcal{A}}(\boldsymbol{m})$ (see Lemma 2.2.14), there holds

$$a := \liminf_{n\to\infty} \mathcal{F}^u(f_n) > -\infty.$$

Furthermore, since $\mathcal{F}^u$ is well-defined on $\mathcal{A}$, it follows that $a < +\infty$. Hence a is finite, and arguing as in Step 6 of Lemma 2.2.11 (ii), there exists a subsequence that realises the limit inferior. In other words, there exists a subsequence $\{f_{n_k}\}_{k\in\mathbb{N}}$ such that

$$\lim_{k\to\infty} \mathcal{F}^u(f_{n_k}) = a = \liminf_{n\to\infty} \mathcal{F}^u(f_n). \tag{2.27}$$

By Mazur's lemma (see, e.g., Corollary 3.8 and Exercise 3.4 in [15]), there exists a sequence $\{g_j\}_{j\in\mathbb{N}}$ with

$$g_j \in \operatorname{conv}\left(\bigcup_{k\ge j} \{f_{n_k}\}\right) \quad \text{for all } j \in \mathbb{N} \tag{2.28}$$

such that $g_j \to f$ strongly in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$; here, $\operatorname{conv}(\cdot)$ denotes the convex hull. Hence, for all $j \in \mathbb{N}$, there exist $N_j \in \mathbb{N}$, a finite subset

$$\{f_{n_{k_i}}\}_{i=1}^{N_j} \subset \{f_{n_k}\}_{k=j}^{\infty},$$

and nonnegative weights $\alpha_{j,i} \ge 0$ with

$$\sum_{i=1}^{N_j} \alpha_{j,i} = 1 \qquad \text{for all } j \in \mathbb{N}, \tag{2.29}$$

such that

$$g_j = \sum_{i=1}^{N_j} \alpha_{j,i} f_{n_{k_i}}.$$

By convexity of $\mathcal{A}$ (see Lemma 2.2.13), and since $\{f_n\}_{n\in\mathbb{N}} \subset \mathcal{A}$, we have $g_j \in \mathcal{A}$ for all $j \in \mathbb{N}$. Furthermore, by convexity of $\mathcal{F}^u$ on $\mathcal{A}$ (see Lemma 2.2.14), we have

$$\mathcal{F}^u(g_j) \le \sum_{i=1}^{N_j} \alpha_{j,i} \mathcal{F}^u(f_{n_{k_i}}) \qquad \text{for all } j \in \mathbb{N}. \tag{2.30}$$

Since $g_j \to f$ strongly in $L^1(\mathbb{T}^d \times \mathbb{R}^d)$, and $\mathcal{A}$ is strongly closed (see Lemma 2.2.13), we have $f \in \mathcal{A}$. Moreover, since $\mathcal{F}^u$ is strongly lower semicontinuous on $\mathcal{A}$ by Lemma 2.2.14 (ii), we deduce that

$$\mathcal{F}^u(f) \le \liminf_{j\to\infty} \mathcal{F}^u(g_j).$$

It remains to prove that

$$\liminf_{j\to\infty} \mathcal{F}^u(g_j) \le a = \liminf_{n\to\infty} \mathcal{F}^u(f_n).$$

Let $\varepsilon > 0$ be arbitrary. By (2.27), there exists $K \in \mathbb{N}$ such that

$$\mathcal{F}^u(f_{n_k}) \le a + \varepsilon \qquad \text{for all } k \ge K. \tag{2.31}$$

Let $j \ge K$. Then, g_j is a convex combination of elements $f_{n_{k_i}}$ with $k_i \ge j \ge K$ (see (2.28)). Using this, together with (2.30) and (2.31), we obtain

$$\mathcal{F}^u(g_j) \le \sum_{i=1}^{N_j} \alpha_{j,i} \mathcal{F}^u(f_{n_{k_i}}) \le \sum_{i=1}^{N_j} \alpha_{j,i}(a+\varepsilon) = a + \varepsilon \qquad \text{for all } j \ge K,$$

where the last equality uses (2.29). Taking the limit inferior yields

$$\liminf_{j\to\infty} \mathcal{F}^u(g_j) \le a + \varepsilon = \liminf_{n\to\infty} \mathcal{F}^u(f_n) + \varepsilon.$$

Letting $\varepsilon \to 0$ yields the desired inequality. By arbitrariness of the sequence $\{f_n\}$, we conclude that $\mathcal{F}^u$ is weakly sequentially lower semicontinuous. □

2.2.4 Existence and Uniqueness of Free Energy Solutions

We establish the existence and uniqueness of free energy solutions to the stationary VPFP system by applying the direct method of the calculus of variations. In particular, we justify restricting the analysis to $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ with sufficiently large $C_\rho, C_\mathcal{E}, C_K > 0$, showing that no generality is lost.

Proposition 2.2.16. *Let $\boldsymbol{m} > 0$ and $u \in H^1 \cap L^\infty(\mathbb{T}^d)$.*

(*i*) *For any $C_\rho, C_\mathcal{E}, C_K$ satisfying (2.14), $\mathcal{F}^u$ attains the infimum on $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$, and the minimiser is unique.*

(*ii*) *There exist large enough constants $\bar{C}_\rho, \bar{C}_\mathcal{E}, \bar{C}_K > 0$ such that*

$$\inf_{f \in \tilde{\mathcal{A}}(\boldsymbol{m})} \mathcal{F}^u(f) = \inf_{f \in \mathcal{A}(\bar{C}_\rho, \bar{C}_\mathcal{E}, \bar{C}_K; \boldsymbol{m})} \mathcal{F}^u(f).$$

In particular, $\mathcal{F}^u$ attains the infimum on $\tilde{\mathcal{A}}(\boldsymbol{m})$, and the minimiser is unique.

Proof of (*i*). By Lemma 2.2.14, $\mathcal{F}^u$ is well-defined and bounded from below on $\tilde{\mathcal{A}}(\boldsymbol{m}) \supset \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$. Therefore,

$$-\infty < \alpha := \inf_{f \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})} \mathcal{F}^u(f) < +\infty.$$

By contradiction, suppose that α is not attained. For any $k \in \mathbb{N}$ with $k > 1$, set

$$A_k := \left\{ f \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m}) : \mathcal{F}^u(f) > \alpha + \frac{1}{k} \right\}.$$

Then, A_k is weakly open for all $k \in \mathbb{N}$ by weak lower semicontinuity of $\mathcal{F}^u$ on $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ (see Lemma 2.2.14). In particular, since the infimum is not attained, $\{A_k\}_{k\in\mathbb{N}}$ is a weakly open cover of $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$. By weak compactness of $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ (see Lemma 2.2.13), there exists a finite subcover. In particular, since $\{A_k\}_{k\in\mathbb{N}}$ is an increasing sequence, there exists $k_0 \in \mathbb{N}$ such that $A_{k_0} = \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$, However, this yields a contradiction, as

$$\alpha = \inf_{f \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})} \mathcal{F}^u(f) = \inf_{f \in \bar{A}_{k_0}} \mathcal{F}^u(f) \ge \alpha + \frac{1}{k_0} > \alpha.$$

Hence, the infimum is attained. Moreover, the minimiser is unique. Indeed, suppose by contradiction that f_1, $f_2 \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ with $f_1 \neq f_2$ are both minimisers, and let $\theta \in (0,1)$ be arbitrary. Since $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ is convex (see Lemma 2.2.13), $\theta f_1 + (1-\theta) f_2 \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$. However, the strict convexity of $\mathcal{F}^u$ (see Lemma 2.2.14) yields a contradiction:

$$\alpha \leq \mathcal{F}^u(\theta f_1 + (1-\theta) f_2) < \theta \mathcal{F}^u(f_1) + (1-\theta)\mathcal{F}^u(f_2) = \alpha.$$

Therefore, the minimiser is unique. □

Proof of (ii). By Lemma 2.2.14, $\mathcal{F}^u$ is well-defined and bounded from below on $\tilde{\mathcal{A}}(\boldsymbol{m})$. Hence, the following quantity is well-defined:

$$-\infty < \inf_{g \in \tilde{\mathcal{A}}(\boldsymbol{m})} \mathcal{F}^u(g) < +\infty.$$

For any $C_\rho, C_\mathcal{E}, C_K$ satisfying (2.14), by Lemma 2.2.13 there holds $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m}) \neq \emptyset$, and the trivial inclusion $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m}) \subset \tilde{\mathcal{A}}(\boldsymbol{m})$ thus yields

$$\inf_{g \in \tilde{\mathcal{A}}(\boldsymbol{m})} \mathcal{F}^u(g) \leq \inf_{g \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})} \mathcal{F}^u(g).$$

For the converse inequality, let $\{f_n\}_{n\in\mathbb{N}} \subset \tilde{\mathcal{A}}(\boldsymbol{m})$ be an arbitrary minimising sequence, that is,

$$\lim_{n\to\infty} \mathcal{F}^u(f_n) = \inf_{g \in \tilde{\mathcal{A}}(\boldsymbol{m})} \mathcal{F}^u(g).$$

Then, by definition of $\mathcal{F}^u$, the sequences $\{\mathcal{E}(f_n)\}_{n\in\mathbb{N}}$, $\{E^K(f_n)\}_{n\in\mathbb{N}}$, and $\{E^D(\varphi_{f_n})\}_{n\in\mathbb{N}}$ are bounded. Hence, there exist $\bar{C}_\mathcal{E}, \bar{C}_K, \tilde{C}_\rho \in \mathbb{R}$ such that

$$\begin{aligned}
\sup_{n\in\mathbb{N}} \int_{\mathbb{T}^d \times \mathbb{R}^d} f_n \log f_n \mathrm{d}x\mathrm{d}v &\leq \bar{C}_\mathcal{E}, \\
\sup_{n\in\mathbb{N}} \int_{\mathbb{T}^d \times \mathbb{R}^d} |v|^2 f_n \mathrm{d}x\mathrm{d}v &\leq \bar{C}_K, \\
\sup_{n\in\mathbb{N}} \int_{\mathbb{T}^d} |\nabla \varphi_{f_n}|^2 \mathrm{d}x &\leq \tilde{C}_\rho.
\end{aligned}$$

By Lemma 2.1.6, the last bound implies that

$$\sup_{n\in\mathbb{N}} \|\rho_{f_n} - \rho_0\|^2_{H^{-1}(\mathbb{T}^d)} \leq 2\tilde{C}_\rho.$$

In particular, using the triangle inequality and that ρ_0 is a constant, there exists $\bar{C}_\rho \geq 0$ such that

$$\sup_{n\in\mathbb{N}} \|\rho_{f_n}\|_{H^{-1}(\mathbb{T}^d)} \leq \bar{C}_\rho.$$

Therefore, the minimising sequence satisfies $\{f_n\}_{n\in\mathbb{N}} \subset \mathcal{A}(\bar{C}_\rho, \bar{C}_\mathcal{E}, \bar{C}_K; \boldsymbol{m})$, which implies that

$$\inf_{g \in \mathcal{A}(\bar{C}_\rho, \bar{C}_\mathcal{E}, \bar{C}_K; \boldsymbol{m})} \mathcal{F}^u(g) \leq \lim_{n\to\infty} \mathcal{F}^u(f_n) = \inf_{g \in \tilde{\mathcal{A}}(\boldsymbol{m})} \mathcal{F}^u(g).$$

Without loss of generality, we may assume that $\bar{C}_\rho, \bar{C}_\mathcal{E}, \bar{C}_K$ satisfy (2.14). Hence, by part (i) $\mathcal{F}^u$ attains the infimum on $\mathcal{A}(\bar{C}_\rho, \bar{C}_\mathcal{E}, \bar{C}_K; \boldsymbol{m})$. Letting $f \in \mathcal{A}(\bar{C}_\rho, \bar{C}_\mathcal{E}, \bar{C}_K; \boldsymbol{m})$ be the unique minimiser of $\mathcal{F}^u$ on $\mathcal{A}(\bar{C}_\rho, \bar{C}_\mathcal{E}, \bar{C}_K; \boldsymbol{m})$, the previous inequality yields

$$\mathcal{F}^u(f) \leq \inf_{g \in \tilde{\mathcal{A}}(\boldsymbol{m})} \mathcal{F}^u(g).$$

Consequently, $\mathcal{F}^u$ attains the infimum on $\tilde{\mathcal{A}}(\boldsymbol{m})$. Moreover, the minimiser of $\mathcal{F}^u$ on $\tilde{\mathcal{A}}(\boldsymbol{m})$ is unique. Indeed, suppose by contradiction that $f_1, f_2 \in \tilde{\mathcal{A}}(\boldsymbol{m})$ are both minimisers. Then, there exist $C_\rho, C_\mathcal{E}, C_K \in \mathbb{R}$ such that $f_1, f_2 \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$. Without loss of generality, we may assume that $C_\rho, C_\mathcal{E}, C_K$ satisfy (2.14). However, this contradicts the uniqueness of the minimiser of $\mathcal{F}^u$ on $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$. □

2.2.4.1 Unconfined Free Energy Solutions

We compute the unique free energy solution of (2.7) in the absence of an external potential, that is, when $u = 0$.

Proposition 2.2.17. *Let $\boldsymbol{m} > 0$ and $u = 0$. The unique minimiser of $\mathcal{F}^u$ on $\tilde{\mathcal{A}}(\boldsymbol{m})$ is given by* (2.13)*, and the corresponding stationary self-consistent potential is $\varphi_\star = 0$.*

Proof. Let $f_\star \in \tilde{\mathcal{A}}(\boldsymbol{m})$ be the unique minimiser of $\mathcal{F}^u$, and define its spatial density by

$$\rho_\star(x) := \int_{\mathbb{R}^d} f_\star(x,v)\mathrm{d}v, \qquad x \in \mathbb{T}^d.$$

For any $(x,v) \in \mathbb{T}^d \times \mathbb{R}^d$, set

$$\zeta(x,v) := \begin{cases} \frac{f_\star(x,v)}{\rho_\star(x)} & \text{if } \rho_\star(x) > 0, \\ 0 & \text{otherwise.} \end{cases}$$

The function ζ is nonnegative a.e. and Lebesgue measurable on $\mathbb{T}^d \times \mathbb{R}^d$. Indeed, $\rho_\star(x) = \int_{\mathbb{R}^d} f_\star(x,v)\,\mathrm{d}v$ is nonnegative a.e. and measurable as the integral of a nonnegative measurable function over v. By measurability of $\rho_\star$, the set $\{\rho_\star > 0\} \times \mathbb{R}^d$ is therefore measurable. On this set, $f_\star(x,v)/\rho_\star(x)$ is the quotient of two nonnegative measurable functions with nonzero denominator, hence nonnegative and measurable. On the complement $\{\rho_\star = 0\} \times \mathbb{R}^d$, ζ is defined as the constant zero, which is measurable. Combining these two measurable pieces shows that ζ is nonnegative a.e. and measurable on the whole domain $\mathbb{T}^d \times \mathbb{R}^d$. Moreover, for a.e. $(x,v) \in \mathbb{T}^d \times \mathbb{R}^d$, there holds

$$f_\star(x,v) = \rho_\star(x)\zeta(x,v).$$

Indeed, if $\rho_\star(x) > 0$ for some $x \in \mathbb{T}^d$, then the equality follows by definition of ζ. Conversely, if $\rho_\star(x) = 0$ for some $x \in \mathbb{T}^d$, then $\int_{\mathbb{R}^d} f_\star(x,v)\mathrm{d}v = 0$. By nonnegativity of $f_\star$, we infer that $f_\star(x,\cdot) = 0$ a.e., and thus the previous identity follows. Integrating such identity over the velocity space yields

$$\rho_\star(x) = \int_{\mathbb{R}^d} f_\star(x,v)\mathrm{d}v = \rho_\star(x)\int_{\mathbb{R}^d} \zeta(x,v)\mathrm{d}v \implies \int_{\mathbb{R}^d} \zeta(x,v)\mathrm{d}v = 1 \quad \text{for a.e. } x \in \mathbb{T}^d \text{ s.t. } \rho_\star(x) > 0.$$

If instead $x \in \mathbb{T}^d$ is such that $\rho_\star(x) = 0$, then $\zeta(x,v) = 0$ for all $v \in \mathbb{R}^d$ by definition. Therefore, we have

$$\int_{\mathbb{R}^d} \zeta(x,v)\mathrm{d}v = 0 \quad \text{for all } x \in \mathbb{T}^d \text{ s.t. } \rho_\star(x) = 0.$$

By Proposition 2.2.16, $f_\star \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ for large enough $C_\rho, C_\mathcal{E}, C_K > 0$. By Lemma 2.2.10, it follows that $f_\star \log f_\star \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$. Therefore, Fubini's theorem yields

$$\begin{aligned}
\mathcal{F}^u(f_\star) &= \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star \log f_\star \mathrm{d}x\mathrm{d}v + \frac{1}{2}\int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 f_\star \mathrm{d}x\mathrm{d}v + \frac{1}{2}\int_{\mathbb{T}^d} |\nabla_x \varphi_\star|^2 \mathrm{d}x \\
&= \int_{\mathbb{T}^d\times\mathbb{R}^d} (\rho_\star\zeta)\log(\rho_\star\zeta)\mathrm{d}x\mathrm{d}v + \frac{1}{2}\int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 \rho_\star\zeta \mathrm{d}x\mathrm{d}v + \frac{1}{2}\int_{\mathbb{T}^d} |\nabla_x \varphi_\star|^2 \mathrm{d}x \\
&= \int_{\mathbb{T}^d} \rho_\star \log \rho_\star \int_{\mathbb{R}^d} \zeta(x,v)\mathrm{d}v\mathrm{d}x + \int_{\mathbb{T}^d} \rho_\star \int_{\mathbb{R}^d} \left(\zeta\log\zeta + \frac{1}{2}|v|^2\zeta\right)\mathrm{d}v\mathrm{d}x + \frac{1}{2}\int_{\mathbb{T}^d} |\nabla_x\varphi_\star|^2\mathrm{d}x.
\end{aligned}$$

Here, $\varphi_\star$ is the unique zero-mean solution to the Poisson equation with forcing term $\rho_\star - \rho_0$. Adopting the standard convention $0\log 0 = 0$, the first term can be rewritten as

$$\begin{aligned}
\int_{\mathbb{T}^d} \rho_\star \log\rho_\star \int_{\mathbb{R}^d} \zeta(x,v)\mathrm{d}v\mathrm{d}x &= \int_{\{\rho_\star>0\}} \rho_\star\log\rho_\star \int_{\mathbb{R}^d} \zeta(x,v)\mathrm{d}v\mathrm{d}x + \int_{\{\rho_\star=0\}} \rho_\star\log\rho_\star\int_{\mathbb{R}^d}\zeta(x,v)\mathrm{d}v\mathrm{d}x \\
&= \int_{\{\rho_\star>0\}} \rho_\star\log\rho_\star\mathrm{d}x,
\end{aligned}$$

and thus

$$\mathcal{F}^u(f_\star) = \int_{\mathbb{T}^d} \rho_\star \log \rho_\star \mathrm{d}x + \int_{\mathbb{T}^d} \rho_\star \int_{\mathbb{R}^d} \left(\zeta \log \zeta + \frac{1}{2}|v|^2 \zeta \right) \mathrm{d}v\mathrm{d}x + \frac{1}{2} \int_{\mathbb{T}^d} |\nabla_x \varphi_\star|^2 \mathrm{d}x. \tag{2.32}$$

Using Lemma 2.2.9 with $f = \zeta$ and $\gamma = \mathrm{M}(v)$ yields

$$\zeta \log \zeta = \zeta \log \left(\frac{\zeta}{\mathrm{M}(v)} \right) + \zeta \log(\mathrm{M}(v)) \geq \zeta - \mathrm{M}(v) - \zeta \log \left((2\pi)^{\frac{d}{2}} \right) - \frac{1}{2}|v|^2 \zeta.$$

Therefore, by nonnegativity of $\rho_\star$, we compute

$$\begin{aligned} \int_{\mathbb{T}^d} \rho_\star \int_{\mathbb{R}^d} \left(\zeta \log \zeta + \frac{1}{2}|v|^2 \zeta \right) \mathrm{d}v\mathrm{d}x &= \int_{\{\rho_\star > 0\}} \rho_\star \int_{\mathbb{R}^d} \left(\zeta \log \zeta + \frac{1}{2}|v|^2 \zeta \right) \mathrm{d}v\mathrm{d}x \\ &\geq \int_{\{\rho_\star > 0\}} \rho_\star \int_{\mathbb{R}^d} \left(\zeta - \mathrm{M}(v) - \zeta \log \left((2\pi)^{\frac{d}{2}} \right) \right) \mathrm{d}v\mathrm{d}x = \boldsymbol{m} \log \left((2\pi)^{\frac{d}{2}} \right), \end{aligned}$$

where the last equality uses the mass constraint $\int_{\mathbb{T}^d} \rho_\star(x)\mathrm{d}x = \boldsymbol{m}$ together with

$$\int_{\mathbb{R}^d} \zeta(x,v)\mathrm{d}v = \int_{\mathbb{R}^d} \mathrm{M}(v)\mathrm{d}v = 1 \quad \text{for a.e. } x \in \mathbb{T}^d \text{ s.t. } \rho_\star(x) > 0.$$

Substituting this in (2.32), we obtain

$$\mathcal{F}^u(f_\star) \geq \int_{\mathbb{T}^d} \rho_\star \log \rho_\star \mathrm{d}x + \frac{d}{2} \boldsymbol{m} \log(2\pi) + \frac{1}{2} \int_{\mathbb{T}^d} |\nabla_x \varphi_\star|^2 \mathrm{d}x.$$

Since the other addenda do not depend on ζ, and $\mathrm{M}(v)$ achieves equality in the above lower bound, by minimality of $f_\star$ we infer that the minimiser has a Maxwellian distribution in the velocity space, that is, $f_\star(x,v) = \rho_\star(x)\mathrm{M}(v)$ a.e. Using this, the free energy reduces to a functional of the spatial density alone:

$$\mathcal{F}^u(f_\star) = \int_{\mathbb{T}^d} \rho_\star \log \rho_\star \mathrm{d}x + \frac{d}{2} \boldsymbol{m} \log(2\pi) + \frac{1}{2} \int_{\mathbb{T}^d} |\nabla_x \varphi_\star|^2 \mathrm{d}x.$$

We claim that $\rho_\star(x) = \frac{\boldsymbol{m}}{|\mathbb{T}^d|}$ minimises the above expression. Indeed, the unique zero-mean solution $\varphi_\star$ to the Poisson equation with forcing $\rho_\star - \rho_0 = 0$ (that is, the Laplace equation) is $\varphi_\star = 0$. Hence, the Dirichlet energy is zero, i.e., its minimum. Furthermore, by Jensen's inequality

$$\frac{1}{|\mathbb{T}^d|} \int_{\mathbb{T}^d} \rho_\star \log \rho_\star \mathrm{d}x \geq \left(\frac{1}{|\mathbb{T}^d|} \int_{\mathbb{T}^d} \rho_\star \mathrm{d}x \right) \log \left(\frac{1}{|\mathbb{T}^d|} \int_{\mathbb{T}^d} \rho_\star \mathrm{d}x \right) = \frac{\boldsymbol{m}}{|\mathbb{T}^d|} \log \left(\frac{\boldsymbol{m}}{|\mathbb{T}^d|} \right).$$

Since $\rho_\star = \frac{\boldsymbol{m}}{|\mathbb{T}^d|}$ achieves the equality, it is the unique minimiser. □

2.2.5 Necessity and Sufficiency of the Euler–Lagrange Equation for Optimality

We characterise the unique minimiser of the free energy functional variationally by means of the first variation method (see, e.g., [32] for an introduction).

Theorem 2.2.18. *Let $\boldsymbol{m} > 0$ and $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. The unique minimiser of $\mathcal{F}^u$ on $\tilde{\mathcal{A}}(\boldsymbol{m})$ is given by*

$$f_\star(x,v) = \boldsymbol{m} \frac{e^{-u(x)-\varphi_\star(x)}}{\int_{\mathbb{T}^d} e^{-u(y)-\varphi_\star(y)}\mathrm{d}y} \frac{e^{-\frac{|v|^2}{2}}}{(2\pi)^{\frac{d}{2}}} = \rho_\star(x)\mathrm{M}(v), \tag{2.33}$$

where $\varphi_\star$ is a weak solution to the Poisson–Boltzmann–Emden equation

$$-\Delta \varphi_\star(x) = \boldsymbol{m} \frac{e^{-u(x)-\varphi_\star(x)}}{\int_{\mathbb{T}^d} e^{-u(y)-\varphi_\star(y)}\mathrm{d}y} - \rho_0. \tag{2.34}$$

The proof is organised as follows. In Claim 2.2.19, we prove that the minimiser is strictly positive a.e. via a contradiction argument. Using this, we introduce a class of admissible perturbations of the minimiser Claim 2.2.20 and characterise the associated perturbation maps in Claim 2.2.21. Finally, using the energy identity established in Claim 2.2.22, we derive the Euler–Lagrange equation satisfied by the minimiser and deduce the Poisson–Boltzmann–Emden equation for the corresponding self-consistent potential.

Claim 2.2.19. *The unique minimiser of $\mathcal{F}^u$ on $\tilde{\mathcal{A}}(\boldsymbol{m})$ is strictly positive a.e.*

Proof. Let $f_\star$ be the minimiser of $\mathcal{F}^u$ on $\tilde{\mathcal{A}}(\boldsymbol{m})$, whose existence and uniqueness follow from Proposition 2.2.16. Define its positivity set by

$$A := \{(x,v) \in \mathbb{T}^d \times \mathbb{R}^d : f_\star(x,v) > 0\} = f_\star^{-1}((0,+\infty)),$$

where A is measurable by measurability of $f_\star$. Since $f_\star \geq 0$ a.e. and

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} f_\star \mathrm{d}x\mathrm{d}v = \boldsymbol{m} > 0,$$

it follows that $|A| > 0$. It suffices to prove that $|A^{\mathrm{c}}| = 0$, that is, up to null sets, the positivity set A is equal to the whole domain $\mathbb{T}^d \times \mathbb{R}^d$. To this end, we argue by contradiction, and construct an admissible perturbation of the minimiser that increases the mass on A^{c} and decreases it on A. Such perturbation strictly decreases the value of the free energy functional, contradicting the minimality of $f_\star$.

Step 1. Admissible Perturbation. Assume by contradiction that $|A^{\mathrm{c}}| > 0$, that is, $f_\star = 0$ on a (possibly unbounded) set of positive measure. By definition, there holds

$$A^{\mathrm{c}} = \{(x,v) \in \mathbb{T}^d \times \mathbb{R}^d : f_\star(x,v) = 0\} = f_\star^{-1}(\{0\}).$$

Since $f_\star$ is measurable, the set A^{c} is measurable. Hence, the inner regularity of the Lebesgue measure (see [10]) yields the identity

$$|A^{\mathrm{c}}| = \sup\{|K| : K \subset A^{\mathrm{c}},\ K \text{ compact}\}.$$

Since $|A^{\mathrm{c}}| > 0$, there exists a compact subset $K \subset A^{\mathrm{c}}$ such that $0 < |K| < +\infty$. Fix any such K, and define

$$g(x,v) := \boldsymbol{m} \frac{e^{-\frac{|v|^2}{2}}}{\int_K e^{-\frac{|z|^2}{2}} \mathrm{d}w\mathrm{d}z} \chi_K(x,v),$$

where χ_K is the characteristic function of K, and $\int_K e^{-\frac{|z|^2}{2}} \mathrm{d}w\mathrm{d}z > 0$ since $|K| > 0$ and $e^{-\frac{|z|^2}{2}} > 0$. We claim that, for sufficiently large $C_\rho, C_\mathcal{E}, C_K$, both g and $f_\star$ belong to $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$. Indeed, by Proposition 2.2.16, there holds $f_\star \in \mathcal{A}(\bar{C}_\rho, \bar{C}_\mathcal{E}, \bar{C}_K; \boldsymbol{m})$ for large enough $\bar{C}_\rho, \bar{C}_\mathcal{E}, \bar{C}_K$. Therefore, it remains to prove that $g \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ for some constants $C_\rho \geq \bar{C}_\rho$, $C_\mathcal{E} \geq \bar{C}_\mathcal{E}$ and $C_K \geq \bar{C}_K$. The nonnegativity of g is immediate, and the mass constraint is satisfied by construction. Moreover, the spatial density of g satisfies

$$\rho_g = \int_{\mathbb{R}^d} g \mathrm{d}v = \frac{\boldsymbol{m}}{\int_K e^{-\frac{|z|^2}{2}} \mathrm{d}z\mathrm{d}w} \int_{\mathbb{R}^d} e^{-\frac{|v|^2}{2}} \chi_K(x,v) \mathrm{d}v.$$

In particular, ρ_g is bounded. Indeed, since $0 \leq \chi_K \leq 1$ for all $(x,v) \in \mathbb{T}^d \times \mathbb{R}^d$, we estimate

$$0 \leq \rho_g \leq \frac{\boldsymbol{m}}{\int_K e^{-\frac{|z|^2}{2}} \mathrm{d}z\mathrm{d}w} \int_{\mathbb{R}^d} e^{-\frac{|v|^2}{2}} \mathrm{d}v = (2\pi)^{\frac{d}{2}} \frac{\boldsymbol{m}}{\int_K e^{-\frac{|z|^2}{2}} \mathrm{d}z\mathrm{d}w} \quad \text{a.e.}$$

Since $L^\infty(\mathbb{T}^d) \hookrightarrow H^{-1}(\mathbb{T}^d)$, it follows that $\rho_g \in H^{-1}(\mathbb{T}^d)$. We next compute

$$E^K(g) = \frac{1}{2} \frac{\boldsymbol{m}}{\int_K e^{-\frac{|v|^2}{2}} \mathrm{d}x\mathrm{d}v} \int_K |v|^2 e^{-\frac{|v|^2}{2}} \mathrm{d}x\mathrm{d}v$$

and

$$\begin{aligned}\mathcal{E}(g) &= \int_K \boldsymbol{m}\frac{e^{-\frac{|v|^2}{2}}}{\int_K e^{-\frac{|v|^2}{2}}\mathrm{d}x\mathrm{d}v}\log\left(\boldsymbol{m}\frac{e^{-\frac{|v|^2}{2}}}{\int_K e^{-\frac{|v|^2}{2}}\mathrm{d}x\mathrm{d}v}\right)\mathrm{d}x\mathrm{d}v \\ &= \boldsymbol{m}\log\left(\frac{\boldsymbol{m}}{\int_K e^{-\frac{|v|^2}{2}}\mathrm{d}x\mathrm{d}v}\right) - \frac{\boldsymbol{m}}{2\int_K e^{-\frac{|v|^2}{2}}\mathrm{d}x\mathrm{d}v}\int_K |v|^2 e^{-\frac{|v|^2}{2}}\mathrm{d}x\mathrm{d}v.\end{aligned}$$

Fubini's theorem yields the estimate

$$0 < \int_K |v|^2 e^{-\frac{|v|^2}{2}}\mathrm{d}x\mathrm{d}v \leq \int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 e^{-\frac{|v|^2}{2}}\mathrm{d}x\mathrm{d}v \leq |\mathbb{T}^d|\int_{\mathbb{R}^d} |v|^2 e^{-\frac{|v|^2}{2}}\mathrm{d}v,$$

where the right-hand side is finite since $e^{-\frac{|v|^2}{2}}$ has finite moments. Thus, both the kinetic energy and the entropy of g are finite. Consequently, $g \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ for suitable constants $C_\rho, C_\mathcal{E}, C_K$, which can be chosen such that $C_\rho \geq \bar{C}_\rho$, $C_\mathcal{E} \geq \bar{C}_\mathcal{E}$ and $C_K \geq \bar{C}_K$.

Step 2. Perturbation Map. For $\varepsilon \in [0,1]$, we define

$$f_\varepsilon := (1-\varepsilon)f_\star + \varepsilon g.$$

By the convexity of $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ (see Lemma 2.2.13), $f_\varepsilon \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ for all $\varepsilon \in [0,1]$. Therefore, the perturbation map $\Phi_g : [0,1] \to \mathbb{R}$ given by

$$\Phi_g(\varepsilon) := \mathcal{F}^u(f_\varepsilon)$$

is well-defined. We next compute $\Phi_g(\varepsilon)$ explicitly for any $\varepsilon \in [0,1]$. To this end, we treat each term separately.

Entropy. Since $f_\star$ is supported on A and g is supported on $K \subset A^\mathrm{c}$, we obtain

$$\begin{aligned}\mathcal{E}(f_\varepsilon) &= \int_A f_\varepsilon \log f_\varepsilon \mathrm{d}x\mathrm{d}v + \int_K f_\varepsilon \log f_\varepsilon \\ &= \int_A (1-\varepsilon)f_\star \log\left((1-\varepsilon)f_\star\right)\mathrm{d}x\mathrm{d}v + \int_K \varepsilon g \log(\varepsilon g) \\ &= (1-\varepsilon)\log(1-\varepsilon)\boldsymbol{m} + (1-\varepsilon)\mathcal{E}(f_\star) + \boldsymbol{m}\varepsilon\log\varepsilon + \varepsilon\mathcal{E}(g).\end{aligned}$$

Kinetic and Potential energy. By linearity of the kinetic and potential energy, we have

$$\begin{aligned}E^K(f_\varepsilon) &= (1-\varepsilon)E^K(f_\star) + \varepsilon E^K(g), \\ E^u(f_\varepsilon) &= (1-\varepsilon)E^u(f_\star) + \varepsilon E^u(g).\end{aligned}$$

Dirichlet energy. By linearity of the inverse Laplacian (see Proposition 2.1.4), we write

$$\varphi_{f_\varepsilon} = (-\Delta)^{-1}(\rho_{f_\varepsilon} - \rho_0) = (1-\varepsilon)(-\Delta)^{-1}(\rho_\star - \rho_0) + \varepsilon(-\Delta)^{-1}(\rho_g - \rho_0) = (1-\varepsilon)\varphi_\star + \varepsilon\varphi_g.$$

We therefore compute

$$\begin{aligned}E^D(\varphi_{f_\varepsilon}) &= \frac{1}{2}\int_{\mathbb{T}^d} |\nabla\varphi_{f_\varepsilon}|^2\mathrm{d}x \\ &= \frac{1}{2}\int_{\mathbb{T}^d} |(1-\varepsilon)\nabla\varphi_\star + \varepsilon\nabla\varphi_g|^2\mathrm{d}x \\ &= \frac{(1-\varepsilon)^2}{2}\int_{\mathbb{T}^d} |\nabla\varphi_\star|^2\mathrm{d}x + \varepsilon(1-\varepsilon)\int_{\mathbb{T}^d} \nabla\varphi_\star\cdot\nabla\varphi_g\mathrm{d}x + \frac{\varepsilon^2}{2}\int_{\mathbb{T}^d} |\nabla\varphi_g|^2\mathrm{d}x \\ &= \frac{1}{2}(1-\varepsilon)^2 E^D(\varphi_\star) + \varepsilon(1-\varepsilon)\int_{\mathbb{T}^d} \nabla\varphi_\star\cdot\nabla\varphi_g\mathrm{d}x + \frac{\varepsilon^2}{2}E^D(\varphi_g).\end{aligned}$$

We claim that the second term in the last identity is finite. Indeed, since $\varphi_\star, \varphi_g \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$, Hölder's inequality yields

$$\int_{\mathbb{T}^d} |\nabla\varphi_\star \cdot \nabla\varphi_g| \mathrm{d}x \leq \sum_{i=1}^d \left\| \frac{\partial\varphi_\star}{\partial x_i} \right\|_{L^2(\mathbb{T}^d)} \left\| \frac{\partial\varphi_g}{\partial x_i} \right\|_{L^2(\mathbb{T}^d)} < +\infty.$$

Combining the above calculations, for any $\varepsilon \in [0,1]$ there holds

$$\begin{aligned}\Phi_g(\varepsilon) = (1-\varepsilon)\log(1-\varepsilon)\boldsymbol{m} + (1-\varepsilon)\mathcal{E}(f_\star) + \boldsymbol{m}\varepsilon\log\varepsilon + \varepsilon\mathcal{E}(g) + (1-\varepsilon)E^K(f_\star) + \varepsilon E^K(g) \\ + (1-\varepsilon)E^u(f_\star) + \varepsilon E^u(g) + \frac{1}{2}(1-\varepsilon)^2 E^D(\varphi_\star) + \varepsilon(1-\varepsilon)\int_{\mathbb{T}^d} \nabla\varphi_\star \cdot \nabla\varphi_g \mathrm{d}x + \frac{\varepsilon^2}{2} E^D(\varphi_g).\end{aligned}$$

Furthermore, for any $\varepsilon \in (0,1)$ we compute

$$\begin{aligned}\Phi_g'(\varepsilon) = -\boldsymbol{m}(1+\log(1-\varepsilon)) - \mathcal{E}(f_\star) + \boldsymbol{m}(1+\log\varepsilon) + \mathcal{E}(g) - E^K(f_\star) + E^K(g) \\ - E^u(f_\star) + E^u(g) - (1-\varepsilon)E^D(\varphi_\star) + (1-2\varepsilon)\int_{\mathbb{T}^d} \nabla\varphi_\star \cdot \nabla\varphi_g \mathrm{d}x + \varepsilon E^D(\varphi_g).\end{aligned}$$

By the boundedness of all addenda except the term $\boldsymbol{m}(1+\log\varepsilon)$, we thus conclude that

$$\lim_{\varepsilon\downarrow 0} \Phi'(\varepsilon) = -\infty.$$

However this leads to a contradiction with the minimality of $f_\star$, as it implies that for small $\varepsilon > 0$ there holds $f_\varepsilon \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m}) \subset \tilde{\mathcal{A}}(\boldsymbol{m})$ and

$$\mathcal{F}^u(f_\star + \varepsilon g) = \Phi(\varepsilon) < \Phi(0) = \mathcal{F}^u(f_\star).$$

We therefore conclude that $|A^{\mathrm{c}}| = 0$. □

Claim 2.2.20. *Let $f_\star$ be the unique minimiser of $\mathcal{F}^u$ on $\tilde{\mathcal{A}}(\boldsymbol{m})$. For any test function $\psi \in \mathcal{C}_c^\infty(\mathbb{T}^d \times \mathbb{R}^d)$, there exist $\tilde{\varepsilon}_\psi > 0$ and costants $C_\rho, C_\mathcal{E}, C_K$ such that the perturbed function*

$$f_\varepsilon(x,v) := Z_\psi(\varepsilon) f_\star(x,v)\big(1+\varepsilon\psi(x,v)\big), \qquad Z_\psi(\varepsilon) := \frac{\boldsymbol{m}}{\int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star(1+\varepsilon\psi)\mathrm{d}x\mathrm{d}v}, \tag{2.35}$$

is well-defined and belongs to $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ for all $\varepsilon \in (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi)$. Furthermore, the bounds

$$\frac{1}{e} \leq 1 + \varepsilon\psi \leq \frac{2e-1}{e} \ \ a.e., \qquad \frac{e}{2e-1} \leq Z_\psi(\varepsilon) \leq e, \tag{2.36}$$

hold for all $\varepsilon \in (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi)$, and

$$\rho_{f_\star\psi} = \int_{\mathbb{R}^d} f_\star\psi \mathrm{d}v \in H^{-1}(\mathbb{T}^d). \tag{2.37}$$

Proof. If $\psi = 0$, the claim is trivial. Therefore, assume without loss of generality that $\psi \in \mathcal{C}_c^\infty(\mathbb{T}^d\times\mathbb{R}^d)\setminus\{0\}$, and set

$$\tilde{\varepsilon}_\psi := \frac{1}{\|\psi\|_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)}} \frac{e-1}{e}.$$

Since $\psi \neq 0$, we have $\tilde{\varepsilon}_\psi > 0$. We first show (2.36). To this end, for all $\varepsilon \in (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi)$ we estimate

$$\frac{1}{e} = 1 - \tilde{\varepsilon}_\psi\|\psi\|_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)} \leq 1 + \varepsilon\psi \leq 1 + \tilde{\varepsilon}_\psi\|\psi\|_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)} = \frac{2e-1}{e}.$$

Combining this with the nonnegativity of $f_\star$ yields

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star(1+\varepsilon\psi)\mathrm{d}x\mathrm{d}v \geq \frac{1}{e}\int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star \mathrm{d}x\mathrm{d}v = \frac{\boldsymbol{m}}{e} > 0$$

and

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star(1+\varepsilon\psi)\mathrm{d}x\mathrm{d}v \le \frac{2e-1}{e}\int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star \mathrm{d}x\mathrm{d}v = \boldsymbol{m}\frac{2e-1}{e}.$$

By definition of $Z_\psi(\varepsilon)$, this implies

$$\frac{e}{2e-1} \le Z_\psi(\varepsilon) \le e.$$

We next show that, for any $\varepsilon \in (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi)$, the perturbed function f_ε in (2.35) satisfies (2.37) and belongs to $\tilde{\mathcal{A}}(\boldsymbol{m})$. To this end, let $\varepsilon \in (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi)$ be arbitrary. We treat each constraint separately.

Nonnegativity. Using (2.36) and the nonnegativity of $f_\star$, we have

$$f_\varepsilon \ge \frac{e}{2e-1}\frac{1}{e}f_\star = \frac{f_\star}{2e-1} \ge 0 \quad \text{a.e.}$$

Mass constraint. The mass constraint is satisfied by construction.

Spatial density constraint. We compute

$$\rho_{f_\varepsilon} = \int_{\mathbb{R}^d} f_\varepsilon \mathrm{d}v = Z_\psi(\varepsilon)\left(\int_{\mathbb{R}^d} f_\star \mathrm{d}v + \varepsilon\int_{\mathbb{R}^d} f_\star\psi \mathrm{d}v\right) = Z_\psi(\varepsilon)\left(\rho_\star + \varepsilon\rho_{f_\star\psi}\right). \tag{2.38}$$

Since $\rho_\star \in H^{-1}(\mathbb{T}^d)$ and the normalisation factor Z_ψ is bounded, it suffices to show (2.37). To this end, define

$$T_{\rho_{f_\star\psi}} : \mathcal{C}^\infty(\mathbb{T}^d) \to \mathbb{R},\ \xi \mapsto \int_{\mathbb{T}^d} \rho_{f_\star\psi}\xi \mathrm{d}x.$$

We first verify that $T_{\rho_{f_\star\psi}}$ is well-defined. For any $\xi \in \mathcal{C}^\infty(\mathbb{T}^d)$, we have

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} |f_\star\psi\xi|\mathrm{d}x\mathrm{d}v \le \|\psi\|_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)}\|\xi\|_{L^\infty(\mathbb{T}^d)}\int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star \mathrm{d}x\mathrm{d}v < +\infty,$$

so that $f_\star\psi\xi \in L^1(\mathbb{T}^d\times\mathbb{R}^d)$. Hence, by Fubini's theorem,

$$T_{\rho_{f_\star\psi}}(\xi) = \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star\psi\xi \mathrm{d}x\mathrm{d}v$$

is well-defined. Moreover, using again Fubini's theorem and the non-negativity of $f_\star$, we obtain

$$\begin{aligned}
|T_{\rho_{f_\star\psi}}(\xi)| &\le \int_{\mathbb{T}^d\times\mathbb{R}^d} |f_\star\psi\xi|\mathrm{d}x\mathrm{d}v \\
&\le \|\psi\|_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)}\int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star|\xi|\mathrm{d}x\mathrm{d}v \\
&= \|\psi\|_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)}\int_{\mathbb{T}^d} \rho_\star|\xi|\mathrm{d}x \\
&= \|\psi\|_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)}\langle\rho_\star, |\xi|\rangle_{H^{-1}(\mathbb{T}^d),H^1(\mathbb{T}^d)}.
\end{aligned}$$

Since $\rho_\star$ is the spatial density of the unique minimiser of $\mathcal{F}^u$ on $\tilde{\mathcal{A}}(\boldsymbol{m})$, there holds $\rho_\star \in H^{-1}(\mathbb{T}^d)$. Moreover, using that $\xi \in \mathcal{C}^\infty(\mathbb{T}^d)$ and $\mathcal{C}^\infty(\mathbb{T}^d) \hookrightarrow H^1(\mathbb{T}^d)$, Claim 2.2.5 and Remark 2.2.4, yield

$$\int_{\mathbb{T}^d} \rho_\star|\xi|\mathrm{d}x = \langle\rho_\star, |\xi|\rangle_{H^{-1}(\mathbb{T}^d),H^1(\mathbb{T}^d)} \le \|\rho_\star\|_{H^{-1}(\mathbb{T}^d)}\|\xi\|_{H^1(\mathbb{T}^d)}.$$

Combining the above estimates, we conclude that

$$|T_{\rho_{f_\star\psi}}(\xi)| \le \|\psi\|_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)}\|\rho_\star\|_{H^{-1}(\mathbb{T}^d)}\|\xi\|_{H^1(\mathbb{T}^d)} \qquad \forall \xi \in \mathcal{C}^\infty(\mathbb{T}^d).$$

Therefore, $T_{\rho_{f_\star\psi}}$ is continuous on $\mathcal{C}^\infty(\mathbb{T}^d)$ with respect to the $H^1(\mathbb{T}^d)$ norm. By Friedrichs' Lemma (see Corollary A.2.2), $\mathcal{C}^\infty(\mathbb{T}^d)$ is dense in $H^1(\mathbb{T}^d)$. Therefore, $T_{\rho_{f_\star\psi}}$ extends uniquely to a continuous linear functional on $H^1(\mathbb{T}^d)$, that is, $T_{\rho_{f_\star\psi}} \in H^{-1}(\mathbb{T}^d)$. Identifying such distribution with $\rho_{f_\star\psi}$, from (2.36) and (2.38), we conclude that

$$\|\rho_{f_\varepsilon}\|_{H^{-1}(\mathbb{T}^d)} \leq e\left(\|\rho_\star\|_{H^{-1}(\mathbb{T}^d)} + \tilde{\varepsilon}_\psi \|\rho_{f_\star\psi}\|_{H^{-1}(\mathbb{T}^d)}\right),$$

and the upper bound is uniform in $\varepsilon \in (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi)$.

Finite entropy. We compute

$$\begin{aligned}
\mathcal{E}(f_\varepsilon) &= \int_{\mathbb{T}^d\times\mathbb{R}^d} \left(Z_\psi(\varepsilon) f_\star(1+\varepsilon\psi)\right) \log\left(Z_\psi(\varepsilon) f_\star(1+\varepsilon\psi)\right) \mathrm{d}x\mathrm{d}v \\
&= \int_{\mathbb{T}^d\times\mathbb{R}^d} Z_\psi(\varepsilon)(1+\varepsilon\psi) f_\star \log f_\star \mathrm{d}x\mathrm{d}v \\
&\qquad + \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star(1+\varepsilon\psi) Z_\psi(\varepsilon) \log\left(Z_\psi(\varepsilon)\right) \mathrm{d}x\mathrm{d}v \\
&\qquad\quad + \int_{\mathbb{T}^d\times\mathbb{R}^d} Z_\psi(\varepsilon) f_\star(1+\varepsilon\psi) \log(1+\varepsilon\psi) \mathrm{d}x\mathrm{d}v.
\end{aligned} \tag{2.39}$$

By Proposition 2.2.17, $f_\star \in \mathcal{A}(\bar{C}_\rho, \bar{C}_\mathcal{E}, \bar{C}_K; \boldsymbol{m})$ for large enough $\bar{C}_\rho, \bar{C}_\mathcal{E}, \bar{C}_K$. Therefore, by Lemma 2.2.10 $f_\star \log f_\star \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$. Combining this with the bounds in (2.36), the first term satisfies

$$\left| \int_{\mathbb{T}^d\times\mathbb{R}^d} Z_\psi(\varepsilon)\left(1+\varepsilon\psi\right) f_\star \log f_\star \mathrm{d}x\mathrm{d}v \right| \leq (2e-1)\|f_\star \log f_\star\|_{L^1(\mathbb{T}^d\times\mathbb{R}^d)}.$$

For the second term, the bounds (2.36) imply that $|\log(Z_\psi(\varepsilon))| \leq 1$, and hence we estimate

$$\left| \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star(1+\varepsilon\psi) Z_\psi(\varepsilon) \log\left(Z_\psi(\varepsilon)\right) \mathrm{d}x\mathrm{d}v \right| \leq \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star(1+\varepsilon\psi) Z_\psi(\varepsilon) \mathrm{d}x\mathrm{d}v = \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\varepsilon \mathrm{d}x\mathrm{d}v = \boldsymbol{m}.$$

For the last term, from (2.36) and the monotonicity of $t \mapsto t\log t$ on $t \geq \frac{1}{e}$, we obtain

$$\frac{1}{e}\log\frac{1}{e} \leq (1+\varepsilon\psi)\log(1+\varepsilon\psi) \leq \frac{2e-1}{e}\log\frac{2e-1}{e}.$$

Combining this with the nonnegativity of $f_\star$ and the bounds in (2.36) yields

$$\frac{1}{2e-1}\log\frac{1}{e}\boldsymbol{m} \leq \int_A f_\star\left(1+\varepsilon\psi\right)\log\left(1+\varepsilon\psi\right)\mathrm{d}x\mathrm{d}v \leq (2e-1)\log\frac{2e-1}{e}\boldsymbol{m},$$

Therefore, $\mathcal{E}(f_\varepsilon)$ is well-defined and finite:

$$|\mathcal{E}(f_\varepsilon)| \leq (2e-1)\|f_\star \log f_\star\|_{L^1(\mathbb{T}^d\times\mathbb{R}^d)} + \boldsymbol{m} + (2e-1)\log\frac{2e-1}{e}\boldsymbol{m}.$$

Moreover, the upper bound is uniform in $\varepsilon \in (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi)$.

Finite kinetic energy. Using the nonnegativity of $f_\star$ and (2.36) yields

$$0 \leq E^K(f_\varepsilon) \leq (2e-1)E^K(f_\star),$$

and the upper bound is uniform in $\varepsilon \in (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi)$.

Therefore, $f_\varepsilon \in \tilde{\mathcal{A}}(\boldsymbol{m})$ for all $|\varepsilon| < \tilde{\varepsilon}_\psi$. In particular, since the upper bounds on the $H^{-1}(\mathbb{T}^d)$ norm of the spatial density, the entropy and the kinetic energy are uniform in ε, it follows that, for large enough $C_\rho, C_\mathcal{E}, C_K$, $f_\varepsilon \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ for all $|\varepsilon| < \tilde{\varepsilon}$. □

Claim 2.2.21. *Let $f_\star$ be the unique minimiser of $\mathcal{F}^u$ on $\tilde{\mathcal{A}}(\boldsymbol{m})$. For all $\psi \in \mathcal{C}_c^\infty(\mathbb{T}^d \times \mathbb{R}^d)$, let $\tilde{\varepsilon}_\psi$ and $C_\rho, C_\mathcal{E}, C_K$ be as in Claim 2.2.20, and define the perturbation map $\Phi_\psi : (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi) \to \mathbb{R}$ by*

$$\Phi_\psi(\varepsilon) := \mathcal{F}^u(f_\varepsilon) = \mathcal{F}^u\big(Z_\psi(\varepsilon) f_\star(1+\varepsilon\psi)\big), \tag{2.40}$$

where Z_ψ is as in (2.35). Then,

(i) $Z_\psi, \Phi_\psi \in \mathcal{C}((-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi))$;

(ii) $Z_\psi \in \mathcal{C}^1((-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi))$, *and Φ_ψ is differentiable at* 0.

Proof of (i). The map Φ_ψ is well-defined since $\mathcal{F}^u$ is well-defined on $\tilde{\mathcal{A}}(\boldsymbol{m})$ by Lemma 2.2.14, and $f_\varepsilon \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ for all $\varepsilon \in (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi)$ by Claim 2.2.20. Let next $\varepsilon \in (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi)$ be arbitrary, and let $\{\varepsilon_k\}_{k\in\mathbb{N}} \subset (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi)$ be such that $\varepsilon_k \to \varepsilon$. We use Lebesgue's dominated convergence theorem (see, e.g., Theorem 2.8.1 in [10]) to show that

$$\lim_{k\to\infty} Z_\psi(\varepsilon_k) = Z_\psi(\varepsilon) \qquad \text{and} \qquad \lim_{k\to\infty} \Phi_\psi(\varepsilon_k) = \Phi_\psi(\varepsilon). \tag{2.41}$$

We first consider $Z_\psi(\varepsilon_k)$. Since $\varepsilon_k \to \varepsilon$, we have

$$f_\star(1+\varepsilon_k\psi) \to f_\star(1+\varepsilon\psi) \quad \text{a.e.}$$

Furthermore, from (2.36), we have

$$|f_\star(1+\varepsilon_k\psi)| \le \frac{2e-1}{e} f_\star \in L^1(\mathbb{T}^d \times \mathbb{R}^d) \quad \text{for all } k \in \mathbb{N},$$

and similarly for ε. Therefore, the dominated convergence theorem yields

$$\lim_{k\to\infty} \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star(1+\varepsilon_k\psi)\mathrm{d}x\mathrm{d}v = \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star(1+\varepsilon\psi)\mathrm{d}x\mathrm{d}v,$$

and hence $Z_\psi(\varepsilon_k) \to Z_\psi(\varepsilon)$. We next address each addendum of $\mathcal{F}^u$ separately. From (2.39), we have

$$\begin{aligned}\mathcal{E}(f_{\varepsilon_k}) = &\int_{\mathbb{T}^d\times\mathbb{R}^d} Z_\psi(\varepsilon_k)(1+\varepsilon_k\psi) f_\star \log f_\star \mathrm{d}x\mathrm{d}v \\ &+ \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star(1+\varepsilon_k\psi) Z_\psi(\varepsilon_k) \log\left(Z_\psi(\varepsilon_k)\right) \mathrm{d}x\mathrm{d}v \\ &+ \int_{\mathbb{T}^d\times\mathbb{R}^d} Z_\psi(\varepsilon_k) f_\star(1+\varepsilon_k\psi)\log(1+\varepsilon_k\psi)\mathrm{d}x\mathrm{d}v,\end{aligned} \tag{2.42}$$

and similarly for $\mathcal{E}(f_\varepsilon)$. Since $\varepsilon_k \to \varepsilon$ and $Z_\psi(\varepsilon_k) \to Z_\psi(\varepsilon)$, for a.e. $(x,v) \in \mathbb{T}^d \times \mathbb{R}^d$ we have

$$\begin{aligned} Z_\psi(\varepsilon_k)(1+\varepsilon_k\psi) f_\star \log f_\star &\to Z_\psi(\varepsilon)(1+\varepsilon\psi) f_\star \log f_\star, \\ f_\star(1+\varepsilon_k\psi) Z_\psi(\varepsilon_k) \log\left(Z_\psi(\varepsilon_k)\right) &\to f_\star(1+\varepsilon\psi) Z_\psi(\varepsilon) \log\left(Z_\psi(\varepsilon)\right), \\ Z_\psi(\varepsilon_k) f_\star(1+\varepsilon_k\psi)\log(1+\varepsilon_k\psi) &\to Z_\psi(\varepsilon) f_\star(1+\varepsilon\psi)\log(1+\varepsilon\psi). \end{aligned}$$

Furthermore, using (2.36), together with the monotonicity of $t \mapsto t\log t$, we estimate

$$\begin{aligned} \left|Z_\psi(\varepsilon_k)(1+\varepsilon_k\psi) f_\star \log f_\star\right| &\le (2e-1)|f_\star \log f_\star| \in L^1(\mathbb{T}^d\times\mathbb{R}^d), \\ \left|f_\star(1+\varepsilon_k\psi) Z_\psi(\varepsilon_k) \log\left(Z_\psi(\varepsilon_k)\right)\right| &\le (2e-1) f_\star \in L^1(\mathbb{T}^d\times\mathbb{R}^d), \\ \left|Z_\psi(\varepsilon_k) f_\star(1+\varepsilon_k\psi)\log(1+\varepsilon_k\psi)\right| &\le (2e-1)\log\frac{2e-1}{e} f_\star \in L^1(\mathbb{T}^d\times\mathbb{R}^d), \end{aligned}$$

and similarly for ε. Therefore, taking the limit as $k \to \infty$ in (2.42), the dominated convergence theorem yields

$$\lim_{k\to\infty} \mathcal{E}(f_{\varepsilon_k}) = \mathcal{E}(f_\varepsilon).$$

For the potential and kinetic energy, continuity is immediate. Indeed, since $\varepsilon_k \to \varepsilon$ and $Z_\psi(\varepsilon_k) \to Z_\psi(\varepsilon)$, by linearity it follows that

$$E^u(f_{\varepsilon_k}) = Z_\psi(\varepsilon_k)\Big(E^u(f_\star) + \varepsilon_k E^u(f_\star\psi)\Big) \to Z_\psi(\varepsilon)\Big(E^u(f_\star) + \varepsilon E^u(f_\star\psi)\Big) = E^u(f_\varepsilon),$$

and similarly

$$E^K(f_{\varepsilon_k}) = Z_\psi(\varepsilon_k)\Big(E^K(f_\star) + \varepsilon_k E^K(f_\star\psi)\Big) \to Z_\psi(\varepsilon)\Big(E^K(f_\star) + \varepsilon E^K(f_\star\psi)\Big) = E^K(f_\varepsilon),$$

where we used that $E^u(f_\star\psi)$ and $E^K(f_\star\psi)$ are well-defined since $\psi \in \mathcal{C}^\infty_c(\mathbb{T}^d \times \mathbb{R}^d)$. We finally address the Dirichlet energy. By linearity of the map $f \mapsto \rho_f$, the triangle inequality yields

$$\|\rho_{f_{\varepsilon_k}} - \rho_{f_\varepsilon}\|_{H^{-1}(\mathbb{T}^d)} \le |Z_\psi(\varepsilon_k) - Z_\psi(\varepsilon)|\|\rho_\star\|_{H^{-1}(\mathbb{T}^d)} + |Z_\psi(\varepsilon_k)\varepsilon_k - Z_\psi(\varepsilon)\varepsilon|\|\rho_{f_\star\psi}\|_{H^{-1}(\mathbb{T}^d)},$$

where we used that $\rho_{f_\star\psi} \in H^{-1}(\mathbb{T}^d)$ (see Claim 2.2.20). Since $\varepsilon_k \to \varepsilon$ and $Z_\psi(\varepsilon_k) \to Z_\psi(\varepsilon)$, we deduce that $\rho_{f_{\varepsilon_k}} \to \rho_{f_\varepsilon}$ strongly in $H^{-1}(\mathbb{T}^d)$. Since $f_\varepsilon, f_{\varepsilon_k} \in \tilde{\mathcal{A}}(\boldsymbol{m})$ for all $k \in \mathbb{N}$, this implies that

$$\rho_{f_{\varepsilon_k}} - \rho_0 \to \rho_{f_\varepsilon} - \rho_0 \quad \text{strongly in } H^{-1}_{\mathrm{zm}}(\mathbb{T}^d).$$

By continuity of the inverse Laplacian $(-\Delta)^{-1} : H^{-1}_{\mathrm{zm}}(\mathbb{T}^d) \to H^1_{\mathrm{zm}}(\mathbb{T}^d)$ (see Proposition 2.1.4), we have

$$\varphi_{f_{\varepsilon_k}} = (-\Delta)^{-1}(\rho_{f_{\varepsilon_k}} - \rho_0) \to (-\Delta)^{-1}\left(\rho_{f_\varepsilon} - \rho_0\right) = \varphi_{f_\varepsilon} \quad \text{strongly in } H^1_{\mathrm{zm}}(\mathbb{T}^d).$$

The definition of the Sobolev norm thus yields

$$E^D(\varphi_{f_{\varepsilon_k}}) = \frac{1}{2}\|\nabla\varphi_{f_{\varepsilon_k}}\|^2_{L^2(\mathbb{T}^d)^d} \to \frac{1}{2}\|\nabla\varphi_{f_\varepsilon}\|^2_{L^2(\mathbb{T}^d)^d} = E^D(\varphi_{f_\varepsilon}).$$

Combining the continuity of Z_ψ and the four addenda in Φ_ψ along the sequence $\{\varepsilon_k\}_{k\in\mathbb{N}}$, we conclude (2.41). By arbitrariness of the sequence and of $\varepsilon \in (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi)$, it follows that Z_ψ and Φ_ψ are continuous. □

Proof of (ii). We apply the dominated convergence theorem to show that Z_ψ is continuously differentiable and Φ_ψ is differentiable at 0. Each term is treated separately.

- **Normalisation Factor** Z_ψ**.** Let $\varepsilon \in (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi) \setminus \{0\}$ be arbitrary. We compute

$$\frac{\int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star(1+\varepsilon\psi)\mathrm{d}x\mathrm{d}v - \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star \mathrm{d}x\mathrm{d}v}{\varepsilon} = \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star\psi \mathrm{d}x\mathrm{d}v.$$

  Therefore, applying the quotient rule yields

$$\frac{\mathrm{d}Z_\psi}{\mathrm{d}\varepsilon}(\varepsilon) = -\frac{Z_\psi(\varepsilon)^2}{\boldsymbol{m}} \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star\psi \mathrm{d}x\mathrm{d}v.$$

  Since Z_ψ is continuous by part (i), it follows that $Z'_\psi(\varepsilon)$ is well-defined and continuous. In particular, using (2.36) yields

$$\|Z'_\psi\|_{L^\infty((-\tilde{\varepsilon}_\psi,\tilde{\varepsilon}_\psi))} \le \frac{e^2}{\boldsymbol{m}}\|f_\star\psi\|_{L^1(\mathbb{T}^d\times\mathbb{R}^d)} < +\infty.$$

  Furthermore, using $Z_\psi(0) = 1$ yields

$$\frac{\mathrm{d}Z_\psi}{\mathrm{d}\varepsilon}(0) = -\frac{1}{\boldsymbol{m}} \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star\psi \mathrm{d}x\mathrm{d}v. \tag{2.43}$$

- **Entropy.** For any $\varepsilon \in (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi) \setminus \{0\}$, applying the mean value theorem to $t \mapsto t \log t$ yields

$$\begin{aligned}\frac{f_\varepsilon \log f_\varepsilon - f_\star \log f_\star}{\varepsilon} &= (1 + \log f_{\xi\varepsilon}) \frac{f_\varepsilon - f_\star}{\varepsilon} \\ &= \Big(1 + \log(Z_\psi(\varepsilon)) + \log\big(f_\star(1 + \xi\varepsilon\psi)\big)\Big)\left(\frac{Z_\psi(\varepsilon) - 1}{\varepsilon} f_\star + Z_\psi(\varepsilon) f_\star \psi\right),\end{aligned}$$

where $\xi = \xi(x, v) \in (0, 1)$. Since $Z_\psi \in \mathcal{C}^1((-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi))$ and $Z_\psi(0) = 1$, we obtain

$$\lim_{\varepsilon \to 0} \frac{f_\varepsilon \log f_\varepsilon - f_\star \log f_\star}{\varepsilon} = (1 + \log f_\star)\left(\frac{\mathrm{d} Z_\psi}{\mathrm{d}\varepsilon}(0) f_\star + f_\star \psi\right) \quad \text{a.e.}$$

Next, using the monotonicity of the logarithm and (2.36), we have

$$|\log(Z_\psi(\varepsilon))| \leq 1, \qquad |\log(1 + \varepsilon\psi)| \leq 1.$$

We therefore estimate

$$\begin{aligned}&\left|\Big(1 + \log(Z_\psi(\varepsilon)) + \log\big(f_\star(1 + \xi\varepsilon\psi)\big)\Big)\left(\frac{Z_\psi(\varepsilon) - 1}{\varepsilon} f_\star + Z_\psi(\varepsilon) f_\star \psi\right)\right| \\ &\qquad \leq (3 + |\log f_\star|)\left(\|Z_\psi'\|_{L^\infty((-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi))} f_\star + f_\star |\psi|\right) =: h\end{aligned}$$

and

$$\left|(1 + \log f_\star)\left(\frac{\mathrm{d} Z_\psi}{\mathrm{d}\varepsilon}(0) f_\star + f_\star \psi\right)\right| \leq (1 + |\log f_\star|)\left(\|Z_\psi'\|_{L^\infty((-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi))} f_\star + f_\star |\psi|\right) \leq h.$$

Since $f_\star, f_\star \log f_\star \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$ and $\psi \in \mathcal{C}_c^\infty(\mathbb{T}^d \times \mathbb{R}^d)$, it follows that $h \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$. Thus, by dominated convergence, we conclude that

$$\lim_{\varepsilon \to 0} \frac{\mathcal{E}(f_\varepsilon) - \mathcal{E}(f_\star)}{\varepsilon} = \int_{\mathbb{T}^d \times \mathbb{R}^d} (1 + \log f_\star)\left(\frac{\mathrm{d} Z_\psi}{\mathrm{d}\varepsilon}(0) f_\star + f_\star \psi\right) \mathrm{d}x \mathrm{d}v. \tag{2.44}$$

- **Kinetic and Potential Energy.** By the linearity of the kinetic and potential energy, we compute

$$\begin{aligned}\lim_{\varepsilon \to 0} \frac{E^K(f_\varepsilon) - E^K(f_\star)}{\varepsilon} &= \lim_{\varepsilon \to 0}\left(\frac{Z_\psi(\varepsilon) - 1}{\varepsilon} E^K(f_\star) + Z_\psi(\varepsilon) E^K(f_\star \psi)\right) \\ &= \frac{\mathrm{d} Z_\psi}{\mathrm{d}\varepsilon}(0) E^K(f_\star) + E^K(f_\star \psi),\end{aligned} \tag{2.45}$$

and similarly

$$\lim_{\varepsilon \to 0} \frac{E^u(f_\varepsilon) - E^u(f_\star)}{\varepsilon} = \frac{\mathrm{d} Z_\psi}{\mathrm{d}\varepsilon}(0) E^u(f_\star) + E^u(f_\star \psi). \tag{2.46}$$

- **Dirichlet Energy.** Let $\varepsilon \in (-\tilde{\varepsilon}_\psi, \tilde{\varepsilon}_\psi) \setminus \{0\}$ be arbitrary. By Claim 2.2.20, $f_\varepsilon, f_\star \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$. Therefore,

$$\rho_{f_\varepsilon} - \rho_0,\ \rho_\star - \rho_0 \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$$

and

$$\varphi_{f_\varepsilon} = (-\Delta)^{-1}(\rho_{f_\varepsilon} - \rho_0),\ \varphi_\star = (-\Delta)^{-1}(\rho_\star - \rho_0) \in H^1_{\mathrm{zm}}(\mathbb{T}^d).$$

Using this and the energy identity (2.6) for the Dirichlet energy of the solutions to the Poisson equation, we compute

$$\begin{aligned}\frac{E^D(\varphi_{f_\varepsilon}) - E^D(\varphi_\star)}{\varepsilon} &= \frac{1}{2} \frac{\langle \rho_{f_\varepsilon}, \varphi_{f_\varepsilon} \rangle - \langle \rho_\star, \varphi_\star \rangle}{\varepsilon} \\ &= \frac{1}{2} \frac{\langle \rho_{f_\varepsilon}, \varphi_{f_\varepsilon} - \varphi_\star \rangle + \langle \rho_{f_\varepsilon}, \varphi_\star \rangle - \langle \rho_\star, \varphi_\star \rangle}{\varepsilon} \\ &= \frac{1}{2}\left(\left\langle \rho_{f_\varepsilon}, \frac{\varphi_{f_\varepsilon} - \varphi_\star}{\varepsilon} \right\rangle + \left\langle \frac{\rho_{f_\varepsilon} - \rho_\star}{\varepsilon}, \varphi_\star \right\rangle\right),\end{aligned} \tag{2.47}$$

where we used the bilinearity of the duality pairing $\langle\cdot,\cdot\rangle_{H^{-1}(\mathbb{T}^d),H^1(\mathbb{T}^d)}$. We next write

$$\rho_{f_\varepsilon} = Z_\psi(\varepsilon)\left(\rho_\star + \varepsilon\rho_{f_\star\psi}\right),$$

and we estimate

$$\|\rho_{f_\varepsilon} - \rho_\star\|_{H^{-1}(\mathbb{T}^d)} \leq |Z_\psi(\varepsilon) - 1|\|\rho_\star\|_{H^{-1}(\mathbb{T}^d)} + |Z_\psi(\varepsilon)\varepsilon|\|\rho_{f_\star\psi}\|_{H^{-1}(\mathbb{T}^d)},$$

where we used the triangle inequality as well as $\rho_{f_\star\psi} \in H^{-1}(\mathbb{T}^d)$ (see Claim 2.2.20). Since $Z_\psi(\varepsilon) \to 1$ as $\varepsilon \to 0$, we deduce that

$$\rho_{f_\varepsilon} \xrightarrow{\varepsilon\to 0} \rho_\star \quad \text{strongly in } H^{-1}(\mathbb{T}^d).$$

Similarly, and since $\rho_{f_\varepsilon} - \rho_0, \rho_\star - \rho_0 \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$, we have

$$\frac{\rho_{f_\varepsilon} - \rho_\star}{\varepsilon} = \frac{Z_\psi(\varepsilon) - 1}{\varepsilon}\rho_\star + Z_\psi(\varepsilon)\rho_{f_\star\psi} \xrightarrow{\varepsilon\to 0} \frac{\mathrm{d}Z_\psi}{\mathrm{d}\varepsilon}(0)\rho_\star + \rho_{f_\star\psi} \quad \text{strongly in } H^{-1}_{\mathrm{zm}}(\mathbb{T}^d).$$

Since the inverse Laplacian is a linear isomorphism of Banach spaces, this implies that

$$\frac{\varphi_{f_\varepsilon} - \varphi_\star}{\varepsilon} = (-\Delta)^{-1}\left(\frac{\rho_{f_\varepsilon} - \rho_\star}{\varepsilon}\right) \xrightarrow{\varepsilon\to 0} (-\Delta)^{-1}\left(\frac{\mathrm{d}Z_\psi}{\mathrm{d}\varepsilon}(0)\rho_\star + \rho_{f_\star\psi}\right) \quad \text{strongly in } H^1_{\mathrm{zm}}(\mathbb{T}^d).$$

Using again the linearity of the inverse Laplacian yields

$$(-\Delta)^{-1}\left(\frac{\mathrm{d}Z_\psi}{\mathrm{d}\varepsilon}(0)\rho_\star + \rho_{f_\star\psi}\right) = \frac{\mathrm{d}Z_\psi}{\mathrm{d}\varepsilon}(0)\varphi_\star + \varphi_{f_\star\psi}.$$

Here,

$$\varphi_{f_\star\psi} = (-\Delta)^{-1}\left(\rho_{f_\star\psi} + \frac{\mathrm{d}Z_\psi}{\mathrm{d}\varepsilon}(0)\rho_0\right) \in H^1_{\mathrm{zm}}(\mathbb{T}^d) \tag{2.48}$$

is well-defined, since $\rho_{f_\star\psi} \in H^{-1}(\mathbb{T}^d)$, and by Fubini's theorem

$$\int_{\mathbb{T}^d}\left(\rho_{f_\star\psi} + \frac{\mathrm{d}Z_\psi}{\mathrm{d}\varepsilon}(0)\rho_0\right)\mathrm{d}x = \int_{\mathbb{T}^d\times\mathbb{R}^d}(f_\star\psi - f_\star\psi)\mathrm{d}x\mathrm{d}v = 0, \tag{2.49}$$

where we substituted (2.43) and $\rho_0 = \frac{\boldsymbol{m}}{|\mathbb{T}^d|}$. Using the convergence established above, we pass to the limit as $\varepsilon \to 0$ in (2.47). By the continuity of the duality pairing, this yields

$$\begin{aligned}\lim_{\varepsilon\to 0}\frac{E^D(\varphi_{f_\varepsilon}) - E^D(\varphi_\star)}{\varepsilon} &= \frac{1}{2}\left(\left\langle\rho_\star, \frac{\mathrm{d}Z_\psi}{\mathrm{d}\varepsilon}(0)\varphi_\star + \varphi_{f_\star\psi}\right\rangle + \left\langle\frac{\mathrm{d}Z_\psi}{\mathrm{d}\varepsilon}(0)\rho_\star + \rho_{f_\star\psi}, \varphi_\star\right\rangle\right)\\ &= \frac{\mathrm{d}Z_\psi}{\mathrm{d}\varepsilon}(0)\langle\rho_\star,\varphi_\star\rangle + \frac{1}{2}(\langle\rho_\star,\varphi_{f_\star\psi}\rangle + \langle\rho_{f_\star\psi},\varphi_\star\rangle).\end{aligned} \tag{2.50}$$

Combining (2.44), (2.45), (2.46) and (2.50) yields

$$\begin{aligned}\Phi'_\psi(0) = \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star\left(1 + \log f_\star + \frac{|v|^2}{2} + u\right)\left(\psi + \frac{\mathrm{d}Z_\psi}{\mathrm{d}\varepsilon}(0)\right)\mathrm{d}x\mathrm{d}v\\ + \frac{\mathrm{d}Z_\psi}{\mathrm{d}\varepsilon}(0)\langle\rho_\star,\varphi_\star\rangle + \frac{1}{2}(\langle\rho_\star,\varphi_{f_\star\psi}\rangle + \langle\rho_{f_\star\psi},\varphi_\star\rangle)\end{aligned} \tag{2.51}$$

and concludes the proof. □

Claim 2.2.22. *Let $\psi \in \mathcal{C}^\infty_c(\mathbb{T}^d\times\mathbb{R}^d)$. Furthermore, let $\rho_{f_\star\psi}$ and $\varphi_{f_\star\psi}$ be as in* (2.37) *and* (2.48)*, respectively. Then,*

$$\langle\rho_{f_\star\psi},\varphi_\star\rangle_{H^{-1}(\mathbb{T}^d),H^1(\mathbb{T}^d)} = \langle\rho_\star,\varphi_{f_\star\psi}\rangle_{H^{-1}(\mathbb{T}^d),H^1(\mathbb{T}^d)}.$$

Proof. Since $\varphi_\star, \varphi_{f_\star\psi}$ have zero mean, and $\rho_0, Z'_\psi(0)\rho_0$ are constant, we have

$$\langle \rho_{f_\star\psi}, \varphi_\star\rangle = \left\langle \rho_{f_\star\psi} + Z'_\psi(0)\rho_0, \varphi_\star\right\rangle \qquad \text{and} \qquad \langle \rho_\star, \varphi_{f_\star\psi}\rangle = \langle \rho_\star - \rho_0, \varphi_{f_\star\psi}\rangle\,.$$

For notational convenience, we set

$$\rho_g := \rho_{f_\star\psi} + Z'_\psi(0)\rho_0,$$

where $\rho_g \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$, since $\rho_{f_\star\psi} \in H^{-1}(\mathbb{T}^d)$ by Claim 2.2.20 and ρ_g has zero mean – see(2.49). Then, the claim is equivalent to

$$\langle \rho_g, \varphi_\star\rangle = \langle \rho_\star - \rho_0, \varphi_{f_\star\psi}\rangle.$$

Since ρ_g has zero mean, its zero-th Fourier mode is zero. For $M \in \mathbb{N}$, with $M > 0$, let

$$S_M(\rho_g)(x) := \sum_{0<|k|\le M} \widehat{\rho_g}_k\, e^{ik\cdot x}, \qquad \text{where } \widehat{\rho_g}_k := \frac{1}{\sqrt{|\mathbb{T}^d|}}\langle \rho_g, e^{-ik\cdot x}\rangle,$$

denote the M-th Fourier partial sum of ρ_g. Since $\rho_g \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$, we have (see Lemma A.2.1)

$$S_M(\rho_g) \xrightarrow{M\to\infty} \rho_g \quad \text{strongly in } H^{-1}_{\mathrm{zm}}(\mathbb{T}^d).$$

Next, arguing as in Lemma 2.1.3 (see (2.3)), there holds

$$\varphi_\star(x) = \sum_{k\,\in\,\mathbb{Z}^d\setminus\{0\}} \frac{\widehat{\rho_\star - \rho_0}_k}{|k|^2} e^{ik\cdot x}, \quad \widehat{\rho_\star - \rho_0}_k := \frac{1}{\sqrt{|\mathbb{T}^d|}}\int_{\mathbb{T}^d} (\rho_\star(x) - \rho_0) e^{-ik\cdot x}\mathrm{d}x,$$

where we used that $\rho_\star - \rho_0$ has zero mean, hence its zero-th Fourier mode is zero. Denoting by

$$S_M(\varphi_\star)(x) := \sum_{0<|k|\le M} \frac{\widehat{\rho_\star - \rho_0}_k}{|k|^2}\, e^{ik\cdot x}$$

the M-th Fourier partial sum of $\varphi_\star$, there holds

$$S_M(\varphi_\star) \xrightarrow{M\to\infty} \varphi_\star \quad \text{strongly in } H^1_{\mathrm{zm}}(\mathbb{T}^d).$$

Therefore, by continuity of the duality pairing, we have

$$\lim_{M\to\infty}\int_{\mathbb{T}^d} S_M(\rho_g) S_M(\varphi_\star)\mathrm{d}x = \lim_{M\to\infty}\langle S_M(\rho_g), S_M(\varphi_\star)\rangle = \langle \rho_g, \varphi_\star\rangle.$$

Conversely, for any $M > 0$ we compute

$$\begin{aligned}
\int_{\mathbb{T}^d} S_M(\rho_g) S_M(\varphi_\star)\mathrm{d}x &= \sum_{0<|j|\le M}\sum_{0<|k|\le M}\int_{\mathbb{T}^d} \widehat{\rho_g}_j \frac{\widehat{\rho_\star - \rho_0}_k}{|k|^2}\, e^{i(j+k)\cdot x}\mathrm{d}x \\
&= \sum_{0<|k|\le M}\int_{\mathbb{T}^d} \widehat{\rho_g}_k \frac{\widehat{\rho_\star - \rho_0}_{-k}}{|k|^2}\mathrm{d}x \\
&= \sum_{0<|k|\le M}\int_{\mathbb{T}^d} \frac{\widehat{\rho_g}_k}{|k|^2}\, \widehat{\rho_\star - \rho_0}_{-k}\mathrm{d}x \\
&= \sum_{0<|j|\le M}\sum_{0<|k|\le M}\int_{\mathbb{T}^d} \frac{\widehat{\rho_g}_j}{|j|^2}\, \widehat{\rho_\star - \rho_0}_k\, e^{i(j+k)\cdot x}\mathrm{d}x,
\end{aligned} \tag{2.52}$$

where the second and the last identity follow from

$$\int_{\mathbb{T}^d} e^{i(j+k)\cdot x}\mathrm{d}x = 0 \qquad \Longleftrightarrow \qquad j + k = 0.$$

Arguing as for ρ_g and $\varphi_\star$, for any $M>0$ the maps

$$\sum_{0<|j|\leq M} \frac{\widehat{\rho_g}_j}{|j|^2}\, e^{ij\cdot x} = S_M(\varphi_{f_\star\psi}) \qquad \text{and} \qquad \sum_{0<|k|\leq M} \widehat{\rho_\star-\rho_0}_k e^{ik\cdot x} = S_M(\rho_\star-\rho_0)$$

denote the M-th Fourier partial sums of $\varphi_{f_\star\psi}\in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ and $\rho_\star-\rho_0\in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$, respectively. Furthermore,

$$S_M(\varphi_{f_\star\psi})\to\varphi_{f_\star\psi} \qquad \text{and} \qquad S_M(\rho_\star-\rho_0)\to\rho_\star-\rho_0$$

strongly in $H^1_{\mathrm{zm}}(\mathbb{T}^d)$ and $H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$, respectively. Using this in (2.52), the continuity of the duality pairing yields

$$\begin{aligned}\lim_{M\to\infty}\int_{\mathbb{T}^d} S_M(\rho_g)S_M(\varphi_\star)\mathrm{d}x &= \lim_{M\to\infty}\int_{\mathbb{T}^d} S_M(\varphi_{f_\star\psi})S_M(\rho_\star-\rho_0)\mathrm{d}x\\ &= \lim_{M\to\infty}\langle S_M(\rho_\star-\rho_0), S_M(\varphi_{f_\star\psi})\rangle = \langle\rho_\star-\rho_0,\varphi_{f_\star\psi}\rangle.\end{aligned}$$

By uniqueness of the limit, we thus conclude that

$$\langle\rho_g,\varphi_\star\rangle = \langle\rho_\star-\rho_0,\varphi_{f_\star\psi}\rangle. \qquad \square$$

Proof of Theorem 2.2.18. Let $\psi\in\mathcal{C}^\infty_c(\mathbb{T}^d\times\mathbb{R}^d)$ be an arbitrary test function, and let Φ_ψ be the corresponding perturbation map (2.40). Then Φ_ψ is continuous on $(-\tilde\varepsilon_\psi,\tilde\varepsilon_\psi)$ and differentiable at 0 by Claim 2.2.21 and attains the minimum at 0 by minimality of $f_\star$ for $\mathcal{F}^u$. We therefore deduce that $\Phi'_\psi(0)=0$. Using the expression (2.51) of $\Phi'_\psi(0)$, together with the energy identity in Claim 2.2.22, yields

$$\Phi'_\psi(0) = \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star\left(1+\log f_\star+\frac{|v|^2}{2}+u\right)\left(\psi+\frac{\mathrm{d}Z_\psi}{\mathrm{d}\varepsilon}(0)\right)\mathrm{d}x\mathrm{d}v + \frac{\mathrm{d}Z_\psi}{\mathrm{d}\varepsilon}(0)\langle\rho_\star,\varphi_\star\rangle + \langle\rho_{f_\star\psi},\varphi_\star\rangle = 0. \quad (2.53)$$

Step 1. We justify the identities

$$\langle\rho_\star,\varphi_\star\rangle = \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star\varphi_\star\mathrm{d}x\mathrm{d}v \qquad \text{and} \qquad \langle\rho_{f_\star\psi},\varphi_\star\rangle = \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star\psi\varphi_\star\mathrm{d}x\mathrm{d}v. \quad (2.54)$$

We argue for $\langle\rho_{f_\star\psi},\varphi_\star\rangle$; the proof for $\langle\rho_\star,\varphi_\star\rangle$ is similar. Define the distribution

$$T_{f_\star\psi}:\mathcal{C}^\infty(\mathbb{T}^d)\to\mathbb{R},\ \xi\mapsto\langle\rho_{f_\star\psi},\xi\rangle.$$

Since $\rho_{f_\star\psi}\in H^{-1}(\mathbb{T}^d)$ by Claim 2.2.20, the functional $T_{f_\star\psi}$ is well-defined and extends uniquely to a continuous linear functional on $H^1(\mathbb{T}^d)$. On the other hand, since $f_\star\in L^1(\mathbb{T}^d\times\mathbb{R}^d)$ and $\psi\in\mathcal{C}^\infty_c(\mathbb{T}^d\times\mathbb{R}^d)$, we have

$$\rho_{f_\star\psi}(x) = \int_{\mathbb{R}^d} f_\star(x,v)\psi(x,v)\mathrm{d}v\in L^1(\mathbb{T}^d).$$

Thus, for any $\xi\in\mathcal{C}^\infty(\mathbb{T}^d)\subset L^\infty(\mathbb{T}^d)$, Fubini's theorem yields

$$T_{f_\star\psi}(\xi) = \langle\rho_{f_\star\psi},\xi\rangle = \int_{\mathbb{T}^d}\left(\int_{\mathbb{R}^d} f_\star\psi\mathrm{d}v\right)\xi(x)\mathrm{d}x = \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star(x,v)\psi(x,v)\xi(x)\mathrm{d}x\mathrm{d}v.$$

Since $T_{f_\star\psi}$ is continuous on $H^1(\mathbb{T}^d)$, the above identity extends by continuity to all $\xi\in H^1(\mathbb{T}^d)$ by density of $\mathcal{C}^\infty(\mathbb{T}^d)$ in $H^1(\mathbb{T}^d)$ (see Friedrichs' Lemma – Corollary A.2.2). In particular, for $\varphi_\star\in H^1(\mathbb{T}^d)$,

$$\langle\rho_{f_\star\psi},\varphi_\star\rangle = \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star(x,v)\psi(x,v)\varphi_\star(x)\mathrm{d}x\mathrm{d}v.$$

Step 2. Substituting the identities (2.54) and the expression (2.43) of $Z'_\psi(0)$ in (2.53) yields

$$\Phi'_\psi(0) = \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star \left(1 + \log f_\star + \frac{|v|^2}{2} + u + \varphi_\star\right)\left(\psi - \frac{1}{\boldsymbol{m}}\int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star\psi \mathrm{d}x\mathrm{d}v\right)\mathrm{d}x\mathrm{d}v = 0. \tag{2.55}$$

Define next

$$H(x,v) := 1 + \log f_\star(x,v) + \frac{|v|^2}{2} + u(x) + \varphi_\star(x),$$

and

$$\begin{aligned}\lambda := \frac{1}{\boldsymbol{m}}\int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star H \mathrm{d}x\mathrm{d}v &= \frac{1}{\boldsymbol{m}}\int_{\mathbb{T}^d\times\mathbb{R}^d}\left(f_\star + f_\star\log f_\star + \frac{|v|^2}{2}f_\star + uf_\star + \varphi_\star f_\star\right)\mathrm{d}x\mathrm{d}v\\ &= \frac{1}{\boldsymbol{m}}\left(\boldsymbol{m} + \mathcal{E}(f_\star) + E^K(f_\star) + E^u(f_\star) + \langle\rho_\star,\varphi_\star\rangle\right) \in \mathbb{R},\end{aligned}$$

where we used the first identity in (2.54) for the last integrand. Substituting this in (2.55), we obtain

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star H\psi \mathrm{d}x\mathrm{d}v - \lambda\int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star\psi\mathrm{d}x\mathrm{d}v = 0.$$

Rearranging this identity, by arbitrariness of $\psi \in \mathcal{C}_c^\infty(\mathbb{T}^d\times\mathbb{R}^d)$ we obtain the Euler–Lagrange equation

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star(H-\lambda)\psi\mathrm{d}x\mathrm{d}v = 0 \qquad \text{for all } \psi\in\mathcal{C}_c^\infty(\mathbb{T}^d\times\mathbb{R}^d),$$

that is, $f_\star(H-\lambda) = 0$ in $\mathcal{D}'(\mathbb{T}^d\times\mathbb{R}^d)$.

Step 3. We prove that $f_\star(H-\lambda)\in L^1(\mathbb{T}^d\times\mathbb{R}^d)$. Since $\lambda\in\mathbb{R}$ and $f_\star\in L^1(\mathbb{T}^d\times\mathbb{R}^d)$, it remains to prove that $f_\star H\in L^1(\mathbb{T}^d\times\mathbb{R}^d)$. We write

$$f_\star H = f_\star + f_\star\log f_\star + \frac{|v|^2}{2}f_\star + uf_\star + \varphi_\star f_\star,$$

and we consider each term separately. By Proposition 2.2.16, $f_\star\in\mathcal{A}(\bar{C}_\rho,\bar{C}_\mathcal{E},\bar{C}_K;\boldsymbol{m})$ for large enough $\bar{C}_\rho,\bar{C}_\mathcal{E},\bar{C}_K$. Hence, $f_\star$ and $f_\star\log f_\star$ belong to $L^1(\mathbb{T}^d\times\mathbb{R}^d)$ by definition of $\mathcal{A}$ and Lemma 2.2.10. Moreover, since the kinetic energy of $f_\star$ is finite by definition of $\mathcal{A}$ and $f_\star, \frac{|v|^2}{2}\geq 0$ a.e., it follows that

$$\frac{|v|^2}{2}f_\star\in L^1(\mathbb{T}^d\times\mathbb{R}^d).$$

Additionally, since $u\in L^\infty(\mathbb{T}^d)$ and $f_\star$ is integrable, there holds $uf_\star\in L^1(\mathbb{T}^d\times\mathbb{R}^d)$. It remains to consider the last term. Since $\varphi_\star\in H^1(\mathbb{T}^d)$, by Claim 2.2.5, there holds $|\varphi_\star|\in H^1(\mathbb{T}^d)$. Combining this with the nonnegativity of $f_\star$, we obtain

$$\int_{\mathbb{T}^d}\rho_\star\int_{\mathbb{R}^d} f_\star|\varphi_\star|\mathrm{d}v\mathrm{d}x = \int_{\mathbb{T}^d}\rho_\star|\varphi_\star|\mathrm{d}x = \langle\rho_\star,|\varphi_\star|\rangle_{H^{-1}(\mathbb{T}^d),H^1(\mathbb{T}^d)} < +\infty,$$

where we used that $\rho_\star\in H^{-1}(\mathbb{T}^d)$ by definition of $\mathcal{A}$. Therefore, Tonelli's theorem (see, e.g., Theorem 3.4.5 in [10]) yields $f_\star\varphi_\star\in L^1(\mathbb{T}^d\times\mathbb{R}^d)$.

Step 4. Since $f_\star(H-\lambda)=0$ in $\mathcal{D}'(\mathbb{T}^d\times\mathbb{R}^d)$ by Step 2 and $f_\star(H-\lambda)\in L^1(\mathbb{T}^d\times\mathbb{R}^d)$ by Step 3, it follows that

$$f_\star(H-\lambda) = 0 \quad \text{a.e.}$$

By Claim 2.2.19, $f_\star > 0$ a.e. Therefore, we conclude that $H=\lambda$ a.e., that is,

$$1 + \log f_\star + \frac{|v|^2}{2} + u + \varphi_\star = \lambda \quad \text{a.e..}$$

Equivalently, we have

$$f_\star(x,v) = e^{\lambda-1}\, e^{-u(x)-\varphi_\star(x)}\, e^{-\frac{|v|^2}{2}} \quad \text{a.e..}$$

Since $f_\star \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$, the right-hand side in the above identity is also integrable. Therefore, Fubini's theorem yields

$$\boldsymbol{m} = \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star \mathrm{d}x\mathrm{d}v = e^{\lambda-1}\int_{\mathbb{T}^d} e^{-u(x)-\varphi_\star(x)}\mathrm{d}x \int_{\mathbb{R}^d} e^{-\frac{|v|^2}{2}}\mathrm{d}v = e^{\lambda-1}\int_{\mathbb{T}^d} e^{-u(x)-\varphi_\star(x)}\mathrm{d}x\,(2\pi)^{\frac{d}{2}}.$$

Since $u, \varphi_\star \in H^1(\mathbb{T}^d)$, the two maps are finite a.e., hence $e^{-u-\varphi_\star} > 0$ a.e. by positivity of the exponential. Consequently, we conclude that

$$e^{\lambda-1} = \frac{\boldsymbol{m}}{\int_{\mathbb{T}^d} e^{-u(x)-\varphi_\star(x)}\mathrm{d}x\,(2\pi)^{\frac{d}{2}}}.$$

Substituting this expression into the previous Euler–Lagrange equation yields

$$f_\star(x,v) = \boldsymbol{m}\frac{e^{-u(x)-\varphi_\star(x)}}{\int_{\mathbb{T}^d} e^{-u(y)-\varphi_\star(y)}\mathrm{d}y}\frac{e^{-\frac{|v|^2}{2}}}{(2\pi)^{\frac{d}{2}}} = \rho_\star(x)\mathrm{M}(v) \qquad \text{a.e..}$$

Furthermore, $\varphi_\star$ is a weak solution to the Poisson–Boltzmann–Emden equation

$$-\Delta\varphi_\star(x) = \boldsymbol{m}\frac{e^{-u(x)-\varphi_\star(x)}}{\int_{\mathbb{T}^d} e^{-u(y)-\varphi_\star(y)}\mathrm{d}y} - \rho_0.$$

This concludes the proof. □

The following result establishes the converse of the previous theorem. In particular, it shows that the Euler–Lagrange equation (2.33) and the associated Poisson–Boltzmann–Emden equation (2.34) are sufficient for optimality.

Theorem 2.2.23. *Let $\boldsymbol{m} > 0$ and $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. Assume that the pair $(f,\varphi) \in L^1(\mathbb{T}^d\times\mathbb{R}^d)\times H^1_{\mathrm{zm}}(\mathbb{T}^d)$ satisfies (2.33)-(2.34). Then, f is the minimiser of $\mathcal{F}^u$ on $\tilde{\mathcal{A}}(\boldsymbol{m})$.*

The proof of the above theorem relies on a preliminary result, which we establish first.

Claim 2.2.24. *Let $g \in L^1(\mathbb{T}^d\times\mathbb{R}^d)$ be such that $g\log g \in L^1(\mathbb{T}^d\times\mathbb{R}^d)$ and*

$$g(x,v) = \rho_g(x)\mathrm{M}(v),$$

where $\mathrm{M}(v)$ is the d-dimensional Maxwellian and $\rho_g \geq 0$ a.e. Then, $\rho_g, \rho_g\log\rho_g \in L^1(\mathbb{T}^d)$.

Proof. We compute

$$g\log g = \mathrm{M}(v)\rho_g\log\rho_g + \rho_g\mathrm{M}(v)\log(\mathrm{M}(v)),$$

which implies that

$$\mathrm{M}(v)\rho_g\log\rho_g = g\log g - \rho_g\mathrm{M}(v)\log(\mathrm{M}(v)). \tag{2.56}$$

Since $g \in L^1(\mathbb{T}^d\times\mathbb{R}^d)$, by Tonelli's theorem $\rho_g = \int_{\mathbb{R}^d} g\mathrm{d}v \in L^1(\mathbb{T}^d)$. Furthermore, we have

$$\int_{\mathbb{R}^d} |\mathrm{M}(v)\log(\mathrm{M}(v))|\mathrm{d}v \leq \frac{d}{2}\log(2\pi)\int_{\mathbb{R}^d}\mathrm{M}(v)\mathrm{d}v + \frac{1}{2}\int_{\mathbb{R}^d}|v|^2\mathrm{M}(v) = \frac{1}{2}\left(d\log(2\pi)+d\right) < +\infty.$$

Combining this with $\rho_g \in L^1(\mathbb{T}^d)$ yields

$$\int_{\mathbb{T}^d}\rho_g\mathrm{d}x\int_{\mathbb{R}^d}|\mathrm{M}(v)\log(\mathrm{M}(v))|\mathrm{d}v < +\infty.$$

Therefore, by Tonelli's theorem we have $\rho_g \mathrm{M}(v)\log(\mathrm{M}(v)) \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$. Since $g\log g \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$ by assumption, from (2.56) it follows that $\mathrm{M}(v)\rho_g \log \rho_g \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$. By Fubini, we thus conclude that

$$\rho_g \log \rho_g = \int_{\mathbb{R}^d} \mathrm{M}(v)\rho_g \log \rho_g \mathrm{d}v \in L^1(\mathbb{T}^d)$$

as wished. □

Proof of Theorem 2.2.23. Let $f_\star \in \tilde{\mathcal{A}}(\boldsymbol{m})$ be the unique minimiser of $\mathcal{F}^u$ on $\tilde{\mathcal{A}}(\boldsymbol{m})$, and let $\varphi_\star \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ denote the corresponding self-consistent potential. We prove that $f \in \tilde{\mathcal{A}}(\boldsymbol{m})$ and

$$\mathcal{F}^u(f) \leq \mathcal{F}^u(f_\star). \tag{2.57}$$

The reverse inequality holds trivially by the minimality of $f_\star$. Hence, equality follows, and the claim is a direct consequence of the uniqueness of the minimiser.

The proof is organised as follows. We first show that f is an admissible state, that is, $f \in \tilde{\mathcal{A}}(\boldsymbol{m})$. Then, we consider perturbations of f given by convex combinations with $f_\star$, and we characterise the corresponding perturbation map. Finally, the convexity of the free energy functional allows to conclude (2.57).

Step 1. f **is an admissible state.** Using (2.33), we write

$$f(x,v) = \rho(x)\mathrm{M}(v) = \boldsymbol{m}\frac{e^{-u(x)-\varphi(x)}}{\int_{\mathbb{T}^d} e^{-u(y)-\varphi(y)}\mathrm{d}y}\frac{e^{-\frac{|v|^2}{2}}}{(2\pi)^{\frac{d}{2}}},$$

where $\varphi \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ is a weak solution to the Poisson–Botltzmann–Emden equation (2.34). Since the Laplacian is an isomorphism of Banach spaces (see Proposition 2.1.4), we have

$$-\Delta\varphi = \rho - \rho_0 \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d),$$

and hence $\rho = -\Delta\varphi + \rho_0 \in H^{-1}(\mathbb{T}^d)$. By integrability f, Fubini's theorem yields

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} f\mathrm{d}x\mathrm{d}v = \int_{\mathbb{T}^d} \rho(x)\mathrm{d}x \int_{\mathbb{R}^d} \mathrm{M}(v)\mathrm{d}v = \boldsymbol{m},$$

which proves that f satisfies the mass constraint. Furthermore, $f > 0$ a.e. Indeed, since $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ and $\varphi \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$, the two maps are finite a.e. Hence, the strict positivity of f follows from the strict positivity of the exponential $t \mapsto e^{-t}$ on $\mathbb{R}$ and of the Maxwellian $\mathrm{M}(v)$. For the kinetic energy, since $f \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$, Fubini's theorem yields

$$\rho = \int_{\mathbb{R}^d} f\mathrm{d}v \in L^1(\mathbb{T}^d).$$

Since the Maxwellian distribution has finite moments, Tonelli's theorem yields

$$E^K(f) = \frac{1}{2}\int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2\rho(x)\mathrm{M}(v)\mathrm{d}x\mathrm{d}v = \frac{\boldsymbol{m}}{2}\int_{\mathbb{R}^d} |v|^2\mathrm{M}(v)\mathrm{d}v < +\infty.$$

It remains to establish the finiteness of the entropy. To this end, we write

$$f\log f = \big(\rho(x)\mathrm{M}(v)\big)\log\big(\rho(x)\mathrm{M}(v)\big) = \mathrm{M}(v)\rho(x)\log\big(\rho(x)\big) + \rho(x)\mathrm{M}(v)\log(\mathrm{M}(v)),$$

and we consider each term separately. For the second term, we estimate

$$\int_{\mathbb{R}^d} |\mathrm{M}(v)\log(\mathrm{M}(v))|\mathrm{d}v \leq \frac{d}{2}\log(2\pi)\int_{\mathbb{R}^d}\mathrm{M}(v)\mathrm{d}v + \frac{1}{2}\int_{\mathbb{R}^d}|v|^2\mathrm{M}(v)\mathrm{d}v,$$

where the right-hand side is finite, since the Maxwellian has finite zero-th and second moments. Therefore, $\mathrm{M}(v)\log\mathrm{M}(v) \in L^1(\mathbb{R}^d)$, and since $\rho \in L^1(\mathbb{T}^d)$, by Fubini's theorem it follows that $\rho(x)\mathrm{M}(v)\log(\mathrm{M}(v)) \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$. It remains to consider the first term. To this end, we show that $\rho\log\rho \in L^1(\mathbb{T}^d)$, so that by Fubini's theorem we conclude that $\mathrm{M}(v)\rho\log\rho \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$. Using the Euler–Lagrange representation of ρ, we estimate

$$|\rho\log\rho| \leq \rho\left|\log\left(\frac{\boldsymbol{m}}{\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}y}\right)\right| + \rho|\varphi+u|.$$

Since $\rho \in L^1(\mathbb{T}^d)$, the first term is integrable. For the second term, since $u, \varphi \in H^1(\mathbb{T}^d)$, by Claim 2.2.5 there holds $|u+\varphi| \in H^1(\mathbb{T}^d)$. Hence, using that $\rho \in H^{-1}(\mathbb{T}^d)$, we conclude that

$$\int_{\mathbb{T}^d} \rho|\varphi+u|\mathrm{d}x = \langle \rho, |u+\varphi|\rangle_{H^{-1}(\mathbb{T}^d),H^1(\mathbb{T}^d)} < +\infty.$$

Consequently, $\rho|\varphi+u| \in L^1(\mathbb{T}^d)$, and hence we conclude that $\rho\log\rho \in L^1(\mathbb{T}^d)$. From the previous reasoning, we conclude that $f\log f$ is the sum of two integrable functions, hence it is integrable. In particular, the entropy of f is finite, and we conclude that $f \in \tilde{\mathcal{A}}(\boldsymbol{m})$.

Step 2. Perturbations of f. Define the perturbation

$$f_\varepsilon := (1-\varepsilon)f + f_\star, \qquad \varepsilon \in [0,1],$$

and the corresponding perturbation map

$$\Phi_{f_\star} : [0,1] \to \mathbb{R},\ \varepsilon \mapsto \mathcal{F}^u(f_\varepsilon).$$

Since $f, f_\star \in \tilde{\mathcal{A}}(\boldsymbol{m})$, there exist constants $C_\rho, C_\mathcal{E}, C_K$ such that $f, f_\star \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$; without loss of generality, we may assume that $C_\rho, C_\mathcal{E}, C_K$ satisfy (2.14). Therefore,

$$f_\varepsilon \in \mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m}) \quad \text{for all } \varepsilon \in [0,1]. \tag{2.58}$$

Indeed, f_ε is a convex combination of elements in $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$, and the latter set is convex by Lemma 2.2.13. Since $\mathcal{F}^u$ is well-defined on $\tilde{\mathcal{A}}(\boldsymbol{m})$ by lemma 2.2.14, it thus follows that $\Phi_{f_\star}$ is well-defined. Consequently, (2.57) is equivalent to

$$\Phi_{f_\star}(0) \leq \Phi_{f_\star}(1). \tag{2.59}$$

Since $f_\star$ is the unique minimiser of $\mathcal{F}^u$ on $\tilde{\mathcal{A}}(\boldsymbol{m})$, by Theorem 2.2.18 we have

$$f_\star(x,v) = \rho_\star(x)\mathrm{M}(v) = \boldsymbol{m}\frac{e^{-u(x)-\varphi_\star(x)}}{\int_{\mathbb{T}^d} e^{-u(y)-\varphi_\star(y)}\mathrm{d}y}\frac{e^{-\frac{|v|^2}{2}}}{(2\pi)^{\frac{d}{2}}},$$

where $\varphi_\star \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ is a weak solution to the Poisson–Boltzmann–Emden equation (2.34). Consequently, for all $\varepsilon \in [0,1]$, there holds

$$f_\varepsilon(x,v) = (1-\varepsilon)\rho(x)\mathrm{M}(v) + \varepsilon\rho_\star(x)\mathrm{M}(v) = [(1-\varepsilon)\rho(x) + \varepsilon\rho_\star(x)]\,\mathrm{M}(v) = \rho_{f_\varepsilon}(x)\mathrm{M}(v), \tag{2.60}$$

where $\rho_{f_\varepsilon} \geq 0$ a.e., since it is the convex combination of a.e. nonnegative functions. From (2.58), for all $\varepsilon \in [0,1]$ we have $f_\varepsilon, f_\varepsilon\log f_\varepsilon \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$ by definition of $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$ and by Lemma 2.2.10, respectively. Combining this with (2.60), by Claim 2.2.24 we have

$$\rho_{f_\varepsilon}, \rho_{f_\varepsilon}\log\rho_{f_\varepsilon} \in L^1(\mathbb{T}^d) \quad \text{for all } \varepsilon \in [0,1]. \tag{2.61}$$

By Fubini's theorem, we therefore compute

$$\begin{aligned}\mathcal{E}(f_\varepsilon) &= \int_{\mathbb{T}^d\times\mathbb{R}^d} \left(\rho_{f_\varepsilon}\mathrm{M}(v)\right)\log\left(\rho_{f_\varepsilon}\mathrm{M}(v)\right)\mathrm{d}x\mathrm{d}v \\ &= \int_{\mathbb{R}^d}\mathrm{M}(v)\mathrm{d}v\int_{\mathbb{T}^d}\rho_{f_\varepsilon}\log\rho_{f_\varepsilon}\mathrm{d}x + \int_{\mathbb{T}^d}\rho_{f_\varepsilon}\mathrm{d}x\int_{\mathbb{R}^d}\mathrm{M}(v)\log(\mathrm{M}(v))\mathrm{d}v, \\ &= \int_{\mathbb{T}^d}\rho_{f_\varepsilon}\log\rho_{f_\varepsilon}\mathrm{d}x - \frac{\boldsymbol{m}}{2}\left(d\log(2\pi)+d\right),\end{aligned}$$

where we used the mass constraint $\int_{\mathbb{T}^d}\rho_{f_\varepsilon}\mathrm{d}x = \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\varepsilon \mathrm{d}x\mathrm{d}v = \boldsymbol{m}$. Similarly, we have

$$E^K(f_\varepsilon) = \frac{1}{2}\int_{\mathbb{T}^d\times\mathbb{R}^d}|v|^2\rho_{f_\varepsilon}\mathrm{M}(v)\mathrm{d}x\mathrm{d}v = \frac{1}{2}\int_{\mathbb{T}^d}\rho_{f_\varepsilon}\mathrm{d}x\int_{\mathbb{R}^d}|v|^2\mathrm{M}(v)\mathrm{d}v = \frac{d}{2}\boldsymbol{m},$$

and

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} u\rho_{f_\varepsilon}\mathrm{M}(v)\mathrm{d}x\mathrm{d}v = \int_{\mathbb{T}^d}u\rho_{f_\varepsilon}\mathrm{d}x\int_{\mathbb{R}^d}\mathrm{M}(v)\mathrm{d}v = \int_{\mathbb{T}^d}u\rho_{f_\varepsilon}\mathrm{d}x.$$

Combining the above calculations, we obtain

$$\Phi_{f_\star}(\varepsilon) = \int_{\mathbb{T}^d}\rho_{f_\varepsilon}\log\rho_{f_\varepsilon}\mathrm{d}x + \int_{\mathbb{T}^d}u\rho_{f_\varepsilon}\mathrm{d}x + \frac{1}{2}\int_{\mathbb{T}^d}|\nabla\varphi_{f_\varepsilon}|^2\mathrm{d}x - \frac{\boldsymbol{m}}{2}d\log(2\pi). \tag{2.62}$$

Consequently, (2.59) – and thus (2.57) – is equivalent to

$$\Phi_{\rho_\star}(0) \le \Phi_{\rho_\star}(1),$$

where the reduced perturbation map $\Phi_{\rho_\star}:[0,1]\to\mathbb{R}$ is given by

$$\Phi_{\rho_\star}(\varepsilon) := \int_{\mathbb{T}^d}\rho_{f_\varepsilon}\log\rho_{f_\varepsilon}\mathrm{d}x + \int_{\mathbb{T}^d}u\rho_{f_\varepsilon}\mathrm{d}x + \frac{1}{2}\int_{\mathbb{T}^d}|\nabla\varphi_{f_\varepsilon}|^2\mathrm{d}x. \tag{2.63}$$

Step 3. We establish fundamental properties of the reduced perturbation map $\Phi_{\rho_\star}$ in (2.63).

Claim 2.2.25.

(*i*) $\Phi_{\rho_\star}$ *is strictly convex.*

(*ii*) $\Phi_{\rho_\star}\in\mathcal{C}([0,1])$.

(*iii*) $\Phi_{\rho_\star}$ *is right-differentiable at* 0.

Proof of Claim 2.2.25 (*i*). By Lemma 2.2.14, $\mathcal{F}^u$ is strictly convex on $\mathcal{A}(C_\rho, C_\mathcal{E}, C_K; \boldsymbol{m})$. From (2.58), it thus follows that $\Phi_{f_\star}(\varepsilon) = \mathcal{F}^u(f_\varepsilon)$ is strictly convex. Since additionally $\Phi_{f_\star}$ and $\Phi_{\rho_\star}$ only differ by a constant – see (2.62) and (2.63) – we conclude that $\Phi_{\rho_\star}$ is strictly convex. □

Proof of Claim 2.2.25 (*ii*). We consider each term separately. For the entropy term, using (2.60), there holds $\rho_{f_\varepsilon} = (1-\varepsilon)\rho + \varepsilon\rho_\star$. Hence, the convexity of the map $t\mapsto|t\log t|$ yields

$$|\rho_{f_\varepsilon}\log\rho_{f_\varepsilon}| \le (1-\varepsilon)|\rho\log\rho| + \varepsilon|\rho_\star\log\rho_\star| \le |\rho\log\rho| + |\rho_\star\log\rho_\star| =: h \qquad \text{for all } \varepsilon\in[0,1].$$

Since $\rho\log\rho, \rho_\star\log\rho_\star \in L^1(\mathbb{T}^d)$ (see (2.61)), it follows that $h\in L^1(\mathbb{T}^d)$. Now, let $\varepsilon\in[0,1]$ be arbitrary, and let $\{\varepsilon_k\}_{k\in\mathbb{N}}\subset[0,1]$ be any sequence such that $\varepsilon_k\to\varepsilon$. Then, $\rho_{f_{\varepsilon_k}}\to\rho_{f_\varepsilon}$ a.e. Hence, using the above domination, the dominated convergence theorem yields

$$\lim_{k\to\infty}\int_{\mathbb{T}^d}\rho_{f_{\varepsilon_k}}\log\rho_{f_{\varepsilon_k}}\mathrm{d}x = \int_{\mathbb{T}^d}\rho_{f_\varepsilon}\log\rho_{f_\varepsilon}\mathrm{d}x.$$

For the potential energy, we use linearity:

$$\lim_{k\to\infty}\int_{\mathbb{T}^d} u\rho_{f_{\varepsilon_k}}\mathrm{d}x = \lim_{k\to\infty}\left((1-\varepsilon_k)\int_{\mathbb{T}^d} u\rho\mathrm{d}x + \varepsilon_k\int_{\mathbb{T}^d} u\rho_\star\mathrm{d}x\right) = \int_{\mathbb{T}^d} u\rho_{f_\varepsilon}\mathrm{d}x.$$

We finally address the Dirichlet energy. By linearity of the map $f\mapsto\rho_f$, the triangle inequality yields

$$\|\rho_{f_{\varepsilon_k}}-\rho_{f_\varepsilon}\|_{H^{-1}(\mathbb{T}^d)} \le |\varepsilon_k-\varepsilon|\big(\|\rho\|_{H^{-1}(\mathbb{T}^d)}+\|\rho_\star\|_{H^{-1}(\mathbb{T}^d)}\big)\to 0.$$

Therefore, $\rho_{f_{\varepsilon_k}}\to\rho_{f_\varepsilon}$ strongly in $H^{-1}(\mathbb{T}^d)$, and in particular

$$\rho_{f_{\varepsilon_k}}-\rho_0\to\rho_{f_\varepsilon}-\rho_0 \quad \text{strongly in } H^{-1}_{\mathrm{zm}}(\mathbb{T}^d).$$

By continuity of the inverse Laplacian $(-\Delta)^{-1}: H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)\to H^1_{\mathrm{zm}}(\mathbb{T}^d)$ (see Proposition 2.1.4), we have

$$\varphi_{f_{\varepsilon_k}} = (-\Delta)^{-1}(\rho_{f_{\varepsilon_k}}-\rho_0)\to(-\Delta)^{-1}\left(\rho_{f_\varepsilon}-\rho_0\right)=\varphi_{f_\varepsilon} \quad \text{strongly in } H^1_{\mathrm{zm}}(\mathbb{T}^d).$$

The continuity of the Sobolev norm thus yields

$$E^D(\varphi_{f_{\varepsilon_k}}) = \frac{1}{2}\|\nabla\varphi_{f_{\varepsilon_k}}\|^2_{L^2(\mathbb{T}^d)^d}\to\frac{1}{2}\|\nabla\varphi_{f_\varepsilon}\|^2_{L^2(\mathbb{T}^d)^d}=E^D(\varphi_{f_\varepsilon}).$$

Therefore, $\Phi_{\rho_\star}$ is continuous along the sequence $\{\varepsilon_k\}$. By arbitrariness of the sequence and ε, it follows that $\Phi_{\rho_\star}$ is continuous. □

Proof of Claim 2.2.25 (*iii*). We aim to show that the following limit exists:

$$\lim_{\varepsilon\downarrow 0}\frac{\Phi_{\rho_\star}(\varepsilon)-\Phi_{\rho_\star}(0)}{\varepsilon}.$$

We treat each addendum separately.

Entropy. Let $\varepsilon\in(0,1]$ be arbitrary. Applying the mean value theorem to the map $\varepsilon\mapsto\rho_{f_\varepsilon}\log\rho_{f_\varepsilon}$, there exists $\theta=\theta(x)\in(0,1)$ such that

$$\frac{\rho_{f_\varepsilon}\log\rho_{f_\varepsilon}-\rho\log\rho}{\varepsilon} = (1+\log\rho_{f_{\theta\varepsilon}})\,\frac{\rho_{f_\varepsilon}-\rho}{\varepsilon} = \left(1+\log\left((1-\theta\varepsilon)\rho+\theta\varepsilon\rho_\star\right)\right)(\rho_\star-\rho).$$

Letting $\varepsilon\downarrow 0$, we obtain

$$\frac{\rho_{f_\varepsilon}\log\rho_{f_\varepsilon}-\rho\log\rho}{\varepsilon}\to(1+\log\rho)\,(\rho_\star-\rho) \quad \text{a.e.}$$

We next show that, for all sufficiently small $\varepsilon\in[0,1]$, there exists $h\in L^1(\mathbb{T}^d)$ such that

$$\left|\left(1+\log\left((1-\varepsilon)\rho+\varepsilon\rho_\star\right)\right)(\rho_\star-\rho)\right|\le h \quad \text{a.e..}$$

Once this domination is established, the dominated convergence theorem implies that

$$\lim_{\varepsilon\downarrow 0}\int_{\mathbb{T}^d}\frac{\rho_{f_\varepsilon}\log\rho_{f_\varepsilon}-\rho\log\rho}{\varepsilon}\mathrm{d}x=\int_{\mathbb{T}^d}(1+\log\rho)\,(\rho_\star-\rho)\mathrm{d}x. \tag{2.64}$$

Let $\varepsilon\in\left[0,\frac{1}{2}\right]$ be arbitrary. We estimate

$$\left|\left(1+\log\left((1-\varepsilon)\rho+\varepsilon\rho_\star\right)\right)(\rho_\star-\rho)\right|\le\rho_\star+\rho+(\rho_\star+\rho)\left|\log\left((1-\varepsilon)\rho+\varepsilon\rho_\star\right)\right|,$$

where we used that $\rho_\star,\rho>0$ a.e. We now address the term $\rho_\star\log\left((1-\varepsilon)\rho+\varepsilon\rho_\star\right)$. By the triangle inequality, we write

$$\left|\rho_\star\log\left((1-\varepsilon)\rho+\varepsilon\rho_\star\right)\right|\le\left|\rho_\star\log\rho_\star\right|+\left|\rho_\star\log\left((1-\varepsilon)\frac{\rho}{\rho_\star}+\varepsilon\right)\right|.$$

On the set $\{x \in \mathbb{T}^d : \rho(x) < \rho_\star(x)\}$, using that $\varepsilon \in \left[0, \frac{1}{2}\right]$, we have

$$\frac{1}{2}\frac{\rho}{\rho_\star} \leq (1-\varepsilon)\frac{\rho}{\rho_\star} + \varepsilon \leq (1-\varepsilon) + \varepsilon = 1,$$

and hence

$$\left|\rho_\star \log\left((1-\varepsilon)\frac{\rho}{\rho_\star} + \varepsilon\right)\right| \leq \rho_\star \left(\log 2 + |\log \rho_\star| + |\log \rho|\right).$$

Conversely, on the set $\{x \in \mathbb{T}^d : \rho(x) \geq \rho_\star(x)\}$ we have

$$1 \leq (1-\varepsilon)\frac{\rho}{\rho_\star} + \varepsilon \leq 1 + \frac{\rho}{\rho_\star},$$

and hence, using the standard estimate $\log(1+x) \leq x$ for all $x > 0$ yields

$$\left|\rho_\star \log\left((1-\varepsilon)\frac{\rho}{\rho_\star} + \varepsilon\right)\right| \leq \rho.$$

Combining the previous estimates, we obtain

$$\left|\rho_\star \log\left((1-\varepsilon)\frac{\rho}{\rho_\star} + \varepsilon\right)\right| \leq \rho_\star \left(\log 2 + |\log \rho_\star| + |\log \rho|\right) + \rho \quad \text{a.e.}$$

With a similar argument, we obtain the estimate

$$\left|\rho \log\left((1-\varepsilon)\rho + \varepsilon\rho_\star\right)\right| \leq |\rho \log \rho| + \rho \log 2 + \rho_\star \quad \text{a.e.}$$

Combining the previous calculations, we obtain

$$\left|\left(1 + \log\left((1-\varepsilon)\rho + \varepsilon\rho_\star\right)\right)(\rho_\star - \rho)\right| \leq C(\rho + \rho_\star + |\rho_\star \log \rho_\star| + |\rho \log \rho| + |\rho_\star \log \rho|) =: h \quad \text{a.e.},$$

for some $C \in \mathbb{R}$. From (2.61), all terms except $\rho_\star \log \rho$ are integrable. Therefore, to conclude that $h \in L^1(\mathbb{T}^d)$, it suffices to show that $\rho_\star \log \rho \in L^1(\mathbb{T}^d)$. Since $f(x) = \rho(x)\mathrm{M}(v)$ satisfies the Euler–Lagrange equation (2.33), we write

$$\rho_\star \log \rho = \rho_\star \log\left(\frac{\boldsymbol{m}}{\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}y}\right) - (u+\varphi)\rho_\star,$$

where $\varphi \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ solves the Poisson–Boltzmann–Emden equation (2.34). Since $\rho_\star \in L^1(\mathbb{T}^d)$, it remains to show that $(u+\varphi)\rho_\star \in L^1(\mathbb{T}^d)$. To this end, since $\rho_\star \in H^{-1}(\mathbb{T}^d)$ and $u, \varphi \in H^1(\mathbb{T}^d)$, we have

$$\int_{\mathbb{T}^d} |u+\varphi|\rho_\star \mathrm{d}x = \langle \rho_\star, |u+\varphi| \rangle_{H^{-1}(\mathbb{T}^d), H^1(\mathbb{T}^d)} < +\infty,$$

where we used that $|u+\varphi| \in H^1(\mathbb{T}^d)$. Hence, $|u+\varphi|\rho_\star \in L^1(\mathbb{T}^d)$, and thus $|\rho_\star \log \rho| \in L^1(\mathbb{T}^d)$ as wished. This implies that $h \in L^1(\mathbb{T}^d)$, and the application of the dominated convergence theorem in (2.64) is justified.

Potential Energy. For any $\varepsilon \in (0, 1]$, there holds

$$\int_{\mathbb{T}^d} u \frac{\rho_{f_\varepsilon} - \rho}{\varepsilon} \mathrm{d}x = \int_{\mathbb{T}^d} u(\rho_\star - \rho)\mathrm{d}x. \tag{2.65}$$

Hence, the potential term is right-differentiable at 0.

Dirichlet Energy. Let $\varepsilon \in (0,1]$ be arbitrary. Using the energy identity (2.6), we compute

$$\begin{aligned}\frac{E^D(\varphi_{f_\varepsilon}) - E^D(\varphi)}{\varepsilon} &= \frac{1}{2}\frac{\langle \rho_{f_\varepsilon}, \varphi_{f_\varepsilon}\rangle - \langle \rho, \varphi\rangle}{\varepsilon} \\ &= \frac{1}{2}\frac{\langle \rho_{f_\varepsilon}, \varphi_{f_\varepsilon} - \varphi\rangle + \langle \rho_{f_\varepsilon}, \varphi\rangle - \langle \rho, \varphi\rangle}{\varepsilon} \\ &= \frac{1}{2}\left(\left\langle \rho_{f_\varepsilon}, \frac{\varphi_{f_\varepsilon} - \varphi}{\varepsilon}\right\rangle + \left\langle \frac{\rho_{f_\varepsilon} - \rho}{\varepsilon}, \varphi\right\rangle\right),\end{aligned}$$

Furthermore, by linearity of the map $\varepsilon \mapsto \rho_{f_\varepsilon} = (1-\varepsilon)\rho + \varepsilon\rho_\star$, there holds

$$\rho_{f_\varepsilon} \xrightarrow{\varepsilon\to 0} \rho \quad \text{strongly in } H^{-1}(\mathbb{T}^d) \quad \text{and} \quad \frac{\rho_{f_\varepsilon} - \rho}{\varepsilon} \xrightarrow{\varepsilon\to 0} \rho_\star - \rho \quad \text{strongly in } H^{-1}_{\mathrm{zm}}(\mathbb{T}^d).$$

Since the inverse Laplacian is an isomorphism of Banach spaces, this implies that

$$\frac{\varphi_{f_\varepsilon} - \varphi}{\varepsilon} \to \varphi_\star - \varphi \quad \text{strongly in } H^1_{\mathrm{zm}}(\mathbb{T}^d).$$

Hence, the continuity of the duality pairing $\langle\cdot,\cdot\rangle_{H^{-1}(\mathbb{T}^d),H^1(\mathbb{T}^d)}$ yields

$$\lim_{\varepsilon\downarrow 0}\frac{E^D(\varphi_{f_\varepsilon}) - E^D(\varphi)}{\varepsilon} = \frac{1}{2}\left(\langle \rho, \varphi_\star - \varphi\rangle + \langle \rho_\star - \rho, \varphi\rangle\right) = \langle \rho_\star - \rho, \varphi\rangle, \tag{2.66}$$

where the last equality follows from the self-adjointness of the inverse Laplacian – the proof follows the same steps as in Claim 2.2.22, and we omit the details.
Combining (2.64), (2.65) and (2.66) yields

$$\Phi'_{\rho_\star}(0) := \lim_{\varepsilon\downarrow 0}\frac{\Phi_{\rho_\star}(\varepsilon) - \Phi_{\rho_\star}(0)}{\varepsilon} = \int_{\mathbb{T}^d}(\rho_\star - \rho)(1 + \log\rho + u + \varphi)\mathrm{d}x, \tag{2.67}$$

concluding the proof of Claim 2.2.25. □

Step 4. Using the Euler–Lagrange representation of ρ in (2.67) yields

$$\begin{aligned}\Phi'_{\rho_\star}(0) &= \int_{\mathbb{T}^d}(\rho_\star - \rho)\left(1 + \log\left(\boldsymbol{m}\frac{e^{-u-\varphi}}{\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}y}\right) + u + \varphi\right)\mathrm{d}x \\ &= \int_{\mathbb{T}^d}(\rho_\star - \rho)\left(1 + \log\left(\frac{\boldsymbol{m}}{\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}y}\right)\right)\mathrm{d}x = 0,\end{aligned}$$

where the last equality follows from the mass constraint:

$$\int_{\mathbb{T}^d}\rho_\star \mathrm{d}x = \int_{\mathbb{T}^d}\rho\mathrm{d}x = \boldsymbol{m}.$$

Since $\Phi_{\rho_\star}$ is convex, continuous, and right-differentiable at 0 by Claim 2.2.25, there holds

$$\Phi_{\rho_\star}(\varepsilon) \geq \Phi_{\rho_\star}(0) + \Phi'_{\rho_\star}(0)\varepsilon = \Phi_{\rho_\star}(0) \quad \text{for all } \varepsilon \in [0,1].$$

From the derivations in Step 2, this implies (2.57) and concludes the proof. □

Below is an immediate consequence of the preceding theorems.

Corollary 2.2.26. *Let $\boldsymbol{m} > 0$ and $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. Then the Poisson–Boltzmann–Emden equation (2.34) admits, in $H^1_{\mathrm{zm}}(\mathbb{T}^d)$, a unique weak solution.*

Proof. Existence follows from Theorem 2.2.18. Indeed, the unique minimiser of $\mathcal{F}^u$ admits the Euler–Lagrange representation (2.33), for which the associated self-consistent potential $\varphi_\star \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ satisfies the Poisson–Boltzmann–Emden equation. To prove uniqueness, let $\varphi_1, \varphi_2 \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ be two solutions to the Poisson–Boltzmann–Emden equation with external potential u. Define

$$f_i(x,v) := \boldsymbol{m}\frac{e^{-u(x)-\varphi_i(x)}}{\int_{\mathbb{T}^d} e^{-u(y)-\varphi_i(y)}\mathrm{d}y}\mathrm{M}(v), \qquad i \in \{1,2\}.$$

By Theorem 2.2.23, each pair (f_i, φ_i) yields a minimiser of $\mathcal{F}^u$ on $\tilde{\mathcal{A}}(\boldsymbol{m})$. Since the minimiser is unique, $f_1 = f_2$ almost everywhere. Hence their spatial densities coincide, $\rho_{f_1} = \rho_{f_2}$, and therefore

$$-\Delta(\varphi_1 - \varphi_2) = \rho_{f_1} - \rho_{f_2} = 0.$$

Since both potentials have zero mean, uniqueness of the zero-mean solution to the Poisson equation gives $\varphi_1 = \varphi_2$. □

Remark 2.2.27. *The above argument provides an alternative route to the existence and uniqueness theory for the Poisson–Boltzmann–Emden equation (see, e.g., [20] for a variational approach and [12] for a Schauder's fixed-point argument in dimensions $d \leq 3$). Rather than analysing the nonlinear elliptic problem directly, existence and uniqueness are obtained from the variational theory of the free energy functional.*

2.2.6 Characterisation of Free Energy Solutions as Weak Solutions

We conclude this discussion by identifying conditions under which the free energy solutions of the stationary VPFP system also qualify as weak stationary solutions in the sense of Definition 2.2.1. For a free energy stationary solution $(f_\star, \varphi_\star) \in \tilde{\mathcal{A}}(\boldsymbol{m}) \times H^1_{\mathrm{zm}}(\mathbb{T}^d)$, all requirements in the definition of a weak stationary solution are guaranteed by the free energy formulation except for the integrability condition

$$\rho_\star(\nabla u + \nabla\varphi_\star) \in L^1(\mathbb{T}^d)^d.$$

A sufficient condition ensuring this integrability property is $\rho_\star \in L^2(\mathbb{T}^d)$. Indeed, by Hölder's inequality,

$$\int_{\mathbb{T}^d} \rho_\star|\nabla u + \nabla\varphi_\star|\,\mathrm{d}x \leq \|\rho_\star\|_{L^2(\mathbb{T}^d)}\|\nabla u + \nabla\varphi_\star\|_{L^2(\mathbb{T}^d)^d}.$$

Since $u, \varphi_\star \in H^1(\mathbb{T}^d)$, the second factor on the right-hand side is finite, and hence the desired integrability condition follows.

In dimensions $d \leq 2$, the condition $\rho_\star \in L^2(\mathbb{T}^d)$ follows from the exponential integrability properties of $H^1(\mathbb{T}^d)$. More precisely, the Euler–Lagrange representation of $\rho_\star$, combined with the Sobolev embeddings in dimension one and the Trudinger–Moser inequality in dimension two, yields sufficient control of the exponential nonlinearity $e^{-u-\varphi_\star}$ to ensure its square integrability (and, in fact, integrability to higher powers). In dimensions $d \geq 3$, however, the same argument no longer applies, as the available Sobolev embeddings do not provide the required control of the exponential nonlinearity. Consequently, we impose an additional assumption. The following result formalises this observation.

Lemma 2.2.28. *Let $\boldsymbol{m} > 0$ and $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. Let $(f_\star, \varphi_\star) \in L^1(\mathbb{T}^d \times \mathbb{R}^d) \times H^1_{\mathrm{zm}}(\mathbb{T}^d)$ be the unique free energy solution to the stationary* VPFP *system* (2.7) *in the sense of Definition* 2.2.7. *Whenever $d \geq 3$, assume in addition that*

$$\varphi_\star \in W^{1,d}_{\mathrm{zm}}(\mathbb{T}^d).$$

Then, the corresponding stationary spatial density satisfies

$$\rho_\star \in L^p(\mathbb{T}^d) \quad \textit{for all } 1 \leq p < +\infty.$$

If $d = 1$, then additionally $\rho_\star \in \mathcal{C}(\mathbb{T}^1)$.

The proof of Lemma 2.2.28 in the case $d \geq 2$ relies on a preliminary result, which we establish first.

Claim 2.2.29. *Let $1 \leq p < +\infty$ and $\alpha > 1$. Then, there exists a constant $c = c(p, \alpha) > 0$ such that*

$$pt \leq t^\alpha + c \quad \text{for all } t \geq 0,$$

and the optimal c is given by

$$c = \left(\frac{p}{\alpha}\right)^{\frac{\alpha}{\alpha-1}} (\alpha - 1).$$

Proof. Define the function $g : [0, +\infty) \mapsto \mathbb{R}$ by

$$g(t) = \frac{pt}{t^\alpha + c}.$$

It suffices to prove that there exists $c > 0$ such that $g(t) \leq 1$ for all $t \geq 0$. To this end, note that $g \in \mathcal{C}^1([0, +\infty))$, and its first derivative is given by

$$g'(t) = \frac{p(t^\alpha + c) - p\alpha t^\alpha}{(t^\alpha + c)^2}.$$

Straightforward calculations yield

$$g'(\tilde{t}) = 0 \quad \iff \quad \tilde{t} = \left(\frac{p}{\alpha - 1}\right)^{\frac{1}{\alpha}}.$$

Moreover, $g'(t) > 0$ for $0 \leq t < \tilde{t}$ and $g'(t) < 0$ for $t > \tilde{t}$. Therefore, g attains its maximum at $\tilde{t}$. Substituting $\tilde{t}$, straightforward calculations yield

$$g(\tilde{t}) \leq 1 \quad \iff \quad c \geq \left(\frac{p}{\alpha}\right)^{\frac{\alpha}{\alpha-1}} (\alpha - 1),$$

which concludes the proof. □

Proof of Lemma 2.2.28. By Theorem 2.2.18, $\rho_\star$ has the Euler–Lagrange representation

$$\rho_\star(x) := \boldsymbol{m} \frac{e^{-u(x)-\varphi_\star(x)}}{\int_{\mathbb{T}^d} e^{-u(y)-\varphi_\star(y)} \mathrm{d}y},$$

where $\varphi_\star \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ solves the Poisson–Boltzmann–Emden equation (2.34). We first consider the case $d = 1$. By Morrey's theorem A.1.3, the embedding

$$H^1(\mathbb{T}^1) \hookrightarrow \mathcal{C}(\mathbb{T}^1)$$

is continuous. Therefore, $u, \varphi_\star \in \mathcal{C}(\mathbb{T}^1)$, and hence the previous Euler–Lagrange representation yields $\rho_\star \in \mathcal{C}(\mathbb{T}^1)$. Since $|\mathbb{T}^1| < +\infty$, this implies in particular that $\rho_\star \in L^p(\mathbb{T}^1)$ for all $1 \leq p < +\infty$. We next consider the case $d \geq 2$. By Jensen's inequality (see, e.g., Theorem 2.12.19 in [10]), we have

$$\int_{\mathbb{T}^d} e^{-u(y)-\varphi_\star(y)} \mathrm{d}y \geq |\mathbb{T}^d| e^{\frac{1}{|\mathbb{T}^d|} \int_{\mathbb{T}^d} (-u-\varphi_\star) \mathrm{d}y} \geq |\mathbb{T}^d| e^{-\|u\|_{L^\infty(\mathbb{T}^d)}},$$

where the last inequality uses that $\varphi_\star$ has zero mean. Using this bound in the Euler–Lagrange representation of $\rho_\star$, together with $u \in L^\infty(\mathbb{T}^d)$, we obtain

$$\rho_\star(x) \leq \frac{\boldsymbol{m}}{|\mathbb{T}^d|} e^{2\|u\|_{L^\infty(\mathbb{T}^d)}} e^{-\varphi_\star(x)} \quad \text{a.e.}$$

Therefore, it suffices to show that $e^{-\varphi_\star} \in L^p(\mathbb{T}^d)$ for all $1 \leq p < +\infty$. To this end, fix any such p. Applying Claim 2.2.29 with $\alpha = \frac{d}{d-1} > 1$, together with the monotonicity of the map $t \mapsto e^t$, we obtain

$$|e^{-\varphi_\star}|^p \leq e^{p|\varphi_\star|} \leq e^c \, e^{|\varphi_\star|^{\frac{d}{d-1}}} \quad \text{a.e.}$$

In dimension $d = 2$, then the definition $H^1(\mathbb{T}^d) = W^{1,2}(\mathbb{T}^d)$ yields $\varphi_\star \in W^{1,d}_{\mathrm{zm}}(\mathbb{T}^2)$. In dimensions $d \geq 3$ this property is instead assumed. Therefore, whenever $d \geq 2$ there holds $\varphi_\star \in W^{1,d}_{\mathrm{zm}}(\mathbb{T}^d)$, and the Trudinger inequality (see, e.g., Remark 9.16 in [15]) then yields

$$\int_{\mathbb{T}^d} e^{|\varphi_\star|^{\frac{d}{d-1}}} \mathrm{d}x < +\infty \quad \text{for all } d \geq 2.$$

From the previous bound, we deduce that $e^{-\varphi_\star} \in L^p(\mathbb{T}^d)$. By arbitrariness of $1 \leq p < +\infty$, we conclude that $\rho_\star \in L^p(\mathbb{T}^d)$ for all $1 \leq p < +\infty$. □

The lemma above provides the starting point for the next chapter, in which we analyse how the free energy solutions of the stationary VPFP system depend on the external potential. In particular, it initiates the elliptic bootstrapping argument in Section 3.1 by which higher regularity of the external potential propagates through the coupled nonlinear stationary system $(\rho_\star, \varphi_\star)$. Before turning to this, we conclude with the characterisation of free energy solutions as weak stationary solutions.

Theorem 2.2.30. *Let $\boldsymbol{m} > 0$ and $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. Let $(f_\star, \varphi_\star) \in \tilde{\mathcal{A}}(\boldsymbol{m}) \times H^1_{\mathrm{zm}}(\mathbb{T}^d)$ be the unique free energy solution to the stationary* VPFP *system* (2.7) *in the sense of Definition* 2.2.7. *Whenever $d \geq 3$, assume in addition that*

$$\varphi_\star \in W^{1,d}_{\mathrm{zm}}(\mathbb{T}^d).$$

Then, $(f_\star, \varphi_\star)$ is also a weak stationary solution to (2.7) *in the sense of Definition* 2.2.1.

Proof. By definition of $\tilde{\mathcal{A}}(\boldsymbol{m})$, there holds $f_\star \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$ and $\int_{\mathbb{T}^d \times \mathbb{R}^d} f_\star \mathrm{d}x\mathrm{d}v = \boldsymbol{m}$. Furthermore, Lemma 2.2.28 yields $\rho_\star \in L^2(\mathbb{T}^d)$, and hence $\rho_\star(\nabla_x u + \nabla_x \varphi_\star) \in L^1(\mathbb{T}^d)^d$ by Hölder's inequality. Therefore, it remains to prove that the pair $(f_\star, \varphi_\star)$ satisfies the weak formulations (2.8). By Theorem 2.2.18, $f_\star$ satisfies the Euler–Lagrange equation (2.33). In particular, $f_\star$ is Maxwellian in the velocity space. Using that $\nabla_v \mathrm{M}(v) - v\mathrm{M}(v) = 0$, we therefore have

$$\nabla_v \cdot (\nabla_v f_\star + v f_\star) = 0.$$

Hence, integration by parts in $v \in \mathbb{R}^d$ yields

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} f_\star(-\Delta_v \psi + v \cdot \nabla_v \psi) \mathrm{d}x\mathrm{d}v = -\int_{\mathbb{T}^d \times \mathbb{R}^d} \nabla_v \cdot (\nabla_v f_\star + v f_\star)\psi \mathrm{d}x\mathrm{d}v = 0.$$

Next, using the Sobolev chain rule (see Lemmas 3.1.1 and 3.1.4) yields

$$\nabla_x \rho_\star = -\rho_\star(\nabla_x u + \nabla_x \varphi_\star) \in L^1(\mathbb{T}^d).$$

Therefore, by Fubini's theorem, for any test function $\psi \in \mathcal{C}^\infty_c(\mathbb{T}^d \times \mathbb{R}^d)$ we have

$$\begin{aligned}
&\int_{\mathbb{T}^d \times \mathbb{R}^d} \big[(v f_\star) \cdot \nabla_x \psi - f_\star(\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v \psi\big] \mathrm{d}x\mathrm{d}v \\
&\quad = \int_{\mathbb{T}^d \times \mathbb{R}^d} (\rho_\star(x)\mathrm{M}(v)v) \cdot \nabla_x \psi \, \mathrm{d}x\mathrm{d}v - \int_{\mathbb{T}^d \times \mathbb{R}^d} \rho_\star(x)(\nabla_x u + \nabla_x \varphi_\star) \cdot \big(v\mathrm{M}(v)\psi\big) \, \mathrm{d}x\mathrm{d}v \\
&\quad = \int_{\mathbb{T}^d \times \mathbb{R}^d} (\rho_\star(x)\mathrm{M}(v)v) \cdot \nabla_x \psi \, \mathrm{d}x\mathrm{d}v + \int_{\mathbb{T}^d \times \mathbb{R}^d} \nabla_x \rho_\star \cdot \big(v\mathrm{M}(v)\psi\big) \, \mathrm{d}x\mathrm{d}v \\
&\quad = \int_{\mathbb{T}^d \times \mathbb{R}^d} (\rho_\star(x)\mathrm{M}(v)v) \cdot \nabla_x \psi \, \mathrm{d}x\mathrm{d}v - \int_{\mathbb{T}^d \times \mathbb{R}^d} (\rho_\star(x)\mathrm{M}(v)v) \cdot \nabla_x \psi \, \mathrm{d}x\mathrm{d}v = 0,
\end{aligned}$$

where the first equality uses integration by parts in v (the boundary terms vanish since ψ is compactly supported), while the third equality uses integration by parts in $x \in \mathbb{T}^d$ and the periodic boundary conditions. Hence, the pair $(f_\star, \varphi_\star)$ satisfies (2.8a). Since additionally $\varphi_\star$ solves the Poisson–Boltzmann–Emden equation (2.34) in the weak sense, it follows that $(f_\star, \varphi_\star)$ is a weak solution to (2.7) in the sense of Definition 2.2.1. □

Remark 2.2.31. *In the recent work of [12], the existence and uniqueness of H^2_{zm} solutions to the Poisson–Boltzmann–Emden equation were established in dimensions $d \leq 3$ via a Schauder fixed-point argument. Since $H^2(\mathbb{T}^3) \hookrightarrow W^{1,3}(\mathbb{T}^3)$ continuously, the additional assumption in the preceding Lemma 2.2.28 and Theorem 2.2.30 in dimension $d = 3$ is redundant. In other words, free energy stationary solutions can be identified with weak stationary solutions of the VPFP system in all physically relevant dimensions without additional assumptions.*

Chapter 3

Regularity of the Free Energy Solution Operators for the Stationary VPFP System

As discussed in Chapter 2, for any $\boldsymbol{m} > 0$ and external potential $u \in H^1 \cap L^\infty(\mathbb{T}^d)$, the stationary VPFP system on the torus admits a unique free energy solution $(f_\star(u), \varphi_\star(u)) \in L^1(\mathbb{T}^d \times \mathbb{R}^d) \times H^1_{\mathrm{zm}}(\mathbb{T}^d)$ in the sense of Definition 2.2.7; the notation $f_\star(u), \varphi_\star(u)$ is used to emphasise the dependence of the free energy solution on the confining potential. The corresponding stationary state is characterised by the Euler–Lagrange relation established in Theorem 2.2.18, namely

$$f_\star(u)(x, v) = \rho_\star(u)(x)\mathrm{M}(v),$$

where

$$\rho_\star(u)(x) = \boldsymbol{m}\frac{e^{-u(x)-\varphi_\star(u)(x)}}{\int_{\mathbb{T}^d} e^{-u(y)-\varphi_\star(u)(y)}\mathrm{d}y}, \tag{3.1}$$

and $\varphi_\star(u)$ solves the Poisson–Boltzmann–Emden equation

$$-\Delta\varphi_\star(u)(x) = \rho_\star(u)(x) - \rho_0. \tag{3.2}$$

The main objective of this chapter is to study how the maps $\rho_\star(u)$ and $\varphi_\star(u)$ depend on the external potential $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. In particular, we investigate their regularity, continuity, and differentiability properties as nonlinear mappings of u, and develop an operator-theoretic framework to describe this dependence. Throughout, we let the mass of the system $\boldsymbol{m} > 0$ be arbitrary.

The chapter is organised as follows. In Section 3.1, elliptic bootstrapping is employed to propagate regularity from the external potential to the stationary free energy solution. We then introduce an operator-theoretic formulation for the mappings $u \mapsto \rho_\star(u)$ and $u \mapsto \varphi_\star(u)$ in Section 3.2. This framework enables us to study the continuity properties of these mappings in suitable functional spaces in Section 3.3. Finally, we conclude this chapter by establishing the Fréchet differentiability of the above operators in low dimensions in Section 3.4.

The regularity theory developed in this chapter provides the analytical basis for the inverse problem studied in the following chapter, where we derive a variational characterisation of confinement and construct an optimal external potential $u_\star$ that generates or approximates a prescribed stationary spatial density.

3.1 Regularity of the Solution Maps via Elliptic Bootstrapping

In this section, we use elliptic bootstrapping to establish improved regularity properties of the solution maps $\rho_\star(u)$ and $\varphi_\star(u)$, starting from the assumption $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. Our goal is to describe how the regularity

of the external potential propagates through the coupled nonlinear system and to identify conditions under which higher regularity can be obtained. We treat the cases $d \in \{1, 2\}$ and $d \geq 3$ separately.

By Lemma 2.2.28, the assumption $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ already yields an initial step in the elliptic bootstrapping argument for dimensions $d = 1, 2$.

Lemma 3.1.1. *Let $d = 1, 2$. For any $u \in H^1 \cap L^\infty(\mathbb{T}^d)$, there holds*

$$\rho_\star(u) \in H^1 \cap L^p(\mathbb{T}^d) \qquad \text{and} \qquad \varphi_\star(u) \in W^{3,2}_{\mathrm{zm}} \cap W^{2,p}_{\mathrm{zm}} \cap \mathcal{C}^1(\mathbb{T}^d)$$

for all $1 \leq p < +\infty$. Moreover, if $d = 1$, then $\rho_\star(u) \in \mathcal{C}(\mathbb{T}^1)$ and $\varphi_\star(u) \in \mathcal{C}^2(\mathbb{T}^1)$.

The proof of Lemma 3.1.1 relies on a preliminary result, which we establish first.

Claim 3.1.2. *Let $d \in \mathbb{N}$, $d \geq 1$ be arbitrary. For any $w \in W^{1,q} \cap L^\infty(\mathbb{T}^d)$, $1 \leq q \leq +\infty$, there holds*

$$e^{-w} \in W^{1,q}(\mathbb{T}^d) \qquad \text{with} \qquad \nabla(e^{-w}) = -e^{-w}\nabla w \quad \text{a.e.}$$

Proof. For $k \geq 0$, define the map $g_k : \mathbb{R} \to \mathbb{R}$ by

$$g_k(t) := \min\{e^{-t}, k\}.$$

Then, g_k is globally Lipschitz, since it is the truncation of the smooth function $t \mapsto e^{-t}$, whose derivative is bounded on the region where $e^{-t} \leq k$. Next, let k be chosen sufficiently large so that

$$k > e^{\|w\|_{L^\infty(\mathbb{T}^d)}}.$$

This choice of k ensures that $g_k(w(x)) = e^{-w(x)}$ a.e. Since the map g_k is globally Lipschitz and $w \in W^{1,q}(\mathbb{T}^d)$, the Sobolev chain rule (see, e.g., Theorem 2.1.11 in [54]) yields

$$e^{-w} = g_k(w) \in W^{1,q}(\mathbb{T}^d),$$

and, for a.e. $x \in \mathbb{T}^d$,

$$\nabla(e^{-w})(x) = \nabla(g_k(w(x))) = g_k'(w(x))\nabla(w(x)) = -e^{-w(x)}\nabla w(x),$$

where the last identity uses that $g_k(w) = e^{-w}$. □

Proof of Lemma 3.1.1. Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ be arbitrary.

Step 1. $\rho_\star(u) \in L^p(\mathbb{T}^d)$ and $\varphi_\star(u) \in W^{2,p} \cap \mathcal{C}^1(\mathbb{T}^d)$ for all $1 \leq p < \infty$.
By Lemma 2.2.28, $\rho_\star(u) \in L^p(\mathbb{T}^d)$ for all $1 \leq p < +\infty$, and hence $\rho_\star(u) - \rho_0 \in L^p_{\mathrm{zm}}(\mathbb{T}^d)$ for all $1 \leq p < +\infty$. Consequently, the elliptic regularity for the torus (see Proposition 2.1.5) yields

$$\varphi_\star(u) = (-\Delta)^{-1}(\rho_\star(u) - \rho_0) \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d) \quad \text{for all } 1 < p < +\infty.$$

Since $|\mathbb{T}^d| < +\infty$, the embedding $W^{2,p}(\mathbb{T}^d) \hookrightarrow W^{2,1}(\mathbb{T}^d)$ is continuous for all $1 < p < +\infty$, and thus $\varphi_\star(u) \in W^{2,1}_{\mathrm{zm}}(\mathbb{T}^d)$. Finally, the embedding $W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d) \hookrightarrow \mathcal{C}^1(\mathbb{T}^d)$ is continuous for all $p > d$ by the Sobolev embedding theorem (see Corollary A.1.4), and hence we conclude that $\varphi_\star(u) \in \mathcal{C}^1(\mathbb{T}^d)$.

Step 2. $\rho_\star(u) \in H^1(\mathbb{T}^d)$ and $\varphi_\star(u) \in W^{3,2}_{\mathrm{zm}}(\mathbb{T}^d)$.
From the Euler–Lagrange representation (3.1), $\rho_\star(u)$ is given by

$$\rho_\star(u)(x) = \boldsymbol{m}\frac{e^{-u(x)-\varphi_\star(u)(x)}}{\int_{\mathbb{T}^d} e^{-u(y)-\varphi_\star(u)(y)}\mathrm{d}y}.$$

By Step 1, we have in particular $\varphi_\star(u) \in H^1 \cap \mathcal{C}^1(\mathbb{T}^d)$. The continuity of the embedding $\mathcal{C}^1(\mathbb{T}^d) \hookrightarrow L^\infty(\mathbb{T}^d)$ thus yields

$$\varphi_\star(u) \in H^1 \cap L^\infty(\mathbb{T}^d).$$

Combining this with the assumption that $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ gives

$$u + \varphi_\star(u) \in H^1 \cap L^\infty(\mathbb{T}^d).$$

Since the multiplication by a constant is a smooth operation in Sobolev spaces, by Claim 3.1.2 it thus follows that $\rho_\star(u) \in H^1(\mathbb{T}^d)$. By the elliptic regularity on the torus, $\rho_\star(u) \in H^1(\mathbb{T}^d)$ implies that $\varphi_\star(u) \in W^{3,2}_{\mathrm{zm}}(\mathbb{T}^d)$.

Step 3. Case $d = 1$.
By the Sobolev embedding theorem, the embedding $H^1(\mathbb{T}^1) \hookrightarrow \mathcal{C}(\mathbb{T}^1)$ is continuous. Since $\rho_\star(u) \in H^1(\mathbb{T}^1)$ by Step 2, it follows that $\rho_\star(u) \in \mathcal{C}(\mathbb{T}^1)$. Moreover, from the continuity of the embedding $W^{3,2}(\mathbb{T}^1) \hookrightarrow \mathcal{C}^2(\mathbb{T}^1)$, we conclude that $\varphi_\star(u) \in \mathcal{C}^2(\mathbb{T}^1)$. □

Corollary 3.1.3. *Let $d = 1, 2$. For any $u \in W^{1,s}(\mathbb{T}^d)$ with $2 < s < +\infty$, there holds*

$$\rho_\star(u) \in W^{1,s} \cap \mathcal{C}(\mathbb{T}^d) \qquad \text{and} \qquad \varphi_\star(u) \in W^{3,s}_{\mathrm{zm}} \cap \mathcal{C}^2(\mathbb{T}^d).$$

Proof. Since $s > 2 \geq d$, the Sobolev embedding theorem yields $W^{1,s}(\mathbb{T}^d) \hookrightarrow \mathcal{C}(\mathbb{T}^d)$ with continuous injection. Hence, the continuity of the embeddings $W^{1,s}(\mathbb{T}^d) \hookrightarrow H^1(\mathbb{T}^d)$ and $\mathcal{C}(\mathbb{T}^d) \hookrightarrow L^\infty(\mathbb{T}^d)$ gives

$$u \in H^1 \cap L^\infty(\mathbb{T}^d).$$

Therefore, $\rho_\star(u)$ and $\varphi_\star(u)$ are well-defined, and by Lemma 3.1.1 there holds $\varphi_\star(u) \in W^{2,p}_{\mathrm{zm}} \cap \mathcal{C}^1(\mathbb{T}^d)$ for all $1 \leq p < +\infty$. In particular, we have

$$\varphi_\star(u) \in W^{1,s} \cap L^\infty(\mathbb{T}^d).$$

Since both u and $\varphi_\star(u)$ belong to $W^{1,s} \cap L^\infty(\mathbb{T}^d)$, the Euler–Lagrange representation (3.1), together with Claim 3.1.2, implies that

$$\rho_\star(u) \in W^{1,s}(\mathbb{T}^d),$$

where we used that the multiplication by a constant is a smooth operation on Sobolev spaces. Elliptic regularity on the torus thus yields $\varphi_\star(u) \in W^{3,s}_{\mathrm{zm}}(\mathbb{T}^d)$. Using again the continuity of the embedding $W^{1,s}(\mathbb{T}^d) \hookrightarrow \mathcal{C}(\mathbb{T}^d)$, we conclude that $\rho_\star(u) \in \mathcal{C}(\mathbb{T}^d)$ and $\varphi_\star(u) \in \mathcal{C}^2(\mathbb{T}^d)$. □

In dimensions $d \geq 3$, Lemma 2.2.28 holds under the additional assumption $\varphi_\star(u) \in W^{1,d}_{\mathrm{zm}}(\mathbb{T}^d)$. Nevertheless, once this lemma is established, the elliptic bootstrapping argument follows the same strategy as in Lemma 3.1.1. For completeness, we provide the full details below.

Lemma 3.1.4. *Let $d \geq 3$, $s_1 \in [2, +\infty)$ and $s_2 \in [d, +\infty)$ be arbitrary. Assume that $u \in W^{1,s_1} \cap L^\infty(\mathbb{T}^d)$ is such that $\varphi_\star(u) \in W^{1,s_2}_{\mathrm{zm}}(\mathbb{T}^d)$. Then, letting $s := \min\{s_1, s_2\}$,*

$$\rho_\star(u) \in W^{1,s} \cap L^p(\mathbb{T}^d) \qquad \text{and} \qquad \varphi_\star(u) \in W^{3,s}_{\mathrm{zm}} \cap W^{2,p}_{\mathrm{zm}} \cap \mathcal{C}^1(\mathbb{T}^d)$$

for all $1 \leq p < +\infty$. Moreover, if $s > d$, then $\rho_\star(u) \in \mathcal{C}(\mathbb{T}^d)$ and $\varphi_\star(u) \in \mathcal{C}^2(\mathbb{T}^d)$.

Proof. Let $u \in W^{1,s_1} \cap L^\infty(\mathbb{T}^d)$ be arbitrary. Since $s_1 \geq 2$, the Sobolev embedding $W^{1,s_1}(\mathbb{T}^d) \hookrightarrow H^1(\mathbb{T}^d)$ is continuous, hence

$$u \in H^1 \cap L^\infty(\mathbb{T}^d),$$

so that $\rho_\star(u), \varphi_\star(u)$ are well-defined. Moreover, since $s_2 \geq d$, we similarly obtain $\varphi_\star(u) \in W^{1,d}_{\mathrm{zm}}(\mathbb{T}^d)$. Applying Lemma 2.2.28, we thus deduce that

$$\rho_\star(u) \in L^p(\mathbb{T}^d) \quad \text{for all } 1 \leq p < +\infty.$$

By elliptic regularity on the torus, this implies

$$\varphi_\star(u) \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d) \quad \text{for all } 1 \le p < +\infty.$$

Choosing $p > d$, the Sobolev embedding $W^{2,p}(\mathbb{T}^d) \hookrightarrow \mathcal{C}^1(\mathbb{T}^d)$ then yields

$$\varphi_\star(u) \in \mathcal{C}^1(\mathbb{T}^d).$$

Hence, we have in particular $\varphi_\star(u) \in W^{1,s} \cap \mathcal{C}^1(\mathbb{T}^d)$. Thus, the continuity of the embedding $\mathcal{C}^1(\mathbb{T}^d) \hookrightarrow L^\infty(\mathbb{T}^d)$ yields

$$\varphi_\star(u) \in W^{1,s} \cap L^\infty(\mathbb{T}^d).$$

Combining this with $u \in W^{1,s_1} \cap L^\infty(\mathbb{T}^d)$, we obtain

$$u + \varphi_\star(u) \in W^{1,s} \cap L^\infty(\mathbb{T}^d).$$

Therefore, by Claim 3.1.2, the Sobolev chain rule implies $\rho_\star(u) \in W^{1,s}(\mathbb{T}^d)$. Applying elliptic regularity once more yields $\varphi_\star(u) \in W^{3,s}_{\mathrm{zm}}(\mathbb{T}^d)$. Finally, if $s > d$, then the embedding $W^{1,s}(\mathbb{T}^d) \hookrightarrow \mathcal{C}(\mathbb{T}^d)$ is continuous. Since $\rho_\star(u) \in W^{1,s}(\mathbb{T}^d)$ and $\varphi_\star(u) \in W^{3,s}(\mathbb{T}^d)$, it follows that $\rho_\star(u) \in \mathcal{C}(\mathbb{T}^d)$ and $\varphi_\star(u) \in \mathcal{C}^2(\mathbb{T}^d)$. □

We next extend the elliptic bootstrapping argument to arbitrary dimensions under higher regularity assumptions on the external potential.

Proposition 3.1.5. *Let $d \in \mathbb{N}$, $d \ge 1$ be arbitrary. Moreover, let $n \in \mathbb{N}$ be such that $n \ge 2$, and choose $d < s < +\infty$. Let $u \in W^{n,s}(\mathbb{T}^d)$, and assume that, if $d \ge 3$, there holds $\varphi_\star(u) \in W^{1,d+\varepsilon}_{\mathrm{zm}}(\mathbb{T}^d)$ for some $\varepsilon > 0$. Then,*

$$\rho_\star(u) \in W^{n,s} \cap \mathcal{C}^{n-1}(\mathbb{T}^d) \qquad \text{and} \qquad \varphi_\star(u) \in W^{n+2,s}_{\mathrm{zm}} \cap \mathcal{C}^{n+1}(\mathbb{T}^d).$$

If, in addition, $u \in \mathcal{C}^\infty(\mathbb{T}^d)$, then $\rho_\star(u), \varphi_\star(u) \in \mathcal{C}^\infty(\mathbb{T}^d)$.

Proof. We first collect the regularity properties of u, $\rho_\star(u)$ and $\varphi_\star(u)$ established in the previous results. We then perform an initial bootstrapping argument yielding the result in the case $n = 2$. Finally, the general conclusion follows by induction on n.

Step 1. Regularity of u, $\rho_\star(u)$ and $\varphi_\star(u)$. Let $u \in W^{n,s}(\mathbb{T}^d)$ be arbitrary. Since $s > d$, the Sobolev embedding theorem yields the continuous embedding

$$W^{n,s}(\mathbb{T}^d) \hookrightarrow \mathcal{C}^{n-1}(\mathbb{T}^d).$$

This implies that $u \in \mathcal{C}^{n-1}(\mathbb{T}^d)$, and since $n \ge 2$, it follows that $u \in \mathcal{C}^1(\mathbb{T}^d)$. The compactness of $\mathbb{T}^d$ yields the continuous embedding $\mathcal{C}^1 \hookrightarrow W^{1,p}(\mathbb{T}^d)$ for all $1 \le p \le +\infty$, and thus we obtain

$$u \in W^{1,q}(\mathbb{T}^d) \quad \text{for all } 1 \le q \le +\infty.$$

Assume first $d \in \{1, 2\}$. Then, by Corollary 3.1.3,

$$\rho_\star(u) \in W^{1,p} \cap \mathcal{C}(\mathbb{T}^d) \quad \text{and} \quad \varphi_\star(u) \in W^{3,p} \cap \mathcal{C}^2(\mathbb{T}^d) \quad \text{for all } 1 \le p < +\infty.$$

Consider next the case $d \ge 3$, and let $\varepsilon > 0$ be such that $\varphi_\star(u) \in W^{1,d+\varepsilon}_{\mathrm{zm}}(\mathbb{T}^d)$. The regularity of u established above implies in particular that

$$u \in W^{1,d+\varepsilon}(\mathbb{T}^d).$$

Since $u, \varphi_\star(u) \in W^{1,d+\varepsilon}(\mathbb{T}^d)$, Lemma 3.1.4 yields

$$\rho_\star(u) \in W^{1,d+\varepsilon} \cap \mathcal{C}(\mathbb{T}^d) \quad \text{and} \quad \varphi_\star(u) \in W^{3,d+\varepsilon} \cap \mathcal{C}^2(\mathbb{T}^d).$$

Finally, let d be arbitrary. The Euler–Lagrange representation (3.1), together with Claim 3.1.2, implies that

$$\nabla\rho_\star(u)(x) = -\rho_\star(u)(x)\big(\nabla u(x) + \nabla\varphi_\star(u)(x)\big) \quad \text{for a.e. } x \in \mathbb{T}^d, \tag{3.3}$$

where we used that the multiplication by a constant is a smooth operation on Sobolev spaces.

Step 2. First Bootstrapping Argument. We first show that $\rho_\star(u) \in W^{2,s}(\mathbb{T}^d)$ and $\varphi_\star(u) \in W^{4,s}_{\mathrm{zm}}(\mathbb{T}^d)$. To this end, recalling (3.3), we write for every $1 \le i \le d$,

$$\frac{\partial \rho_\star(u)}{\partial x_i} = -\rho_\star(u)\left(\frac{\partial u}{\partial x_i} + \frac{\partial \varphi_\star(u)}{\partial x_i}\right) = \rho_\star(u) P_{\{x_i\}}, \qquad P_{\{x_i\}} := -\frac{\partial u}{\partial x_i} - \frac{\partial \varphi_\star(u)}{\partial x_i}. \tag{3.4}$$

Fix $1 \le i \le d$. By Step 1,

$$\rho_\star(u) \in W^{1,d+\varepsilon} \cap \mathcal{C}(\mathbb{T}^d), \qquad u \in W^{n,s} \cap \mathcal{C}^{n-1}(\mathbb{T}^d), \qquad \varphi_\star(u) \in \mathcal{C}^2(\mathbb{T}^d).$$

Since $\mathbb{T}^d$ is compact, the embeddings

$$\mathcal{C}(\mathbb{T}^d) \hookrightarrow L^\infty(\mathbb{T}^d) \qquad \text{and} \qquad \mathcal{C}^2(\mathbb{T}^d) \hookrightarrow W^{2,d+\varepsilon} \cap L^\infty(\mathbb{T}^d)$$

are continuous. Hence, setting $r := \min\{s, d+\varepsilon\}$ and recalling that $n \ge 2$, we obtain

$$\rho_\star(u), \frac{\partial u}{\partial x_i}, \frac{\partial \varphi_\star(u)}{\partial x_i} \in W^{1,r} \cap L^\infty(\mathbb{T}^d),$$

and therefore

$$P_{\{x_i\}} \in W^{1,r} \cap L^\infty(\mathbb{T}^d),$$

with

$$\frac{\partial P_{\{x_i\}}}{\partial x_j} = -\frac{\partial^2 u}{\partial x_j \partial x_i} - \frac{\partial^2 \varphi_\star(u)}{\partial x_j \partial x_i}, \qquad 1 \le j \le d.$$

Applying the Sobolev product rule (see, e.g., Proposition 9.4 in [15]) yields

$$\frac{\partial \rho_\star(u)}{\partial x_i} \in W^{1,r} \cap L^\infty(\mathbb{T}^d),$$

and, for every $1 \le j \le d$,

$$\frac{\partial^2 \rho_\star(u)}{\partial x_j \partial x_i} = \frac{\partial \rho_\star(u)}{\partial x_j} P_{\{x_i\}} + \rho_\star(u) \frac{\partial P_{\{x_i\}}}{\partial x_j} = \rho_\star(u) P_{\{x_j, x_i\}},$$

where

$$P_{\{x_j,x_i\}} := P_{\{x_j\}} P_{\{x_i\}} + \frac{\partial P_{\{x_i\}}}{\partial x_j}.$$

Since i is arbitrary, we conclude that $\rho_\star(u) \in W^{2,r}(\mathbb{T}^d)$. Elliptic regularity then implies $\varphi_\star(u) \in W^{4,r}_{\mathrm{zm}}(\mathbb{T}^d)$. Because $r > d$, the Sobolev embedding theorem yields the continuous embedding $W^{1,r}(\mathbb{T}^d) \hookrightarrow \mathcal{C}(\mathbb{T}^d)$, so that

$$\rho_\star(u) \in \mathcal{C}^1(\mathbb{T}^d), \qquad \varphi_\star(u) \in \mathcal{C}^3(\mathbb{T}^d).$$

Combining this with the continuous embedding

$$\mathcal{C}^1(\mathbb{T}^d) \hookrightarrow W^{1,s} \cap L^\infty(\mathbb{T}^d),$$

and using that $u \in W^{2,s}(\mathbb{T}^d)$, we infer that for every $1 \le i \le d$,

$$\rho_\star(u), \frac{\partial u}{\partial x_i}, \frac{\partial \varphi_\star(u)}{\partial x_i} \in W^{1,s} \cap L^\infty(\mathbb{T}^d).$$

Consequently,

$$P_{\{x_i\}} \in W^{1,s} \cap L^\infty(\mathbb{T}^d).$$

Applying once again the Sobolev product rule gives

$$\frac{\partial \rho_\star(u)}{\partial x_i} \in W^{1,s}(\mathbb{T}^d).$$

Since i is arbitrary, it follows that $\rho_\star(u) \in W^{2,s}(\mathbb{T}^d)$, and elliptic regularity therefore yields $\varphi_\star(u) \in W^{4,s}_{\mathrm{zm}}(\mathbb{T}^d)$. This concludes the proof for the case $n = 2$. We now conclude Step 2 by proving that, whenever $n \geq 3$,

$$P_{\{x_i,x_j\}} \in W^{\min\{n-2,2\},s} \cap \mathcal{C}^{\min\{n-3,1\}}(\mathbb{T}^d), \qquad 1 \leq i,j \leq d.$$

Fix $1 \leq j \leq d$ and recall that

$$P_{\{x_j\}} := -\frac{\partial u}{\partial x_j} - \frac{\partial \varphi_\star(u)}{\partial x_j}.$$

By Step 1, $u \in W^{n,s} \cap \mathcal{C}^{n-1}(\mathbb{T}^d)$. Moreover, since $\varphi_\star(u) \in W^{4,s} \cap \mathcal{C}^3(\mathbb{T}^d)$, it follows that

$$P_{\{x_j\}} \in W^{\min\{n-1,3\},s} \cap \mathcal{C}^{\min\{n-2,2\}}(\mathbb{T}^d),$$

and

$$\frac{\partial P_{\{x_j\}}}{\partial x_i} \in W^{\min\{n-2,2\},s} \cap \mathcal{C}^{\min\{n-3,1\}}(\mathbb{T}^d), \qquad 1 \leq i \leq d.$$

Since $\min\{n-2,2\} \geq 1$ for $n \geq 3$, we have

$$P_{\{x_j\}} \in \mathcal{C}^1(\mathbb{T}^d) \subset L^\infty(\mathbb{T}^d).$$

As j is arbitrary, the Sobolev product rule implies that

$$P_{\{x_i\}}P_{\{x_j\}} \in W^{\min\{n-1,3\},s} \cap \mathcal{C}^{\min\{n-2,2\}}(\mathbb{T}^d), \qquad 1 \leq i,j \leq d.$$

Therefore, by the definition of $P_{\{x_i,x_j\}}$,

$$P_{\{x_i,x_j\}} = P_{\{x_i\}}P_{\{x_j\}} + \frac{\partial P_{\{x_j\}}}{\partial x_i} \in W^{\min\{n-2,2\},s} \cap \mathcal{C}^{\min\{n-3,1\}}(\mathbb{T}^d),$$

as wished.

Step 3. Induction Argument. Let $n \geq 3$, and assume that for some $k \in \mathbb{N}$, $2 \leq k < n$,

$$\rho_\star(u) \in W^{k,s} \cap \mathcal{C}^{k-1}(\mathbb{T}^d) \qquad \text{and} \qquad \varphi_\star(u) \in W^{k+2,s}_{\mathrm{zm}} \cap \mathcal{C}^{k+1}(\mathbb{T}^d),$$

and that, for arbitrary $1 \leq i_1, \cdots, i_k \leq d$,

$$\frac{\partial^k \rho_\star(u)}{\partial x_{i_1} \cdots \partial x_{i_k}} = \rho_\star(u) P_{\{x_{i_1},\cdots,x_{i_k}\}}, \tag{3.5}$$

where

$$P_{\{x_{i_1},\cdots,x_{i_k}\}} \in W^{\min\{n-k,2\},s} \cap \mathcal{C}^{\min\{n-k-1,1\}}(\mathbb{T}^d).$$

Observe that the case $k = 2$ was established in Step 2. We now prove that

$$\rho_\star(u) \in W^{k+1,s} \cap \mathcal{C}^k(\mathbb{T}^d) \qquad \text{and} \qquad \varphi_\star(u) \in W^{k+3,s}_{\mathrm{zm}} \cap \mathcal{C}^{k+2}(\mathbb{T}^d).$$

Moreover, if $n > k+1$, then for every $1 \leq i_1, \cdots, i_{k+1} \leq d$,

$$\frac{\partial^{k+1} \rho_\star(u)}{\partial x_{i_1} \cdots \partial x_{i_{k+1}}} = \rho_\star(u) P_{\{x_{i_1},\cdots,x_{i_{k+1}}\}},$$

where

$$P_{\{x_{i_1},\cdots,x_{i_{k+1}}\}} \in W^{\min\{n-k-1,2\},s} \cap \mathcal{C}^{\min\{n-k-2,1\}}(\mathbb{T}^d). \tag{3.6}$$

We first establish that $\rho_\star(u) \in W^{k+1,s}(\mathbb{T}^d)$. Let $1 \leq i_2, \cdots, i_{k+1} \leq d$ be arbitrary. Since $2 \leq k < n$, by induction hypothesis we have

$$\rho_\star(u), P_{\{x_{i_2},\cdots,x_{i_{k+1}}\}} \in W^{1,s} \cap \mathcal{C}(\mathbb{T}^d).$$

The Sobolev product rule, together with (3.5), then implies that

$$\frac{\partial^k \rho_\star(u)}{\partial x_{i_2} \cdots \partial x_{i_{k+1}}} \in W^{1,s}(\mathbb{T}^d) \cap L^\infty(\mathbb{T}^d)$$

with

$$\frac{\partial^{k+1} \rho_\star(u)}{\partial x_{i_1} \cdots \partial x_{i_{k+1}}} = \frac{\partial \rho_\star(u)}{\partial x_{i_1}} P_{\{x_{i_2}, \cdots, x_{i_{k+1}}\}} + \rho_\star(u) \frac{\partial P_{\{x_{i_2}, \cdots, x_{i_{k+1}}\}}}{\partial x_{i_1}}, \qquad 1 \leq i_1 \leq d.$$

Using (3.4), we write

$$\frac{\partial^{k+1} \rho_\star(u)}{\partial x_{i_1} \cdots \partial x_{i_{k+1}}} = \rho_\star(u) P_{\{x_{i_1}, \cdots, x_{i_{k+1}}\}},$$

where

$$P_{\{x_{i_1}, \cdots, x_{i_{k+1}}\}} := P_{\{x_{i_1}\}} P_{\{x_{i_2}, \cdots, x_{i_{k+1}}\}} + \frac{\partial P_{\{x_{i_2}, \cdots, x_{i_{k+1}}\}}}{\partial x_{i_1}}.$$

By arbitrariness of $1 \leq i_1, \cdots, i_{k+1} \leq d$, it follows that $\rho_\star(u) \in W^{k+1,s}(\mathbb{T}^d)$. Elliptic regularity then yields $\varphi_\star(u) \in W^{k+3,s}_{\mathrm{zm}}(\mathbb{T}^d)$. Since $s > d$, the continuity of the embedding $W^{1,s}(\mathbb{T}^d) \hookrightarrow \mathcal{C}(\mathbb{T}^d)$ gives

$$\rho_\star(u) \in \mathcal{C}^k(\mathbb{T}^d), \qquad \varphi_\star(u) \in \mathcal{C}^{k+2}(\mathbb{T}^d).$$

It remains to prove (3.6) whenever $n > k+1$. We proceed via a secondary induction argument.

- **Base Step.** Let $1 \leq i \leq d$ be arbitrary. Then, by definition

  $$P_{\{x_i\}} = -\frac{\partial u}{\partial x_i} - \frac{\partial \varphi_\star(u)}{\partial x_i}.$$

  By Step 1, $u \in W^{n,s} \cap \mathcal{C}^{n-1}(\mathbb{T}^d)$. Since $\varphi_\star(u) \in W^{k+3,s}_{\mathrm{zm}} \cap \mathcal{C}^{k+2}(\mathbb{T}^d)$, it follows that

  $$P_{\{x_i\}} \in W^{\min\{n-1,k+2\},s} \cap \mathcal{C}^{\min\{n-2,k+1\}}(\mathbb{T}^d).$$

- **Induction Argument.** Let $1 \leq j < k+1$, and assume that, for all $t \in \mathbb{N}$, $1 \leq t \leq j$,

  $$P_{\{x_{i_1}, \cdots, x_{i_t}\}} \in W^{\min\{n-t,k+3-t\},s} \cap \mathcal{C}^{\min\{n-1-t,k+2-t\}}(\mathbb{T}^d), \quad 1 \leq i_1, \cdots, i_t \leq d.$$

  We prove that

  $$P_{\{x_{i_1}, \cdots, x_{i_{j+1}}\}} \in W^{\min\{n-j-1,k+2-j\},s} \cap \mathcal{C}^{\min\{n-2-j,k+1-j\}}(\mathbb{T}^d), \quad 1 \leq i_1, \cdots, i_{j+1} \leq d. \tag{3.7}$$

  For any $1 \leq i_1, \cdots, i_{j+1} \leq d$, it holds

  $$P_{\{x_{i_1}, \cdots, x_{i_{j+1}}\}} = P_{\{x_{i_1}\}} P_{\{x_{i_2}, \cdots, x_{i_{j+1}}\}} + \frac{\partial P_{\{x_{i_2}, \cdots, x_{i_{j+1}}\}}}{\partial x_{i_1}}.$$

  The induction hypothesis yields

  $$P_{\{x_{i_1}\}} \in W^{\min\{n-1,k+2\},s} \cap \mathcal{C}^{\min\{n-2,k+1\}}(\mathbb{T}^d)$$

  and

  $$P_{\{x_{i_2}, \cdots, x_{i_{j+1}}\}} \in W^{\min\{n-j,k+3-j\},s} \cap \mathcal{C}^{\min\{n-1-j,k+2-j\}}(\mathbb{T}^d).$$

  Using that $1 \leq j < k+1 < n$, the Sobolev product rule then implies

  $$P_{\{x_{i_1}\}} P_{\{x_{i_2}, \cdots, x_{i_{j+1}}\}} \in W^{\min\{n-j,k+3-j\},s} \cap \mathcal{C}^{\min\{n-1-j,k+2-j\}}(\mathbb{T}^d).$$

  Furthermore, we have

  $$\frac{\partial P_{\{x_{i_2}, \cdots, x_{i_{j+1}}\}}}{\partial x_{i_1}} \in W^{\min\{n-j-1,k+2-j\},s} \cap \mathcal{C}^{\min\{n-2-j,k+1-j\}}(\mathbb{T}^d).$$

  Consequently, the definition of $P_{\{x_{i_1}, \cdots, x_{i_{j+1}}\}}$ yields (3.7) as claimed.

Letting $j = k$ in (3.7) yields (3.6) and concludes the induction argument. In particular, if $u \in \mathcal{C}^\infty(\mathbb{T}^d)$, then $\rho_\star(u), \varphi_\star(u) \in \mathcal{C}^\infty(\mathbb{T}^d)$. This concludes the proof. □

3.2 Free Energy Solution Operators

In this section, we develop a functional-analytic framework to study the stability (i.e., continuity) and Fréchet differentiability properties of the solution mappings

$$u \mapsto \rho_\star(u) \quad \text{and} \quad u \mapsto \varphi_\star(u), \qquad u \in H^1 \cap L^\infty(\mathbb{T}^d).$$

To quantify the regularity of the spatial density and exploit elliptic regularity for $\varphi_\star(u)$, we restrict the class of admissible external potentials according to the integrability of $\rho_\star(u)$. Specifically, for any $1 \leq p < +\infty$ and $\boldsymbol{m} > 0$, we define the set

$$\tilde{\mathcal{U}}(p, \boldsymbol{m}) := \left\{ u \in H^1 \cap L^\infty(\mathbb{T}^d) : \rho_\star(u) \in L^p(\mathbb{T}^d) \right\}.$$

Then, $\tilde{\mathcal{U}}(p, \boldsymbol{m})$ is nonempty. Indeed, letting $u = 0$, by Proposition 2.2.17 there holds

$$\rho_\star(0)(x) = \frac{\boldsymbol{m}}{|\mathbb{T}^d|} \in \mathcal{C}^\infty(\mathbb{T}^d) \subset L^p(\mathbb{T}^d) \quad \text{for all } 1 \leq p < +\infty. \tag{3.8}$$

Furthermore, for any $\boldsymbol{m} > 0$, $\{\tilde{\mathcal{U}}(p, \boldsymbol{m})\}_{p \in [1, +\infty)}$ is nonincreasing with respect to p. If $p = 1$, then $\varphi_\star(u) \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ (see Chapter 2). If $1 < p < +\infty$, then the isomorphism property of the inverse Laplacian (see Proposition 2.1.5)

$$(-\Delta)^{-1} : L^p_{\mathrm{zm}} \to W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$$

yields $\varphi_\star(u) \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$ for any potential $u \in \tilde{\mathcal{U}}(p, \boldsymbol{m})$. Accordingly, we define the *free energy solution operators* as follows. If $p = 1$, define

$$\Gamma^1_\rho : \tilde{\mathcal{U}}(1, \boldsymbol{m}) \to L^1(\mathbb{T}^d)\,,\ u \mapsto \rho_\star(u) \quad \text{and} \quad \Gamma^1_\varphi : \tilde{\mathcal{U}}(1, \boldsymbol{m}) \to H^1_{\mathrm{zm}}(\mathbb{T}^d)\,,\ u \mapsto \varphi_\star(u).$$

If $1 < p < +\infty$, define

$$\Gamma^p_\rho : \tilde{\mathcal{U}}(p, \boldsymbol{m}) \to L^p(\mathbb{T}^d)\,,\ u \mapsto \rho_\star(u) \quad \text{and} \quad \Gamma^p_\varphi : \tilde{\mathcal{U}}(p, \boldsymbol{m}) \to W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)\,,\ u \mapsto \varphi_\star(u).$$

In other words, Γ^p_ρ and Γ^p_φ map the external potentials in $\tilde{\mathcal{U}}(p, \boldsymbol{m})$ to the corresponding free energy spatial density and self-consistent potential, respectively. Henceforth, we shall refer to Γ^p_ρ and Γ^p_φ simply as solution operators. Moreover, we define the *free energy entropy operator* (simply, entropy operator) as

$$\Gamma_{\mathcal{E}} : \tilde{\mathcal{U}}(1, \boldsymbol{m}) \to L^1(\mathbb{T}^d),\ u \mapsto \rho_\star(u) \log\left(\rho_\star(u)\right). \tag{3.10}$$

In Proposition 3.3.6, it will be shown that $\Gamma_{\mathcal{E}}$ is well-defined.

Remark 3.2.1. *Since $\rho_\star(u) \in L^1(\mathbb{T}^d)$ and $\varphi_\star(u) \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ for all $u \in H^1 \cap L^\infty(\mathbb{T}^d)$, there holds $\tilde{\mathcal{U}}(1, \boldsymbol{m}) = H^1 \cap L^\infty(\mathbb{T}^d)$ for all d. Furthermore, if $d = 1, 2$, then $\tilde{\mathcal{U}}(p, \boldsymbol{m}) = H^1 \cap L^\infty(\mathbb{T}^d)$ for all $p \in [1, +\infty)$ by Lemma 2.2.28.*

For compactness and stability considerations, we further restrict to subclasses of external potentials with uniform control on the density and the self-consistent potential. To this end, let $\boldsymbol{m} > 0$ and $C_\varphi, C_\rho \geq 0$ be given. If $p = 1$, we define

$$\mathcal{U}(C_\varphi, C_\rho; 1, \boldsymbol{m}) := \left\{ u \in \tilde{\mathcal{U}}(1, \boldsymbol{m}) : \|\varphi_\star(u)\|_{H^1(\mathbb{T}^d)} \leq C_\varphi\,,\ \|\rho_\star(u)\log(\rho_\star(u))\|_{L^1(\mathbb{T}^d)} \leq C_\rho \right\}. \tag{3.11a}$$

If $1 < p < +\infty$, we define

$$\mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m}) := \left\{ u \in \tilde{\mathcal{U}}(p, \boldsymbol{m}) : \|\varphi_\star(u)\|_{H^1(\mathbb{T}^d)} \leq C_\varphi\,,\ \|\rho_\star(u)\|_{L^p(\mathbb{T}^d)} \leq C_\rho \right\}. \tag{3.11b}$$

For any $1 \le p < +\infty$, there exists $C_\varphi, C_\rho \ge 0$ such that $\mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m}) \neq \emptyset$. Indeed, choosing $u = 0$ yields the constant density $\rho_\star(0)$ (see (3.8)) and $\varphi_\star(0) = 0$ (see Proposition 2.2.17). Hence, for large enough $C_\varphi, C_\rho \ge 0$, the sets are nonempty. Moreover, for any $\boldsymbol{m} > 0$ and $1 \le p < +\infty$, one has

$$\bigcup_{C_\varphi, C_\rho \ge 0} \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m}) = \tilde{\mathcal{U}}(p, \boldsymbol{m}).$$

Indeed, for $1 < p < +\infty$ this follows immediately from the definition. For the case $p = 1$, since $f_\star(u) \in \tilde{\mathcal{A}}(\boldsymbol{m})$, the definition of $\tilde{\mathcal{A}}(\boldsymbol{m})$ (see (2.9)), together with Claim 2.2.24, implies the previous identity.

Remark 3.2.2. *In the case $p = 1$, uniform control of $\rho_\star(u)$ in $L^1(\mathbb{T}^d)$ is not sufficient to enable compactness arguments, since $L^1(\mathbb{T}^d)$ is nonreflexive. In this setting, uniform control of the entropy $\rho_\star(u)\log\rho_\star(u)$ yields equi-integrability of the spatial density, while uniform control of the self-consistent potential in $H^1(\mathbb{T}^d)$ provides the compactness needed for the elliptic part of the system.*

Remark 3.2.3. *In the case $1 < p < +\infty$, if $p \ge \frac{2d}{d+2}$, then the Sobolev embedding $W^{2,p}(\mathbb{T}^d) \hookrightarrow H^1(\mathbb{T}^d)$ is continuous (see Corollary A.1.4). Consequently, the continuity of the inverse Laplacian $(-\Delta)^{-1} : L^p_{\mathrm{zm}}(\mathbb{T}^d) \to W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$, together with the uniform control of $\rho_\star(u)$ in $L^p(\mathbb{T}^d)$, implies a uniform bound of $\varphi_\star(u)$ in $H^1(\mathbb{T}^d)$. Therefore, in this range, the $H^1(\mathbb{T}^d)$-control of $\varphi_\star(u)$ in the definition of $\mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$ is redundant. We nevertheless impose it to avoid the distinction between the cases $1 < p < \frac{2d}{d+2}$ and $p \ge \frac{2d}{d+2}$.*

To study continuity and regularity properties of the solution operators, we fix a natural Banach structure on the space of potentials. In particular, we endow the space $H^1 \cap L^\infty(\mathbb{T}^d)$ with the topology induced by the norm

$$\|u\|_{H^1 \cap L^\infty(\mathbb{T}^d)} = \sqrt{\|u\|^2_{H^1(\mathbb{T}^d)} + \|u\|^2_{L^\infty(\mathbb{T}^d)}}\,, \tag{3.12}$$

and $\tilde{\mathcal{U}}(p, \boldsymbol{m})$, $\mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$ with the corresponding subspace topology. We remark that the space $H^1 \cap L^\infty(\mathbb{T}^d)$ endowed with the above norm is a Banach space.

The solution operators in (3.9) are not injective. To see this, let u_1 , $u_2 \in H^1 \cap L^\infty(\mathbb{T}^d)$ be such that, e.g., $\rho_\star(u_1) = \rho_\star(u_2)$ – the reasoning for $\varphi_\star$ is similar. Then, $\varphi_\star(u_1) - \varphi_\star(u_2)$ solves the Laplace equation in the weak sense: for all $\psi \in \mathcal{C}^\infty(\mathbb{T}^d)$,

$$\int_{\mathbb{T}^d} \nabla(\varphi_\star(u_1)(x) - \varphi_\star(u_2)(x)) \cdot \nabla\psi(x)\mathrm{d}x = \int_{\mathbb{T}^d} (\rho_\star(u_1)(x) - \rho_\star(u_2))\psi(x)\mathrm{d}x = 0\,.$$

By arbitrariness of $\psi \in \mathcal{C}^\infty(\mathbb{T}^d)$, we conclude that

$$\nabla(\varphi_\star(u_1)(x) - \varphi_\star(u_2)(x)) = 0 \quad \Longrightarrow \quad \varphi_\star(u_1)(x) = \varphi_\star(u_2)(x) + c \quad \text{a.e.}$$

for some $c \in \mathbb{R}$. By definition of weak stationary solution to (VPFP-CP), $\varphi_\star(u_1)$ and $\varphi_\star(u_2)$ have both zero mean, and thus $c = 0$, that is,

$$\varphi_\star(u_1)(x) = \varphi_\star(u_2)(x) \quad \text{a.e.}$$

From (3.1) and since $\rho_\star(u_1) = \rho_\star(u_2)$ a.e., we infer that

$$\frac{e^{-u_1(x)-\varphi_\star(u_1)(x)}}{\int_{\mathbb{T}^d} e^{-u_1(x)-\varphi_\star(u_1)(x)}\mathrm{d}x} = \frac{e^{-u_2(x)-\varphi_\star(u_2)(x)}}{\int_{\mathbb{T}^d} e^{-u_2(x)-\varphi_\star(u_2)(x)}\mathrm{d}x}\,.$$

Letting a denote the quotient

$$a := \frac{\int_{\mathbb{T}^d} e^{-u_1(x)-\varphi_\star(u_1)(x)}\mathrm{d}x}{\int_{\mathbb{T}^d} e^{-u_2(x)-\varphi_\star(u_2)(x)}\mathrm{d}x} > 0\,,$$

from the previous equality we obtain

$$-u_1(x) - \varphi_\star(u_1)(x) = -u_2(x) - \varphi_\star(u_2)(x) + \log a\,.$$

Since $\varphi_\star(u_1) = \varphi_\star(u_2)$, we thus infer that

$$u_1(x) = u_2(x) + b$$

for some $b \in \mathbb{R}$. Therefore, the previous maps are injective only up to fixing additive constants in the external potential. A possible solution is to restrict the potentials to the zero mean subset of $H^1 \cap L^\infty(\mathbb{T}^d)$, that is, to the set

$$H^1_{\mathrm{zm}} \cap L^\infty(\mathbb{T}^d) = \left\{ u \in H^1 \cap L^\infty(\mathbb{T}^d) : \int_{\mathbb{T}^d} u(x)\mathrm{d}x = 0 \right\}.$$

3.3 Continuity of the Solution Operators

We establish fundamental continuity properties of the free energy solution operators Γ^p_ρ and Γ^p_φ defined in (3.9).

Theorem 3.3.1. *Let $\boldsymbol{m} > 0$ and $1 \le p < +\infty$. Let $C_\varphi, C_\rho \ge 0$ be arbitrary.*

- *The solution operator Γ^p_φ is weakly sequentially continuous on $\mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$, and Γ^1_ρ is weak-to-strong sequentially continuous on $\mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$.*
- *If $1 < p < +\infty$, the solution operators Γ^1_φ and Γ^p_ρ are weakly sequentially continuous on $\mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$.*

Proof. We prove both statements simultaneously. First, we combine the compactness arguments developed in Claims 3.3.2 and 3.3.3, which yield the existence of subsequences such that the corresponding solutions converge weakly in the relevant spaces. We then apply the Lebesgue–Vitali theorem in Claim 3.3.4, which upgrades the convergence of the stationary densities to strong convergence in $L^1(\mathbb{T}^d)$. Finally, we identify the weak and strong limits with the solution of the limiting Poisson–Boltzmann–Emden system in Claim 3.3.5, which concludes the proof and establishes the claimed sequential continuity properties.

Let $1 \le p < +\infty$, and assume without loss of generality that $C_\varphi, C_\rho \ge 0$ are sufficiently large so that $\mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m}) \neq \emptyset$. Let $\{u_n\}_{n\in\mathbb{N}} \subset \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$ be such that $u_n \rightharpoonup u$ weakly in $H^1 \cap L^\infty(\mathbb{T}^d)$. We aim to prove that, for every $1 \le p < +\infty$,

$$\varphi_\star(u_n) \rightharpoonup \varphi_\star(u) \quad \text{weakly in } H^1_{\mathrm{zm}}(\mathbb{T}^d) \qquad \text{and} \qquad \rho_\star(u_n) \to \rho_\star(u) \quad \text{strongly in } L^1(\mathbb{T}^d),$$

and that, if in addition $1 < p < +\infty$,

$$\begin{aligned} &\varphi_\star(u_n) \rightharpoonup \varphi_\star(u) \quad \text{weakly in } W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d), \quad \text{and} \\ &\rho_\star(u_n) \rightharpoonup \rho_\star(u) \quad \text{weakly in } L^p(\mathbb{T}^d). \end{aligned}$$

Since $H^1 \cap L^\infty(\mathbb{T}^d)$, equipped with the norm (3.12), is a Banach space, the weak limit u also belongs to $H^1 \cap L^\infty(\mathbb{T}^d)$. In particular, $\rho_\star(u)$ and $\varphi_\star(u)$ are well-defined.

In the sequel, let the sequence $\{u_n\}$ be as above.

Claim 3.3.2. *For any $1 \le p < +\infty$, there exists a subsequence $\{u_{n_k}\}_{k\in\mathbb{N}} \subset \{u_n\}_{n\in\mathbb{N}}$ such that*

$$\left(\int_{\mathbb{T}^d} e^{-u_{n_k}(x) - \varphi_\star(u_{n_k})(x)} \mathrm{d}x \right)^{-1} =: \frac{1}{Z_{n_k}} \to a$$

for some $a \ge 0$.

Proof of Claim 3.3.2. Since $u_n \rightharpoonup u$ weakly in $H^1 \cap L^\infty(\mathbb{T}^d)$, the sequence is bounded in $H^1 \cap L^\infty(\mathbb{T}^d)$ (see, e.g., Proposition 3.5 in [15]). By definition of the $H^1 \cap L^\infty(\mathbb{T}^d)$ norm adopted (3.12), the sequence is in particular bounded in $H^1(\mathbb{T}^d)$, that is, there exists $R_u \geq 0$ such that $\|u_n\|_{H^1(\mathbb{T}^d)} \leq R_u$ for all $n \in \mathbb{N}$. Hence, by Hölder's inequality [15, Thm 4.6],

$$\|u_n\|_{L^1(\mathbb{T}^d)} \leq \sqrt{|\mathbb{T}^d|}\|u_n\|_{L^2(\mathbb{T}^d)} \leq \sqrt{|\mathbb{T}^d|}\|u_n\|_{H^1(\mathbb{T}^d)} \leq \sqrt{|\mathbb{T}^d|}R_u\,.$$

Furthermore, by Jensen's inequality [10, Thm 2.12.19],

$$\frac{1}{|\mathbb{T}^d|}\int_{\mathbb{T}^d} e^{-u_n-\varphi_\star(u_n)}\mathrm{d}x \geq e^{\frac{1}{|\mathbb{T}^d|}\int_{\mathbb{T}^d}(-u_n-\varphi_\star(u_n))\mathrm{d}x}\,. \tag{3.14}$$

Since, $\varphi_\star(w)$ has zero mean for any $w \in H^1 \cap L^\infty(\mathbb{T}^d)$, from (3.14) and the previous bound it follows that

$$Z_n := \int_{\mathbb{T}^d} e^{-u_n-\varphi_\star(u_n)}\mathrm{d}x \geq |\mathbb{T}^d|e^{-\frac{R_u}{\sqrt{|\mathbb{T}^d|}}} =: c > 0\,.$$

Therefore, the sequence $\{Z_n^{-1}\}_{n\in\mathbb{N}}$ satisfies

$$\frac{1}{Z_n} \in \left(0, \frac{1}{c}\right] \subset \left[0, \frac{1}{c}\right] \quad \text{for all } n \in \mathbb{N}\,.$$

By compactness of $[0, c^{-1}] \subset \mathbb{R}$, there exist a subsequence $\{u_{n_k}\}_{n_k\in\mathbb{N}} \subset \{u_n\}_{n\in\mathbb{N}}$ and $a \in [0, c^{-1}]$ such that

$$\frac{1}{Z_{n_k}} \to a\,,$$

as wished. □

Claim 3.3.3. *For any $1 \leq p < +\infty$, there exist a subsequence $\{u_{n_k}\}_{k\in\mathbb{N}} \subset \{u_n\}_{n\in\mathbb{N}}$ and a map $\varphi_\star \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ such that*

$$u_{n_k} \to u \quad a.e., \qquad \varphi_\star(u_{n_k}) \to \varphi_\star \quad a.e.,$$

and

$$\varphi_\star(u_{n_k}) \rightharpoonup \varphi_\star \quad \textit{weakly in } H^1_{\mathrm{zm}}(\mathbb{T}^d).$$

If in addition $1 < p < +\infty$, then, up to extraction of a further subsequence,

$$\varphi_\star(u_{n_k}) \rightharpoonup \varphi_\star \ \textit{ weakly in } W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d),$$

and there exists $\rho_\star \in L^p(\mathbb{T}^d)$ such that

$$\rho_\star(u_{n_k}) \rightharpoonup \rho_\star \quad \textit{weakly in } L^p(\mathbb{T}^d).$$

Proof of Claim 3.3.3. Let $1 \leq p < +\infty$ be arbitrary. Since the embedding $H^1 \cap L^\infty(\mathbb{T}^d) \hookrightarrow H^1(\mathbb{T}^d)$ is continuous and $u_n \rightharpoonup u$ weakly in $H^1 \cap L^\infty(\mathbb{T}^d)$, it follows that $u_n \rightharpoonup u$ weakly in $H^1(\mathbb{T}^d)$. In particular, the sequence is bounded in $H^1(\mathbb{T}^d)$ [15, Prop 3.5], and therefore in $L^2(\mathbb{T}^d)$. By the Rellich–Kondrachov theorem A.1.5, the embedding

$$H^1(\mathbb{T}^d) \hookrightarrow L^2(\mathbb{T}^d)$$

is compact, and hence $\{u_n\}_{n\in\mathbb{N}}$ is relatively compact in $L^2(\mathbb{T}^d)$. Consequently, there exists a subsequence $\{u_{n_1}\}_{n_1\in\mathbb{N}} \subset \{u_n\}_{n\in\mathbb{N}}$ that converges strongly in $L^2(\mathbb{T}^d)$. Since $\{u_n\}$ converges weakly to u in $H^1(\mathbb{T}^d)$, and hence in $L^2(\mathbb{T}^d)$, while the subsequence $\{u_{n_1}\}$ converges strongly (hence also weakly) in $L^2(\mathbb{T}^d)$, uniqueness of weak limits implies that the limit is u:

$$u_{n_1} \to u \quad \text{strongly in } L^2(\mathbb{T}^d).$$

Up to a further subsequence, $u_{n_1} \to u$ a.e. (see, e.g., Theorem 4.9 in [15]).

Next, since $\{u_n\}_{n\in\mathbb{N}} \subset \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$, by definition of the latter set the sequence $\{\varphi_\star(u_n)\}$ is bounded in $H^1_{\mathrm{zm}}(\mathbb{T}^d)$. Since this holds in particular for the sequence $\{\varphi_\star(u_{n_1})\}$ and $H^1_{\mathrm{zm}}(\mathbb{T}^d)$ is reflexive (see Lemma A.2.3), there exists a subsequence $\{u_{n_2}\}_{n_2\in\mathbb{N}} \subset \{u_{n_1}\}_{n_1\in\mathbb{N}}$ such that

$$\varphi_\star(u_{n_2}) \rightharpoonup \varphi_\star \ \text{ weakly in } \ H^1_{\mathrm{zm}}(\mathbb{T}^d)$$

for some $\varphi_\star \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ (see, e.g., Theorem 3.18 in [15]). In particular, the sequence $\{\varphi_\star(u_{n_2})\}_{n_2\in\mathbb{N}}$ is bounded in $H^1(\mathbb{T}^d)$. Hence, using again the compactness of the embedding $H^1(\mathbb{T}^d) \hookrightarrow L^2(\mathbb{T}^d)$, and arguing as for $\{u_n\}$, there exists a further subsequence $\{u_{n_3}\}_{n_3\in\mathbb{N}} \subset \{u_{n_2}\}_{n_2\in\mathbb{N}}$ such that

$$\varphi_\star(u_{n_3}) \to \varphi_\star \quad \text{a.e.}$$

Collecting the above convergences, the first part of the claim holds along the subsequence $\{u_{n_3}\}_{n_3\in\mathbb{N}}$.

We now let $1 < p < +\infty$ and turn to the second part. Since $\{u_n\}_{n\in\mathbb{N}} \subset \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$, by definition of the latter set the sequence $\{\rho_\star(u_n)\}$ is bounded in $L^p(\mathbb{T}^d)$. Since this holds in particular for the sequence $\{\rho_\star(u_{n_3})\}$ and $L^p(\mathbb{T}^d)$ is reflexive [15, Thm 4.10], there exists a subsequence $\{u_{n_4}\}_{n_4\in\mathbb{N}} \subset \{u_{n_3}\}_{n_3\in\mathbb{N}}$ such that

$$\rho_\star(u_{n_4}) \rightharpoonup \rho_\star \ \text{ weakly in } \ L^p(\mathbb{T}^d)$$

for some $\rho_\star \in L^p(\mathbb{T}^d)$ [15, Thm 3.18]. Furthermore, from the definition of $\rho_\star(\cdot)$ and the boundedness of $\{\rho_\star(u_n)\}$ in $L^p(\mathbb{T}^d)$, it follows that $\{\rho_\star(u_n) - \rho_0\}_{n\in\mathbb{N}}$ is bounded in $L^p_{\mathrm{zm}}(\mathbb{T}^d)$. By the continuity of the inverse Laplacian $(-\Delta)^{-1} : L^p_{\mathrm{zm}}(\mathbb{T}^d) \to W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$ (see Proposition 2.1.5), we deduce that the sequence

$$\{\varphi_\star(u_n)\} = \{(-\Delta)^{-1}(\rho_\star(u_n) - \rho_0)\}$$

is bounded in $W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$. Since this holds in particular for the sequence $\{\varphi_\star(u_{n_4})\}$, and $W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$ is reflexive – as it is a closed linear subspace of the reflexive space $W^{2,p}(\mathbb{T}^d)$ – there exists a further subsequence $\{u_{n_5}\}_{n_5\in\mathbb{N}} \subset \{u_{n_4}\}_{n_4\in\mathbb{N}}$ such that

$$\varphi_\star(u_{n_5}) \rightharpoonup \tilde{\varphi}_\star \ \text{ weakly in } W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$$

for some $\tilde{\varphi}_\star \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$. By the Rellich–Kondrachov theorem A.1.5, the embedding $W^{2,p}(\mathbb{T}^d) \hookrightarrow W^{1,p}(\mathbb{T}^d)$ is compact. Hence, up to a further subsequence (not relabelled) we have

$$\varphi_\star(u_{n_5}) \to \tilde{\varphi}_\star \ \text{ strongly in } W^{1,p}(\mathbb{T}^d),$$

and thus, extracting a subsequence if necessary, we have

$$\varphi_\star(u_{n_5}) \to \tilde{\varphi}_\star \quad \text{a.e.}$$

Combining this with the a.e. convergence $\varphi_\star(u_{n_3}) \to \varphi_\star$, the inclusion $\{u_{n_5}\}_{n_5\in\mathbb{N}} \subset \{u_{n_3}\}_{n_3\in\mathbb{N}}$ implies that $\tilde{\varphi}_\star = \varphi_\star$ a.e. Consequently, and collecting the above convergences, along the subsequence $\{u_{n_5}\}_{n_5\in\mathbb{N}}$ all the stated properties hold. □

Claim 3.3.4. *For any $1 \le p < +\infty$, let $\{u_{n_k}\}_{k\in\mathbb{N}} \subset \{u_n\}_{n\in\mathbb{N}}$ be as in Claim 3.3.3, so that $u_{n_k} \to u$ a.e. and $\varphi_\star(u_{n_k}) \rightharpoonup \varphi_\star$ weakly (in the appropriate Sobolev space) and a.e. Then, up to a subsequence,*

$$\rho_\star(u_{n_k}) \to \rho_\star \ \text{ strongly in } \ L^1(\mathbb{T}^d),$$

where

$$\rho_\star(x) = \boldsymbol{m}\frac{e^{-u(x)-\varphi_\star(x)}}{\int_{\mathbb{T}^d} e^{-u(y)-\varphi_\star(y)}\mathrm{d}y}.$$

Furthermore, if $1 < p < +\infty$, $\rho_\star(u_{n_k}) \rightharpoonup \rho_\star$ weakly in $L^p(\mathbb{T}^d)$.

Proof of Claim 3.3.4. Let $\{u_{n_k}\}_{k\in\mathbb{N}}$ be a subsequence satisfying the conclusions of both Claim 3.3.2 and Claim 3.3.3. We first prove that $\{\rho_\star(u_{n_k})\}$ is uniformly integrable for all $1 \le p < +\infty$, and then apply the Lebesgue–Vitali theorem.

Consider first the case $p = 1$, and let $R > 1$ be arbitrary. For all $k \in \mathbb{N}$, on the set $\{|\rho_\star(u_{n_k})| > R\} = \{\rho_\star(u_{n_k}) > R\}$ we estimate

$$\rho_\star(u_{n_k}) \le \frac{\rho_\star(u_{n_k})\log(\rho_\star(u_{n_k}))}{\log R}.$$

Since $\{u_{n_k}\}_{k\in\mathbb{N}} \subset \mathcal{U}(C_\varphi, C_\rho; 1, \boldsymbol{m})$, by definition of the latter set (3.11a), we therefore have

$$\begin{aligned}\int_{\{\rho_\star(u_{n_k})>R\}} \rho_\star(u_{n_k})(x)\mathrm{d}x &\le \frac{1}{\log R}\int_{\{\rho_\star(u_{n_k})>R\}} \rho_\star(u_{n_k})\log(\rho_\star(u_{n_k}))\mathrm{d}x\\ &\le \frac{1}{\log R}\|\rho_\star(u_{n_k})\log(\rho_\star(u_{n_k}))\|_{L^1(\mathbb{T}^d)} \le \frac{C_\rho}{\log R}.\end{aligned}$$

By arbitrariness of $k \in \mathbb{N}$ and $R > 1$, we deduce that

$$\lim_{R\to\infty}\sup_{k\in\mathbb{N}}\int_{\{\rho_\star(u_{n_k})>R\}} \rho_\star(u_{n_k})(x)\mathrm{d}x = 0\,,$$

that is, $\{\rho_\star(u_{n_k})\}$ is uniformly integrable in the sense of Definition 4.5.1 in [10].

We next consider the case $1 < p < +\infty$. Let $k \in \mathbb{N}$ and $R > 0$ be arbitrary. On the set $\{|\rho_\star(u_{n_k})| > R\} = \{\rho_\star(u_{n_k}) > R\}$ we estimate

$$\rho_\star(u_{n_k}) \le \frac{\rho_\star(u_{n_k})^p}{R^{p-1}}.$$

Since $\{u_{n_k}\}_{k\in\mathbb{N}} \subset \mathcal{U}(C_\rho; p, \boldsymbol{m})$, by definition of the latter set (3.11b) we therefore have

$$\begin{aligned}\int_{\{\rho_\star(u_{n_k})>R\}} \rho_\star(u_{n_k})(x)\mathrm{d}x &\le \frac{1}{R^{p-1}}\int_{\{\rho_\star(u_{n_k})>R\}} \rho_\star(u_{n_k})^p\mathrm{d}x\\ &\le \frac{1}{R^{p-1}}\|\rho_\star(u_{n_k})\|^p_{L^p(\mathbb{T}^d)} \le \frac{C_\rho^p}{R^{p-1}}.\end{aligned}$$

By arbitrariness of $k \in \mathbb{N}$ and $R > 0$, we deduce that

$$\lim_{R\to\infty}\sup_{k\in\mathbb{N}}\int_{\{\rho_\star(u_{n_k})>R\}} \rho_\star(u_{n_k})(x)\mathrm{d}x = 0\,,$$

that is, $\{\rho_\star(u_{n_k})\}$ is uniformly integrable.

For any $1 \le p < +\infty$, using the definition of $\rho_\star(\cdot)$ (3.1), we write

$$\rho_\star(u_{n_k})(x) = \boldsymbol{m}\,\frac{e^{-u_{n_k}(x)-\varphi_\star(u_{n_k})(x)}}{\int_{\mathbb{T}^d} e^{-u_{n_k}(y)-\varphi_\star(u_{n_k})(y)}\mathrm{d}y} = \frac{\boldsymbol{m}}{Z_{n_k}}\,e^{-u_{n_k}(x)-\varphi_\star(u_{n_k})(x)}.$$

By Claim 3.3.2, there exists $a \ge 0$ such that

$$\frac{1}{Z_{n_k}} \to a\,.$$

Furthermore, by Claim 3.3.3, we have $u_{n_k} \to u$ a.e. and $\varphi_\star(u_{n_k}) \to \varphi_\star$ a.e. Consequently,

$$\rho_\star(u_{n_k})(x) \to \rho_\star(x) := \boldsymbol{m}\,a\,e^{-u(x)-\varphi_\star(x)} \quad \text{for a.e. } x \in \mathbb{T}^d.$$

Since a.e. convergence on a finite measure space implies convergence in measure (see, e.g., Theorem 2.2.3 in [10]), it follows that $\rho_\star(u_{n_k}) \to \rho_\star$ in measure. Since $\rho_\star(u_{n_k}) \to \rho_\star$ in measure and the sequence is

uniformly integrable by the previous reasoning, the Lebesgue–Vitali theorem (see, e.g., Theorem 4.5.4 in [10]) implies

$$\rho_\star(u_{n_k}) \to \rho_\star \quad \text{strongly in } L^1(\mathbb{T}^d)\,.$$

In particular,

$$\|\rho_\star\|_{L^1(\mathbb{T}^d)} = \lim_{k\to\infty} \|\rho_\star(u_{n_k})\|_{L^1(\mathbb{T}^d)} = \boldsymbol{m}.$$

Using the definition of $\rho_\star$, this implies

$$\boldsymbol{m}\, a \int_{\mathbb{T}^d} e^{-u(x)-\varphi_\star(x)}\,\mathrm{d}x = \boldsymbol{m},$$

and hence

$$a = \frac{1}{\int_{\mathbb{T}^d} e^{-u(x)-\varphi_\star(x)}\,\mathrm{d}x}.$$

Therefore,

$$\rho_\star(x) = \boldsymbol{m}\frac{e^{-u(x)-\varphi_\star(x)}}{\int_{\mathbb{T}^d} e^{-u(y)-\varphi_\star(y)}\,\mathrm{d}y}.$$

For the last statement, let $1 < p < +\infty$. By Claim 3.3.3, there exists $\tilde{\rho}_\star \in L^p(\mathbb{T}^d)$ such that

$$\rho_\star(u_{n_k}) \rightharpoonup \tilde{\rho}_\star \quad \text{weakly in } L^p(\mathbb{T}^d).$$

Since the embedding $L^p(\mathbb{T}^d) \hookrightarrow L^1(\mathbb{T}^d)$ is continuous, weak convergence in $L^p(\mathbb{T}^d)$ implies weak convergence in $L^1(\mathbb{T}^d)$, and hence

$$\rho_\star(u_{n_k}) \rightharpoonup \tilde{\rho}_\star \quad \text{weakly in } L^1(\mathbb{T}^d).$$

On the other hand, the previous reasoning yields $\rho_\star(u_{n_k}) \to \rho_\star$ strongly (hence weakly) in $L^1(\mathbb{T}^d)$. The uniqueness of weak limits implies $\tilde{\rho}_\star = \rho_\star$ a.e. □

Claim 3.3.5. *Let $1 \le p < +\infty$, and let $\varphi_\star, \rho_\star$ be as in Claim* 3.3.4. *Then, $\varphi_\star$ is a weak solution, in the sense of Definition* 2.1.1, *to the Poisson–Boltzmann–Emden equation with external potential $u \in H^1 \cap L^\infty(\mathbb{T}^d)$, that is,*

$$-\Delta\varphi_\star = \boldsymbol{m}\,\frac{e^{-u-\varphi_\star}}{\int_{\mathbb{T}^d} e^{-u-\varphi_\star}\mathrm{d}y} - \rho_0,$$

and $\rho_\star \in H^{-1}(\mathbb{T}^d)$. In particular, $\rho_\star = \rho_\star(u)$ an $\varphi_\star = \varphi_\star(u)$.

Proof of Claim 3.3.5. Let $\{u_{n_k}\}_{k\in\mathbb{N}} \subset \{u_n\}_{n\in\mathbb{N}}$ be as in Claim 3.3.4, so that

$$\varphi_\star(u_{n_k}) \rightharpoonup \varphi_\star \quad \text{weakly in } H^1_{\mathrm{zm}}(\mathbb{T}^d)$$

and

$$\rho_\star(u_{n_k}) \to \rho_\star = \boldsymbol{m}\frac{e^{-u-\varphi_\star}}{\int_{\mathbb{T}^d} e^{-u-\varphi_\star}\mathrm{d}y} \quad \text{strongly in } L^1(\mathbb{T}^d).$$

Since $\{u_{n_k}\}_{k\in\mathbb{N}} \subset \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$, by definition of the latter set, the sequence $\{\varphi_\star(u_{n_k})\}_{k\in\mathbb{N}}$ is bounded in $H^1_{\mathrm{zm}}(\mathbb{T}^d)$. Therefore, by continuity of the Laplacian

$$-\Delta : H^1_{\mathrm{zm}}(\mathbb{T}^d) \to H^{-1}_{\mathrm{zm}}(\mathbb{T}^d),$$

and using the distributional identity

$$\rho_\star(u_{n_k}) - \rho_0 = -\Delta\varphi_\star(u_{n_k}), \quad k \in \mathbb{N},$$

the sequence $\{\rho_\star(u_{n_k}) - \rho_0\}_{k\in\mathbb{N}}$ is bounded in $H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$, and hence

$$\sup_{k\in\mathbb{N}} \|\rho_\star(u_{n_k})\|_{H^{-1}(\mathbb{T}^d)} \le C$$

for some $C \geq 0$. Define next the distribution $T_{\rho_\star} \in \mathcal{D}'(\mathbb{T}^d)$ by

$$T_{\rho_\star}(\psi) := \int_{\mathbb{T}^d} \rho_\star \psi \mathrm{d}x, \qquad \psi \in \mathcal{C}^\infty(\mathbb{T}^d).$$

Then, $T_{\rho_\star}$ is well-defined since $\rho_\star \in L^1(\mathbb{T}^d)$. Furthermore, $T_{\rho_\star}$ is continuous with respect to the $H^1(\mathbb{T}^d)$ topology. Indeed, since $\rho_\star(u_{n_k}) \to \rho_\star$ strongly (hence weakly) in $L^1(\mathbb{T}^d)$, for all $\psi \in \mathcal{C}^\infty(\mathbb{T}^d)$ we have

$$\begin{aligned}\int_{\mathbb{T}^d} \rho_\star \psi \mathrm{d}x = \lim_{k\to\infty} \int_{\mathbb{T}^d} \rho_\star(u_{n_k})\psi \mathrm{d}x &= \lim_{k\to\infty} \langle \rho_\star(u_{n_k}), \psi\rangle_{H^{-1}(\mathbb{T}^d),H^1(\mathbb{T}^d)} \\ &\leq \limsup_{k\to\infty} \|\rho_\star(u_{n_k})\|_{H^{-1}(\mathbb{T}^d)} \|\psi\|_{H^1(\mathbb{T}^d)} \leq C\|\psi\|_{H^1(\mathbb{T}^d)}.\end{aligned}$$

where we used that $\rho_\star(u_{n_k}) \in L^1 \cap H^{-1}(\mathbb{T}^d)$ for all $k \in \mathbb{N}$ by definition of $\rho_\star(\cdot)$. By Friedrichs' Lemma (see Corollary A.2.2), $\mathcal{C}^\infty(\mathbb{T}^d)$ is dense in $H^1(\mathbb{T}^d)$, and hence $T_{\rho_\star}$ extends continuously to a distribution in $H^{-1}(\mathbb{T}^d)$. Identifying such distribution with $\rho_\star$ yields

$$\rho_\star \in L^1 \cap H^{-1}(\mathbb{T}^d).$$

Furthermore, $\rho_\star(u_{n_k}) \rightharpoonup \rho_\star$ weakly in $H^{-1}(\mathbb{T}^d)$, and hence $\rho_\star(u_{n_k}) - \rho_0 \rightharpoonup \rho_\star - \rho_0$ weakly in $H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$. On the other hand, we have

$$(-\Delta)^{-1}(\rho_\star(u_{n_k}) - \rho_0) = \varphi_\star(u_{n_k}) \rightharpoonup \varphi_\star \quad \text{weakly in } H^1_{\mathrm{zm}}(\mathbb{T}^d).$$

Consequently, since the Laplacian is an isomorphism of Banach spaces, uniqueness of the weak limits implies the distributional identity

$$-\Delta\varphi_\star = \rho_\star - \rho_0,$$

that is, $\varphi_\star \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ is the unique zero-mean solution to the Poisson equation with forcing term $\rho_\star - \rho_0 \in H^{-1}_{\mathrm{zm}}(\mathbb{T}^d)$. Substituting the expression of $\rho_\star$ yields the first conclusion.

We now turn to the second part. Define

$$f_\star(x,v) := \rho_\star(x)\mathrm{M}(v),$$

where $\mathrm{M}(v)$ is the normalised d-dimensional Maxwellian. Then, since $\rho_\star \in L^1(\mathbb{T}^d)$ and $\mathrm{M}(\cdot) \in L^1(\mathbb{R}^d)$, Fubini's theorem implies that $f_\star \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$. Moreover, from the previous reasoning, $\varphi_\star \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ is the unique zero-mean solution to the Poisson–Boltzmann–Emden equation (3.2) with external potential $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. Therefore, by Theorem 2.2.23 $(f_\star, \varphi_\star)$ is the unique free energy solution to the stationary VPFP system with external potential $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. We therefore conclude that $\rho_\star = \rho_\star(u)$ and $\varphi_\star = \varphi_\star(u)$ as wished. □

We now conclude the proof of Theorem 3.3.1. Let $\{u_{n_k}\}_{k\in\mathbb{N}} \subset \{u_n\}_{n\in\mathbb{N}}$, $\rho_\star$ and $\varphi_\star$ be as in Claim 3.3.4. Then, from Claim 3.3.5 it follows that $\rho_\star = \rho_\star(u)$ and $\varphi_\star = \varphi_\star(u)$. Hence, by Claims 3.3.3 and 3.3.4, for all $1 \leq p < +\infty$ we have

$$\varphi_\star(u_{n_k}) \rightharpoonup \varphi_\star(u) \quad \text{weakly in } H^1_{\mathrm{zm}}(\mathbb{T}^d) \qquad \text{and} \qquad \rho_\star(u_{n_k}) \to \rho_\star(u) \quad \text{strongly in } L^1(\mathbb{T}^d),$$

and, if $1 < p < +\infty$,

$$\begin{aligned}&\varphi_\star(u_{n_k}) \rightharpoonup \varphi_\star(u) \quad \text{weakly in } W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d), \quad \text{and} \\ &\rho_\star(u_{n_k}) \rightharpoonup \rho_\star(u) \quad \text{weakly in } L^p(\mathbb{T}^d).\end{aligned}$$

By the same argument, any subsequence of $\{u_n\}_{n\in\mathbb{N}}$ has a further subsequence for which the desired convergence holds. This implies (3.13). The claim follows by arbitrariness of $\{u_n\}$. □

Let $\Gamma_{\mathcal{E}}$ denote the entropy operator defined in (3.10).

Proposition 3.3.6. *Let $\boldsymbol{m} > 0$.*

(*i*) *$\Gamma_{\mathcal{E}}$ is well-defined.*

(*ii*) *For any $1 < p < +\infty$ and $C_\varphi, C_\rho \geq 0$, $\Gamma_{\mathcal{E}}$ is weak-to-strong sequentially continuous on $\mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$.*

Proof of Proposition 3.3.6 (*i*). Let $u \in \tilde{\mathcal{U}}(1, \boldsymbol{m})$ be arbitrary. Then, by definition of solution operators,

$$f_\star(u)(x, v) := \rho_\star(u)(x)\mathrm{M}(v) \in \tilde{\mathcal{A}}(\boldsymbol{m}),$$

where the admissible set of states $\tilde{\mathcal{A}}(\boldsymbol{m})$ is as in (2.9). By definition of the latter set and Lemma 2.2.10, we have

$$f_\star(u), f_\star(u)\log(f_\star(u)) \in L^1(\mathbb{T}^d \times \mathbb{R}^d).$$

Since $\rho_\star(u) \geq 0$ a.e. (see (3.1)), from Claim 2.2.24 it follows that $\rho_\star(u)\log(\rho_\star(u)) \in L^1(\mathbb{T}^d)$. Hence, $\Gamma_{\mathcal{E}}$ is well-defined. □

The proof of Proposition 3.3.6 (*ii*) relies on several auxiliary results, which we present first.

Claim 3.3.7. *Let $1 < p < +\infty$ be arbitrary. Then, there exists $m > 0$ such that*

$$|x \log x| \leq m(1 + x^p) \quad \textit{for all } x \geq 0.$$

Proof of Claim 3.3.7. Define the function $g : (0, +\infty) \to \mathbb{R}$ by

$$g(x) := \frac{x \log x}{x^p}.$$

Using the change of variables $x = e^t$, $t \in \mathbb{R}$, yields

$$\tilde{g}(t) := g(e^t) = te^{(1-p)t}.$$

Then, $\tilde{g} \in \mathcal{C}^1(\mathbb{R})$, and its first derivative is given by

$$\tilde{g}'(t) = e^{(1-p)t}(1 + (1-p)t).$$

Straightforward calculations yield

$$\tilde{g}'(t) = 0 \quad \Longleftrightarrow \quad t = \frac{1}{p-1} > 0,$$

and

$$\tilde{g}'(t) > 0 \quad \text{if } t < \frac{1}{p-1}, \qquad \tilde{g}'(t) < 0 \quad \text{if } t > \frac{1}{p-1}.$$

Consequently, $\tilde{g}$ attains its maximum at $t = (p-1)^{-1}$ with value

$$\tilde{g}(t) = \frac{1}{e(p-1)} =: m_p.$$

We thus deduce that

$$g(x) = \frac{x \log x}{x^p} \leq m_p \quad \text{for all } x > 0.$$

Combining this with the standard estimate $x \log x \geq -\frac{1}{e}$ for all $x \geq 0$ yields

$$|x \log x| \leq m_p\, x^p + \frac{1}{e} \qquad \text{for all } x > 0.$$

Using the standard convention $0 \log 0 = 0$, the above estimate extends to $x \geq 0$. Hence, the claim follows with any $m \geq \max\{m_p, e^{-1}\}$. □

Claim 3.3.8. *Let $\boldsymbol{m} > 0$, $1 < p < +\infty$ and $C_\varphi, C_\rho \geq 0$ be arbitrary. Then, the collection*

$$\{\Gamma\mathcal{E}(u)\}_{u \in \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})}$$

is relatively weakly compact in $L^1(\mathbb{T}^d)$.

Proof of Claim 3.3.8. We apply the Dunford–Pettis theorem (see, e.g., Theorem 4.30 in [15]). Without loss of generality, we assume that $C_\varphi, C_\rho > 0$ are chosen such that $\mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m}) \neq \emptyset$.

Let $u \in \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$. Since $\mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m}) \subset \tilde{\mathcal{U}}(1, \boldsymbol{m})$, $\Gamma\mathcal{E}(u) = \rho_\star(u)\log(\rho_\star(u)) \in L^1(\mathbb{T}^d)$ by Proposition 3.3.6 (i). Moreover, by Claim 3.3.7 there exists $m_p > 0$ such that

$$|\rho_\star(u)\log(\rho_\star(u))| \leq m_p(1 + \rho_\star(u)^p) \quad \text{a.e.}$$

Since $\|\rho_\star(u)\|_{L^p(\mathbb{T}^d)} \leq C_\rho$ by definition of $\mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$, integrating the previous estimate yields

$$\int_{\mathbb{T}^d} |\rho_\star(u)\log(\rho_\star(u))| \mathrm{d}x \leq m_p(|\mathbb{T}^d| + C_\rho^p).$$

By arbitrariness of $u \in \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$, the collection $\{\Gamma\mathcal{E}(u)\}_{u \in \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})}$ is uniformly bounded in $L^1(\mathbb{T}^d)$.

We next prove equi-integrability. Fix $1 < q < p$. Let $\varepsilon > 0$ be arbitrary, and let $A \subset \mathbb{T}^d$ be any Lebesgue measurable set such that $|A| < \delta$, where $\delta > 0$. By Claim 3.3.7, there exists $m_q > 0$ such that

$$|\rho_\star(u)\log(\rho_\star(u))| \leq m_q(1 + \rho_\star(u)^q) \quad \text{a.e.}$$

Since the embedding $L^p(\mathbb{T}^d) \hookrightarrow L^q(\mathbb{T}^d)$ is continuous and $\rho_\star(u) \in L^p(\mathbb{T}^d)$, it follows that $\rho_\star(u)^q \in L^1(\mathbb{T}^d)$. Hence, integrating the above identity over A yields

$$\int_A |\rho_\star(u)\log(\rho_\star(u))| \mathrm{d}x \leq m_q|A| + m_q \int_A \rho_\star(u)^q \mathrm{d}x \leq m_q|A| + m_q \|\rho_\star(u)\|_{L^p(\mathbb{T}^d)}^{\frac{p}{q}} |A|^{\frac{p-q}{q}},$$

where the last estimate follows from Hölder's inequality [15, Thm 4.6]. Choosing

$$0 < \delta < \max\left\{ \frac{\varepsilon}{2m_q}, \left(\frac{\varepsilon}{2m_q C_\rho^{\frac{p}{q}}} \right)^{\frac{p}{p-q}} \right\}$$

ensures that

$$\int_A |\rho_\star(u)\log(\rho_\star(u))| \mathrm{d}x < m_q\delta + m_q C_\rho^{\frac{p}{q}} \delta^{\frac{p-q}{q}} < \varepsilon$$

uniformly in $u \in \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$. By arbitrariness of $|A| < \delta$ and $\varepsilon > 0$, equi-integrability follows.

Since $\mathbb{T}^d$ has finite measure, tightness is immediate. Indeed, for all $\varepsilon > 0$, it suffices to let $X = \mathbb{T}^d$, where $|X| < +\infty$, to obtain

$$\int_{\mathbb{T}^d \setminus X} |\rho_\star(u)\log(\rho_\star(u))| \mathrm{d}x = 0 < \varepsilon \quad \text{for all } u \in \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m}).$$

By boundedness, equi-integrability and tightness, the Dunford–Pettis theorem implies that the set $\{\Gamma\mathcal{E}(u)\}_{u \in \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})}$ has compact closure in the weak topology of $L^1(\mathbb{T}^d)$. □

Proof of Proposition 3.3.6 (ii). Let $\{u_n\}_{n \in \mathbb{N}} \subset \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$ be such that $u_n \rightharpoonup u$ weakly in $H^1 \cap L^\infty(\mathbb{T}^d)$. We aim to prove that

$$\rho_\star(u_n)\log(\rho_\star(u_n)) \to \rho_\star(u)\log(\rho_\star(u)) \quad \text{strongly in } L^1(\mathbb{T}^d). \tag{3.15}$$

Since $H^1 \cap L^\infty(\mathbb{T}^d)$ is a Banach space, the weak limit u also belongs to $H^1 \cap L^\infty(\mathbb{T}^d)$. Moreover, by the weak sequential continuity of the solution operators established in Theorem 3.3.1, there holds [15, Prop 3.5]

$$\|\rho_\star(u)\|_{L^p(\mathbb{T}^d)} \leq \liminf_{n\to\infty} \|\rho_\star(u_n)\|_{L^p(\mathbb{T}^d)} \leq C_\rho, \qquad \|\varphi_\star(u)\|_{H^1(\mathbb{T}^d)} \leq \liminf_{n\to\infty} \|\varphi_\star(u_n)\|_{H^1(\mathbb{T}^d)} \leq C_\varphi.$$

Hence, $u \in \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$, and therefore Proposition 3.3.6 (i) yields $\rho_\star(u)\log\rho_\star(u) \in L^1(\mathbb{T}^d)$. Next, by Claim 3.3.8 and the Dunford–Pettis theorem, the sequence

$$\{\rho_\star(u_n)\log(\rho_\star(u_n))\}_{n\in\mathbb{N}}$$

is bounded in $L^1(\mathbb{T}^d)$ and has uniformly absolutely continuous integrals in the sense of Definition 4.5.2 in [10]. Therefore, the above sequence is uniformly integrable (see, e.g., Proposition 4.5.3 in [10]). Moreover, by Theorem 3.3.1, we have

$$\rho_\star(u_n) \to \rho_\star(u) \quad \text{strongly in } L^1(\mathbb{T}^d).$$

Hence, there exists a subsequence $\{u_{n_k}\}_{k\in\mathbb{N}} \subset \{u_n\}_{n\in\mathbb{N}}$ such that (see [15, Thm 4.9]) $\rho_\star(u_{n_k}) \to \rho_\star(u)$ a.e., and thus

$$\rho_\star(u_{n_k})\log(\rho_\star(u_{n_k})) \to \rho_\star(u)\log(\rho_\star(u)) \quad \text{for a.e. } x \in \mathbb{T}^d.$$

Since a.e. convergence on a finite measure space implies convergence in measure [10, Thm 2.2.3], it follows that

$$\rho_\star(u_{n_k})\log(\rho_\star(u_{n_k})) \to \rho_\star(u)\log(\rho_\star(u)) \quad \text{in measure.}$$

Since the sequence $\{\rho_\star(u_{n_k})\log(\rho_\star(u_{n_k}))\}$ converges in measure to $\rho_\star(u)\log(\rho_\star(u))$ and is uniformly integrable, the Lebesgue–Vitali theorem (see, e.g., [10, Thm 4.5.4]) implies that

$$\rho_\star(u_{n_k})\log(\rho_\star(u_{n_k})) \to \rho_\star(u)\log(\rho_\star(u)) \quad \text{strongly in } L^1(\mathbb{T}^d).$$

By the same argument, any subsequence of $\{u_n\}_{n\in\mathbb{N}}$ has a further subsequence for which the desired convergence holds. This implies (3.15). The claim follows by arbitrariness of $\{u_n\}$. □

The following result builds on Proposition 3.3.6 and will be employed in the next chapter to establish the well-posedness of the variational characterisation of confinement.

Lemma 3.3.9. *Let $\boldsymbol{m} > 0$ and $1 \leq p < +\infty$. Then, for every $C_\varphi, C_\rho \geq 0$, the functional*

$$\mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m}) \to \mathbb{R}, \ u \mapsto \int_{\mathbb{T}^d} \rho_\star(u)\log(\rho_\star(u))\mathrm{d}x$$

is weakly sequentially lower semicontinuous. Furthermore, if $1 < p < +\infty$, the functional is weakly sequentially continuous.

Proof. Since $\mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m}) \subset \tilde{\mathcal{U}}(1, \boldsymbol{m})$ for all $1 \leq p < +\infty$ and $C_\varphi, C_\rho \geq 0$, and $\Gamma_{\mathcal{E}}$ is well-defined on $\tilde{\mathcal{U}}(1, \boldsymbol{m})$ by Proposition 3.3.6 (i), the functional is well-defined for all $1 \leq p < +\infty$ and $C_\varphi, C_\rho \geq 0$. Without loss of generality, we assume that C_φ, C_ρ are chosen such that $\mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m}) \neq \emptyset$.

For the second claim, let $1 < p < +\infty$ be arbitrary. Let $\{u_n\}_{n\in\mathbb{N}} \subset \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$ be such that $u_n \rightharpoonup u$ weakly in $H^1 \cap L^\infty(\mathbb{T}^d)$. Then, Proposition 3.3.6 yields

$$\rho_\star(u_n)\log(\rho_\star(u_n)) \to \rho_\star(u)\log(\rho_\star(u)) \quad \text{strongly (hence weakly) in } L^1(\mathbb{T}^d).$$

Hence,

$$\lim_{n\to\infty} \int_{\mathbb{T}^d} \rho_\star(u_n)\log(\rho_\star(u_n))\mathrm{d}x = \int_{\mathbb{T}^d} \rho_\star(u)\log(\rho_\star(u))\mathrm{d}x.$$

The result follows by arbitrariness of $\{u_n\}$.

We next turn to the first claim. Let $1 \le p < +\infty$ be arbitrary, and let $\{u_n\}_{n\in\mathbb{N}} \subset \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$ be such that $u_n \rightharpoonup u$ weakly in $H^1 \cap L^\infty(\mathbb{T}^d)$. We aim to show that

$$\int_{\mathbb{T}^d} \rho_\star(u)\log(\rho_\star(u))\mathrm{d}x \le \liminf_{n\to\infty} \int_{\mathbb{T}^d} \rho_\star(u_n)\log(\rho_\star(u_n))\mathrm{d}x \tag{3.16}$$

To this end, for any $0 < \varepsilon < 1 < k$, define the upper-lower truncation operator $T_{\varepsilon,k} : L^1(\mathbb{T}^d) \to L^\infty(\mathbb{T}^d)$ by

$$T_{\varepsilon,k}(f(x)) := \begin{cases} f(x) & \text{if } \varepsilon \le |f(x)| \le k, \\ \varepsilon & \text{if } |f(x)| < \varepsilon, \\ k & \text{if } |f(x)| > k. \end{cases}$$

Fix $0 < \varepsilon < 1 < k$. Applying the standard tangent line inequality

$$t\log t \ge t_0 \log t_0 + (1+\log t_0)(t-t_0) = t - t_0 + t\log t_0\,, \qquad t, t_0 \ge 0,$$

with $t = \rho_\star(u_n)$ and $t_0 = T_{\varepsilon,k}(\rho_\star(u))$, we obtain

$$\rho_\star(u_n)\log(\rho_\star(u_n)) \ge \rho_\star(u_n) - T_{\varepsilon,k}(\rho_\star(u)) + \rho_\star(u_n)\log\left(T_{\varepsilon,k}(\rho_\star(u))\right) \quad \text{a.e.}$$

Since $\rho_\star(u) \in L^1(\mathbb{T}^d)$, the definition of $T_{\varepsilon,k}(\rho_\star(u))$ yields

$$T_{\varepsilon,k}(\rho_\star(u)) \in L^\infty(\mathbb{T}^d) \subset L^1(\mathbb{T}^d).$$

Moreover, since $\log\varepsilon \le \log\left(T_{\varepsilon,k}(\rho_\star(u))\right) \le \log k$ and $\rho_\star(u_n) \in L^1(\mathbb{T}^d)$, we deduce that

$$\rho_\star(u_n)\log\left(T_{\varepsilon,k}(\rho_\star(u))\right) \in L^1(\mathbb{T}^d).$$

Hence, every term in the previous estimate is integrable, and the monotonicity of the integral thus yields

$$\int_{\mathbb{T}^d} \rho_\star(u_n)\log(\rho_\star(u_n))\mathrm{d}x \ge \int_{\mathbb{T}^d} \rho_\star(u_n)\mathrm{d}x - \int_{\mathbb{T}^d} T_{\varepsilon,k}(\rho_\star(u))\mathrm{d}x + \int_{\mathbb{T}^d} \rho_\star(u_n)\log\left(T_{\varepsilon,k}(\rho_\star(u))\right)\mathrm{d}x\,.$$

Since $\{u_n\}_{n\in\mathbb{N}} \subset \mathcal{U}(C_\varphi, C_\rho; p, \boldsymbol{m})$ and $u_n \rightharpoonup u$ weakly in $H^1 \cap L^\infty(\mathbb{T}^d)$, by Theorem 3.3.1 we have

$$\rho_\star(u_n) \to \rho_\star(u) \quad \text{strongly (hence weakly) in } L^1(\mathbb{T}^d).$$

Since additionally $\log\left(T_{\varepsilon,k}(\rho_\star(u))\right) \in L^\infty(\mathbb{T}^d)$, passing to the limit inferior we obtain

$$\liminf_{n\to\infty} \int_{\mathbb{T}^d} \rho_\star(u_n)\log(\rho_\star(u_n))\mathrm{d}x \ge \int_{\mathbb{T}^d} \rho_\star(u)\mathrm{d}x - \int_{\mathbb{T}^d} T_{\varepsilon,k}(\rho_\star(u))\mathrm{d}x + \int_{\mathbb{T}^d} \rho_\star(u)\log\left(T_{\varepsilon,k}(\rho_\star(u))\right)\mathrm{d}x\,. \tag{3.17}$$

Next, for all $0 < \varepsilon < 1 < k$, there holds

$$0 \le T_{\varepsilon,k}(\rho_\star(u)) \le 1 + \rho_\star(u) \in L^1(\mathbb{T}^d).$$

Moreover,

$$T_{\varepsilon,k}(\rho_\star(u)) \to \rho_\star(u) \quad \text{a.e. as } \varepsilon \to 0 \text{ and } k \to \infty.$$

Therefore, the dominated convergence theorem [10, Thm 2.8.1] yields

$$\lim_{\varepsilon\to 0}\lim_{k\to\infty} \int_{\mathbb{T}^d} T_{\varepsilon,k}(\rho_\star(u))\mathrm{d}x = \int_{\mathbb{T}^d} \rho_\star(u)\mathrm{d}x.$$

Consequently, taking the limit as $k \to \infty$ and $\varepsilon \to 0$ in (3.17) yields

$$\liminf_{n\to\infty} \int_{\mathbb{T}^d} \rho_\star(u_n)\log(\rho_\star(u_n))\mathrm{d}x \ge \limsup_{\varepsilon\to 0}\limsup_{k\to\infty} \int_{\mathbb{T}^d} \rho_\star(u)\log\left(T_{\varepsilon,k}(\rho_\star(u))\right)\mathrm{d}x\,.$$

To conclude (3.16), it suffices to show that

$$\lim_{\varepsilon\to 0}\lim_{k\to\infty}\int_{\mathbb{T}^d}\rho_\star(u)\log\left(T_{\varepsilon,k}(\rho_\star(u))\right)\mathrm{d}x=\int_{\mathbb{T}^d}\rho_\star(u)\log(\rho_\star(u))\mathrm{d}x\,. \tag{3.18}$$

To this end, for any $0<\varepsilon<1$ and $k>1$, we split the integral as follows:

$$\begin{aligned}\int_{\mathbb{T}^d}\rho_\star(u)\log\left(T_{\varepsilon,k}(\rho_\star(u))\right)\mathrm{d}x&=\int_{\{\rho_\star(u)<1\}}\rho_\star(u)\log\left(T_{\varepsilon,k}(\rho_\star(u))\right)\mathrm{d}x+\int_{\{\rho_\star(u)\geq 1\}}\rho_\star(u)\log\left(T_{\varepsilon,k}(\rho_\star(u))\right)\mathrm{d}x\\&=\int_{\{\rho_\star(u)<1\}}\rho_\star(u)\log\max\{\rho_\star(u),\varepsilon\}\mathrm{d}x+\int_{\{\rho_\star(u)\geq 1\}}\rho_\star(u)\log\min\{\rho_\star(u),k\}\mathrm{d}x\,.\end{aligned}$$

For any $1<k_1\leq k_2$, there holds

$$0\leq\rho_\star(u)\log\left(\min\{\rho_\star(u),k_1\}\right)\leq\rho_\star(u)\log\left(\min\{\rho_\star(u),k_2\}\right)\quad\text{a.e. on }\{\rho_\star(u)\geq 1\}.$$

Hence, by the monotone convergence theorem [10, Thm 2.8.2],

$$\lim_{\varepsilon\to 0}\lim_{k\to\infty}\int_{\{\rho_\star(u)\geq 1\}}\rho_\star(u)\log\left(\min\{\rho_\star(u),k\}\right)\mathrm{d}x=\int_{\{\rho_\star(u)\geq 1\}}\rho_\star(u)\log\left(\rho_\star(u)\right)\mathrm{d}x\,,$$

Similarly, for any $\varepsilon_1\geq\varepsilon_2>0$, there holds

$$0\leq\rho_\star(u)\left|\log\left(\max\{\rho_\star(u),\varepsilon_1\}\right)\right|\leq\rho_\star(u)\left|\log\left(\max\{\rho_\star(u),\varepsilon_2\}\right)\right|\quad\text{a.e. on }\{\rho_\star(u)<1\}.$$

Hence, the monotone convergence theorem yields

$$\begin{aligned}\lim_{\varepsilon\to 0}\lim_{k\to\infty}-\int_{\{\rho_\star(u)<1\}}\rho_\star(u)\log\left(\max\{\rho_\star(u),\varepsilon\}\right)\mathrm{d}x&=\lim_{\varepsilon\to 0}\int_{\{\rho_\star(u)<1\}}\rho_\star(u)\left|\log\left(\max\{\rho_\star(u),\varepsilon\}\right)\right|\mathrm{d}x\\&=\int_{\{\rho_\star(u)<1\}}\rho_\star(u)\left|\log(\rho_\star(u))\right|\mathrm{d}x\\&=-\int_{\{\rho_\star(u)<1\}}\rho_\star(u)\log(\rho_\star(u))\mathrm{d}x\end{aligned}$$

Combining the previous limits yields (3.18) as wished. □

3.4 Fréchet Differentiability of Solution Operators in Low Dimensions

In this section, we establish the continuous Fréchet differentiability of Γ_ρ^p and Γ_φ^p in low dimensions ($d=1,2$) for a suitable choice of $1\leq p<+\infty$, using an implicit argument similar to that of Theorem 3.1 in [11]. First, we establish the continuous Fréchet differentiability of Γ_φ^p. We then deduce the corresponding property for Γ_ρ^p by applying the Fréchet chain rule (see, e.g., Theorem 2.2.1 in [18]) to the Euler–Lagrange representation of $\rho_\star(u)$.

The restriction to low dimensions is dictated by the regularity theory established previously. When $d=1,2$, Lemma 2.2.28 yields the improved regularity

$$\rho_\star(u)\in L^p(\mathbb{T}^d)\quad\text{and}\quad\varphi_\star(u)\in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)\quad\text{for all }1\leq p<+\infty,\ u\in H^1\cap L^\infty(\mathbb{T}^d).$$

Hence, the natural domain of the free energy solution operators Γ_ρ^p and Γ_φ^p is the fixed space $\tilde{\mathcal{U}}(p,\boldsymbol{m})=H^1\cap L^\infty(\mathbb{T}^d)$ for all $1\leq p<+\infty$ (see also Remark 3.2.1). Conversely, when $d\geq 3$, Lemma 2.2.28 yields the above regularity only under the additional assumption that $\varphi_\star(u)\in W^{1,d}_{\mathrm{zm}}(\mathbb{T}^d)$. However, since the Poisson–Boltzmann–Emden equation (3.2) is highly nonlinear in $\varphi_\star(u)$, the set of all $u\in H^1\cap L^\infty(\mathbb{T}^d)$ for which this additional regularity holds is not expected to be stable under linear perturbations and therefore does not naturally form a linear space. Consequently, we restrict our analysis to the case $d=1,2$, where

the solution operators are defined on a fixed Banach space. Extending these results to higher dimensions would require a different functional framework and is left for future work.

For any spatial dimension d and $1 \leq p < +\infty$, we introduce the spaces

$$X^p := \left(H^1 \cap L^\infty(\mathbb{T}^d)\right) \times W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d) \qquad \text{and} \qquad Y^p := L^p_{\mathrm{zm}}(\mathbb{T}^d),$$

where $H^1 \cap L^\infty(\mathbb{T}^d)$ is endowed with the norm defined in (3.12), and X^p with the associated product norm

$$\|(u,\varphi)\|_{X^p} := \|u\|_{H^1\cap L^\infty(\mathbb{T}^d)} + \|\varphi\|_{W^{2,p}(\mathbb{T}^d)}.$$

We then define the Poisson–Boltzmann–Emden operator $F^p : X^p \to Y^p$ by

$$F^p(u,\varphi) := \Delta\varphi + \left(\boldsymbol{m}\frac{e^{-u-\varphi}}{\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}x} - \rho_0\right).$$

By definition, when $d = 1, 2$, $\varphi_\star(u)$ satisfies

$$F^p(u,\varphi_\star(u)) = 0 \qquad \text{for all } u \in H^1 \cap L^\infty(\mathbb{T}^d).$$

To establish the continuous Fréchet differentiability of Γ^p_φ when $d = 1, 2$, we will therefore apply the implicit function theorem (see, e.g., Theorem 10.2.1 in [21]).

The remainder of this section is organised as follows. In Subsection 3.4.1, we prove several intermediate results. Building on this, Subsection 3.4.2 establishes that F^p is continuously Fréchet differentiable with respect to both u and φ for a suitable choice of p, and its Fréchet derivative with respect to φ is a Banach space isomorphism. Finally, Subsection 3.4.3 proves the continuous Fréchet differentiability of Γ^p_φ and Γ^p_ρ when $d = 1, 2$.

3.4.1 Preliminary Results

We begin with a fundamental observation.

Claim 3.4.1. *Let $d \geq 1$ and $p \in [1, +\infty)$ be such that $p > \frac{d}{2}$. Then, $e^{-u-\varphi} \in L^\infty(\mathbb{T}^d)$ for all $(u,\varphi) \in X^p$.*

Proof. Let $(u,\varphi) \in X^p$ be arbitrary. Since $p > \frac{d}{2}$, the Sobolev embedding theorem (Corollary A.1.4) yields the continuous embedding

$$W^{2,p}(\mathbb{T}^d) \hookrightarrow \mathcal{C}(\mathbb{T}^d).$$

In particular, this implies that $\varphi \in \mathcal{C}(\mathbb{T}^d)$. By compactness of $\mathbb{T}^d$, there holds $\mathcal{C}(\mathbb{T}^d) \hookrightarrow L^\infty(\mathbb{T}^d)$. Combining this with the boundedness of u yields $e^{-u-\varphi} \in L^\infty(\mathbb{T}^d)$ as wished. □

The preceding result implies that the maps

$$G^p_1 : X^p \to L^p(\mathbb{T}^d),\ (u,\varphi) \mapsto e^{-u-\varphi} \qquad \text{and} \qquad G^p_2 : X^p \to \mathbb{R},\ (u,\varphi) \mapsto \int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}x$$

are well-defined whenever $p > \frac{d}{2}$. In the remainder of this subsection, we prove that they are continuously Fréchet differentiable. Combining this with the Fréchet chain rule, in the next subsection we establish the continuous Fréchet differentiability of F^p. We proceed by treating each mapping separately in the following two lemmas.

Lemma 3.4.2. *Let $d \geq 1$ and $p \in [1, +\infty)$ be such that $p > \frac{d}{2}$. Then, $G^p_1 \in \mathcal{C}^1(X^p, L^p(\mathbb{T}^d))$ with Fréchet derivative*

$$DG^p_1(u,\varphi) : X^p \to L^p(\mathbb{T}^d),\ (h,k) \mapsto -e^{-u-\varphi}(h+k)$$

for all $(u,\varphi) \in X^p$.

The proof of Lemma 3.4.2 relies on a preliminary result, which we establish first.

Claim 3.4.3. *Let $d \geq 1$ and $p \in [1, +\infty)$ be such that $p > \frac{d}{2}$. For any $(u, \varphi) \in X^p$, there holds*

$$\lim_{\|h\|_{H^1 \cap L^\infty(\mathbb{T}^d)} \to 0} \frac{\|e^{-u-\varphi}(e^{-h} - 1 + h)\|_{L^p(\mathbb{T}^d)}}{\|h\|_{H^1 \cap L^\infty(\mathbb{T}^d)}} = 0$$

and

$$\lim_{\|k\|_{W^{2,p}(\mathbb{T}^d)} \to 0} \frac{\|e^{-u-\varphi}(e^{-k} - 1 + k)\|_{L^p(\mathbb{T}^d)}}{\|k\|_{W^{2,p}(\mathbb{T}^d)}} = 0.$$

Proof. Let $(u, \varphi) \in X^p$ and let $h : \mathbb{T}^d \to \mathbb{R}$ be arbitrary. We begin with the Taylor expansion of the scalar function

$$\mathbb{R} \to \mathbb{R}, \qquad t \mapsto e^{-t} - 1 + t,$$

at $t = 0$, with remainder in Lagrange form:

$$e^{-t} - 1 + t = \frac{t^2}{2} e^{-\xi t}, \qquad \xi \in (0, 1).$$

Applied pointwise with $t = h(x)$, this yields

$$|e^{-h} - 1 + h| \leq \frac{1}{2} e^{|h|} h^2 \quad \text{a.e. on } \mathbb{T}^d. \tag{3.19}$$

First limit - $H^1 \cap L^\infty(\mathbb{T}^d)$. Let $h \in H^1 \cap L^\infty(\mathbb{T}^d) \setminus \{0\}$ be arbitrary. Using (3.19) yields

$$\|e^{-h} - 1 + h\|_{L^p(\mathbb{T}^d)} \leq |\mathbb{T}^d|^{\frac{1}{p}} e^{\|h\|_{L^\infty(\mathbb{T}^d)}} \frac{\|h\|^2_{L^\infty(\mathbb{T}^d)}}{2}.$$

Since $(u, \varphi) \in X^p$ with $p > \frac{d}{2}$, Claim 3.4.1 implies that $e^{-u-\varphi} \in L^\infty(\mathbb{T}^d)$. Combining this with the above estimate, we obtain

$$\begin{aligned} \frac{\|e^{-u-\varphi}(e^{-h} - 1 + h)\|_{L^p(\mathbb{T}^d)}}{\|h\|_{H^1 \cap L^\infty(\mathbb{T}^d)}} &\leq |\mathbb{T}^d|^{\frac{1}{p}} \|e^{-u-\varphi}\|_{L^\infty(\mathbb{T}^d)} \frac{e^{\|h\|_{L^\infty(\mathbb{T}^d)}}}{2} \frac{\|h\|^2_{L^\infty(\mathbb{T}^d)}}{\|h\|_{H^1 \cap L^\infty(\mathbb{T}^d)}} \\ &\leq |\mathbb{T}^d|^{\frac{1}{p}} \|e^{-u-\varphi}\|_{L^\infty(\mathbb{T}^d)} \frac{e^{\|h\|_{H^1 \cap L^\infty(\mathbb{T}^d)}}}{2} \|h\|_{H^1 \cap L^\infty(\mathbb{T}^d)}, \end{aligned}$$

where, in the last inequality, we used that $\|h\|_{L^\infty(\mathbb{T}^d)} \leq \|h\|_{H^1 \cap L^\infty(\mathbb{T}^d)}$. Letting $h \to 0$ in $H^1 \cap L^\infty(\mathbb{T}^d)$ yields the first limit.

Second limit - $W^{2,p}(\mathbb{T}^d)$. Let $k \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d) \setminus \{0\}$ be arbitrary. Since $p > \frac{d}{2}$, the embedding

$$W^{2,p}(\mathbb{T}^d) \hookrightarrow L^\infty(\mathbb{T}^d)$$

is continuous, hence $k \in L^\infty(\mathbb{T}^d)$ and $\|k\|_{L^\infty(\mathbb{T}^d)} \leq C \|k\|_{W^{2,p}(\mathbb{T}^d)}$ for some $C \geq 0$. Using (3.19), we obtain

$$|e^{-k} - 1 + k| \leq \frac{1}{2} e^{\|k\|_{L^\infty(\mathbb{T}^d)}} \|k\|_{L^\infty(\mathbb{T}^d)} |k| \leq \frac{C}{2} e^{C\|k\|_{W^{2,p}(\mathbb{T}^d)}} \|k\|_{W^{2,p}(\mathbb{T}^d)} |k|.$$

Taking the $L^p(\mathbb{T}^d)$-norm yields

$$\|e^{-k} - 1 + k\|_{L^p(\mathbb{T}^d)} \leq \frac{C}{2} e^{C\|k\|_{W^{2,p}(\mathbb{T}^d)}} \|k\|_{W^{2,p}(\mathbb{T}^d)} \|k\|_{L^p(\mathbb{T}^d)} \leq \frac{C}{2} e^{C\|k\|_{W^{2,p}(\mathbb{T}^d)}} \|k\|^2_{W^{2,p}(\mathbb{T}^d)}.$$

Finally, since $e^{-u-\varphi} \in L^\infty(\mathbb{T}^d)$, there holds

$$\frac{\|e^{-u-\varphi}(e^{-k}-1+k)\|_{L^p(\mathbb{T}^d)}}{\|k\|_{W^{2,p}(\mathbb{T}^d)}} \leq \frac{C}{2} e^{C\|k\|_{W^{2,p}(\mathbb{T}^d)}} \|k\|_{W^{2,p}(\mathbb{T}^d)}.$$

Letting $k \to 0$ in $W^{2,p}(\mathbb{T}^d)$ concludes the proof. □

Proof of Lemma 3.4.2. We first prove that G_1^p is Fréchet differentiable with respect to both arguments, and then establish that the Fréchet derivatives are continuous.

Step 1. G_1^p is Fréchet differentiable with respect to u. We aim to prove that, for all $(u,\varphi) \in X^p$, there exists a bounded linear operator

$$D_u G_1^p(u,\varphi) \in \mathcal{L}(H^1 \cap L^\infty(\mathbb{T}^d), L^p(\mathbb{T}^d))$$

such that

$$\lim_{\|h\|_{H^1\cap L^\infty(\mathbb{T}^d)} \to 0} \frac{\|G_1^p(u+h,\varphi) - G_1^p(u,\varphi) - D_u G_1^p(u,\varphi)[h]\|_{L^p(\mathbb{T}^d)}}{\|h\|_{H^1\cap L^\infty(\mathbb{T}^d)}} = 0. \tag{3.20}$$

Fix $(u,\varphi) \in X^p$, for which Claim 3.4.1 implies that $e^{-u-\varphi} \in L^\infty(\mathbb{T}^d)$. For all $h \in H^1 \cap L^\infty(\mathbb{T}^d)$, we set

$$D_u G_1^p(u,\varphi)[h] := -e^{-u-\varphi} h.$$

Then, $D_u G_1^p(u,\varphi)$ is linear and bounded. Linearity is immediate. For boundedness, we use the continuity of the embeddings $H^1 \cap L^\infty(\mathbb{T}^d) \hookrightarrow L^p(\mathbb{T}^d)$ and $H^1 \cap L^\infty(\mathbb{T}^d) \hookrightarrow L^\infty(\mathbb{T}^d)$ to justify the estimate

$$\begin{aligned}\|D_u G_1^p(u,\varphi)[h]\|_{L^p(\mathbb{T}^d)} \leq \|e^{-u-\varphi}\|_{L^\infty(\mathbb{T}^d)} \|h\|_{L^p(\mathbb{T}^d)} &\leq \|e^{-u-\varphi}\|_{L^\infty(\mathbb{T}^d)} |\mathbb{T}^d|^{\frac{1}{p}} \|h\|_{L^\infty(\mathbb{T}^d)} \\ &\leq \|e^{-u-\varphi}\|_{L^\infty(\mathbb{T}^d)} |\mathbb{T}^d|^{\frac{1}{p}} \|h\|_{H^1\cap L^\infty(\mathbb{T}^d)}.\end{aligned}$$

We next let $h \in H^1 \cap L^\infty(\mathbb{T}^d) \setminus \{0\}$ be arbitrary and compute

$$\begin{aligned}\frac{\|G_1^p(u+h,\varphi) - G_1^p(u,\varphi) - D_u G_1^p(u,\varphi)[h]\|_{L^p(\mathbb{T}^d)}}{\|h\|_{H^1\cap L^\infty(\mathbb{T}^d)}} &= \frac{\|e^{-u-h-\varphi} - e^{-u-\varphi} + e^{-u-\varphi} h\|_{L^p(\mathbb{T}^d)}}{\|h\|_{H^1\cap L^\infty(\mathbb{T}^d)}} \\ &= \frac{\|e^{-u-\varphi}(e^{-h}-1+h)\|_{L^p(\mathbb{T}^d)}}{\|h\|_{H^1\cap L^\infty(\mathbb{T}^d)}}.\end{aligned}$$

Therefore, (3.20) follows from Claim 3.4.3. By arbitrariness of $(u,\varphi) \in X^p$, G_1^p is Fréchet differentiable with respect to u for all $\varphi \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$.

Step 2. G_1^p is continuously Fréchet differentiable with respect to φ. Fix $(u,\varphi) \in X^p$. For any $k \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$, we set

$$D_\varphi(u,\varphi)[k] := -e^{-u-\varphi} k.$$

Then, $D_\varphi(u,\varphi) \in \mathcal{L}(W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d), L^p(\mathbb{T}^d))$. Indeed, linearity is immediate. For boundedness, we recall that $e^{-u-\varphi} \in L^\infty(\mathbb{T}^d)$ by Claim 3.4.1 and estimate

$$\|D_\varphi G_1^p(u,\varphi)[k]\|_{L^p(\mathbb{T}^d)} \leq \|e^{-u-\varphi}\|_{L^\infty(\mathbb{T}^d)} \|k\|_{L^p(\mathbb{T}^d)} \leq \|e^{-u-\varphi}\|_{L^\infty(\mathbb{T}^d)} \|k\|_{W^{2,p}(\mathbb{T}^d)}.$$

We next let $k \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d) \setminus \{0\}$ be arbitrary and compute

$$\frac{\|G_1^p(u,\varphi+k) - G_1^p(u,\varphi) - D_\varphi G_1^p(u,\varphi)[k]\|_{L^p(\mathbb{T}^d)}}{\|k\|_{W^{2,p}(\mathbb{T}^d)}} = \frac{\|e^{-u-\varphi}(e^{-k}-1+k)\|_{L^p(\mathbb{T}^d)}}{\|k\|_{W^{2,p}(\mathbb{T}^d)}}.$$

By Claim 3.4.3, we therefore conclude that $D_\varphi G_1^p(u,\varphi)$ is the Fréchet derivative of G_1^p with respect to φ at (u,φ). By arbitrariness of $(u,\varphi) \in X^p$, we conclude that G_1^p is Fréchet differentiable with respect to φ for all $u \in H^1 \cap L^\infty(\mathbb{T}^d)$.

Step 3. The Fréchet derivatives are continuous. For any $(u,\varphi),(h,k)\in X^p$, we set

$$DG_1^p(u,\varphi)[h,k] := D_uG_1^p(u,\varphi)[h] + D_\varphi G_1^p(u,\varphi)[k] = -e^{-u-\varphi}(h+k).$$

By the product space structure of X^p and the previous steps, there holds

$$DG_1^p(u,\varphi)\in\mathcal{L}(X^p,L^p(\mathbb{T}^d)).$$

Let next $\{(u_n,\varphi_n)\}_{n\in\mathbb{N}}\subset X^p$ be such that $(u_n,\varphi_n)\to(u,\varphi)$ strongly in X^p. We prove that

$$DG_1^p(u_n,\varphi_n)\to DG_1^p(u,\varphi)\quad\text{strongly in }\mathcal{L}(X^p,L^p(\mathbb{T}^d)). \tag{3.21}$$

By the continuous embeddings $H^1\cap L^\infty(\mathbb{T}^d)\hookrightarrow L^\infty(\mathbb{T}^d)$ and $W^{2,p}(\mathbb{T}^d)\hookrightarrow L^\infty(\mathbb{T}^d)$, we have

$$u_n\to u\quad\text{and}\quad\varphi_n\to\varphi\quad\text{strongly in }L^\infty(\mathbb{T}^d). \tag{3.22}$$

In particular, there exists $M\geq 0$ such that

$$\|u_n\|_{L^\infty(\mathbb{T}^d)},\ \|\varphi_n\|_{L^\infty(\mathbb{T}^d)},\ \|u\|_{L^\infty(\mathbb{T}^d)},\ \|\varphi\|_{L^\infty(\mathbb{T}^d)}\leq M\quad\text{for all } n\in\mathbb{N}.$$

Therefore, using the mean value theorem in the form

$$|e^{-a}-e^{-b}|\leq e^{\max\{|a|,|b|\}}|a-b|,$$

applied pointwise with $a=u_n(x)+\varphi_n(x)$ and $b=u(x)+\varphi(x)$, we obtain

$$|e^{-u_n-\varphi_n}-e^{-u-\varphi}|\leq e^{2M}\big(|u_n-u|+|\varphi_n-\varphi|\big)\quad\text{a.e. on }\mathbb{T}^d.$$

This implies that

$$\|e^{-u_n-\varphi_n}-e^{-u-\varphi}\|_{L^\infty(\mathbb{T}^d)}\leq e^{2M}\Big(\|u_n-u\|_{L^\infty(\mathbb{T}^d)}+\|\varphi_n-\varphi\|_{L^\infty(\mathbb{T}^d)}\Big). \tag{3.23}$$

For any $(h,k)\in X^p$, we next compute

$$\begin{aligned}\|DG_1^p(u_n,\varphi_n)[h,k]-DG_1^p(u,\varphi)[h,k]\|_{L^p(\mathbb{T}^d)}&\leq\|e^{-u_n-\varphi_n}-e^{-u-\varphi}\|_{L^\infty(\mathbb{T}^d)}\|h+k\|_{L^p(\mathbb{T}^d)}\\&\leq C\,\|e^{-u_n-\varphi_n}-e^{-u-\varphi}\|_{L^\infty(\mathbb{T}^d)}\|(h,k)\|_{X^p},\end{aligned}$$

where we used the continuous embeddings $H^1\cap L^\infty(\mathbb{T}^d)\hookrightarrow L^p(\mathbb{T}^d)$ and $W^{2,p}(\mathbb{T}^d)\hookrightarrow L^p(\mathbb{T}^d)$, together with the definition of X^p-norm. Taking the supremum over $\|(h,k)\|_{X^p}\leq 1$ yields

$$\|DG_1^p(u_n,\varphi_n)-DG_1^p(u,\varphi)\|_{\mathcal{L}(X^p,L^p(\mathbb{T}^d))}\leq C\,\|e^{-u_n-\varphi_n}-e^{-u-\varphi}\|_{L^\infty(\mathbb{T}^d)}.$$

Using (3.23) and (3.22), we conclude that the right-hand side tends to zero as $n\to\infty$, proving (3.21). By arbitrariness of the sequence $\{(u_n,\varphi_n)\}$, it follows that

$$DG_1^p\in\mathcal{C}(X^p,\mathcal{L}(X^p,L^p(\mathbb{T}^d))).$$

Step 4. Conclusion. By defintion of DG_1^p, and since the projections onto the first and second component of X^p are continuous linear operators, the continuity of DG_1^p implies the continuity of $D_uG_1^p$ and $D_\varphi G_1^p$ on X^p. Hence, G_1^p is continuously Fréchet differentiable with respect to both arguments. Therefore, $G_1^p\in\mathcal{C}^1(X^p,L^p(\mathbb{T}^d))$ (see, e.g., (8.9.1) in [21]) with Fréchet derivative given by DG_1^p. □

We next turn to the second map.

Lemma 3.4.4. *Let $d \geq 1$ and $p \in [1, +\infty)$ be such that $p > \frac{d}{2}$. Then, $G_2^p \in \mathcal{C}^1(X^p, \mathbb{R})$ with Fréchet derivative*

$$DG_2^p(u,\varphi) : X^p \to \mathbb{R},\ (h,k) \mapsto -\int_{\mathbb{T}^d} e^{-u-\varphi}(h+k)\mathrm{d}x$$

for all $(u, \varphi) \in X^p$.

The result follows immediately from the preceding lemma and the Fréchet chain rule. Indeed, defining the continuous linear functional

$$I^p : L^p(\mathbb{T}^d) \to \mathbb{R},\quad f \mapsto \int_{\mathbb{T}^d} f\mathrm{d}x,$$

we have

$$G_2^p = I^p \circ G_1^p.$$

Hence, $G_2^p \in \mathcal{C}^1(X^p, \mathbb{R})$ and

$$DG_2^p(u,\varphi) = I^p \circ DG_1^p(u,\varphi),$$

which yields the stated expression.

3.4.2 Fréchet Differentiability of the Poisson–Boltzmann–Emden Operator

The preceding results yield the first ingredient required for the application of the implicit function theorem, namely the continuous Fréchet differentiability of the Poisson–Boltzmann–Emden operator F^p, as established below.

Lemma 3.4.5. *Let $d \geq 1$ and $p \in [1, +\infty)$ be such that $p > \frac{d}{2}$. Then, $F^p \in \mathcal{C}^1(X^p, Y^p)$, and its Fréchet derivative is given, for all $(u, \varphi) \in X^p$, by*

$$DF^p(u,\varphi)[h,k] = \Delta k - \boldsymbol{m} e^{-u-\varphi}\left(\frac{h+k}{\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}x} - \frac{\int_{\mathbb{T}^d} e^{-u-\varphi}(h+k)\mathrm{d}x}{\left(\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}x\right)^2}\right).$$

Proof. We first verify that F^p is well-defined on X^p. We then use Lemmas 3.4.2 and 3.4.4 to conclude that $F^p \in \mathcal{C}^1(X^p, Y^p)$.

Step 1. F^p is well-defined. Since $e^{-u-\varphi} \in L^\infty(\mathbb{T}^d)$ for all $(u, \varphi) \in X^p$, there holds in particular $e^{-u-\varphi} > 0$ a.e., and thus the normalisation factor is well-defined:

$$\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}x > 0.$$

Moreover, since $|\mathbb{T}^d| < +\infty$, the embedding $L^\infty(\mathbb{T}^d) \hookrightarrow L^q(\mathbb{T}^d)$ is continuous for all $1 \leq q \leq +\infty$. Combining this with the boundedness of u and the continuous embedding $W^{2,p}(\mathbb{T}^d) \hookrightarrow L^\infty(\mathbb{T}^d)$ yields

$$\boldsymbol{m}\frac{e^{-u-\varphi}}{\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}x} \in L^q(\mathbb{T}^d) \quad \text{for all } 1 \leq q \leq +\infty.$$

By definition of ρ_0, we have in particular that

$$\boldsymbol{m}\frac{e^{-u-\varphi}}{\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}x} - \rho_0 \in L^p_{\mathrm{zm}}(\mathbb{T}^d).$$

Next, since $\varphi \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$, Proposition 2.1.5 implies that

$$\Delta\varphi \in L^p_{\mathrm{zm}}(\mathbb{T}^d).$$

We therefore conclude that

$$F^p(u,\varphi) \in L^p_{\mathrm{zm}}(\mathbb{T}^d).$$

Step 2. Continuous Fréchet differentiability. By Lemmas 3.4.2 and 3.4.4, the maps $G_1^p : X^p \to L^p(\mathbb{T}^d)$ and $G_2^p : X^p \to \mathbb{R}$ are continuously Fréchet differentiable. Therefore, by the Fréchet quotient rule, the nonlinear map

$$(u, \varphi) \to \frac{G_1^p(u, \varphi)}{G_2^p(u, \varphi)}$$

appearing in the definition of F^p is continuously Fréchet differentiable. Moreover, since the Laplacian is a bounded linear operator $\Delta : W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d) \to L^p_{\mathrm{zm}}(\mathbb{T}^d)$, it follows that

$$D_\varphi(\Delta\varphi)[k] = \Delta k.$$

Consequently, we conclude that F^p is continuously Fréchet differentiable with respect to both variables. More precisely, using the expressions for DG_1^p and DG_2^p, the partial Fréchet derivatives

$$D_u F^p(u, \varphi) \in \mathcal{L}(H^1 \cap L^\infty(\mathbb{T}^d), Y^p)$$

and

$$D_\varphi F^p(u, \varphi) \in \mathcal{L}(W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d), Y^p)$$

are given by

$$D_u F^p(u, \varphi)[h] = -\boldsymbol{m} e^{-u-\varphi} \left(\frac{h}{\int_{\mathbb{T}^d} e^{-u-\varphi} \mathrm{d}x} - \frac{\int_{\mathbb{T}^d} e^{-u-\varphi} h \mathrm{d}x}{\left(\int_{\mathbb{T}^d} e^{-u-\varphi} \mathrm{d}x\right)^2} \right),$$

and

$$D_\varphi F^p(u, \varphi)[k] = \Delta k - \boldsymbol{m} e^{-u-\varphi} \left(\frac{k}{\int_{\mathbb{T}^d} e^{-u-\varphi} \mathrm{d}x} - \frac{\int_{\mathbb{T}^d} e^{-u-\varphi} k \mathrm{d}x}{\left(\int_{\mathbb{T}^d} e^{-u-\varphi} \mathrm{d}x\right)^2} \right),$$

respectively, where the expression of $D_\varphi F^p$ follows from the Fréchet sum rule (see, e.g., Theorem 2.2.1 in [18]). Since the partial derivatives above are continuous as maps from X^p into the corresponding spaces of bounded linear operators, it follows that $F^p \in \mathcal{C}^1(X^p, Y^p)$ (see, e.g., (8.9.1) in [21]), with Fréchet derivative

$$DF^p(u, \varphi)[h, k] = D_u F^p(u, \varphi)[h] + D_\varphi F^p(u, \varphi)[k],$$

that is,

$$DF^p(u, \varphi)[h, k] = \Delta k - \boldsymbol{m} e^{-u-\varphi} \left(\frac{h + k}{\int_{\mathbb{T}^d} e^{-u-\varphi} \mathrm{d}x} - \frac{\int_{\mathbb{T}^d} e^{-u-\varphi} (h + k) \mathrm{d}x}{\left(\int_{\mathbb{T}^d} e^{-u-\varphi} \mathrm{d}x\right)^2} \right).$$

This concludes the proof. □

The second ingredient required for the application of the implicit function theorem to F^p is the Banach space isomorphism property of the partial derivative $D_\varphi F^p(u, \varphi) : W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d) \to L^p_{\mathrm{zm}}(\mathbb{T}^d)$. Since this operator is bounded and linear for every $(u, \varphi) \in X^p$, it therefore remains to prove that it is invertible. To this end, for an arbitrary $(u, \varphi) \in X^p$, we write

$$D_\varphi F^p(u, \varphi)[k] = (\Delta - K(u, \varphi))[k] = \Delta \left(\mathrm{Id}_{W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)} - \Delta^{-1} K(u, \varphi) \right)[k],$$

where $\mathrm{Id}_{W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)}$ denotes the identity operator on $W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$, and $K : X^p \to \mathcal{L}(W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d), L^p_{\mathrm{zm}}(\mathbb{T}^d))$ is defined by

$$K(u, \varphi)[k] := \boldsymbol{m}(K_1 + K_2)(u, \varphi)[k],$$

with

$$K_1(u, \varphi)[k] := \frac{e^{-u-\varphi} k}{\int_{\mathbb{T}^d} e^{-u-\varphi} \mathrm{d}x},$$

and

$$K_2(u, \varphi)[k] := -\frac{e^{-u-\varphi} \int_{\mathbb{T}^d} e^{-u-\varphi} k \mathrm{d}x}{\left(\int_{\mathbb{T}^d} e^{-u-\varphi} \mathrm{d}x\right)^2}.$$

We note that, although $K_1(u,\varphi)$ and $K_2(u,\varphi)$ are individually operators with values in $L^p(\mathbb{T}^d)$, their sum takes values in the zero-mean space $L^p_{\mathrm{zm}}(\mathbb{T}^d)$. Indeed,

$$\int_{\mathbb{T}^d} K(u,\varphi)[k]\mathrm{d}x = \boldsymbol{m}\frac{\int_{\mathbb{T}^d} e^{-u-\varphi}k\mathrm{d}x}{\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}x} - \boldsymbol{m}\frac{\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}x \int_{\mathbb{T}^d} e^{-u-\varphi}k\mathrm{d}x}{(\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}x)^2} = 0.$$

Consequently, $K(u,\varphi)$ maps $W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$ into $L^p_{\mathrm{zm}}(\mathbb{T}^d)$. Furthermore, by Proposition 2.1.5, the Laplacian is a Banach space isomorphism. In particular,

$$\Delta^{-1} : L^p_{\mathrm{zm}}(\mathbb{T}^d) \to W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$$

is linear and bounded for any $1 < p < +\infty$, and therefore the composition $\Delta^{-1}K(u,\varphi)$ defines a bounded linear operator on $W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$ for every $(u,\varphi) \in X^p$. The invertibility of $D_\varphi F^p(u,\varphi)$ can thus be reduced to the invertibility of

$$\mathrm{Id}_{W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)} - \Delta^{-1}K(u,\varphi).$$

The following result establishes this property.

Lemma 3.4.6. *Let $d \geq 1$ and $p \in (1,+\infty)$ be such that $p > \frac{d}{2}$. Then, for any $(u,\varphi) \in X^p$, there holds $D_\varphi F^p(u,\varphi)^{-1} \in \mathcal{L}(L^p_{\mathrm{zm}}(\mathbb{T}^d), W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d))$.*

Proof. We first show that $\Delta^{-1}K(u,\varphi)$ is a compact operator in Step 1. This allows us to conclude the proof in Step 2 by applying Fredholm's alternative (see, e.g., Theorem 6.6 in [15]; for a more general treatment of Fredholm's theory, see [27]). Throughout the proof, we let $(u,\varphi) \in X^p$ be arbitrary and use that $e^{-u-\varphi} \in L^\infty(\mathbb{T}^d)$ by Claim 3.4.1.

Step 1. For any $k \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$, there holds

$$K_2(u,\varphi)[k] = -\frac{e^{-u-\varphi}\int_{\mathbb{T}^d} e^{-u-\varphi}k\mathrm{d}x}{(\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}x)^2} \in \mathrm{Span}\left\{\frac{e^{-u-\varphi}}{(\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}x)^2}\right\}.$$

Therefore, $K_2(u,\varphi)$ is a finite-rank operator, and thus $K_2(u,\varphi) \in \mathcal{K}(W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d), L^p(\mathbb{T}^d))$ (see, e.g., Section 6.1 in [15]). We next establish that $K_1(u,\varphi) \in \mathcal{K}(W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d), L^p(\mathbb{T}^d))$. To this end, let

$$\{k_n\}_{n\in\mathbb{N}} \subset \overline{B^{W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)}(0,1)}$$

be an arbitrary sequence. Since $W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$ is a closed linear subspace of the reflexive space $W^{2,p}(\mathbb{T}^d)$, it is itself reflexive (see, e.g., Proposition 3.20 in [15]). Moreover, the sequence is bounded by construction. Hence, there exists a subsequence $\{k_{n_1}\}_{n_1\in\mathbb{N}}$ such that $k_{n_1} \rightharpoonup k$ weakly in $W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$ (see, e.g., Theorem 3.18 in [15]). By continuity of the embedding $W^{2,p}(\mathbb{T}^d) \hookrightarrow W^{1,p}(\mathbb{T}^d)$ (see Corollary A.1.4) and by compactness of the embedding $W^{1,p}(\mathbb{T}^d) \hookrightarrow L^p(\mathbb{T}^d)$ (see the Rellich–Kondrachov theorem A.1.5), there exists a subsequence $\{k_{n_2}\}_{n_2\in\mathbb{N}} \subset \{k_{n_1}\}_{n_1\in\mathbb{N}}$ that converges strongly in $L^p_{\mathrm{zm}}(\mathbb{T}^d)$. Since $k_{n_1} \rightharpoonup k$ weakly in $W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$, hence in $L^p_{\mathrm{zm}}(\mathbb{T}^d)$, and $\{k_{n_2}\}_{n_2\in\mathbb{N}}$ converges strongly (hence weakly) in $L^p_{\mathrm{zm}}(\mathbb{T}^d)$, uniqueness of weak limits implies that $k_{n_2} \to k$ strongly in $L^p_{\mathrm{zm}}(\mathbb{T}^d)$. Therefore,

$$\|K_1(u,\varphi)[k_{n_2}] - K_1(u,\varphi)[k]\|_{L^p(\mathbb{T}^d)} \leq \frac{\|e^{-u-\varphi}\|_{L^\infty(\mathbb{T}^d)}}{\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}x}\|k_{n_2} - k\|_{L^p(\mathbb{T}^d)} \to 0.$$

Hence, the sequence $\{K_1(u,\varphi)[k_n]\}_{n\in\mathbb{N}}$ admits a strongly convergent subsequence. By arbitrariness of the sequence $\{k_n\}_{n\in\mathbb{N}}$, it follows that $K_1(u,\varphi) \in \mathcal{K}(W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d), L^p(\mathbb{T}^d))$. Since $K(u,\varphi)$ is the sum of compact operators, it follows that it is itself compact (see, e.g., Theorem 6.1 in [15]). In particular, since the range of $K(u,\varphi)$ satisfies $\mathcal{R}(K(u,\varphi)) \subset L^p_{\mathrm{zm}}(\mathbb{T}^d)$, we conclude that

$$K(u,\varphi) \in \mathcal{K}(W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d), L^p_{\mathrm{zm}}(\mathbb{T}^d)).$$

By Proposition 2.1.5, Δ is a linear isomorphism of Banach spaces. Hence, its inverse Δ^{-1} exists, and it is linear and continuous by the bounded inverse theorem (see, e.g., Corollary 2.7 in [15]), that is,

$$\Delta^{-1} \in \mathcal{L}(L^p_{\mathrm{zm}}(\mathbb{T}^d), W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)).$$

Since the composition of a linear bounded operator with a compact operator is again compact (see, e.g., Proposition 6.3 in [15]), it follows that

$$\Delta^{-1}K(u,\varphi) \in \mathcal{K}(W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)).$$

Step 2. We first prove that

$$\mathrm{Id}_{W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)} - \Delta^{-1}K(u,\varphi)$$

is bijective. By Fredholm's alternative, the latter operator is injective if and only if it is surjective. Therefore, it suffices to show injectivity. Let $k \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$ be arbitrary. Then,

$$\left(\mathrm{Id}_{W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)} - \Delta^{-1}K(u,\varphi)\right)[k] = 0 \qquad \Longleftrightarrow \qquad \Delta^{-1}K(u,\varphi)[k] = k \qquad \Longleftrightarrow \qquad K(u,\varphi)[k] = \Delta k,$$

where the last equivalence uses the isomorphism property of Δ. Multiplying by k both sides of the last equality and integrating yields

$$\int_{\mathbb{T}^d} K(u,\varphi)[k]k\mathrm{d}x = \int_{\mathbb{T}^d} k\Delta k\mathrm{d}x. \tag{3.24}$$

Integrating by parts the right-hand side of the above equality yields

$$\int_{\mathbb{T}^d} k\Delta k\mathrm{d}x = -\int_{\mathbb{T}^d} |\nabla k|^2\mathrm{d}x. \tag{3.25}$$

For the left-hand side, we instead obtain

$$\int_{\mathbb{T}^d} K(u,\varphi)[k]k\mathrm{d}x = \boldsymbol{m}\left(\frac{\int_{\mathbb{T}^d} e^{-u-\varphi}k^2\mathrm{d}x}{\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}x} - \frac{\left(\int_{\mathbb{T}^d} e^{-u-\varphi}k\mathrm{d}x\right)^2}{\left(\int_{\mathbb{T}^d} e^{-u-\varphi}\mathrm{d}x\right)^2}\right). \tag{3.26}$$

Let

$$\mu(u,\varphi)(x) := \frac{e^{-u(x)-\varphi(x)}}{\int_{\mathbb{T}^d} e^{-u(y)-\varphi(y)}\,\mathrm{d}y}.$$

Then $\mu(u,\varphi) \in L^\infty(\mathbb{T}^d)$ is strictly positive a.e., and in particular it defines the density of an absolutely continuous probability measure on $\mathbb{T}^d$. Substituting μ in (3.26) gives

$$\int_{\mathbb{T}^d} K(u,\varphi)[k]k\mathrm{d}x = \boldsymbol{m}\int_{\mathbb{T}^d} k^2\mu(u,\varphi)\mathrm{d}x - \boldsymbol{m}\left(\int_{\mathbb{T}^d} k\mu(u,\varphi)\mathrm{d}x\right)^2 = \boldsymbol{m}\,\mathrm{Var}_{\mu(u,\varphi)}(k),$$

where $\mathrm{Var}_{\mu(u,\varphi)}$ denotes the variance with respect to the probability measure $\mu(u,\varphi)\mathrm{d}x$. Combining the above equality with (3.25) and substituting in (3.24) yields

$$-\int_{\mathbb{T}^d} |\nabla k|^2\mathrm{d}x = \boldsymbol{m}\,\mathrm{Var}_{\mu(u,\varphi)}(k).$$

Since the left-hand side is nonpositive and the right-hand side is nonnegative, we conclude that

$$\int_{\mathbb{T}^d} |\nabla k|^2\mathrm{d}x = 0.$$

Hence, k is constant, and since $k \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$ has zero mean, we conclude that $k = 0$. Hence, the operator $\mathrm{Id}_{W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)} - \Delta^{-1}K(u,\varphi)$ is injective, and thus bijective by Fredholm's alternative. Therefore, its inverse

$$\left(\mathrm{Id}_{W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)} - \Delta^{-1}K(u,\varphi)\right)^{-1}$$

is well-defined. In particular, it is linear and bounded by the bounded inverse theorem. We therefore conclude that

$$D_\varphi F^p(u,\varphi)^{-1}[h] = \left(\Delta\left(\mathrm{Id}_{W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)} - \Delta^{-1}K(u,\varphi)\right)\right)^{-1}[h] = \left(\mathrm{Id}_{W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)} - \Delta^{-1}K(u,\varphi)\right)^{-1}\Delta^{-1}[h]$$

is linear and bounded, since it is the composition of linear bounded operators. □

3.4.3 Application of the Implicit Function Theorem

We conclude this section by proving the continuous Fréchet differentiability of the free energy solution operators Γ_ρ^p and Γ_φ^p in dimensions $d = 1, 2$. The result follows by applying the implicit function theorem to the operator F^p, whose continuous Fréchet differentiability and invertibility properties of the partial derivative with respect to φ were established in the previous subsection.

Theorem 3.4.7. *Let $d = 1, 2$ and $p \in (1, +\infty)$. Then,*

$$\Gamma_\varphi^p \in \mathcal{C}^1\big(H^1 \cap L^\infty(\mathbb{T}^d), W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)\big) \qquad \text{and} \qquad \Gamma_\rho^p \in \mathcal{C}^1\big(H^1 \cap L^\infty(\mathbb{T}^d), L^p(\mathbb{T}^d)\big),$$

and the Fréchet derivatives are given, respectively, by

$$D\Gamma_\varphi^p(u)[h] = -\big(D_\varphi F^p(u, \Gamma_\varphi^p(u))\big)^{-1} D_u F^p(u, \Gamma_\varphi^p(u))[h] \tag{3.27}$$

and

$$D\Gamma_\rho^p(u)[h] = -\Gamma_\rho^p(u)\left(h + D\Gamma_\varphi^p(u)[h] - \int_{\mathbb{T}^d} \Gamma_\rho^p(u)\big(h + D\Gamma_\varphi^p(u)[h]\big)\mathrm{d}x\right) \tag{3.28}$$

for all $u \in H^1 \cap L^\infty(\mathbb{T}^d)$.

Proof. Let $w \in H^1 \cap L^\infty(\mathbb{T}^d)$ be arbitrary. By Theorem 2.2.18, the corresponding free energy stationary self-consistent potential $\varphi_\star(w)$ satisfies the Poisson–Boltzmann–Emden equation (3.2). Moreover, Lemma 2.2.28 yields the improved regularity $\varphi_\star(w) \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$. Consequently, by the definition of F^p, we have

$$F^p(w, \varphi_\star(w)) = 0 \qquad \text{for all } w \in H^1 \cap L^\infty(\mathbb{T}^d). \tag{3.29}$$

Fix now $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. By Lemmas 3.4.5 and 3.4.6, the operator F^p is continuously Fréchet differentiable, and the partial derivative $D_\varphi F^p(u, \varphi_\star(u))$ is a Banach space isomorphism. Hence, the Implicit Function Theorem applies: there exist an open neighbourhood $U \subset H^1 \cap L^\infty(\mathbb{T}^d)$ of u and a unique continuously Fréchet differentiable mapping $g : U \to W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$ such that $g(u) = \varphi_\star(u)$ and, for all $u' \in U$,

$$(u', g(u')) \in X^p, \qquad F^p(u', g(u')) = 0, \qquad \text{and} \qquad Dg(u')[h] = -\big(D_\varphi F^p(u', g(u'))\big)^{-1} D_u F^p(u', g(u'))[h].$$

On the other hand, by (3.29), the mapping $u' \mapsto \varphi_\star(u')$ also satisfies

$$F^p(u', \varphi_\star(u')) = 0 \qquad \text{for all } u' \in U.$$

The uniqueness of the mapping g therefore yields

$$g(u') = \varphi_\star(u') \qquad \text{for all } u' \in U.$$

Hence, $\varphi_\star \in \mathcal{C}^1\big(U, W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)\big)$, with Fréchet derivative

$$D\varphi_\star(u)[h] = -\big(D_\varphi F^p(u, \varphi_\star(u))\big)^{-1} D_u F^p(u, \varphi_\star(u))[h]. \tag{3.30}$$

Since $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ was arbitrary, we conclude that

$$\Gamma_\varphi^p = \varphi_\star \in \mathcal{C}^1\big(H^1 \cap L^\infty(\mathbb{T}^d), W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)\big)$$

with Fréchet derivative given by (3.30) for all $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. It remains to prove the differentiability of Γ_ρ^p. Using the Euler–Lagrange representation of the stationary density and the definitions of G_1^p and G_2^p, we have

$$\rho_\star(u)(x) = \boldsymbol{m}\frac{e^{-u(x)-\varphi_\star(u)(x)}}{\int_{\mathbb{T}^d} e^{-u(y)-\varphi_\star(u)(y)}\mathrm{d}y} = \boldsymbol{m}\frac{G_1^p(u, \varphi_\star(u))}{G_2^p(u, \varphi_\star(u))}.$$

By Lemmas 3.4.2 and 3.4.4, the mappings G_1^p and G_2^p are continuously Fréchet differentiable. Together with the established regularity of $\Gamma_\varphi^p = \varphi_\star$, the Fréchet chain and quotient rules imply that

$$\Gamma_\rho^p = \rho_\star \in \mathcal{C}^1\big(H^1 \cap L^\infty(\mathbb{T}^d), L^p(\mathbb{T}^d)\big),$$

and its Fréchet derivative is given by

$$D\Gamma_\rho^p(u)[h] = -\Gamma_\rho^p(u)\left(h + D\Gamma_\varphi^p(u)[h] - \int_{\mathbb{T}^d} \Gamma_\rho^p(u)(h + D\Gamma_\varphi^p(u)[h])\mathrm{d}x\right)$$

for all $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. □

For spatial dimensions $d \geq 3$, the preceding theorem does not directly apply, since the set

$$\left\{u \in H^1 \cap L^\infty(\mathbb{T}^d) : \varphi_\star(u) \in W^{1,d}_{\mathrm{zm}}(\mathbb{T}^d)\right\}$$

is, in general, neither a linear subspace nor an open subset of $H^1 \cap L^\infty(\mathbb{T}^d)$. Nevertheless, once this obstruction is resolved, the continuous Fréchet differentiability of the solution operators follows by the same argument as above, as shown in the next result. For completeness, we provide the details.

Lemma 3.4.8. *Let $d \geq 3$ and $p \in \left(\frac{d}{2}, +\infty\right)$ be arbitrary. Assume that there exists an open subset $U \subset H^1 \cap L^\infty(\mathbb{T}^d)$ such that $\varphi_\star(u) \in W^{1,d}_{\mathrm{zm}}(\mathbb{T}^d)$ for all $u \in U$. Then,*

$$\Gamma_\varphi^p \in \mathcal{C}^1(U, W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)) \qquad \text{and} \qquad \Gamma_\rho^p \in \mathcal{C}^1(U, L^p(\mathbb{T}^d)),$$

and their Fréchet derivatives are given by (3.27) *and* (3.28)*, respectively, for all $u \in U$.*

Proof. Let $w \in U$ be arbitrary. Since $\varphi_\star(w) \in W^{1,d}_{\mathrm{zm}}(\mathbb{T}^d)$, Lemma 3.1.4 yields $\rho_\star(w) \in L^p(\mathbb{T}^d)$ and $\varphi_\star(w) \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$. Consequently, and by definition of solution operator, there holds $F^p(w, \varphi_\star(w)) = 0$. Next, fix $u \in U$. Since $p > \frac{d}{2}$, by Lemmas 3.4.5 and 3.4.6, the operator F^p is continuously Fréchet differentiable, and the partial derivative $D_\varphi F^p(u, \varphi_\star(u))$ is a Banach space isomorphism. By the implicit function theorem, there thus exist an open neighbourhood $U' \subset H^1 \cap L^\infty(\mathbb{T}^d)$ of u and a unique continuously Fréchet differentiable mapping $g : U' \to W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$ such that $g(u) = \varphi_\star(u)$ and, for all $u' \in U'$,

$$(u', g(u')) \in X^p, \qquad F^p(u', g(u')) = 0, \qquad \text{and} \qquad Dg(u')[h] = -\big(D_\varphi F^p(u', g(u'))\big)^{-1} D_u F^p(u', g(u'))[h].$$

Let now $W := U \cap U'$. Then, the previous reasoning yields

$$F^p(w, \varphi_\star(w)) = 0 \qquad \text{for all } w \in W.$$

The uniqueness of the mapping g therefore yields

$$g(w) = \varphi_\star(w) \qquad \text{for all } w \in W.$$

Hence, $\varphi_\star \in \mathcal{C}^1\big(W, W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)\big)$, with Fréchet derivative

$$D\varphi_\star(w)[h] = -\big(D_\varphi F^p(w, \varphi_\star(w))\big)^{-1} D_u F^p(w, \varphi_\star(w))[h].$$

Since $u \in U$ was arbitrary, we conclude that

$$\Gamma_\varphi^p = \varphi_\star \in \mathcal{C}^1(U, W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d))$$

with Fréchet derivative given by (3.27) for all $u \in U$. Finally, the continuous Fréchet differentiability of Γ_ρ^p follows from the Euler–Lagrange representation of $\rho_\star$ by applying the Fréchet chain, product, and quotient rules, exactly as in the proof of Theorem 3.4.7. This yields the derivative formula (3.28) for all $u \in U$ and concludes the proof. □

Chapter 4

Variational Formulation of the Confinement Problem for the Stationary VPFP System

Chapter 2 established that, for every admissible external potential $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ and prescribed mass $\boldsymbol{m} > 0$, the VPFP system admits a unique free energy stationary solution in the sense of Definition 2.2.7. Consequently, the free energy solution operators

$$u \mapsto \rho_\star(u), \qquad u \mapsto \varphi_\star(u),$$

introduced in (3.9), are well defined. Moreover, as discussed in Section 3.2, restricting the analysis to zero-mean external potentials $u \in H^1_{\mathrm{zm}} \cap L^\infty(\mathbb{T}^d)$ yields injectivity of these operators, so that each admissible external potential determines a unique stationary equilibrium.

The present chapter studies the inverse question. Rather than determining the stationary state associated with a prescribed external potential, we ask whether a desired stationary density can be realised by an appropriate choice of u. More precisely, given a reference density ρ_{ref}, we seek an external potential whose associated free energy stationary density $\rho_\star(u)$ yields, or at least approximates, ρ_{ref}. This is the confinement problem considered throughout the remainder of this chapter.

A first difficulty arising from this objective is that every admissible density can arise as a free energy stationary state, so exact confinement is, in general, impossible. In particular, as shown in the next section, no free energy stationary spatial density can vanish on a set of positive measure, excluding perfectly localised density profiles from the range of the solution operator. This suggests that the inverse problem must be formulated as one of approximate rather than exact confinement.

Motivated by this observation, we formulate confinement as a variational optimisation problem over the space of admissible external potentials. Given a desired spatial density $\rho_{\mathrm{ref}} \in L^1(\mathbb{T}^d)$ with $\rho_{\mathrm{ref}} \geq 0$ almost everywhere, we seek an external potential $u_\star \in H^1 \cap L^\infty(\mathbb{T}^d)$ such that

$$\rho_\star(u_\star) \approx \rho_{\mathrm{ref}}$$

in a suitable metric.

The remainder of this chapter is organised as follows. In Section 4.1, we establish the unattainability of exact confinement for a broad class of prescribed density profiles. Section 4.2 then formulates the approximate confinement problem as an entropy-based variational optimisation problem over the space of admissible external potentials. The well-posedness of this optimisation problem is established in Section 4.3. Finally, Section 4.4 presents numerical experiments illustrating the proposed approach in one spatial dimension.

Throughout this chapter, and unless specified otherwise, the mass of the VPFP system $\boldsymbol{m} > 0$ is fixed but arbitrary.

4.1 Range of Free Energy Solution Operators and Exact Confinement

The solution operators characterise the free energy stationary states that can be generated through admissible external potentials. Understanding the range of these operators is therefore a fundamental prerequisite for the inverse confinement problem. Indeed, a prescribed density can only be achieved exactly if it belongs to the range of the map $u \mapsto \rho_\star(u)$. In this section, we show that this range is strictly smaller than the space of admissible densities by constructing density profiles that cannot occur as stationary densities for any external potential.

The obstruction identified below stems from the exponential representation of free energy stationary spatial densities.

Lemma 4.1.1. *There exists $\rho \in L^1 \cap H^{-1}(\mathbb{T}^d)$ with $\rho \geq 0$ a.e. and $\int_{\mathbb{T}^d} \rho \mathrm{d}x = \boldsymbol{m}$ such that $\rho \neq \rho_\star(u)$ for all $u \in H^1 \cap L^\infty(\mathbb{T}^d)$.*

Proof. Let $\rho : \mathbb{T}^d \to \mathbb{R}$ be given by

$$\rho(x) := \begin{cases} \exp\left(-\frac{1}{1-\left(\frac{|x-x_0|}{a}\right)^p}\right), & \text{if } |x-x_0| < a, \\ 0, & \text{otherwise,} \end{cases} \tag{4.1}$$

where x_0 denotes the center of $\mathbb{T}^d$, $p > 0$ controls the flatness of the core versus the steepness of the edges, and $0 < a < \pi$ is a cutoff radius ensuring that ρ is localised within the torus. Since $\rho \in \mathcal{C}^\infty(\mathbb{T}^d)$, the Poisson equation

$$-\Delta\varphi = \boldsymbol{m}\frac{\rho}{\int_{\mathbb{T}^d} \rho \mathrm{d}y} - \rho_0$$

admits a unique zero-mean solution $\varphi \in \mathcal{C}^\infty(\mathbb{T}^d)$ by the isomorphism property of the inverse Laplacian (see Proposition 2.1.5). Suppose by contradiction that there exists $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ such that $\rho = \rho_\star(u)$. Then, using the Euler–Lagrange representation in Theorem 2.2.18, there holds

$$\frac{\rho(x)}{\int_{\mathbb{T}^d} \rho(y)\mathrm{d}y} = \frac{e^{-u(x)-\varphi_\star(u)(x)}}{\int_{\mathbb{T}^d} e^{-u(x)-\varphi_\star(u)(x)}\mathrm{d}x} \quad \text{a.e.}\,.$$

Let $w(u) := u + \varphi_\star(u)$ and note that $w(u) \in H^1(\mathbb{T}^d)$. Since $\rho(x) = 0$ for all $x \in B^{\mathbb{T}^d}(x_0, a)^{\mathrm{c}}$, from the previous identity it follows that

$$e^{-w(u)(x)} = 0 \quad \text{for a.e. } x \in B^{\mathbb{T}^d}(x_0, a)^{\mathrm{c}}\,.$$

Since $w(u) \in H^1(\mathbb{T}^d)$, it is finite almost everywhere, and therefore $e^{-w(u)(x)} > 0$ for a.e. $x \in \mathbb{T}^d$. However, this contradicts the fact that $|B^{\mathbb{T}^d}(x_0, a)^{\mathrm{c}}| > 0$. Consequently, $\rho \neq \rho_\star(u)$ for all $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. □

The previous argument extends directly to any density profile that vanishes on a set of positive measure, as stated below.

Corollary 4.1.2. *Let $\rho \in L^1 \cap H^{-1}(\mathbb{T}^d)$ be such that $\rho \geq 0$ a.e., $\int_{\mathbb{T}^d} \rho \mathrm{d}x = \boldsymbol{m}$, and $|\mathrm{supp}(\rho)^{\mathrm{c}}| > 0$. Then, $\rho \neq \rho_\star(u)$ for all $u \in H^1 \cap L^\infty(\mathbb{T}^d)$.*

Consequently, the range of the free energy solution operator $u \mapsto \rho_\star(u)$ excludes all density profiles containing vacuum regions; in other words, within the present framework, exact confinement to a prescribed subregion of the domain is unattainable. This obstruction motivates the approximate confinement problem studied in the remainder of this chapter, where the objective is instead to find an external potential whose associated stationary density is as close as possible to a prescribed reference profile.

4.2 Entropy-Based Variational Formulation of Confinement

Inspired by the ideas in [11], we formulate the approximate confinement problem as a variational problem over the space of admissible external potentials. Since exact confinement is, in general, not attainable within this class, we instead introduce an entropy-based functional that quantifies the discrepancy between the free-energy stationary spatial densities and a prescribed reference density.

The following discussion is organised as follows. In Subsection 4.2.1, we define an initial functional that encodes the sought variational formulation of confinement. The limitations and degeneracies of this preliminary approach are then discussed in Subsection 4.2.2. These observations motivate the introduction of a regularised variational problem, which is presented in Subsection 4.2.3.

4.2.1 Unregularised Problem

Let $H^1 \cap L^\infty(\mathbb{T}^d)$ be endowed with the topology induced by the norm

$$\|u\|^2_{H^1\cap L^\infty(\mathbb{T}^d)} = \|u\|^2_{H^1(\mathbb{T}^d)} + \|u\|^2_{L^\infty(\mathbb{T}^d)}\,. \tag{4.2}$$

Moreover, let $\rho_{\mathrm{ref}} \in L^\infty(\mathbb{T}^d)$, with $\rho_{\mathrm{ref}} \geq 0$ a.e., be a prescribed reference density. We consider the functional

$$J : H^1 \cap L^\infty(\mathbb{T}^d) \to \mathbb{R}, \quad u \mapsto \mathcal{H}(\rho_\star(u) \mid \rho_{\mathrm{ref}} + \eta) + \frac{\alpha}{2}\|u\|^2_{H^1\cap L^\infty(\mathbb{T}^d)},$$

where $\alpha > 0$ is a regularisation parameter, and $\mathcal{H}$ denotes the Kullback–Leibler (KL) divergence, also referred to as the relative entropy [48], defined by

$$\mathcal{H}(\rho_\star(u) \mid \rho_{\mathrm{ref}} + \eta) := \int_{\mathbb{T}^d} \rho_\star(u)(x) \log\left(\frac{\rho_\star(u)(x)}{\rho_{\mathrm{ref}}(x) + \eta}\right) \mathrm{d}x\,. \tag{4.3}$$

The parameter $0 < \eta \ll 1$ ensures that the KL divergence is well-defined even when the reference density ρ_{ref} vanishes on a set of positive measure (see Lemma 4.3.1). The relative entropy quantifies the deviation of the spatial density $\rho_\star(u)$ from the prescribed reference density ρ_{ref}. Accordingly, the functional J balances the discrepancy between the free energy stationary spatial density and the target density against the cost associated with the external potential.

Formally, we seek minimisers of J over $H^1 \cap L^\infty(\mathbb{T}^d)$, possibly subject to additional requirements. To complete the formulation, we consider the following class of constraint functionals.

Definition 4.2.1 (**Constraint Functionals and Constraint Levels**).
Let $n_S \in \mathbb{N}$, $n_S \geq 1$. For each $1 \leq i \leq n_S$, let $(k_i, p_i) \in \mathbb{N} \times [1, +\infty)$, $I_i \subset \mathbb{R}$ with $I_i \neq \emptyset$. We define the ***constraint functionals*** *by*

$$S_i : W^{k_i,p_i}(\mathbb{T}^d) \times I_i \to \mathbb{R},$$

and we call $(z_i)_{i=1}^{n_S} \in \prod_{i=1}^{n_S} I_i$ the vector of ***constraint levels****.*

Given a family $\{(S_i, z_i)\}_{i=1}^{n_S}$, $n_S \geq 1$, as in Definition 4.2.1, we consider the inequality constraints

$$S_i(\rho_\star(u), z_i) \leq 0, \qquad 1 \leq i \leq n_S, \tag{4.4}$$

which can be used to encode additional confinement specifications. The same framework can be extended to incorporate constraints on the self-consistent potential $\varphi_\star(u)$ or directly on the external potential u. However, since these extensions follow analogous arguments and do not introduce new conceptual features, we do not pursue them here for the sake of brevity.

We clarify the role of the inequality constraints introduced in (4.4). A first class of constraints enforces spatial mass localisation. For instance, given a measurable set $A \subset \mathbb{T}^d$ and a parameter $0 < z < \boldsymbol{m}$, one may require

$$\int_A \rho_\star(u)(x)\mathrm{d}x \geq \boldsymbol{m} - z,$$

which enforces that at least a fraction $\boldsymbol{m} - z$ of the total mass is concentrated inside the region A. This provides a quantitative formulation of confinement within a prescribed spatial domain. A second class of constraints controls the regularity of the stationary state. For a given a measurable set $A \subset \mathbb{T}^d$ and prescribed parameters $k \in \mathbb{N}$, $1 < p < \infty$, $z > 0$, one may impose

$$\|\rho_\star(u)\|_{W^{k,p}(A)} \leq z.$$

Such constraints can be used to prevent formation of excessively irregular stationary profiles within a region A of the domain. These two examples illustrate that the constraint framework is flexible enough to encode both geometric localisation properties and functional-analytic regularity requirements.

4.2.2 Degeneracies of the Unregularised Formulation

The variational framework introduced above presents several issues.

Lack of Smoothness, Compactness, and Flexibility. The analysis on the space $H^1(\mathbb{T}^d) \cap L^\infty(\mathbb{T}^d)$ introduces several difficulties. First, the regularising term

$$\frac{\alpha}{2}\|u\|^2_{H^1 \cap L^\infty(\mathbb{T}^d)}$$

is non-smooth due to the presence of the $L^\infty(\mathbb{T}^d)$-norm in (4.2), which is not Fréchet differentiable. Combined with the highly nonlinear structure of the problem, arising from both the exponential Euler–Lagrange representation of $\rho_\star(u)$ and the KL divergence, this lack of smoothness may severely affect the stability and performance of numerical optimisation approaches. Second, the space $H^1(\mathbb{T}^d) \cap L^\infty(\mathbb{T}^d)$ is not reflexive, which prevents the direct use of weak compactness arguments that are central to the variational analysis of the optimisation problem. Finally, this choice of admissible space does not provide sufficient flexibility to impose additional regularity on the external potentials. In particular, higher-order regularity may be required to ensure that the optimised potentials possess the smoothness properties expected from physically realisable external fields. To overcome these limitations simultaneously, we restrict the optimisation problem to a smoother reflexive Sobolev space of admissible external potentials. More precisely, the variational problem is formulated over the Sobolev space

$$W^{m,r}(\mathbb{T}^d), \qquad m \in \mathbb{N},\ m \geq 1, \qquad r \in (1, +\infty),$$

for a choice of m and r such that the embedding

$$W^{m,r}(\mathbb{T}^d) \hookrightarrow H^1(\mathbb{T}^d) \cap L^\infty(\mathbb{T}^d)$$

is continuous. Accordingly, the regularising term is replaced by

$$\frac{\alpha}{r}\|u\|^r_{W^{m,r}(\mathbb{T}^d)}.$$

This choice resolves the aforementioned difficulties simultaneously: it replaces the non-smooth L^∞-based regularisation by a smooth Sobolev penalty, restores the reflexivity of the underlying space, and allows the imposed regularity of the admissible potentials to be adjusted through the choice of n. Moreover, the continuous embedding ensures that the optimised potentials remain in $H^1(\mathbb{T}^d) \cap L^\infty(\mathbb{T}^d)$, preserving the regularity required for the definition of $\rho_\star(u)$.

Free Energy Structure vs Weak Stationary Structure. As discussed in Section 2.2.6, in dimensions $d = 1, 2$ every free energy solution to the stationary VPFP system is also a weak stationary solution in the sense of Definition 2.2.1. In contrast, for dimensions $d \geq 3$, this property holds under the additional assumption that

$$\varphi_\star(u) \in W^{1,d}_{\mathrm{zm}}(\mathbb{T}^d).$$

Since the present variational problem aims at approximating prescribed stationary spatial densities for the VPFP system, we therefore assume that admissible external potentials satisfy

$$\varphi_\star(u) \in W^{1,s}(\mathbb{T}^d), \quad s \geq \max\{2, d\}.$$

However, under this additional requirement the previous variational formulation no longer possesses sufficient compactness properties. To recover compactness, we therefore introduce the additional regularising term

$$\frac{\beta}{s}\|\varphi_\star(u)\|^s_{W^{1,s}(\mathbb{T}^d)}, \qquad \beta > 0.$$

Constraint Instability. Let $\{(S_i, z_i)\}_{i=1}^{n_S}$ be a collection of constraint functionals and constraint levels as in Definition 4.2.1. In order for the variational problem to be well-posed, one must ensure that

- the feasible set is nonemtpy, that is, there exist external potentials satisfying (4.4);
- the weak limit $u_\star$ of a minimising sequence $\{u_n\}$ also satisfies the constraints.

To clarify the second requirement, assume that for any minimising sequence, there exists a subsequence $\{u_{n_k}\}$ such that

$$\rho_\star(u_{n_k}) \rightharpoonup \rho_\star(u_\star) \quad \text{weakly in } W^{k_i,p_i}(\mathbb{T}^d), \quad \text{for all } 1 \leq i \leq n_S. \tag{4.5}$$

Then, if $S_i(\cdot, z_i)$ is weakly sequentially lower semicontinuous for all $1 \leq i \leq n_S$, there holds

$$S_i(\rho_\star(u_\star), z_i) \leq \liminf_{k\to\infty} S_i(\rho_\star(u_{n_k}), z_i) \leq 0 \quad \text{for all } \ 1 \leq i \leq n_S.$$

Therefore, $u_\star$ satisfies the inequality constraints. However, the current variational formulation lacks the necessary compactness properties to ensure (4.5) in the general case when $p_i > 1$ or $k_i > 0$ for some i. To overcome this issue, we introduce additional regularisation terms

$$\frac{\gamma_i}{p_i}\|\rho_\star(u)\|^{p_i}_{W^{k_i,p_i}(\mathbb{T}^d)}, \quad \gamma_i > 0,$$

whenever $p_i > 1$ or $k_i > 0$ for some $1 \leq i \leq n_S$, ensuring boundedness of $\rho_\star(u)$ in the corresponding space along minimising sequences. Furthermore, since $W^{k,1}(\mathbb{T}^d)$ is nonreflexive and does not possess a canonical predual space for all $k \in \mathbb{N}$, we additionally require that $p_i > 1$ whenever $k_i > 0$. Under this assumption, the condition ($p_i > 1$ or $k_i > 0$) simplifies to $p_i > 1$.

Additive Constant Degeneracy. The external potential u enters the stationary VPFP system only through its gradient ∇u, and is therefore defined only up to an additive constant. Moreover, as discussed in Section 3.2, the free energy stationary spatial density satisfies

$$\rho_\star(u) = \rho_\star(u + c) \qquad \text{for all } c \in \mathbb{R}.$$

Hence, denoting by

$$\bar{u} := \frac{1}{|\mathbb{T}^d|}\int_{\mathbb{T}^d} u(x)\mathrm{d}x$$

the mean value of u, we obtain

$$\mathcal{H}(\rho_\star(u) \mid \rho_{\mathrm{ref}}) = \mathcal{H}(\rho_\star(u - \bar{u}) \mid \rho_{\mathrm{ref}}).$$

Furthermore, if u satisfies the constraints (4.4), then so does $u - \bar{u}$. Consequently,

$$J(u) - J(u - \bar{u}) = \frac{\alpha}{r}\left(\|u\|^r_{L^r(\mathbb{T}^d)} - \|u - \bar{u}\|^r_{L^r(\mathbb{T}^d)}\right),$$

showing that the optimisation problem contains an artificial minimisation over additive constants which do not affect the dynamics of the VPFP system. Therefore, we further restrict the analysis to zero-mean external potentials, thus removing the degeneracy and yielding a more natural variational formulation.

4.2.3 Regularised Variational Problem

We present a regularised variational problem in light of the preceding discussion. Let $n_S \in \mathbb{N}$, and let $\{(S_i, z_i)\}_{i=1}^{n_S}$ be a collection of constraint functionals and constraint levels as in Definition 4.2.1, with the convention that the collection is empty if $n_S = 0$. We define the sets

$$P_S := \{1 \leq i \leq n_S : p_i > 1\}, \qquad E_S := \{(k_i, p_i) : i \in P_S\}, \tag{4.6}$$

again with the convention that $P_S, E_S = \emptyset$ if $n_S = 0$. For $m \in \mathbb{N}$ with $m \geq 1$, $r \in (1, +\infty)$ and $s \in [2, +\infty)$, let the admissible set of external potentials be

$$\begin{aligned}\mathcal{W}_{\mathrm{reg}}(m, r, s, E_S, \{(S_i, z_i)\}_{i=1}^{n_S}) := \Big\{ u \in W^{m,r}_{\mathrm{zm}}(\mathbb{T}^d) : \varphi_\star(u) \in W^{1,s}_{\mathrm{zm}}(\mathbb{T}^d), \\ \rho_\star(u) \in W^{k,p}(\mathbb{T}^d) \quad \forall\, (k,p) \in E_S, \\ S_i(\rho_\star(u), z_i) \leq 0 \quad \forall\, 1 \leq i \leq n_S \Big\}.\end{aligned}$$

We endow the set $\mathcal{W}_{\mathrm{reg}}$ with the $W^{m,r}(\mathbb{T}^d)$-topology. Moreover, for

$$\rho_{\mathrm{ref}} \in L^\infty(\mathbb{T}^d) \ \text{ with } \ \rho_{\mathrm{ref}} \geq 0 \ \text{ a.e.}, \qquad \eta > 0, \qquad \alpha > 0, \qquad \beta > 0, \qquad \gamma_i > 0 \quad \forall\, i \in P_S,$$

we define the regularised functional $J_{\mathrm{reg}} : W^{m,r}_{\mathrm{zm}}(\mathbb{T}^d) \to \mathbb{R}$ by

$$J_{\mathrm{reg}}(u) := \mathcal{H}(\rho_\star(u) \mid \rho_{\mathrm{ref}} + \eta) + \frac{\alpha}{r}\|u\|^r_{W^{m,r}(\mathbb{T}^d)} + \frac{\beta}{s}\|\varphi_\star(u)\|^s_{W^{1,s}(\mathbb{T}^d)} + \sum_{i \in P_S} \frac{\gamma_i}{p_i}\|\rho_\star(u)\|^{p_i}_{W^{k_i,p_i}(\mathbb{T}^d)}, \tag{4.7}$$

with the understanding that the last term is zero when $P_S = \emptyset$. In the sequel, we show that there exists $u_\star \in \mathcal{W}_{\mathrm{reg}}(m, r, s, E_S, \{(S_i, z_i)\}_{i=1}^{n_S})$ such that

$$J_{\mathrm{reg}}(u_\star) = \inf_{u \in \mathcal{W}_{\mathrm{reg}}\left(m, r, s, E_S, \{(S_i, z_i)\}_{i=1}^{n_S}\right)} J_{\mathrm{reg}}(u),$$

that is, the regularised problem is well-posed. To this end, we assume the following.

Assumption 4.2.2. *Let there hold:*

(*i*) $\eta > 0$ *and* $\rho_{\mathrm{ref}} \in L^\infty(\mathbb{T}^d)$ *with* $\rho_{\mathrm{ref}} \geq 0$ *a.e.*;

(*ii*) $\alpha > 0$ *and* $\beta > 0$;

(*iii*) $m \in \mathbb{N}$, $m \geq 1$, *and* $r \in (1, +\infty)$ *with* $W^{m,r}(\mathbb{T}^d) \hookrightarrow H^1 \cap L^\infty(\mathbb{T}^d)$ *continuously*;

(*iv*) $s \in [2, \infty)$ *with* $s \geq d$;

(*v*) *for* $n_S \in \mathbb{N}$, $\{(S_i, z_i)\}_{i=1}^{n_S}$ *as in Definition* 4.2.1, *and* P_S *as in* (4.6), $S_i(\cdot, z_i)$ *is weakly sequentially lower semicontinuous for all* $1 \leq i \leq n_S$; *moreover,* $p_i \in (1, +\infty)$ *if* $k_i > 0$, *and* $\gamma_i > 0$ *for all* $i \in P_S$;

(*vi*) $\mathcal{W}_{\mathrm{reg}}(m, r, s, E_S, \{(S_i, z_i)\}_{i=1}^{n_S}) \neq \emptyset$.

We note that Assumption 4.2.2 (*vi*) is automatically satisfied when the collection of inequality constraints is empty. Indeed, choosing $u \equiv 0$, Proposition 2.2.17 yields

$$\rho_\star(u) \equiv \frac{\boldsymbol{m}}{|\mathbb{T}^d|} \in \mathcal{C}^\infty(\mathbb{T}^d), \qquad \varphi_\star(u) \equiv 0 \in \mathcal{C}^\infty(\mathbb{T}^d),$$

so that the admissible set is non-empty for all r, s.

We conclude this discussion by remarking that the space $W^{m,r}_{\mathrm{zm}}(\mathbb{T}^d)$ is a closed linear subspace of $W^{m,r}(\mathbb{T}^d)$ and is therefore a Banach space. Moreover, since $r \in (1, +\infty)$, the space $W^{m,r}(\mathbb{T}^d)$ is reflexive; consequently, $W^{m,r}_{\mathrm{zm}}(\mathbb{T}^d)$ is reflexive as well (see, e.g., Proposition 3.20 in [15]). The same reasoning applies to $W^{1,s}_{\mathrm{zm}}(\mathbb{T}^d)$.

4.3 Well-Posedness Analysis

We first establish fundamental properties of the functional J_{reg} in Lemma 4.3.1, and subsequently prove the existence of minimisers on the set of admissible potentials in Proposition 4.3.2.

Lemma 4.3.1. *Let Assumption* 4.2.2 *hold. Then, the functional* J_{reg} *is well-defined and bounded from below on* $\mathcal{W}_{\mathrm{reg}}(m, r, s, E_S, \{(S_i, z_i)\}_{i=1}^{n_S})$.

Proof. We first show that J_{reg} is well-defined and bounded from below on $\mathcal{W}_{\mathrm{reg}}(m, r, s, E_S, \{(S_i, z_i)\}_{i=1}^{n_S})$. By the definition of $\mathcal{W}_{\mathrm{reg}}$, the regularisation terms

$$\frac{\alpha}{r}\|u\|^r_{W^{m,r}(\mathbb{T}^d)}, \quad \frac{\beta}{s}\|\varphi_\star(u)\|^s_{W^{1,s}(\mathbb{T}^d)}, \quad \frac{\gamma_i}{p_i}\|\rho_\star(u)\|^{p_i}_{W^{k_i,p_i}(\mathbb{T}^d)} \quad (i \in P_S),$$

are well-defined and nonnegative for any admissible potential $u \in \mathcal{W}_{\mathrm{reg}}(m, r, s, E_S, \{(S_i, z_i)\}_{i=1}^{n_S})$. It thus remains to verify that the Kullback–Leibler (KL) divergence is well-defined and bounded from below. By Assumption 4.2.2 (iii), it suffices to consider external potentials $u \in H^1_{\mathrm{zm}} \cap L^\infty(\mathbb{T}^d)$. For any such u, there holds $\rho_\star(u) \in L^1(\mathbb{T}^d)$ by definition of free energy solutions to the stationary VPFP system (see Section 3.2). Moreover, Proposition 3.3.6 yields

$$\rho_\star(u) \log\left(\rho_\star(u)\right) \in L^1(\mathbb{T}^d).$$

Since $\rho_{\mathrm{ref}} + \eta \in L^\infty(\mathbb{T}^d)$ and $\rho_{\mathrm{ref}} + \eta \geq \eta > 0$ a.e. by Assumption 4.2.2 (i), the function $\log(\rho_{\mathrm{ref}} + \eta)$ is well-defined and essentially bounded. Indeed,

$$|\log(\rho_{\mathrm{ref}} + \eta)| \leq \max\left\{|\log \eta|, |\log(\|\rho_{\mathrm{ref}}\|_{L^\infty(\mathbb{T}^d)} + \eta)|\right\} \quad \text{a.e.}$$

Hence, $\log(\rho_{\mathrm{ref}} + \eta) \in L^\infty(\mathbb{T}^d)$, and consequently,

$$\rho_\star(u) \log\left(\frac{\rho_\star(u)}{\rho_{\mathrm{ref}} + \eta}\right) = \rho_\star(u) \log \rho_\star(u) - \rho_\star(u) \log(\rho_{\mathrm{ref}} + \eta) \in L^1(\mathbb{T}^d).$$

Therefore, the KL divergence (4.3) is well-defined on $H^1_{\mathrm{zm}} \cap L^\infty(\mathbb{T}^d)$. Moreover, using the standard estimate $x \log x \geq -\frac{1}{e}$ for all $x \geq 0$, we obtain for all $u \in H^1_{\mathrm{zm}} \cap L^\infty(\mathbb{T}^d)$,

$$\begin{aligned}
\mathcal{H}(\rho_\star(u) \mid \rho_{\mathrm{ref}} + \eta) &= \int_{\mathbb{T}^d} \rho_\star(u) \log\left(\frac{\rho_\star(u)}{\rho_{\mathrm{ref}} + \eta}\right) \mathrm{d}x \\
&= \int_{\mathbb{T}^d} \rho_\star(u) \log \rho_\star(u) \, \mathrm{d}x - \int_{\mathbb{T}^d} \rho_\star(u) \log(\rho_{\mathrm{ref}} + \eta) \, \mathrm{d}x \qquad (4.8) \\
&\geq -\frac{|\mathbb{T}^d|}{e} - \boldsymbol{m} \max\left\{|\log \eta|, \, |\log(\|\rho_{\mathrm{ref}}\|_{L^\infty(\mathbb{T}^d)} + \eta)|\right\} > -\infty.
\end{aligned}$$

By definition (4.7), it follows that J_{reg} is well-defined and bounded from below on $\mathcal{W}_{\mathrm{reg}}$. □

Proposition 4.3.2. *Let Assumption* 4.2.2 *hold. Then, there exists* $u_\star \in \mathcal{W}_{\mathrm{reg}}(m, r, s, E_S, \{(S_i, z_i)\}_{i=1}^{n_S})$ *such that*

$$J_{\mathrm{reg}}(u_\star) = \inf_{u \in \mathcal{W}_{\mathrm{reg}}\left(m, r, s, E_S, \{(S_i, z_i)\}_{i=1}^{n_S}\right)} J_{\mathrm{reg}}(u).$$

Proof. We first establish uniform bounds on u_n, $\varphi_\star(u_n)$, $\rho_\star(u_n)$, and the entropy term along an arbitrary minimising sequence for J_{reg}. These bounds yield weak compactness in $W^{m,r}_{\mathrm{zm}}(\mathbb{T}^d)$ and $W^{1,s}_{\mathrm{zm}}(\mathbb{T}^d)$ for the external and self-consistent potentials, respectively. We then prove the weak sequential lower semicontinuity of J_{reg} along a subsequence $\{u_{n_k}\}$, thereby obtaining a minimiser candidate $u_\star$. We conclude the proof by showing that the weak limit $u_\star$ belongs to $\mathcal{W}_{\mathrm{reg}}(m, r, s, E_S, \{(S_i, z_i)\}_{i=1}^{n_S})$.

By Assumption 4.2.2 (vi), there holds $\mathcal{W}_{\mathrm{reg}}(m,r,s,E_S,\{(S_i,z_i)\}_{i=1}^{n_S}) \neq \emptyset$. Together with Lemma 4.3.1, this implies that

$$\inf_{u\in\mathcal{W}_{\mathrm{reg}}\left(m,r,s,E_S,\{(S_i,z_i)\}_{i=1}^{n_S}\right)} J_{\mathrm{reg}}(u) > -\infty,$$

that is, the infimum is well-defined. Hence, there exists a minimising sequence, that is, there exists $\{u_n\}_{n\in\mathbb{N}} \subset \mathcal{W}_{\mathrm{reg}}(m,r,s,E_S,\{(S_i,z_i)\}_{i=1}^{n_S})$ such that

$$\lim_{n\to\infty} J_{\mathrm{reg}}(u_n) = \inf_{u\in\mathcal{W}_{\mathrm{reg}}\left(m,r,s,E_S,\{(S_i,z_i)\}_{i=1}^{n_S}\right)} J_{\mathrm{reg}}(u).$$

Step 1. Uniform Bounds on u_n, $\varphi_\star(u_n)$, $\rho_\star(u_n)$ and the entropy term. Since the infimum is well-defined, the sequence $\{J_{\mathrm{reg}}(u_n)\}_{n\in\mathbb{N}}$ is bounded, that is, there exists $C\in\mathbb{R}$ such that

$$\sup_{n\in\mathbb{N}} J_{\mathrm{reg}}(u_n) \leq C.$$

Using the lower bound (4.8), the definition of J_{reg} yields

$$-c + \frac{\alpha}{r}\|u_n\|^r_{W^{m,r}(\mathbb{T}^d)} + \frac{\beta}{s}\|\varphi_\star(u_n)\|^s_{W^{1,s}(\mathbb{T}^d)} + \sum_{i\in P_S}\frac{\gamma_i}{p_i}\|\rho_\star(u)\|^{p_i}_{W^{k_i,p_i}(\mathbb{T}^d)} \leq J_{\mathrm{reg}}(u_n) \leq C \quad \text{for all } n\in\mathbb{N},$$

where

$$c := \frac{|\mathbb{T}^d|}{e} + \boldsymbol{m}\max\left\{|\log\eta|, |\log(\|\rho_{\mathrm{ref}}\|_{L^\infty(\mathbb{T}^d)}+\eta)|\right\}.$$

Consequently, by nonnegativity of the norm and Assumption 4.2.2 (ii), there holds

$$\|u_n\|_{W^{m,r}} \leq C_u \quad \text{and} \quad \|\varphi_\star(u_n)\|_{W^{1,s}(\mathbb{T}^d)} \leq C_\varphi \quad \text{for all } n\in\mathbb{N},$$

for some $C_u, C_\varphi > 0$. Similarly, if $P_S \neq \emptyset$, by Assumption 4.2.2 (v) there holds $\gamma_i > 0$ for all $i \in P_S$. Hence, by nonnegativity of $\|\cdot\|$, there exist $C_{W_i} \geq 0$, $i\in P_s$, such that

$$\|\rho_\star(u_n)\|_{W^{k_i,p_i}(\mathbb{T}^d)} \leq C_{W_i} \quad \text{for all } n\in\mathbb{N},\ i\in P_S.$$

Moreover, using again the nonnegativity of the norm, for all $n\in\mathbb{N}$ we have

$$\int_{\mathbb{T}^d}\rho_\star(u_n)\log\left(\rho_\star(u_n)\right)\mathrm{d}x - \boldsymbol{m}\max\left\{|\log\eta|, |\log(\|\rho_{\mathrm{ref}}\|_{L^\infty(\mathbb{T}^d)}+\eta)|\right\} \leq \mathcal{H}(\rho_\star(u_n)\mid \rho_{\mathrm{ref}}+\eta) \leq J_{\mathrm{reg}}(u_n) \leq C.$$

Hence, there exists $\tilde{C}\in\mathbb{R}$ such that

$$\int_{\mathbb{T}^d}\rho_\star(u_n)\log\left(\rho_\star(u_n)\right)\mathrm{d}x \leq \tilde{C} \quad \text{for all } n\in\mathbb{N}.$$

We write

$$\begin{aligned}\int_{\mathbb{T}^d}|\rho_\star(u_n)\log\left(\rho_\star(u_n)\right)|\mathrm{d}x &= \int_{\{\rho_\star(u_n)\geq 1\}}\rho_\star(u_n)\log\left(\rho_\star(u_n)\right)\mathrm{d}x - \int_{\{\rho_\star(u_n)<1\}}\rho_\star(u_n)\log\left(\rho_\star(u_n)\right)\mathrm{d}x\\ &= \int_{\mathbb{T}^d}\rho_\star(u_n)\log\left(\rho_\star(u_n)\right)\mathrm{d}x - 2\int_{\{\rho_\star(u_n)<1\}}\rho_\star(u_n)\log\left(\rho_\star(u_n)\right)\mathrm{d}x.\end{aligned}$$

Using the standard estimate $0 \leq -x\log x \leq \frac{1}{e}$ for all $x\geq 0$, together with the previous bound, yields

$$\int_{\mathbb{T}^d}|\rho_\star(u_n)\log\left(\rho_\star(u_n)\right)|\mathrm{d}x \leq \tilde{C} + 2\frac{|\mathbb{T}^d|}{e} =: C_{\mathcal{E}}.$$

Step 2. Weak compactness for $\{u_n\}$, $\{\varphi_\star(u_n)\}$ and $\{\rho_\star(u_n)\}$. By Step 1, the sequence $\{u_n\}_{n\in\mathbb{N}}$ is bounded in $W^{m,r}_{\mathrm{zm}}(\mathbb{T}^d)$. Since $W^{m,r}_{\mathrm{zm}}(\mathbb{T}^d)$ is reflexive, there exist a subsequence $\{u_{n_k}\}_{k\in\mathbb{N}} \subset \{u_n\}_{n\in\mathbb{N}}$ and $u_\star \in W^{m,r}_{\mathrm{zm}}(\mathbb{T}^d)$ such that (see, e.g., Theorem 3.18 in [15])

$$u_{n_k} \rightharpoonup u_\star \quad \text{weakly in } W^{m,r}_{\mathrm{zm}}(\mathbb{T}^d).$$

The continuity of the embedding $W^{m,r}(\mathbb{T}^d) \hookrightarrow H^1 \cap L^\infty(\mathbb{T}^d)$ (as per Assumption 4.2.2 (*iii*)) then implies that

$$u_{n_k} \rightharpoonup u_\star \quad \text{weakly in } H^1_{\mathrm{zm}} \cap L^\infty(\mathbb{T}^d). \tag{4.9}$$

We next address the self-consistent potential. By Step 1, the sequence $\{\varphi_\star(u_{n_k})\}$ is bounded in $W^{1,s}_{\mathrm{zm}}(\mathbb{T}^d)$. Since $W^{1,s}_{\mathrm{zm}}(\mathbb{T}^d)$ is reflexive, there exists a further subsequence (not relabelled) such that

$$\varphi_\star(u_{n_k}) \rightharpoonup \varphi_\star \quad \text{weakly in } W^{1,s}_{\mathrm{zm}}(\mathbb{T}^d).$$

It remains to prove that $\varphi_\star = \varphi_\star(u_\star)$. Since $s \geq 2$, the embedding

$$W^{1,s}_{\mathrm{zm}}(\mathbb{T}^d) \hookrightarrow H^1_{\mathrm{zm}}(\mathbb{T}^d)$$

is continuous. Hence, there holds

$$\varphi_\star(u_{n_k}) \rightharpoonup \varphi_\star \quad \text{weakly in } H^1_{\mathrm{zm}}(\mathbb{T}^d).$$

Since $\{u_{n_k}\} \subset H^1 \cap L^\infty(\mathbb{T}^d)$ by (4.9), the sequence $\{\varphi_\star(u_{n_k})\}$ is bounded in $H^1_{\mathrm{zm}}(\mathbb{T}^d)$ by weak convergence, and the entropy term is uniformly bounded in $L^1(\mathbb{T}^d)$ by Step 1, there exist $\tilde{C}_\varphi, \tilde{C}_\rho \geq 0$ such that

$$\{u_{n_k}\}_{k\in\mathbb{N}} \subset \mathcal{U}(\tilde{C}_\varphi, \tilde{C}_\rho; 1, \boldsymbol{m}),$$

where the set $\mathcal{U}$ is as in (3.11a). Combining this with (4.9), from Theorem 3.3.1 we obtain

$$\varphi_\star(u_{n_k}) \rightharpoonup \varphi_\star(u_\star) \quad \text{weakly in } H^1_{\mathrm{zm}}(\mathbb{T}^d).$$

The uniqueness of weak limits yields $\varphi_\star = \varphi_\star(u_\star)$ as wished.

Finally, we address the spatial density. Since $\{u_{n_k}\}_{k\in\mathbb{N}} \subset \mathcal{U}(\tilde{C}_\varphi, \tilde{C}_\rho; 1, \boldsymbol{m})$ and $u_{n_k} \rightharpoonup u$ weakly in $H^1_{\mathrm{zm}} \cap L^\infty(\mathbb{T}^d)$, Theorem 3.3.1 yields

$$\rho_\star(u_{n_k}) \to \rho_\star(u_\star) \quad \text{strongly (hence weakly) in } L^1(\mathbb{T}^d). \tag{4.10}$$

Moreover, if $P_S \neq \emptyset$, let $i \in P_S$ be arbitrary. By Step 1 the sequence $\{\rho_\star(u_{n_k})\}$ is bounded in $W^{k_i,p_i}(\mathbb{T}^d)$. Since $p_i \in (1, +\infty)$ by Assumption 4.2.2 (*v*), it follows that $W^{k_i,p_i}(\mathbb{T}^d)$ is reflexive. Hence, there exists a subsequence (not relabelled) such that

$$\rho_\star(u_{n_k}) \rightharpoonup \rho_{\star_i} \quad \text{weakly in } W^{k_i,p_i}(\mathbb{T}^d).$$

Since $|\mathbb{T}^d| < +\infty$, the embedding $W^{k_i,p_i}(\mathbb{T}^d) \hookrightarrow L^1(\mathbb{T}^d)$ is continuous, which implies that

$$\rho_\star(u_{n_k}) \rightharpoonup \rho_{\star_i} \quad \text{weakly in } L^1(\mathbb{T}^d).$$

Combining this with (4.10), uniqueness of weak limits yields $\rho_{\star_i} = \rho_\star(u_\star)$. By arbitrariness of $i \in P_S$, we conclude that, whenever $P_S \neq \emptyset$,

$$\rho_\star(u_{n_k}) \rightharpoonup \rho_\star(u_\star) \quad \text{weakly in } W^{k_i,p_i}(\mathbb{T}^d) \quad \text{for all } i \in P_S. \tag{4.11}$$

Step 3. Weak Sequential Lower Semicontinuity of J_{reg} along $\{u_{n_k}\}$. The weak convergence established in Step 2, together with the weak lower semicontinuity of the norm, yields

$$\frac{\alpha}{r}\|u_\star\|^r_{W^{m,r}(\mathbb{T}^d)} \leq \liminf_{k\to\infty} \frac{\alpha}{r}\|u_{n_k}\|^r_{W^{m,r}(\mathbb{T}^d)}, \qquad \frac{\beta}{s}\|\varphi_\star(u_\star)\|^s_{W^{1,s}(\mathbb{T}^d)} \leq \liminf_{k\to\infty} \frac{\beta}{s}\|\varphi_\star(u_{n_k})\|^s_{W^{1,s}(\mathbb{T}^d)}, \tag{4.12}$$

and, if $P_S \neq \emptyset$,

$$\frac{\gamma_i}{p_i}\|\rho_\star(u_\star)\|^{p_i}_{W^{k_i,p_i}(\mathbb{T}^d)} \leq \liminf_{k\to\infty} \frac{\gamma_i}{p_i}\|\rho_\star(u_{n_k})\|^{p_i}_{W^{k_i,p_i}(\mathbb{T}^d)} \quad \text{for all } i \in P_S. \tag{4.13}$$

Therefore, it remains to consider the KL divergence. From Step 2, there holds $u_{n_k} \rightharpoonup u_\star$ weakly in $H^1_{\mathrm{zm}} \cap L^\infty(\mathbb{T}^d)$ and $\{u_{n_k}\}_{k\in\mathbb{N}} \subset \mathcal{U}(\tilde{C}_\varphi, \tilde{C}_\rho; 1, \boldsymbol{m})$. Hence, Lemma 3.3.9 yields

$$\int_{\mathbb{T}^d} \rho_\star(u_\star)\log\big(\rho_\star(u_\star)\big)\mathrm{d}x \leq \liminf_{k\to\infty} \int_{\mathbb{T}^d} \rho_\star(u_{n_k})\log\big(\rho_\star(u_{n_k})\big)\mathrm{d}x.$$

Moreover, since $\log(\rho_{\mathrm{ref}}+\eta) \in L^\infty(\mathbb{T}^d)$ by Assumption 4.2.2 (i), and $\rho_\star(u_{n_k}) \rightharpoonup \rho_\star(u_\star)$ weakly in $L^1(\mathbb{T}^d)$ by Step 2, we obtain

$$\lim_{k\to\infty} \int_{\mathbb{T}^d} \rho_\star(u_{n_k})\log\big(\rho_{\mathrm{ref}}+\eta\big)\mathrm{d}x = \int_{\mathbb{T}^d} \rho_\star(u_\star)\log\big(\rho_{\mathrm{ref}}+\eta\big)\mathrm{d}x.$$

Hence, by definition of KL divergence, there holds

$$\mathcal{H}(\rho_\star(u_\star) \mid \rho_{\mathrm{ref}}+\eta) \leq \liminf_{k\to\infty} \mathcal{H}(\rho_\star(u_{n_k}) \mid \rho_{\mathrm{ref}}+\eta).$$

Combining this with (4.12) and (4.13) yields

$$J_{\mathrm{reg}}(u_\star) \leq \liminf_{k\to\infty} J_{\mathrm{reg}}(u_{n_k}) = \inf_{u\in\mathcal{W}_{\mathrm{reg}}\left(m,r,s,E_S,\{(S_i,z_i)\}_{i=1}^{n_S}\right)} J_{\mathrm{reg}}(u),$$

where the last identity uses that $\{u_{n_k}\}$ is a subsequence of the minimising sequence $\{u_n\}$, hence it is itself minimising.

Step 4. Feasibility of $u_\star$. It remains to show that $u_\star \in \mathcal{W}_{\mathrm{reg}}(m,r,s,E_S,\{(S_i,z_i)\}_{i=1}^{n_S})$. From Step 2, there holds

$$u_\star \in W^{m,r}_{\mathrm{zm}}(\mathbb{T}^d), \qquad \varphi_\star(u_\star) \in W^{1,s}_{\mathrm{zm}}(\mathbb{T}^d).$$

If $n_S = 0$, this concludes the proof. Therefore, assume without loss of generality that $n_S \geq 1$. We prove that

$$S_i(\rho_\star(u_\star), z_i) \leq 0 \quad \text{for all } 1 \leq i \leq n_S.$$

By Assumption 4.2.2 (v), $S_i(\cdot, z_i)$ is weakly sequentially lower semicontinuous for all $1 \leq i \leq n_S$. If $p_i = 1$, then (4.10) yields

$$S_i(\rho_\star(u_\star), z_i) \leq \liminf_{k\to\infty} S_i(\rho_\star(u_{n_k}), z_i) \leq 0,$$

where we used that $\{u_{n_k}\}_{k\in\mathbb{N}} \subset \mathcal{W}_{\mathrm{reg}}(m,r,s,E_S,\{(S_i,z_i)\}_{i=1}^{n_S})$, and thus each element u_{n_k} of the subsequence satisfies the inequality constraints. Similarly, if $p_i > 1$, then $i \in P_S$, and hence (4.11) yields

$$S_i(\rho_\star(u_\star), z_i) \leq \liminf_{k\to\infty} S_i(\rho_\star(u_{n_k}), z_i) \leq 0.$$

Finally, since $\rho_\star(u_\star) \in W^{k_i,p_i}(\mathbb{T}^d)$ for all $i \in P_S$, by definition (4.6) there holds

$$\rho_\star(u_\star) \in W^{k,p}(\mathbb{T}^d) \quad \text{for all } (k,p) \in E_S.$$

This shows that $u_\star \in \mathcal{W}_{\mathrm{reg}}(m,r,s,E_S,\{(S_i,z_i)\}_{i=1}^{n_S})$ and concludes the proof. □

Remark 4.3.3. *As discussed in the previous chapter, restricting the admissible potentials to the zero-mean subspace $W^{m,r}_{\mathrm{zm}}(\mathbb{T}^d)$ guarantees injectivity of the free energy solution operator $u \mapsto \rho_\star(u)$. Nevertheless, injectivity alone is insufficient to establish uniqueness of the minimiser for the variational confinement problem. While the relative entropy is strictly convex and the regularisation terms are convex, the dependence of the density $\rho_\star(u)$ on the potential u need not be strictly convex, thereby leaving open the possibility of multiple minimisers.*

We conclude this section by deriving the integrability properties of the stationary spatial density and its corresponding self-consistent potential associated with the minimisers of the variational confinement problem.

Corollary 4.3.4. *Let Assumption 4.2.2 hold with $r > d$. Then, for any minimiser $u_\star$ of J_{reg} over $\mathcal{W}_{\mathrm{reg}}(m, r, s, E_S, \{(S_i, z_i)\}_{i=1}^{n_S})$, there holds*

$$\rho_\star(u_\star) \in W^{m,\min\{r,s\}}(\mathbb{T}^d) \cap \bigcap_{(k,p)\in E_S} W^{k,p}(\mathbb{T}^d), \qquad \varphi_\star(u_\star) \in W^{m+2,\min\{r,s\}}_{\mathrm{zm}}(\mathbb{T}^d) \cap \bigcap_{(k,p)\in E_S} W^{k+2,p}_{\mathrm{zm}}(\mathbb{T}^d).$$

In particular, if $s > d$, then $\rho_\star(u_\star) \in \mathcal{C}^{m-1}(\mathbb{T}^d)$ and $\varphi_\star(u_\star) \in \mathcal{C}^{m+1}(\mathbb{T}^d)$.

Proof. Let $u_\star$ be a minimiser of J_{reg} over $\mathcal{W}_{\mathrm{reg}}(m, r, s, E_S, \{(S_i, z_i)\}_{i=1}^{n_S})$ of J_{reg}. Then, by definition of $\mathcal{W}_{\mathrm{reg}}$, there holds

$$u_\star \in W^{m,r}_{\mathrm{zm}}(\mathbb{T}^d), \qquad \varphi_\star(u_\star) \in W^{1,s}_{\mathrm{zm}}(\mathbb{T}^d), \qquad \rho_\star(u_\star) \in W^{k,p}(\mathbb{T}^d) \quad \text{for all } (k,p) \in E_S.$$

Applying Lemma 3.1.1, Corollary 3.1.3 (for $d = 1, 2$), and Lemma 3.1.4 (for $d \geq 3$), we obtain

$$\rho_\star(u_\star) \in W^{m,\min\{r,s\}}(\mathbb{T}^d), \qquad \varphi_\star(u_\star) \in W^{m+2,\min\{r,s\}}_{\mathrm{zm}}(\mathbb{T}^d).$$

Moreover, since $p \in (1, +\infty)$ for all $(k,p) \in E_S$, elliptic regularity on the torus (Proposition 2.1.5) yields

$$\varphi_\star(u) \in W^{k+2,p}_{\mathrm{zm}}(\mathbb{T}^d) \quad \text{for all } (k,p) \in E_S.$$

Finally, if $s > d$, then $\min\{r, s\} > d$, and hence the Sobolev embedding theorem implies

$$W^{1,\min\{r,s\}}(\mathbb{T}^d) \hookrightarrow \mathcal{C}(\mathbb{T}^d).$$

Hence, $\rho_\star(u_\star) \in \mathcal{C}^{m-1}(\mathbb{T}^d)$ and $\varphi_\star(u_\star) \in \mathcal{C}^2(\mathbb{T}^d)$. □

4.4 Numerical Study

Having established the well-posedness of the regularised variational problem in the previous section, we now turn to its numerical approximation in the one-dimensional setting. The aim of this section is to construct a computational framework for approximating minimisers of J_{reg} and to assess its performance across a range of confinement configurations. To this end, we adopt a discretise–then–optimise strategy. More precisely, we discretise the spatial domain $\mathbb{T}^1 \cong \mathbb{S}^1$, together with the mappings $\rho_\star(u)$ and $\varphi_\star(u)$, the admissible control variable u, and the regularised objective functional J_{reg}. This leads to a finite-dimensional optimisation problem approximating the original variational formulation. We then derive a first-order optimisation scheme for the resulting discrete problem and study its numerical behaviour. In particular, we consider several prescribed reference densities ρ_{ref} and different choices of the regularisation parameters to evaluate the ability of the method to generate stationary states exhibiting the desired confinement properties. For simplicity, we set the mass of the system to unity, that is, $\boldsymbol{m} = 1$.

This section is organised as follows. We first recall the variational problem in the one-dimensional setting in Subsection 4.4.1 for ease of exposition. Subsection 4.4.2 introduces the spatial and functional discretisation used to obtain a finite-dimensional approximation of the problem. We then derive a numerical optimisation scheme used to compute minimisers of the discretised variational problem in Subsection 4.4.3. Finally, Subsection 4.4.4 presents numerical simulations illustrating the performance of the method and its ability to produce stationary states with prescribed confinement properties.

4.4.1 Variational Problem

We adapt the regularised variational problem from Section 4.2.3 to one spatial dimension. For simplicity, we consider the unconstrained case, i.e. $n_S = 0$, and set $r = 2$, so that admissible controls satisfy $u \in H^1_{\mathrm{zm}}(\mathbb{T}^1)$. By the continuous embeddings $H^1(\mathbb{T}^1) \hookrightarrow \mathcal{C}(\mathbb{T}^1) \hookrightarrow L^\infty(\mathbb{T}^1)$, every such u is bounded. Consequently, Lemma 3.1.1 implies $\varphi_\star(u) \in \mathcal{C}^2(\mathbb{T}^1)$, so that the additional constraint $\varphi_\star(u) \in W^{1,s}_{\mathrm{zm}}(\mathbb{T}^1)$ (for $s \geq \max\{2, d\}$) in the definition of $\mathcal{W}_{\mathrm{reg}}$ is redundant. Thus, for any $\alpha, \eta > 0$ and any a.e. nonnegative $\rho_{\mathrm{ref}} \in L^\infty(\mathbb{T}^1)$, we consider the reduced variational problem

$$\min_{u \in H^1_{\mathrm{zm}}(\mathbb{T}^1)} \left\{ \int_{\mathbb{T}^1} \rho_\star(u) \log\left(\frac{\rho_\star(u)}{\rho_{\mathrm{ref}} + \eta} \right) \mathrm{d}x + \frac{\alpha}{2} \|u\|^2_{H^1(\mathbb{T}^1)} \right\}. \tag{4.14}$$

By Proposition 4.3.2, a minimiser $u_\star \in H^1_{\mathrm{zm}}(\mathbb{T}^1)$ exists. Moreover, since $\varphi_\star(u_\star) \in \mathcal{C}^2(\mathbb{T}^1)$, Corollary 4.3.4 yields $\rho_\star(u_\star) \in \mathcal{C}(\mathbb{T}^1)$.

4.4.2 Discretisation

In one spatial dimension, we identify $\mathbb{T}^1 = \mathbb{R}/(2\pi\mathbb{Z})$. Let $N \in \mathbb{N}$ denote the number of grid points, and define the discretisation step size

$$h := \frac{2\pi}{N}.$$

We approximate the spatial domain by the grid

$$\{\mathbf{x}_i\}_{i=0}^{N-1}, \qquad \mathbf{x}_i = ih \quad \text{for all } 0 \leq i \leq N-1,$$

with indices understood modulo N to enforce periodic boundary conditions. Henceforth, we shall refer to the $\bmod(N)$ indexing as periodic indexing. Accordingly, for all $0 \leq i \leq N-1$, we approximate the admissible external potentials and the corresponding free energy stationary self-consistent potentials and spatial densities, respectively, by

$$\begin{aligned} u \approx \mathbf{u} \in \mathbb{R}^N, \qquad & \mathbf{u}_i := u(\mathbf{x}_i), \\ \varphi_\star(u) \approx \boldsymbol{\varphi}^{\mathbf{u}}_\star \in \mathbb{R}^N, \qquad & (\boldsymbol{\varphi}^{\mathbf{u}}_\star)_i := \varphi_\star(u)(\mathbf{x}_i), \\ \rho_\star(u) \approx \boldsymbol{\rho}^{\mathbf{u}}_\star \in \mathbb{R}^N, \qquad & (\boldsymbol{\rho}^{\mathbf{u}}_\star)_i := \frac{e^{-\mathbf{u}_i - (\boldsymbol{\varphi}^{\mathbf{u}}_\star)_i}}{\sum_{j=0}^{N-1} h e^{-\mathbf{u}_j - (\boldsymbol{\varphi}^{\mathbf{u}}_\star)_j}}. \end{aligned} \tag{4.15}$$

Similarly, for any prescribed reference spatial density $\rho_{\mathrm{ref}} \in L^\infty(\mathbb{T}^1)$ with $\rho_{\mathrm{ref}} \geq 0$ a.e., we approximate

$$\rho_{\mathrm{ref}} \approx \boldsymbol{\rho}_{\mathrm{ref}} \in \mathbb{R}^N, \qquad (\boldsymbol{\rho}_{\mathrm{ref}})_i := \rho_{\mathrm{ref}}(\mathbf{x}_i).$$

To approximate first and second derivatives, we employ standard central finite differences [43]. Let $w : \mathbb{T}^1 \to \mathbb{R}$ be a sufficiently regular map, and let $\mathbf{w} \in \mathbb{R}^N$ be its discrete representation. For the first derivative, we write

$$\frac{\mathrm{d}w}{\mathrm{d}x} \approx D_h \mathbf{w} \in \mathbb{R}^N,$$

with components defined by

$$(D_h \mathbf{w})_i := \frac{\mathbf{w}_{i+1} - \mathbf{w}_{i-1}}{2h} \quad \text{for all } 0 \leq i \leq N-1,$$

and periodic boundary conditions enforced via periodic indexing. Similarly, the second derivative is approximated by

$$\frac{\mathrm{d}^2 w}{\mathrm{d}x^2} \approx D_h^2 \mathbf{w} \in \mathbb{R}^N, \qquad \left(D_h^2 \mathbf{w}\right)_i := \frac{\mathbf{w}_{i-1} - 2\mathbf{w}_i + \mathbf{w}_{i+1}}{h^2}.$$

Consequently, the discretised Poisson–Boltzmann–Emden equation associated with (3.2) becomes an N-dimensional nonlinear system for the unknowns $\{(\boldsymbol{\varphi}^{\mathbf{u}}_\star)_i\}_{i=0}^{N-1}$ given by

$$-\frac{(\boldsymbol{\varphi}^{\mathbf{u}}_\star)_{i-1} - 2(\boldsymbol{\varphi}^{\mathbf{u}}_\star)_i + (\boldsymbol{\varphi}^{\mathbf{u}}_\star)_{i+1}}{h^2} = \frac{e^{-\mathbf{u}_i - (\boldsymbol{\varphi}^{\mathbf{u}}_\star)_i}}{\sum_{j=0}^{N-1} h e^{-\mathbf{u}_j - (\boldsymbol{\varphi}^{\mathbf{u}}_\star)_j}} - \frac{1}{2\pi}. \tag{4.16}$$

In the same spirit, we discretise the regularised cost functional in (4.14), thereby obtaining a mapping $\tilde{J} : \mathbb{R}^N \to \mathbb{R}$ defined by

$$\tilde{J}(\mathbf{u}) := h \sum_{i=0}^{N-1} (\boldsymbol{\rho}^{\mathbf{u}}_\star)_i \log\Big(\frac{(\boldsymbol{\rho}^{\mathbf{u}}_\star)_i}{(\boldsymbol{\rho}_{\mathrm{ref}})_i + \eta} \Big) + \frac{\alpha}{2} h \sum_{i=0}^{N-1} \Big(\mathbf{u}_i^2 + (D_h \mathbf{u})_i^2 \Big), \tag{4.17}$$

where $\boldsymbol{\varphi}^{\mathbf{u}}_\star \in \mathbb{R}^N$ is defined as the solution of the nonlinear system (4.16), and the discrete density $\boldsymbol{\rho}^{\mathbf{u}}_\star \in \mathbb{R}^N$ is given by (4.15).

The above construction yields a fully finite-dimensional approximation of the original variational problem (4.14), which can be summarised as follows:

$$\min_{\mathbf{u} \in \mathbb{R}^N} \left\{ h \sum_{i=0}^{N-1} (\boldsymbol{\rho}^{\mathbf{u}}_\star)_i \log\Big(\frac{(\boldsymbol{\rho}^{\mathbf{u}}_\star)_i}{(\boldsymbol{\rho}_{\mathrm{ref}})_i + \eta} \Big) + \frac{\alpha}{2} h \sum_{i=0}^{N-1} \Big(\mathbf{u}_i^2 + (D_h \mathbf{u})_i^2 \Big) \right\} \tag{4.18a}$$

$$\text{subject to} \quad h \sum_{i=0}^{N-1} \mathbf{u}_i = 0, \tag{4.18b}$$

$$-\frac{(\boldsymbol{\varphi}^{\mathbf{u}}_\star)_{i-1} - 2(\boldsymbol{\varphi}^{\mathbf{u}}_\star)_i + (\boldsymbol{\varphi}^{\mathbf{u}}_\star)_{i+1}}{h^2} = \frac{e^{-\mathbf{u}_i - (\boldsymbol{\varphi}^{\mathbf{u}}_\star)_i}}{\sum_{j=0}^{N-1} h e^{-\mathbf{u}_j - (\boldsymbol{\varphi}^{\mathbf{u}}_\star)_j}} - \frac{1}{2\pi}, \tag{4.18c}$$

$$(\boldsymbol{\rho}^{\mathbf{u}}_\star)_i := \frac{e^{-\mathbf{u}_i - (\boldsymbol{\varphi}^{\mathbf{u}}_\star)_i}}{\sum_{j=0}^{N-1} h e^{-\mathbf{u}_j - (\boldsymbol{\varphi}^{\mathbf{u}}_\star)_j}}. \tag{4.18d}$$

In (4.18a), the minimisation is carried out over $\mathbb{R}^N$, as the discrete H^1-regularisation term ensures coercivity with respect to $\mathbf{u}$, while the zero-mean condition is enforced via (4.18b). The dependence of $\tilde{J}$ on $\mathbf{u}$ is both explicit, through the quadratic regularisation term, and implicit, through the coupled nonlinear system (4.18c)–(4.18d).

4.4.3 Numerical Optimisation Scheme

The finite-dimensional optimisation problem (4.18) is solved using a nested iterative strategy. For a fixed control vector $\mathbf{u}$, the inner iteration computes the associated self-consistent potential by solving the nonlinear system (4.18c). This yields the corresponding discrete density via (4.18d). Given these quantities, the outer iteration updates the control variable using a gradient-based scheme applied to the discretised objective functional (4.18a), while enforcing the zero-mean constraint (4.18b).

4.4.3.1 Inner Solver: Poisson–Boltzmann–Emden Equation

For a fixed control vector $\mathbf{u} \in \mathbb{R}^N$, the associated self-consistent potential $\boldsymbol{\varphi}^{\mathbf{u}}_\star$ is defined as the solution of the nonlinear discrete Poisson–Boltzmann–Emden system (4.16). To compute this solution efficiently, we interpret the equation as a fixed-point problem. Given an iterate $\boldsymbol{\varphi}^{(k)} \in \mathbb{R}^N$, we define the corresponding density

$$\boldsymbol{\rho}^{(k)} := \boldsymbol{\rho}(\mathbf{u}, \boldsymbol{\varphi}^{(k)}),$$

and compute the next iterate by solving the linear Poisson problem

$$-D_h^2 \boldsymbol{\psi}^{(k+1)} = \boldsymbol{\rho}^{(k)} - \frac{1}{2\pi},$$

subject to the zero-mean constraint

$$h \sum_{i=0}^{N-1} \boldsymbol{\psi}_i^{(k+1)} = 0.$$

This Poisson equation is solved in Fourier space using the Fast Fourier Transform (FFT), which diagonalises the discrete Laplacian under periodic boundary conditions. In a pure Picard iteration one would set $\boldsymbol{\varphi}^{(k+1)} = \boldsymbol{\psi}^{(k+1)}$, but this approach may converge slowly or exhibit oscillations. We therefore use a damped and accelerated variant using an Anderson-like scheme (see, e.g., [49] for an introduction), which is illustrated next.

Given a damping parameter $0 < \beta < 1$, we define the damped iterates

$$\tilde{\boldsymbol{\varphi}}^{(k)} := \boldsymbol{\varphi}^{(k)} + \beta\big(\boldsymbol{\psi}^{(k+1)} - \boldsymbol{\varphi}^{(k)}\big).$$

To further accelerate convergence, we apply a history-based Anderson-type mixing. Let $m \in \mathbb{N}$ be the memory depth and set $m_k = \min\{m, k+1\}$. We store the most recent m_k iterates and define, for $j = 0, \ldots, m_k - 1$,

$$\tilde{\boldsymbol{\varphi}}^{(k-j)} := \boldsymbol{\varphi}^{(k-j)} + \beta\big(\boldsymbol{\psi}^{(k-j+1)} - \boldsymbol{\varphi}^{(k-j)}\big).$$

The coefficients $\{\omega_j\}_{j=0}^{m_k-1}$ are then obtained by solving the least-squares problem

$$\min_{\omega_0,\ldots,\omega_{m_k-1}\in\mathbb{R}} \left\| \sum_{j=0}^{m_k-1} \omega_j \big(\boldsymbol{\psi}^{(k-j+1)} - \boldsymbol{\varphi}^{(k-j)}\big) \right\|_2,$$

and the next iterate is defined as

$$\boldsymbol{\varphi}^{(k+1)} = \sum_{j=0}^{m_k-1} \omega_j \tilde{\boldsymbol{\varphi}}^{(k-j)}.$$

The iteration is continued until a prescribed residual tolerance is reached. More precisely, given a tolerance threshold $\varepsilon > 0$, the update is performed until the following condition is satisfied:

$$\|\boldsymbol{\varphi}^{(k+1)} - \boldsymbol{\varphi}^{(k)}\|_2 \leq \varepsilon.$$

The pseudocode of the proposed approach is provided in Algorithm 4.4.1.

Algorithm 4.4.1 Inner Solver: Poisson–Boltzmann–Emden Fixed-Point Iteration with Damped Anderson Mixing

Require: Control $\mathbf{u} \in \mathbb{R}^N$, damping parameter $0 < \beta < 1$, memory depth $m \in \mathbb{N}$, tolerance $\varepsilon > 0$, maximum iterations $K_{\max}$.

Initialisation: $k \leftarrow 0$, $\boldsymbol{\varphi}^{(0)} \leftarrow \mathbf{0}$, Res $\leftarrow +\infty$.

while $k < K_{\max}$ **and** Res $> \varepsilon$ **do**

  Compute density: $\boldsymbol{\rho}^{(k)} := \boldsymbol{\rho}(\mathbf{u}, \boldsymbol{\varphi}^{(k)})$

  Solve Poisson problem via FFT:

$$-D_h^2 \boldsymbol{\psi}^{(k+1)} = \boldsymbol{\rho}^{(k)} - \frac{1}{2\pi}, \qquad h \sum_{i=0}^{N-1} \boldsymbol{\psi}_i^{(k+1)} = 0$$

  Damped update: $\tilde{\boldsymbol{\varphi}}^{(k)} := \boldsymbol{\varphi}^{(k)} + \beta(\boldsymbol{\psi}^{(k+1)} - \boldsymbol{\varphi}^{(k)})$

  Set memory length: $m_k \leftarrow \min\{m, k+1\}$

  Compute coefficients $\{\omega_j\}_{j=0}^{m_k-1}$ by

$$\min_{\omega} \left\| \sum_{j=0}^{m_k-1} \omega_j (\boldsymbol{\psi}^{(k-j+1)} - \boldsymbol{\varphi}^{(k-j)}) \right\|_2$$

  Update iterate

$$\boldsymbol{\varphi}^{(k+1)} := \sum_{j=0}^{m_k-1} \omega_j \tilde{\boldsymbol{\varphi}}^{(k-j)}$$

  Res $\leftarrow \left\| \boldsymbol{\varphi}^{(k+1)} - \boldsymbol{\varphi}^{(k)} \right\|_2$

  $k \leftarrow k+1$

end while

return $\boldsymbol{\varphi}^{(k)}$

4.4.3.2 Outer Optimisation: Preconditioned Gradient Descent with Armijo Line Search

We minimise the objective functional $\tilde{J}$ using a preconditioned gradient descent scheme [45, Chapter 3] of the form

$$\mathbf{u}^{(k+1)} = \mathbf{u}^{(k)} - \tau_k \, \mathcal{P}(\nabla \tilde{J}(\mathbf{u}^{(k)})),$$

where the preconditioner $\mathcal{P}$ acts as a spectral smoothing operator in Fourier space. Specifically, for a discrete vector $\mathbf{w} \in \mathbb{R}^N$, we define

$$\mathcal{P}(\mathbf{w}) := \mathcal{F}^{-1}\left(\frac{\widehat{\mathbf{w}}(k)}{1 + \gamma k^2} \right),$$

where $\mathcal{F}$ denotes the discrete Fourier transform, k are the Fourier modes, $\widehat{\mathbf{w}}(k)$ is the k-th discrete Fourier coefficient of $\mathbf{w}$, and $\gamma > 0$ is a tuning parameter. This choice corresponds to a H^1-type Sobolev gradient preconditioning [44, Chapter 29]: high-frequency components of the gradient are damped, while low-frequency components are preserved. As a result, the effective descent direction is smoother, improving conditioning and stability of the optimisation.

The gradient $\nabla \tilde{J}(\mathbf{u})$ is computed via automatic differentiation and depends implicitly on $\boldsymbol{\varphi}_\star(\mathbf{u})$, obtained by solving the nonlinear PBE system. Each evaluation of $\tilde{J}$ or $\nabla \tilde{J}$ therefore requires the inner fixed-point solver. The step size τ_k is determined by an Armijo backtracking line search [5]. Given the descent direction

$$\mathbf{d}^{(k)} = -\mathcal{P}(\nabla \tilde{J}(\mathbf{u}^{(k)})),$$

we choose the largest $\tau_k = \xi^\ell \tau_0$ with $\xi \in (0,1)$ such that

$$\tilde{J}(\mathbf{u}^{(k)} + \tau_k \mathbf{d}^{(k)}) \leq \tilde{J}(\mathbf{u}^{(k)}) + c\,\tau_k \nabla \tilde{J}(\mathbf{u}^{(k)}) \cdot \mathbf{d}^{(k)}, \tag{4.19}$$

for a fixed $c \in (0,1)$. Here, $\ell \in \mathbb{N}$ is the backtracking index: it counts how many times the initial step size τ_0 is reduced by the factor ξ. In other words, $\ell = 0$ corresponds to the tentative step τ_0, while increasing values of ℓ correspond to progressively smaller step sizes $\xi^\ell \tau_0$ until the Armijo condition (4.19) is satisfied. Each trial step in the line search requires a fresh evaluation of the inner nonlinear solver for $\boldsymbol{\varphi}_\star(\mathbf{u})$, making the method a nested optimisation scheme consisting of an outer preconditioned gradient descent loop coupled with an inner fixed-point iteration for the Poisson–Boltzmann–Emden equation. The pseudocode of the proposed approach is provided in Algorithm 4.4.2.

Algorithm 4.4.2 Outer Solver: Preconditioned Gradient Descent with Armijo Line Search

Require: Step parameters $\tau_0 > 0$, $\xi \in (0,1)$, Armijo constant $c \in (0,1)$, preconditioner parameter $\gamma > 0$, tolerance $\delta > 0$, maximum iterations $N_{\max}$

Initialisation: $k \leftarrow 0$, $\mathbf{u}^{(0)} \leftarrow \mathbf{0}$.
while $k < N_{\max}$ **and** $\|\mathbf{u}^{(k+1)} - \mathbf{u}^{(k)}\|_2 > \delta$ **do**
  Compute $\boldsymbol{\varphi}_\star^{\mathbf{u}^{(k)}}$ using the Inner Solver in Algorithm 4.4.1
  Compute $\boldsymbol{\rho}_\star^{\mathbf{u}^{(k)}}$ using (4.18d)
  Evaluate reduced objective $\tilde{J}(\mathbf{u}^{(k)})$
  Compute gradient $\mathbf{g}^{(k)} := \nabla \tilde{J}(\mathbf{u}^{(k)})$ (via automatic differentiation)
  Apply preconditioner: $\mathbf{p}^{(k)} := \mathcal{P}(\mathbf{g}^{(k)})$.
  Set descent direction: $\mathbf{d}^{(k)} := -\mathbf{p}^{(k)}$.
  Line search initialisation: $\tau \leftarrow \tau_0$, Armijo $\leftarrow \tilde{J}(\mathbf{u}^{(k)} + \tau \mathbf{d}^{(k)}) \leq \tilde{J}(\mathbf{u}^{(k)}) + c\,\tau\,\mathbf{g}^{(k)} \cdot \mathbf{d}^{(k)}$
  while not Armijo **do**
    $\tau \leftarrow \xi \tau$
    Armijo $\leftarrow \tilde{J}(\mathbf{u}^{(k)} + \tau \mathbf{d}^{(k)}) \leq \tilde{J}(\mathbf{u}^{(k)}) + c\,\tau\,\mathbf{g}^{(k)} \cdot \mathbf{d}^{(k)}$
  end while
  Update control: $\mathbf{u}^{(k+1)} := \mathbf{u}^{(k)} + \tau \mathbf{d}^{(k)}$
  Res $\leftarrow \|\mathbf{u}^{(k+1)} - \mathbf{u}^{(k)}\|_2$
  $k \leftarrow k + 1$
end while
return $\mathbf{u}^{(k)}$

4.4.4 Numerical Results

We present numerical experiments designed to validate the proposed variational framework and to assess the robustness of the associated numerical scheme developed in the previous sections. Below, we summarise the main results.

We consider four reference densities designed to test different confinement regimes. The first reference density, ρ_{ref_1}, is defined as in (4.1) and is chosen to enforce maximal localisation within a prescribed region of $\mathbb{T}^1$. The second reference density is given by

$$\rho_{\mathrm{ref}_2}(x) := \frac{2 + \cos(x - \pi)}{\int_{\mathbb{T}^1} (2 + \cos(y - \pi))\,\mathrm{d}y},$$

and is used to assess the ability of the method to reproduce smooth oscillatory confinement profiles. The third reference density is defined by

$$\tilde{\rho}_{\mathrm{ref}_3}(x) := \exp\left(-\frac{(x - \frac{\pi}{2})^2}{2\sigma^2}\right) + \exp\left(-\frac{(x - \frac{3\pi}{2})^2}{2\sigma^2}\right), \qquad \rho_{\mathrm{ref}_3}(x) := \frac{\tilde{\rho}_{\mathrm{ref}_3}(x)}{\int_{\mathbb{T}^1} \tilde{\rho}_{\mathrm{ref}_3}(y)\,\mathrm{d}y},$$

where $\sigma > 0$ controls the width of each mode. This reference density is used to evaluate the performance of the numerical scheme in the presence of bimodal confinement structures. The fourth reference density ρ_{ref_4} is generated as a random smooth profile on $\mathbb{T}^1$. To this end, we first sample a realisation of spatial white noise and then apply a spectral smoothing operator defined in Fourier space by a second-order low-pass filter of the form $(1 + |k|^2)^{-1}$, where k denotes the Fourier mode. The resulting function is transformed back to physical space, taken in absolute value to ensure nonnegativity, and finally normalised to unit mass, yielding a random but smooth confinement profile with controlled spatial regularity.

For each reference density ρ_{ref_i}, $1 \leq i \leq 4$, we test the method across different values of the regularisation parameter α, namely

$$\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}, 0.35, 0.9\}.$$

We discretise $\mathbb{T}^1$ using $N = 128$ grid points, chosen as a power of two to exploit the efficiency of the FFT in the inner solver (Algorithm 4.4.1). The inner solver employs the parameters $\beta = 0.9$, $m = 4$, $\varepsilon = 10^{-6}$, and $K_{\max} = 75$. For the outer optimisation (Algorithm 4.4.2), we set $\tau_0 = 0.5$, $\xi = 0.5$, $c = 10^{-4}$, $\gamma = 0.05$, $\delta = 10^{-5}$, and $N_{\max} = 20000$.

Figure 4.4.1 displays the reference densities ρ_{ref_i}, $1 \leq i \leq 4$, together with the corresponding optimised spatial densities obtained via Algorithm 4.4.2 under different regularisation regimes. Figures 4.4.2 and 4.4.3 show the associated optimal external and self-consistent potentials, respectively. With a slight abuse of notation, we denote the optimal external potentials by $u_\star$, and the associated spatial densities and self-consistent potentials by $\rho_\star(u_\star)$ and $\varphi_\star(u_\star)$, respectively. For small values of the regularisation parameter α, the optimised densities closely match the reference profiles. In contrast, increasing α penalises the control energy, leading to progressively flatter densities and corresponding potentials. As established in Corollary 4.3.4, the spatial densities and self-consistent potentials lie in $\mathcal{C}(\mathbb{T}^1)$ and $\mathcal{C}^2(\mathbb{T}^1)$, respectively, and are in fact smooth.

Figure 4.4.4 shows the evolution of the discretised cost functional $\tilde{J}$ in (4.17) with respect to the iteration counter k. Figure 4.4.5 similarly shows the evolution of $\|\nabla \tilde{J}\|_2$. Across all reference densities and regularisation regimes, the cost functional decreases monotonically and stabilises as the optimisation progresses, while the gradient norm exhibits a corresponding monotone decay towards zero. This indicates convergence of the optimisation scheme towards minimisers of the cost functional.

Conclusion

The numerical experiments demonstrate that the proposed discretise–then–optimise framework provides an effective approach to the inverse problem considered in this chapter. Across a wide range of reference densities and regularisation parameters, the optimisation scheme achieves stable convergence and accurately recovers the prescribed reference densities while controlling the associated effort through an appropriate choice of regularisation.

An important direction for future research is to complement the present discretise–then–optimise approach with an optimise–then–discretise formulation. In particular, leveraging the Fréchet differentiability of the solution operators Γ_ρ^p and Γ_φ^p established in Section 3.4, we aim to derive a first-order optimality system at the continuous level, following the approach of [11].

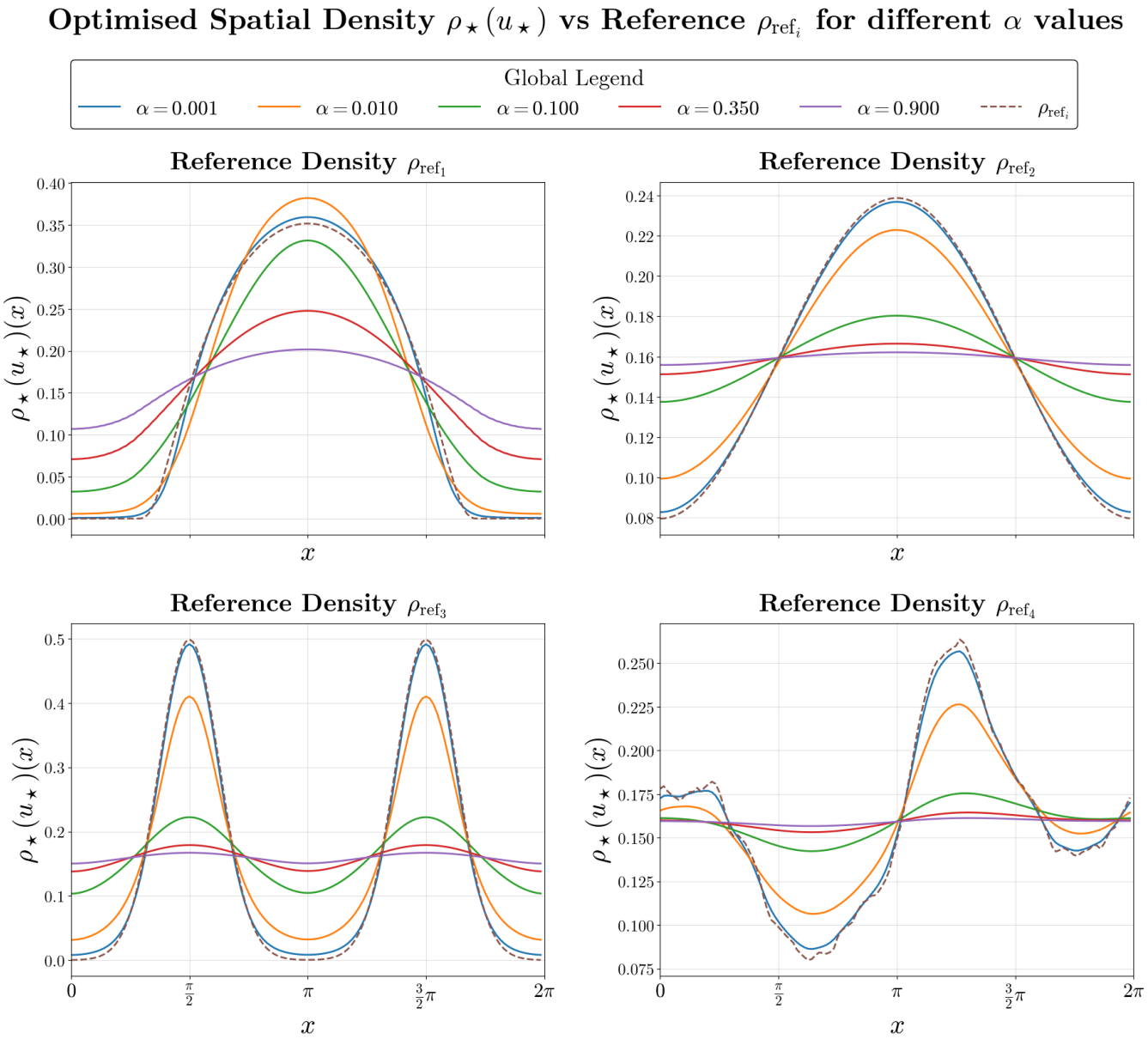


Figure 4.4.1: Reference densities ρ_{ref_i}, $1 \leq i \leq 4$, and corresponding optimal spatial densities $\rho_\star(u_\star)$ for different values of the regularisation parameter α.

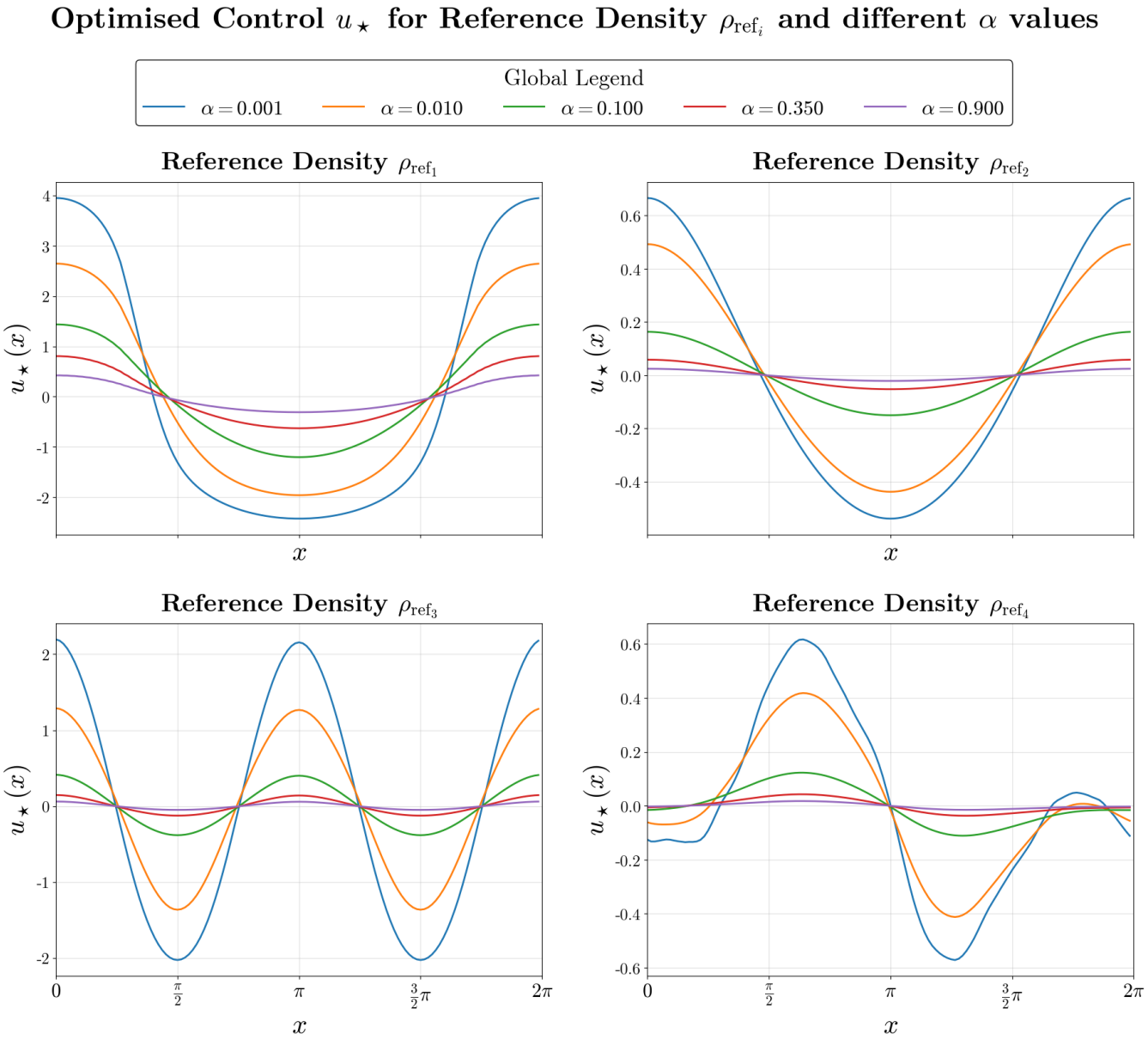


Figure 4.4.2: Optimal external potentials $u_\star$ corresponding to the reference densities ρ_{ref_i} for different values of the regularisation parameter α.

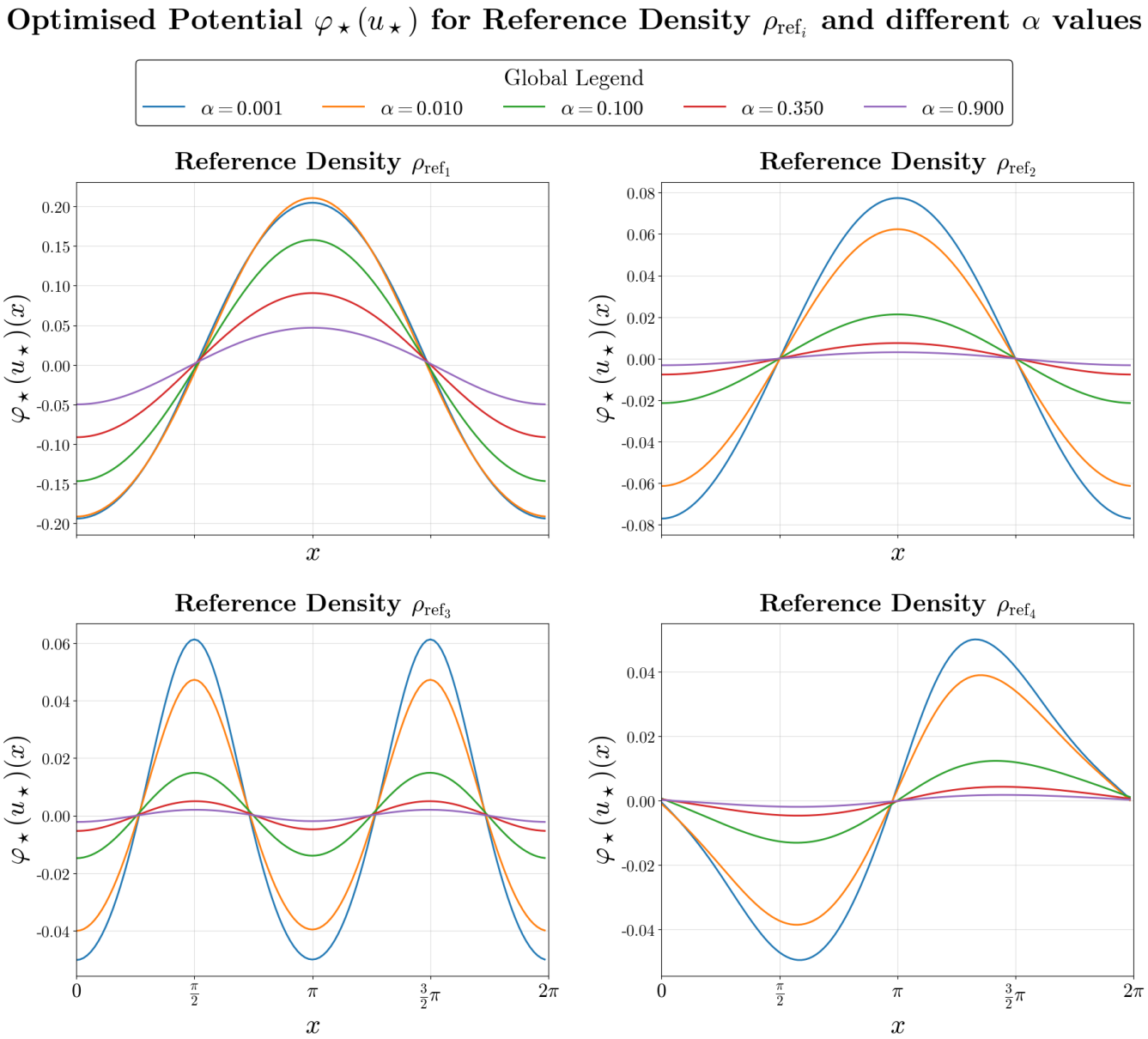


Figure 4.4.3: Optimal self-consistent potentials $\varphi_\star(u_\star)$ corresponding to the reference densities ρ_{ref_i} for different values of the regularisation parameter α.

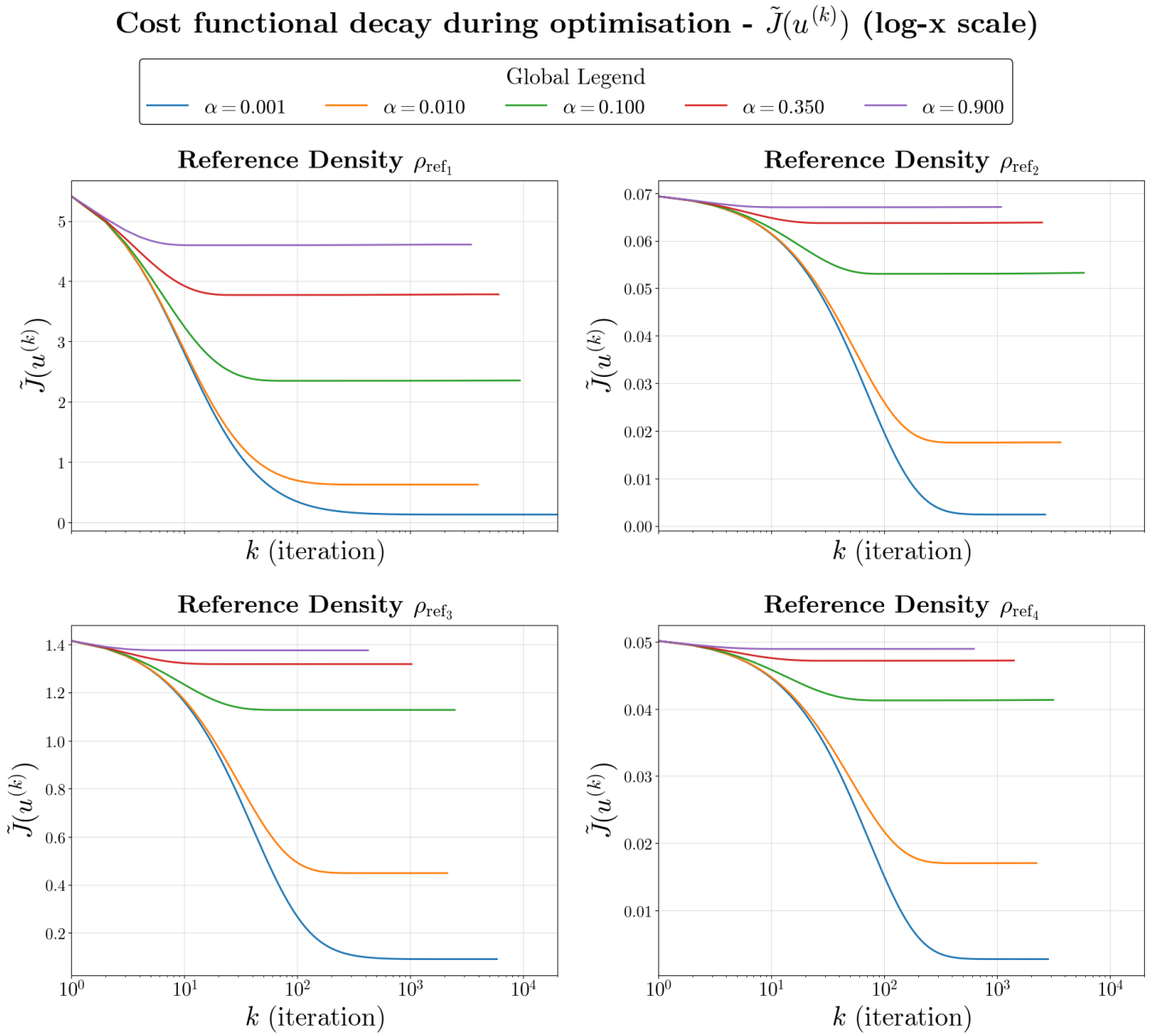


Figure 4.4.4: Cost functional decay $\tilde{J}(\mathbf{u}^{(k)})$ vs iteration counter corresponding to the reference densities ρ_{ref_i} for different values of the regularisation parameter α.

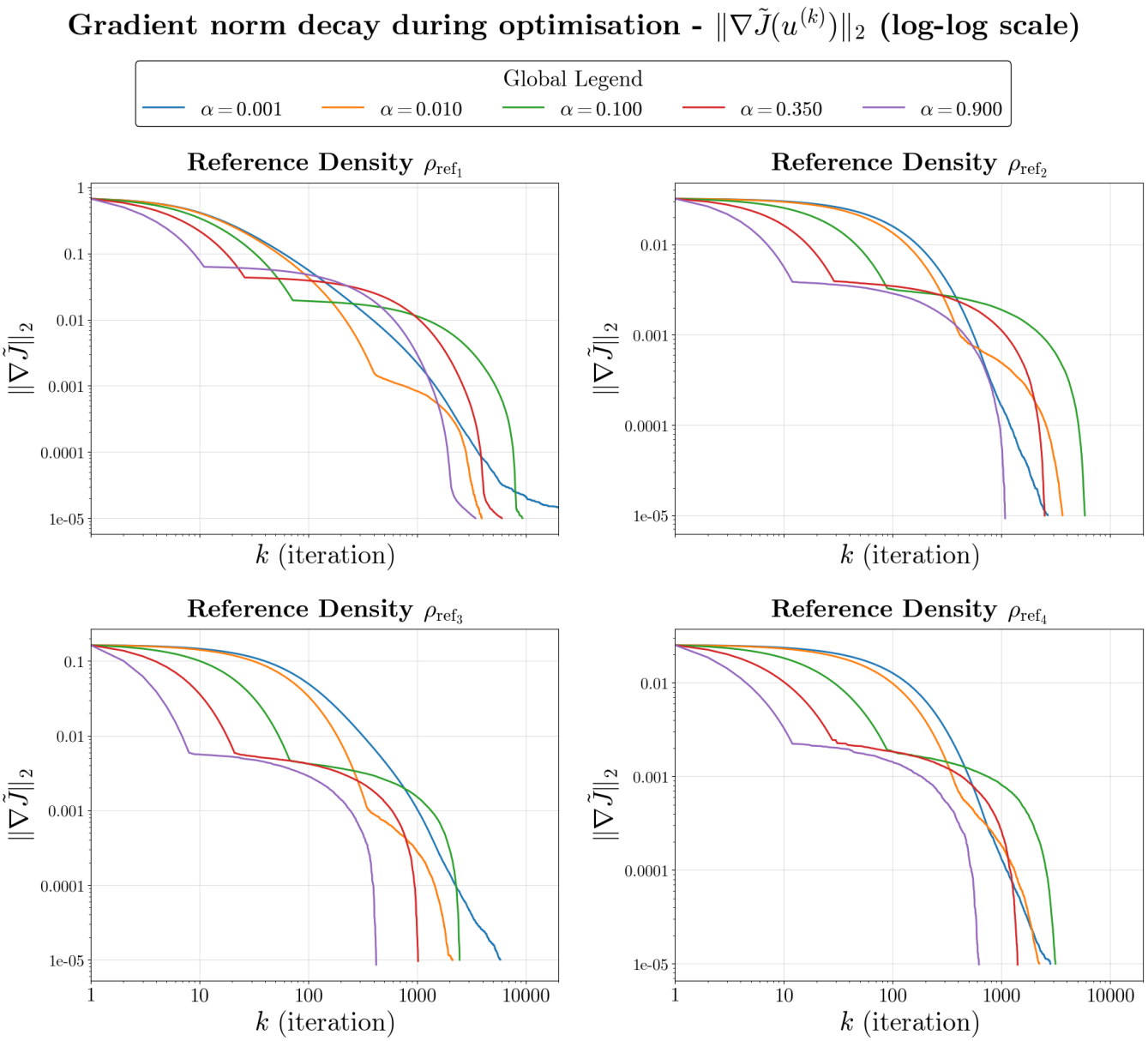


Figure 4.4.5: Graident 2-norm decay $\|\nabla \tilde{J}(\mathbf{u}^{(k)})\|_2$ vs iteration counter corresponding to the reference densities ρ_{ref_i} for different values of the regularisation parameter α.

Chapter 5

Hypocoercivity and Control of Stationary VPFP Equilibria

The confinement problem developed in the previous chapter produces equilibria tailored to prescribed density profiles. The final question is whether these designed equilibria are dynamically meaningful, namely whether nearby solutions converge towards them. To this end, this chapter is devoted to the study of the linearised VPFP system around a free energy stationary solution with prescribed total mass $\boldsymbol{m} > 0$. We decompose the nonlinear solution and the associated self-consistent potential as perturbations of equilibrium and investigate the stability properties of the linearised dynamics.

To obtain quantitative decay estimates, the analytical part of this chapter follows the hypocoercivity framework developed in [2], with the adaptation of the spatial domain from $\mathbb{R}^d$ to the periodic setting $\mathbb{T}^d$ requiring the corresponding treatment of the Poisson equation. Building on this analysis, the chapter investigates numerically how the resulting decay rates depend on the underlying equilibrium and explores control strategies for accelerating convergence towards equilibrium. These numerical experiments suggest two natural directions for future work. The first concerns the dependence of hypocoercive decay rates on the equilibrium state, and consequently on the external potential u. Following the operator-theoretic viewpoint developed in Chapter 3, where the maps $u \mapsto (\rho_\star(u), \varphi_\star(u))$ are analysed as nonlinear solution operators, one may similarly study the decay rate as a nonlinear mapping of u, namely $u \mapsto \lambda(u)$, thereby quantifying how confinement design influences convergence towards equilibrium. The second direction is control-theoretic: while the numerical experiments explore control strategies at the discrete level, a natural continuation is to analyse their effectiveness for the continuous VPFP dynamics and to identify optimal mechanisms for accelerating convergence towards equilibrium.

The remainder of this chapter is organised as follows. In Section 5.1, we derive the linearised VPFP system around an arbitrary free energy stationary solution. Section 5.2 introduces the functional-analytic framework for the linearised problem, including the Hilbert space of admissible perturbations and the associated inner product. In Section 5.3, we then adapt the hypocoercivity analysis of [2] to the periodic setting and establish exponential decay estimates for solutions of the linearised system. Finally, Section 5.4 presents numerical experiments illustrating the dependence of decay rates on the equilibrium state and investigating control strategies to accelerate convergence towards equilibrium.

Since the analysis applies uniformly to all external potentials satisfying the assumptions stated in Sections 5.2 and 5.3, we simplify the notation by writing $f_\star$, $\rho_\star$, and $\varphi_\star$ for the stationary distribution, its spatial density, and the corresponding self-consistent potential, respectively.

5.1 Linearised VPFP System

Proceeding as in [2], we linearise the VPFP system around a free energy stationary solution by considering small perturbations of size $\mathcal{O}(\eta)$, where $0 < \eta \ll 1$. Substituting this ansatz into the coupled VPFP system and retaining only first-order terms in η, we derive a linear kinetic equation for the perturbed state coupled

to a linearised Poisson equation. This yields the linearised VPFP system, which governs the dynamics near equilibrium and serves as the starting point for the subsequent analysis.

Let $(f_\star, \varphi_\star) \in L^1(\mathbb{T}^d \times \mathbb{R}^d) \times H^1_{\mathrm{zm}}(\mathbb{T}^d)$ be the unique free energy solution of the stationary VPFP system with external potential $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ (see Chapter 2). For a small perturbation parameter $0 < \eta \ll 1$, we set

$$f(t,x,v) = f_\star(x,v) + \eta\, f_\star(x,v)\, h(t,x,v), \qquad \varphi(t,x) = \varphi_\star(x) + \eta\, \psi_h(t,x),$$

where $h : [0,\infty) \times \mathbb{T}^d \times \mathbb{R}^d \to \mathbb{R}$ and $\psi_h : [0,\infty) \times \mathbb{T}^d \to \mathbb{R}$ denote the perturbation functions for the state and the self-consistent potential relative to the free energy solution, respectively. Performing the substitution $(f, \varphi) \mapsto (f_\star + \eta f_\star h, \varphi_\star + \eta\psi_h)$ in the kinetic equation (VPFPa) of the VPFP system yields

$$\begin{aligned}\eta f_\star \frac{\partial h}{\partial t} + v \cdot \big((1+\eta h)\nabla_x f_\star + \eta f_\star \nabla_x h\big) - \big(\nabla_x u + \nabla_x \varphi_\star + \eta \nabla_x \psi_h\big) \cdot \big((1+\eta h)\nabla_v f_\star + \eta f_\star \nabla_v h\big) \\ = (1+\eta h)\Delta_v f_\star + 2\eta (\nabla_v f_\star) \cdot (\nabla_v h) + \eta f_\star \Delta_v h + (1+\eta h)\nabla_v \cdot (v f_\star) + \eta f_\star v \cdot \nabla_v h\,.\end{aligned}$$

Since the pair $(f_\star, \varphi_\star)$ solves the stationary kinetic equation (2.7a), the above PDE simplifies to

$$\begin{aligned}\eta f_\star \frac{\partial h}{\partial t} + \eta f_\star v \cdot \nabla_x h - \eta(1+\eta h)\nabla_x \psi_h \cdot \nabla_v f_\star - \eta f_\star \big(\nabla_x u + \nabla_x \varphi_\star + \eta \nabla_x \psi_h\big) \cdot \nabla_v h \\ = 2\eta (\nabla_v f_\star) \cdot (\nabla_v h) + \eta f_\star \Delta_v h + \eta f_\star v \cdot \nabla_v h\,.\end{aligned}$$

By Theorem 2.2.18, there holds

$$f_\star(x,v) = \rho_\star(x)\mathrm{M}(v) = \boldsymbol{m} \frac{e^{-u(x)-\varphi_\star(x)}}{\int_{\mathbb{T}^d} e^{-u(y)-\varphi_\star(y)}\mathrm{d}y} \frac{e^{-\frac{|v|^2}{2}}}{(2\pi)^{\frac{d}{2}}} > 0 \quad \text{for a.e. } (x,v) \in \mathbb{T}^d \times \mathbb{R}^d.$$

Since $\eta > 0$ and $f_\star > 0$ a.e., dividing both sides of the above PDE by $\eta f_\star$ and collecting the $\mathcal{O}(\eta)$ terms on the right-hand side yields

$$\frac{\partial h}{\partial t} + v \cdot \nabla_x h + v \cdot \nabla_x \psi_h - (\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v h + v \cdot \nabla_v h - \Delta_v h = \eta \nabla_x \psi_h \cdot (\nabla_v h - hv),$$

where we used that

$$\nabla_v f_\star = \rho_\star \nabla_v \mathrm{M}(v) = -f_\star v.$$

Formally, in the limit as $\eta \to 0^+$, the $\mathcal{O}(\eta)$ terms can be neglected. Hence, keeping only the $\mathcal{O}(1)$ terms yields the linearised VPFP kinetic equation

$$\frac{\partial h}{\partial t} + v \cdot \nabla_x h + v \cdot \nabla_x \psi_h - (\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v h + v \cdot \nabla_v h - \Delta_v h = 0.$$

Applying next the substitution $(f, \varphi) \mapsto (f_\star + \eta f_\star h, \varphi_\star + \eta\psi_h)$ in the Poisson equation (VPFPb) yields

$$-\Delta_x \varphi_\star - \eta \Delta_x \psi_h = \int_{\mathbb{R}^d} f_\star \mathrm{d}v + \eta \int_{\mathbb{R}^d} h f_\star \mathrm{d}v - \rho_0.$$

Since $\varphi_\star$ solves the stationary Poisson equation (2.7b), the above PDE simplifies to

$$-\Delta_x \psi_h = \int_{\mathbb{R}^d} h f_\star \mathrm{d}v\,.$$

Thus, the linearised VPFP system is

$$\frac{\partial h}{\partial t} + v \cdot \nabla_x h + v \cdot \nabla_x \psi_h - (\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v h = \Delta_v h - v \cdot \nabla_v h, \tag{5.1a}$$

$$-\Delta_x \psi_h = \int_{\mathbb{R}^d} h f_\star \mathrm{d}v. \tag{5.1b}$$

The compatibility condition for the well-posedness of the Poisson equation (see Section 2.1) is then

$$\int_{\mathbb{T}^d}\int_{\mathbb{R}^d} hf_\star \mathrm{d}v\mathrm{d}x = 0 \quad \text{for a.e. } t \in [0,+\infty).$$

The above zero-mean condition additionally ensures the conservation of mass in time. Indeed, if (h,ψ_h) is a sufficiently regular solution to the linearised VPFP system (5.1), then for a.e. $t \in [0,+\infty)$ there holds

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} f(t,x,v)\mathrm{d}x\mathrm{d}v = \int_{\mathbb{T}^d\times\mathbb{R}^d} f_\star(x,v)\mathrm{d}x\mathrm{d}v + \eta\int_{\mathbb{T}^d\times\mathbb{R}^d} h(t,x,v)f_\star(x,v)\mathrm{d}x\mathrm{d}v = \boldsymbol{m}.$$

In the sequel, we will denote by

$$\hat{\rho}_h := \int_{\mathbb{R}^d} hf_\star \mathrm{d}v \qquad \text{and} \qquad \psi_h := (-\Delta)^{-1}\rho_h \tag{5.2}$$

the weighted density and the unique zero-mean solution to the associated Poisson equation (5.1b), for all perturbation functions h belonging to a suitable class $\mathcal{H}$. In the following section, we specify the admissible class $\mathcal{H}$ and establish that ψ_h is well-defined.

5.2 Functional-Analytic Framework

Throughout this chapter, we impose the following.

Assumption 5.2.1. *Let the external potential $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ be such that the free energy stationary spatial density satisfies $\rho_\star \in L^\infty(\mathbb{T}^d)$.*

This condition is used to simplify the linearised analysis, but it is not restrictive for the inverse problem considered in Chapter 4. Indeed, Corollary 4.3.4 shows that, under the additional assumption $\varphi_\star(u_\star) \in W^{1,d+\varepsilon}_{\mathrm{zm}}(\mathbb{T}^d)$ for some $\varepsilon > 0$, every minimiser $u_\star$ generates a stationary state whose spatial density belongs to this admissible class. Finally, the elliptic regularity results established in Section 3.1, together with the previous assumption, imply the following.

Claim 5.2.2. *Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisfy Assumtpion* 5.2.1. *Then, the corresponding free energy stationary spatial density and self-consistent potential satisfy*

$$\rho_\star \in H^1 \cap L^p(\mathbb{T}^d), \qquad \varphi_\star \in W^{3,2}_{\mathrm{zm}} \cap W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d),$$

for all $1 \le p < +\infty$. In particular, there holds

$$\nabla_x \rho_\star = -\rho_\star(\nabla_x u + \nabla_x \varphi_\star) \quad \textit{a.e.}$$

We next consider the following notion.

Definition 5.2.3 (**Invariant Measure**)**.**
Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$, and let $f_\star$ denote the associated free energy stationary state. We define the ***invariant measure*** *as the unique positive finite Radon measure μ on $\mathbb{T}^d \times \mathbb{R}^d$ whose Radon–Nikodym density with respect to the Lebesgue measure is given by $f_\star$.*

In the sequel, we shall often write

$$\mathrm{d}\mu(x,v) = f_\star(x,v)\mathrm{d}x\mathrm{d}v.$$

The following result establishes that the definition of invariant measure is well-posed.

Lemma 5.2.4. *Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$. Then, the corresponding invariant measure μ is well-defined.*

Proof. Since $f_\star \in L^1(\mathbb{T}^d \times \mathbb{R}^d)$ by definition of free energy solution, we define the measure μ on $\mathbb{T}^d \times \mathbb{R}^d$ by

$$\mu(A) := \int_A f_\star \mathrm{d}x\mathrm{d}v, \quad \text{for all Lebesgue measurable } A \subset \mathbb{T}^d \times \mathbb{R}^d.$$

This measure is well-defined, non-negative, and finite, since

$$\mu(\mathbb{T}^d \times \mathbb{R}^d) = \|f_\star\|_{L^1(\mathbb{T}^d \times \mathbb{R}^d)} = \boldsymbol{m} < \infty.$$

Moreover, μ is absolutely continuous with respect to the Lebesgue measure in the sense of Definition 3.2.1 in [10], with Radon–Nikodym density $f_\star$. We therefore write

$$\mathrm{d}\mu(x, v) = f_\star(x, v)\mathrm{d}x\mathrm{d}v.$$

To verify that μ is a Radon measure, note that $\mathbb{T}^d \times \mathbb{R}^d$ is a locally compact Hausdorff space and $f_\star \in L^1_{\mathrm{loc}}(\mathbb{T}^d \times \mathbb{R}^d)$. Hence μ is locally finite and regular, i.e. Radon, by standard results (see [10]).

Finally, uniqueness of the representation follows from the Radon–Nikodym theorem (see, e.g., Theorem 3.2.2 in [10]). Indeed, let ν be any Radon measure absolutely continuous with respect to the Lebesgue measure and with density $f_\star$. Then, by the Radon-Nikodym theorem there holds

$$\nu(A) = \int_A f_\star(x, v)\mathrm{d}x\mathrm{d}v = \mu(A) \quad \text{for all Lebesgue measurable } A \subset \mathbb{T}^d \times \mathbb{R}^d.$$

Hence, $\nu = \mu$. Therefore, μ is the unique positive bounded Radon measure with Radon–Nikodym density $f_\star$ with respect to the Lebesgue measure. □

Proceding as in [2], we define the admissible space of perturbations as follows.

Definition 5.2.5 (**Admissible Perturbations and Inner Product**).
Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisfy Assumption 5.2.1*, and let μ be the corresponding invariant measure. Define the space of admissible perturbations as*

$$\mathcal{H} := \left\{ h \in L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu) : \int_{\mathbb{T}^d \times \mathbb{R}^d} h \mathrm{d}\mu = 0 \right\}.$$

Moreover, define the map $(\cdot, \cdot)_{\mathcal{H}} : \mathcal{H} \times \mathcal{H} \to \mathbb{R}$ by

$$(h_1, h_2)_{\mathcal{H}} := \int_{\mathbb{T}^d \times \mathbb{R}^d} h_1 h_2 \mathrm{d}\mu + \int_{\mathbb{T}^d} \hat{\rho}_{h_1} \psi_{h_2} \mathrm{d}x,$$

where $\hat{\rho}_h, \psi_h$ are as in (5.2).

Lemma 5.2.6. *For any $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisying Assumption* 5.2.1*, $(\mathcal{H}, (\cdot, \cdot)_{\mathcal{H}})$ is a Hilbert space.*

The proof of Lemma 5.2.6 relies on a preliminary result, which we establish first.

Claim 5.2.7. *For any $h \in \mathcal{H}$, there holds $\hat{\rho}_h \in L^2_{\mathrm{zm}}(\mathbb{T}^d)$ and $\psi_h \in H^2_{\mathrm{zm}}(\mathbb{T}^d)$.*

Proof of Claim 5.2.7. Let $h \in \mathcal{H}$ be arbitrary. Since $(\mathbb{T}^d \times \mathbb{R}^d, \mu)$ is a finite measure space, the following is a continuous injection

$$L^1(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu) \hookrightarrow L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu).$$

Hence, $h \in L^1(\mathbb{T}^d \times \mathbb{R}^d; \mu)$. Moreover, since

$$\mathrm{d}\mu(x, v) = f_\star(x, v)\mathrm{d}x\mathrm{d}v,$$

Fubini's theorem [15, Thm 4.5] implies that

$$\hat{\rho}_h = \int_{\mathbb{R}^d} h f_\star \mathrm{d}v \in L^1(\mathbb{T}^d).$$

Next, using the Euler–Lagrange representation $f_\star(x,v) = \rho_\star(x)\mathrm{M}(v)$ (see Theorem 2.2.18), we write

$$\hat{\rho}_h(x) = \int_{\mathbb{R}^d} h(x,v)\rho_\star(x)\mathrm{M}(v)\mathrm{d}v.$$

Since $\rho_\star \in L^\infty(\mathbb{T}^d)$ by Assumption 5.2.1, we estimate

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} |h\rho_\star|^2 \mathrm{M}(v)\mathrm{d}x\mathrm{d}v \le \|\rho_\star\|_{L^\infty(\mathbb{T}^d)} \int_{\mathbb{T}^d\times\mathbb{R}^d} |h|^2 \rho_\star \mathrm{M}(v)\mathrm{d}x\mathrm{d}v$$

where the right-hand side is finite since $h \in L^2(\mathbb{T}^d\times\mathbb{R}^d;\mathrm{d}\mu)$. Hence, by Fubini's theorem,

$$\int_{\mathbb{R}^d} |h\rho_\star|^2 \mathrm{M}(v)\mathrm{d}v \in L^1(\mathbb{T}^d).$$

Since $\mathrm{M}(v)\mathrm{d}v$ is a probability measure on $\mathbb{R}^d$, Jensen's inequality [10, Thm 2.12.19] yields

$$\int_{\mathbb{R}^d} |h(x,v)\rho_\star(x)|^2 \mathrm{M}(v)\mathrm{d}v \ge \left(\int_{\mathbb{R}^d} h(x,v)\rho_\star(x)\mathrm{M}(v)\mathrm{d}v\right)^2 = \hat{\rho}_h^2 \quad \text{for a.e. } x \in \mathbb{T}^d,$$

Integrating the above inequality over $\mathbb{T}^d$, we obtain

$$\int_{\mathbb{T}^d} |\hat{\rho}_h|^2 \mathrm{d}x = \int_{\mathbb{T}^d} \left(\int_{\mathbb{R}^d} h(x,v)\rho_\star(x)\mathrm{M}(v)\mathrm{d}v\right)^2 \mathrm{d}x \le \|\rho_\star\|_{L^\infty(\mathbb{T}^d)} \int_{\mathbb{T}^d\times\mathbb{R}^d} |h|^2 \mathrm{d}\mu < +\infty. \tag{5.3}$$

Hence, $\hat{\rho}_h \in L^2(\mathbb{T}^d)$. In particular, by definition of $\mathcal{H}$, there holds $\hat{\rho}_h \in L^2_{\mathrm{zm}}(\mathbb{T}^d)$. Therefore, elliptic regularity on the torus (see Proposition 2.1.4) yields

$$\psi_h = (-\Delta)^{-1}\hat{\rho}_h \in H^2_{\mathrm{zm}}(\mathbb{T}^d). \qquad \square$$

Proof of Lemma 5.2.6. We first establish that $(\cdot,\cdot)_{\mathcal{H}}$ is an inner product, and then conclude that $\mathcal{H}$ is closed with respect to the norm

$$\|h\|_{\mathcal{H}} := \sqrt{(h,h)_{\mathcal{H}}}, \qquad h \in \mathcal{H}.$$

Step 1. Inner Product. The mapping $(\cdot,\cdot)_{\mathcal{H}}$ is well-defined. Indeed, for any $h_1, h_2 \in \mathcal{H}$, Claim 5.2.7 and the Cauchy–Schwarz inequality (see, e.g., Corollary 2.11.3 in [10]) yield

$$|(h_1,h_2)_{\mathcal{H}}| \le \|h_1\|_{L^2(\mathbb{T}^d\times\mathbb{R}^d;\mathrm{d}\mu)}\|h_2\|_{L^2(\mathbb{T}^d\times\mathbb{R}^d;\mathrm{d}\mu)} + \|\hat{\rho}_{h_1}\|_{L^2(\mathbb{T}^d)}\|\psi_{h_2}\|_{L^2(\mathbb{T}^d)} < +\infty.$$

Moreover, the map

$$\mathcal{H} \to L^2_{\mathrm{zm}}(\mathbb{T}^d),\ h \mapsto \hat{\rho}_h = \int_{\mathbb{R}^d} h f_\star \mathrm{d}v$$

is linear. Since the inverse Laplacian is linear, it follows that the map

$$\mathcal{H} \to H^2_{\mathrm{zm}}(\mathbb{T}^d),\ h \mapsto \psi_h = (-\Delta)^{-1}\hat{\rho}_h,$$

is also linear. Consequently, $(\cdot,\cdot)_{\mathcal{H}}$ is bilinear. Symmetry is immediate:

$$\begin{aligned}(h_1,h_2)_{\mathcal{H}} &= \int_{\mathbb{T}^d\times\mathbb{R}^d} h_1 h_2 \mathrm{d}\mu + \int_{\mathbb{T}^d} \hat{\rho}_{h_1}\psi_{h_2}\mathrm{d}x \\ &= \int_{\mathbb{T}^d\times\mathbb{R}^d} h_2 h_1 \mathrm{d}\mu + \int_{\mathbb{T}^d} \psi_{h_1}\hat{\rho}_{h_2}\mathrm{d}x = (h_1,h_2)_{\mathcal{H}},\end{aligned}$$

where the second identity follows from integration by parts. Finally, another integration by parts argument yields

$$(h,h)_{\mathcal{H}} = \int_{\mathbb{T}^d\times\mathbb{R}^d} |h|^2 \mathrm{d}\mu + \int_{\mathbb{T}^d} |\nabla\psi_h|^2 \mathrm{d}x \ge 0, \tag{5.4}$$

where $(h,h)_{\mathcal{H}} = 0$ if and only if $h = 0$. Hence, $(\cdot,\cdot)_{\mathcal{H}}$ defines an inner product on $\mathcal{H}$.

Step 2. Vector Space Structure and Completeness. The set $\mathcal{H}$ can be easily seen to be a linear vector space. From (5.4) we write

$$\|h\|_{\mathcal{H}}^2 = \|h\|_{L^2(\mathbb{T}^d\times\mathbb{R}^d;\mathrm{d}\mu)}^2 + \|\nabla\psi_h\|_{L^2(\mathbb{T}^d)^d}^2.$$

Next, let $\{h_n\}_{n\in\mathbb{N}} \subset \mathcal{H}$ be a Cauchy sequence with respect to $\|\cdot\|_{\mathcal{H}}$. Then, from the previous identity it follows that $\{h_n\}$ is a Cauchy sequence in $L^2(\mathbb{T}^d\times\mathbb{R}^d;\mathrm{d}\mu)$. Since the latter space is complete, there exists $h \in L^2(\mathbb{T}^d\times\mathbb{R}^d;\mathrm{d}\mu)$ such that

$$h_n \to h \quad \text{strongly in } L^2(\mathbb{T}^d\times\mathbb{R}^d;\mathrm{d}\mu).$$

Combining this with the continuity of the map

$$L^2(\mathbb{T}^d\times\mathbb{R}^d;\mathrm{d}\mu) \to \mathbb{R},\ h \mapsto \int_{\mathbb{T}^d\times\mathbb{R}^d} h\mathrm{d}\mu,$$

it follows that

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} h\mathrm{d}\mu = \lim_{\mathbb{T}^d\times\mathbb{R}^d} h_n\mathrm{d}\mu = 0.$$

Hence, $h \in \mathcal{H}$. By Claim 5.2.7, we thus deduce that $\hat{\rho}_h \in L^2_{\mathrm{zm}}(\mathbb{T}^d)$ and $\psi_h \in H^2_{\mathrm{zm}}(\mathbb{T}^d)$. It remains to prove that

$$\nabla\psi_{h_n} \to \nabla\psi_h \quad \text{strongly in } L^2_{\mathrm{zm}}(\mathbb{T}^d)^d. \tag{5.5}$$

To this end, consider the linear map

$$\mathcal{H} \to L^2_{\mathrm{zm}}(\mathbb{T}^d),\ h \mapsto \hat{\rho}_h.$$

By (5.3), the above map is continuous with respect to the $L^2(\mathbb{T}^d\times\mathbb{R}^d;\mathrm{d}\mu)$ norm. Hence, since $h_n \to h$ in $L^2(\mathbb{T}^d\times\mathbb{R}^d;\mathrm{d}\mu)$, it follows that

$$\hat{\rho}_{h_n} \to \hat{\rho}_h \quad \text{strongly in } L^2_{\mathrm{zm}}(\mathbb{T}^d).$$

Combining this with the continuity of the inverse Laplacian (see Proposition 2.1.4), we obtain

$$\psi_{h_n} \to \psi_h \quad \text{strongly in } H^2_{\mathrm{zm}}(\mathbb{T}^d),$$

which implies (5.5). From the previous reasoning, there holds

$$\|h_n - h\|_{\mathcal{H}} \to 0.$$

By arbitrariness of the Cauchy sequence, we conclude that $\mathcal{H}$ is complete with respect to $\|\cdot\|_{\mathcal{H}}$. □

5.3 Hypocoercivity Analysis

We now study the long-time behaviour of solutions to the linearised VPFP system (5.1) by means of hypocoercivity methods. To this end, we first reformulate (5.1) as an abstract linear evolution equation on the Hilbert space of admissible perturbations $\mathcal{H}$.

Henceforth, we shall assume that $u \in H^1\cap L^\infty(\mathbb{T}^d)$ satisfies Assumption 5.2.1. Let μ be the associated invariant measure in the sense of Definition 5.2.3, and let $(\mathcal{H},(\cdot,\cdot)_{\mathcal{H}})$ denote the Hilbert space of admissible perturbations introduced in Definition 5.2.5. The linearised kinetic equation (5.1a) can then be written in the form

$$\frac{\mathrm{d}h}{\mathrm{d}t} + \mathrm{T}h = \mathrm{L}h,$$

where the transport operator $\mathrm{T} : \mathcal{D}(\mathrm{T}) \to \mathcal{H}$ is defined by

$$\mathrm{T}h := v\cdot\nabla_x h + v\cdot\nabla_x\psi_h - (\nabla_x u + \nabla_x\varphi_\star)\cdot\nabla_v h,$$

and the collision operator $\mathrm{L} : \mathcal{D}(\mathrm{L}) \to \mathcal{H}$ is given by

$$\mathrm{L}h := \Delta_v h - v \cdot \nabla_v h.$$

We define the operators L and T on the common core

$$\mathcal{A} := \left\{ f \in \mathcal{C}_c^\infty(\mathbb{T}^d \times \mathbb{R}^d) : \int_{\mathbb{T}^d \times \mathbb{R}^d} f \, \mathrm{d}\mu = 0 \right\} \subset \mathcal{H}.$$

On this class, all integrations by parts are classical and no boundary terms arise. Standard density arguments in the associated weighted Sobolev spaces allow all identities and estimates to be extended to the natural domains of T and L. Moreover, since T is antisymmetric and L is symmetric on $\mathcal{A}$ (see Lemma 5.3.5), both operators are closable in $\mathcal{H}$ (see Chapter 1 in [50]). For simplicity of exposition, we therefore identify T and L with their closures and do not distinguish between the operators and their closures in the sequel. All computations are performed on the core $\mathcal{A}$, and the resulting identities are understood for the corresponding closed operators.

The hypocoercivity framework used throughout this chapter is based on the approach of [2]. For completeness, we provide a detailed exposition in Appendix B; here, we only recall the main ingredients needed for the subsequent analysis. Let $\Pi : \mathcal{H} \to \mathcal{H}$ denote the orthogonal projection onto $\ker(\mathrm{L})$ and define the auxiliary operator $\mathrm{A} : \mathcal{A} \to \mathcal{H}$ by

$$\mathrm{A}h := (\mathrm{Id}_{\mathcal{H}} + (\mathrm{T}\Pi)^* \mathrm{T}\Pi)^{-1} (\mathrm{T}\Pi)^* h.$$

The exponential stability of the stationary solution to the linearised VPFP system is established through the Lyapunov functional $\mathrm{H}_\delta : \mathcal{A} \to \mathbb{R}$ defined by

$$\mathrm{H}_\delta[h] := \frac{1}{2}\|h\|_{\mathcal{H}}^2 + \delta(\mathrm{A}h, h)_{\mathcal{H}},$$

for a suitable choice of $\delta > 0$. The hypocoercivity theorem requires the following structural assumptions on the transport and collision operators.

Assumption 5.3.1. *There exist positive constants λ_m, λ_M, and C_M such that, for every $f \in \mathcal{A}$, the following properties hold:*

(*i*) T *and* L *are densely defined, with* T *antisymmetric and* L *symmetric.*

(*ii*) *microscopic coercivity:* $-(\mathrm{L}f, f)_{\mathcal{H}} \geq \lambda_m \|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}}^2$;

(*iii*) *macroscopic coercivity:* $\|\mathrm{T}\Pi f\|_{\mathcal{H}}^2 \geq \lambda_M \|\Pi f\|_{\mathcal{H}}^2$;

(*iv*) *parabolic macroscopic dynamics:* $\Pi \mathrm{T} \Pi f = 0$;

(*v*) *bounded auxiliary operators:* $\|\mathrm{AT}(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}} + \|\mathrm{AL}f\|_{\mathcal{H}} \leq C_M \|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}}$.

Under these assumptions, the following hypocoercivity result holds.

Theorem 5.3.2. *Let Assumption* 5.3.1 *hold. Then there exists $\delta_\star > 0$ such that, for every $\delta \in (0, \delta_\star)$ and every sufficiently smooth solution $h : [0, +\infty) \to \mathcal{H}$ of* (5.1)*, there holds*

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{H}_\delta[h](t) \leq -\lambda \mathrm{H}_\delta[h](t),$$

for some constant $\lambda > 0$ depending on δ.

As discussed in Appendix B, combining Theorem 5.3.2 with Grönwall's inequality [4, Lemma (6.1)] yields

$$\mathrm{H}_\delta[h](t) \leq \mathrm{H}_\delta[h_0]e^{-\lambda t}, \qquad \text{for all } t \geq 0,$$

for every sufficiently smooth solution $h : [0, +\infty) \to \mathcal{H}$ with initial datum $h_0 \in \mathcal{H}$. Consequently, Corollary B.2.2 implies that

$$\|h(t)\|_{\mathcal{H}}^2 \leq \frac{1+\delta}{1-\delta}\|h_0\|_{\mathcal{H}}^2 e^{-\lambda t}, \qquad \text{for all } t \geq 0,$$

and therefore $h(t) \to 0$ in $\mathcal{H}$ as $t \to \infty$.

The remainder of this section is devoted to verifying Assumption 5.3.1 for the linearised VPFP system. More precisely, by adapting the analysis of [2] to the present setting, we derive sufficient conditions on the external potential u ensuring that each of the required hypocoercivity properties is satisfied. We conclude this discussion by identifying sufficient conditions under which the optimal external potentials obtained from the inverse problem studied in Chapter 4 give rise to stationary states that are linearly exponentially stable in the sense of Theorem 5.3.2.

5.3.1 Transport and Collision Operators

We begin by establishing two preliminary results, which are then used to verify Assumption 5.3.1 (i).

Claim 5.3.3. *Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisfy Assumption 5.2.1. For any $h \in \mathcal{A}$, there holds $\hat{\rho}_h \in L^p_{\mathrm{zm}}(\mathbb{T}^d)$ and $\psi_h \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$ for all $1 \leq p < +\infty$.*

Proof. Let $h \in \mathcal{A}$ be arbitrary. Since $\mathcal{A} \subset \mathcal{H}$, by Claim 5.2.7 there holds

$$\hat{\rho}_h \in L^2_{\mathrm{zm}}(\mathbb{T}^d), \qquad \psi_h \in H^2_{\mathrm{zm}}(\mathbb{T}^d).$$

Next, using the Euler–Lagrange representation of $f_\star$, together with the identity $\mathrm{d}\mu = f_\star \mathrm{d}x\mathrm{d}v$, yields

$$\mathrm{d}\mu(x,v) = \rho_\star(x)\mathrm{M}(v)\mathrm{d}x\mathrm{d}v,$$

where $\rho_\star \in L^\infty(\mathbb{T}^d)$ by Assumption 5.2.1. Let $1 \leq p < +\infty$ be arbitrary. Since $h \in \mathcal{C}^\infty_c(\mathbb{T}^d \times \mathbb{R}^d) \subset L^p(\mathbb{T}^d)$, applying Jensen's inequality with probability measure $\mathrm{M}(v)\mathrm{d}v$ yields

$$|\hat{\rho}_h|^p \leq \left(\int_{\mathbb{R}^d} |h|\rho_\star \mathrm{M}(v)\mathrm{d}v\right)^p \leq \int_{\mathbb{R}^d} |h|^p \rho_\star^p \mathrm{M}(v)\mathrm{d}v \leq \|h\|^p_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)}\|\rho_\star\|^p_{L^\infty(\mathbb{T}^d)} \int_{\mathbb{R}^d} \mathrm{M}(v)\mathrm{d}v \quad \text{a.e.}$$

Integrating over $\mathbb{T}^d$, we obtain

$$\int_{\mathbb{T}^d} |\hat{\rho}_h|^p \mathrm{d}x \leq |\mathbb{T}^d| \|h\|^p_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)}\|\rho_\star\|^p_{L^\infty(\mathbb{T}^d)} < +\infty.$$

Therefore, $\hat{\rho}_h \in L^p(\mathbb{T}^d)$ for all $1 \leq p < +\infty$. Moreover, by definition of $\mathcal{A}$ and Fubini's theorem,

$$\int_{\mathbb{T}^d} \hat{\rho}_h \mathrm{d}x = \int_{\mathbb{T}^d\times\mathbb{R}^d} h f_\star \mathrm{d}x\mathrm{d}v = 0.$$

Hence, $\hat{\rho}_h \in L^p_{\mathrm{zm}}(\mathbb{T}^d)$ for all $1 \leq p < +\infty$. Elliptic regularity on the torus (see Proposition 2.1.5) finally yields

$$\psi_h = (-\Delta)^{-1}\hat{\rho}_h \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d) \quad \text{for all } 1 \leq p < +\infty.$$

This concludes the proof. □

Lemma 5.3.4. *Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisfy Assumption 5.2.1. Then, $\mathcal{A}$ is dense in $\mathcal{H}$.*

Proof. Let $h \in \mathcal{H}$ be arbitrary. Since $\mathcal{C}_c^\infty(\mathbb{T}^d \times \mathbb{R}^d)$ is dense in $L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)$ (see, e.g., Corollary 4.23 in [15]), there exists a sequence $\{h_n\}_{n\in\mathbb{N}} \subset \mathcal{C}_c^\infty(\mathbb{T}^d \times \mathbb{R}^d)$ such that $h_n \to h$ strongly in $L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)$. For each $n \in \mathbb{N}$, let $K_n := \mathrm{supp}(h_n)$. Since K_n is compact, it is in particular bounded, hence there exists $r_n > 0$ such that

$$K_n \subset \mathbb{T}^d \times \overline{B^{\mathbb{R}^d}(0, r_n)}.$$

Without loss of generality, we may assume that $r_n \geq 1$ for all $n \in \mathbb{N}$. Let next $\gamma_n \in \mathcal{C}_c^\infty(\mathbb{R}^d)$ be such that

$$\gamma_n(\mathbb{R}^d) = [0,1], \qquad \gamma_n(v) = 1 \quad \text{for all } v \in \overline{B^{\mathbb{R}^d}(0, r_n)}, \qquad \gamma_n(v) = 0 \quad \text{for all } v \in B^{\mathbb{R}^d}(0, 2r_n)^{\mathrm{c}}.$$

By construction there holds $K_n \subset \mathbb{T}^d \times \mathrm{supp}(\gamma_n) \subset \mathbb{T}^d \times \overline{B^{\mathbb{R}^d}(0, 2r_n)}$. Moreover, since $r_n \geq 1$, by definition of γ_n there holds

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} \gamma_n \mathrm{d}\mu \geq \int_{\mathbb{T}^d \times \overline{B^{\mathbb{R}^d}(0,1)}} \mathrm{d}\mu = \mu(\mathbb{T}^d \times \overline{B^{\mathbb{R}^d}(0,1)}) > 0. \tag{5.6}$$

Therefore, we define

$$\tilde{h}_n(x,v) := h_n(x,v) - \frac{\int_{\mathbb{T}^d \times \mathbb{R}^d} h_n \mathrm{d}\mu}{\int_{\mathbb{T}^d \times \mathbb{R}^d} \gamma_n \mathrm{d}\mu} \cdot \gamma_n(v), \qquad n \in \mathbb{N}.$$

It is immediate to verify that $\{\tilde{h}_n\}_{n\in\mathbb{N}} \subset \mathcal{A}$. Therefore, it only remains to prove that $\tilde{h}_n \to h$ strongly in $\mathcal{H}$, that is,

$$\|\tilde{h}_n - h\|_{\mathcal{H}}^2 = \|\tilde{h}_n - h\|_{L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)}^2 + \|\nabla \psi_{\tilde{h}_n} - \nabla \psi_h\|_{L^2(\mathbb{T}^d)^d}^2 \to 0 \quad \text{as } n \to \infty. \tag{5.7}$$

We treat each term separately. For the first term, since $h_n \to h$ strongly in $L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)$, by definition of $\tilde{h}_n$ it suffices to prove that

$$\frac{\int_{\mathbb{T}^d \times \mathbb{R}^d} h_n \mathrm{d}\mu}{\int_{\mathbb{T}^d \times \mathbb{R}^d} \gamma_n \mathrm{d}\mu} \gamma_n \to 0 \quad \text{strongly in } L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu).$$

Using (5.6), there holds

$$\left\| \frac{\int_{\mathbb{T}^d \times \mathbb{R}^d} h_n \mathrm{d}\mu}{\int_{\mathbb{T}^d \times \mathbb{R}^d} \gamma_n \mathrm{d}\mu} \gamma_n \right\|_{L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)} \leq \frac{1}{\mu(\mathbb{T}^d \times \overline{B^{\mathbb{R}^d}(0,1)})} \left| \int_{\mathbb{T}^d \times \mathbb{R}^d} h_n \mathrm{d}\mu \right| \qquad \text{for all } n \in \mathbb{N}.$$

Since μ is a bounded measure, the embedding $L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu) \hookrightarrow L^1(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)$ is continuous by Hölder's inequality. Consequently, since h has zero μ-mean and $h_n \to h$ strongly in $L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)$, we deduce that

$$\left| \int_{\mathbb{T}^d \times \mathbb{R}^d} h_n \mathrm{d}\mu \right| = \left| \int_{\mathbb{T}^d \times \mathbb{R}^d} (h_n - h) \mathrm{d}\mu \right| \leq \|h_n - h\|_{L^1(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)} \leq C \cdot \|h_n - h\|_{L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)} \to 0 \quad \text{as } n \to \infty.$$

From the previous reasoning, this implies that

$$\|\tilde{h}_n - h\|_{L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)} \to 0 \qquad \text{as } n \to \infty.$$

We next consider the second term in (5.7). To this end, let the weighted density operator

$$\hat{\Gamma}_\rho : L^2_{\mathrm{zm}}(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu) \to L^2_{\mathrm{zm}}(\mathbb{T}^d)$$

be given by

$$\hat{\Gamma}_\rho(h) := \hat{\rho}_h = \int_{\mathbb{R}^d} h f_\star \mathrm{d}v, \qquad h \in L^2_{\mathrm{zm}}(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu).$$

Then, $\hat{\Gamma}_\rho$ is linear and bounded. To see boundedness, from the Euler–Lagrange identity $f_\star = \rho_\star \mathrm{M}(v)$ and Jensen's inequality applied to the probability space $(\mathbb{R}^d, \mathrm{M}(v)\mathrm{d}v)$, we obtain

$$\|\hat{\Gamma}_\rho(h)\|_{L^2(\mathbb{T}^d)}^2 \leq \int_{\mathbb{T}^d} \left(\int_{\mathbb{R}^d} |h| \rho_\star \mathrm{M}(v) \mathrm{d}v \right)^2 \mathrm{d}x \leq \int_{\mathbb{T}^d} \int_{\mathbb{R}^d} |h|^2 \rho_\star^2 \mathrm{M}(v) \mathrm{d}v \mathrm{d}x \leq \|\rho_\star\|_{L^\infty(\mathbb{T}^d)} \|h\|_{L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)}^2,$$

where the last inequality follows from Fubini's theorem, Assumption 5.2.1, and the identity $\mathrm{d}\mu = f_\star \mathrm{d}x\mathrm{d}v$. Since $\tilde{h}_n \to h$ strongly in $L^2_{\mathrm{zm}}(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)$, the continuity of $\hat{\Gamma}_\rho$ implies that

$$\hat{\rho}_{\tilde{h}_n} \to \hat{\rho}_h \quad \text{strongly in } L^2_{\mathrm{zm}}(\mathbb{T}^d).$$

Since the inverse Laplacian $(-\Delta)^{-1} : L^2_{\mathrm{zm}}(\mathbb{T}^d) \to H^2_{\mathrm{zm}}(\mathbb{T}^d)$ is an isomorphism of Banach spaces (see Proposition 2.1.4), we therefore conclude that

$$\psi_{\tilde{h}_n} \to \psi_h \quad \text{strongly in } H^2_{\mathrm{zm}}(\mathbb{T}^d).$$

This implies in particular that

$$\|\nabla\psi_{\tilde{h}_n} - \nabla\psi_h\|_{L^2(\mathbb{T}^d)^d} \to 0 \quad \text{as } n \to \infty.$$

The claim follows by arbitrariness of $h \in \mathcal{H}$. □

Lemma 5.3.5. *Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisfy Assumption 5.2.1. Then, operators* T *and* L *are densely defined on $\mathcal{H}$, with* T *antisymmetric and* L *symmetric.*

Proof. We first show that T and L are well-defined on $\mathcal{A}$. By the preceding Lemma, this implies that T and L are densely defined on $\mathcal{H}$. We conclude by proving the symmetry properties of the two operators.

Step 1. The transport operator T is well-defined. Let $h \in \mathcal{A}$ be arbitrary. We prove that $\mathrm{T}h \in \mathcal{H}$. Since $h \in \mathcal{C}^\infty_c(\mathbb{T}^d \times \mathbb{R}^d)$, there holds

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} |v \cdot \nabla_x h|^2 \mathrm{d}\mu \leq \|\nabla_x h\|^2_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)^d} \int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 \mathrm{d}\mu = \|\nabla_x h\|^2_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)^d} \int_{\mathbb{T}^d} \rho_\star \mathrm{d}x \int_{\mathbb{R}^d} |v|^2 \mathrm{M}(v)\mathrm{d}v,$$

where the last identity uses $\mathrm{d}\mu = \rho_\star \mathrm{M}(v)\mathrm{d}x\mathrm{d}v$ and Fubini's theorem. Since $\rho_\star \in L^1(\mathbb{T}^d)$ and $\mathrm{M}(v)$ has finite second moment, it follows that $v \cdot \nabla_x h \in L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)$. Next, by Claim 5.3.3, there holds

$$\psi_h \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d) \quad \text{for all } 1 \leq p < +\infty.$$

Choosing $p > d$, the Sobolev embedding theorem (Corollary A.1.4) yields the continuous embedding $W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d) \hookrightarrow \mathcal{C}^1(\mathbb{T}^d)$. We therefore estimate

$$\int_{\mathbb{T}^d\times\mathbb{R}^d} |v \cdot \nabla_x \psi_h|^2 \mathrm{d}\mu \leq \|\nabla_x \psi_h\|^2_{L^\infty(\mathbb{T}^d)^d} \int_{\mathbb{T}^d\times\mathbb{R}^d} |v|^2 \mathrm{d}\mu < +\infty,$$

and thus $v \cdot \nabla_x \psi_h \in L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)$. We turn to the last addendum. By Assumption 5.2.1, there holds $\rho_\star \in L^\infty(\mathbb{T}^d)$, which implies that $f_\star = \rho_\star \mathrm{M} \in L^\infty(\mathbb{T}^d \times \mathbb{R}^d)$. Moreover, elliptic regularity on the torus yields

$$\varphi_\star = (-\Delta)^{-1}(\rho_\star - \rho_0) \in H^2_{\mathrm{zm}}(\mathbb{T}^d).$$

Since $K := \mathrm{supp}(h)$ is compact, it is in particular bounded, and thus there exists $\bar{v} \geq 0$ such that

$$K \subset \mathbb{T}^d \times [-\bar{v}, \bar{v}]^d.$$

Using that $u \in H^1 \cap L^\infty(\mathbb{T}^d)$, we then estimate

$$\begin{aligned}
\int_{\mathbb{T}^d\times\mathbb{R}^d} |(\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v h|^2 \mathrm{d}\mu &= \int_K |(\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v h|^2 f_\star \mathrm{d}x\mathrm{d}v \\
&\leq \|f_\star\|_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)} \|\nabla_v h\|^2_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)^d} \int_{\mathbb{T}^d\times[-\bar{v},\bar{v}]^d} |\nabla_x u + \nabla_x \varphi_\star|^2 \mathrm{d}x\mathrm{d}v \\
&\leq \|f_\star\|_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)} \|\nabla_v h\|^2_{L^\infty(\mathbb{T}^d\times\mathbb{R}^d)^d} \|\nabla_x u + \nabla_x \varphi_\star\|^2_{L^2(\mathbb{T}^d)^d} (2\bar{v})^d,
\end{aligned}$$

where the last inequality follows from Fubini's theorem. This shows that

$$(\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v h \in L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu).$$

Combining this with the previous reasoning, we deduce that $\mathrm{T}h \in L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)$. To conclude that $\mathrm{T}h \in \mathcal{H}$, it remains to show that

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} \mathrm{T}h \mathrm{d}\mu = 0.$$

Since $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisfies Assumption 5.2.1, Claim 5.2.2, together with the Euler–Lagrange representation of $f_\star$, yields

$$\nabla_x f_\star = -f_\star(\nabla_x u + \nabla_x \varphi_\star). \tag{5.8}$$

Combining this with the identity $\mathrm{d}\mu = f_\star \mathrm{d}x\mathrm{d}v$, integration by parts in x yields

$$\begin{aligned}\int_{\mathbb{T}^d \times \mathbb{R}^d} v \cdot \nabla_x h \mathrm{d}\mu = \int_{\mathbb{T}^d \times \mathbb{R}^d} v \cdot \nabla_x h f_\star \mathrm{d}x\mathrm{d}v &= -\int_{\mathbb{T}^d \times \mathbb{R}^d} (vh) \cdot \nabla_x f_\star \mathrm{d}x\mathrm{d}v \\ &= \int_{\mathbb{T}^d \times \mathbb{R}^d} v \cdot (\nabla_x u + \nabla_x \varphi_\star) h \mathrm{d}\mu.\end{aligned}$$

Similarly, using that

$$\nabla_v f_\star = \nabla_v(\rho_\star \mathrm{M}) = -v f_\star, \tag{5.9}$$

integration by parts in v yields

$$-\int_{\mathbb{T}^d \times \mathbb{R}^d} (\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v h \mathrm{d}\mu = -\int_{\mathbb{T}^d \times \mathbb{R}^d} v \cdot (\nabla_x u + \nabla_x \varphi_\star) h \mathrm{d}\mu.$$

Consequently, we have

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} \mathrm{T}h \mathrm{d}\mu = \int_{\mathbb{T}^d \times \mathbb{R}^d} v \cdot \nabla_x \psi_h f_\star \mathrm{d}x\mathrm{d}v = \int_{\mathbb{T}^d} \rho_\star \nabla_x \psi_h \cdot \left(\int_{\mathbb{R}^d} v \mathrm{M}(v) \mathrm{d}v \right) \mathrm{d}x = 0,$$

where the second equality follows from Fubini's theorem.

Step 2. The collision operator L is well-defined. Let $h \in \mathcal{A}$ be arbitrary. Then, $\Delta_v h \in L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)$ since

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} |\Delta_v h|^2 \mathrm{d}\mu \leq \|\Delta_v h\|^2_{L^\infty(\mathbb{T}^d \times \mathbb{R}^d)} \mu(\mathbb{T}^d \times \mathbb{R}^d) < +\infty.$$

For the second term, since $\mathrm{d}\mu = f_\star \mathrm{d}x\mathrm{d}v$, we estimate

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} |v \cdot \nabla_v h|^2 \mathrm{d}\mu \leq \|\nabla_v h\|^2_{L^\infty(\mathbb{T}^d)^d} \int_{\mathbb{T}^d \times \mathbb{R}^d} |v|^2 f_\star \mathrm{d}x\mathrm{d}v,$$

where the right-hand side is finite since $f_\star$ has bounded velocity moments. We therefore conclude that $\mathrm{L}h \in L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)$. Finally, using that $\nabla_v f_\star = -v f_\star$, integration by parts in v yields

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} \Delta_v h \mathrm{d}\mu = \int_{\mathbb{T}^d \times \mathbb{R}^d} \Delta_v h f_\star \mathrm{d}x\mathrm{d}v = \int_{\mathbb{T}^d \times \mathbb{R}^d} \nabla_v h \cdot (v f_\star) \mathrm{d}x\mathrm{d}v = \int_{\mathbb{T}^d \times \mathbb{R}^d} v \cdot \nabla_v h \mathrm{d}\mu.$$

This implies that

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} \mathrm{L}h \mathrm{d}\mu = 0,$$

and hence $\mathrm{L}h \in \mathcal{H}$.

Step 3. T **is antisymmetric.** Let $h_1, h_2 \in \mathcal{A}$ be arbitrary. We aim to prove that

$$(\mathrm{T}h_1, h_2)_{\mathcal{H}} = -(h_1, \mathrm{T}h_2)_{\mathcal{H}}.$$

Recalling that

$$(\mathrm{T}h_1, h_2)_{\mathcal{H}} = \langle \mathrm{T}h_1, h_2 \rangle_{L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)} + \langle \hat{\rho}_{\mathrm{T}h_1}, \psi_{h_2} \rangle_{L^2(\mathbb{T}^d)},$$

we treat each term separately. From (5.8), integration by parts in x yields

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} (v \cdot \nabla_x h_1) h_2 \mathrm{d}\mu = -\int_{\mathbb{T}^d \times \mathbb{R}^d} h_1 (v \cdot \nabla_x h_2) \mathrm{d}\mu + \int_{\mathbb{T}^d \times \mathbb{R}^d} h_1 h_2 v \cdot (\nabla_x u + \nabla_x \varphi_\star) \mathrm{d}\mu.$$

Similarly, using (5.9) and integrating by parts in v, we obtain

$$-\int_{\mathbb{T}^d \times \mathbb{R}^d} (\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v h_1 h_2 \mathrm{d}\mu = \int_{\mathbb{T}^d \times \mathbb{R}^d} h_1 (\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v h_2 \mathrm{d}\mu - \int_{\mathbb{T}^d \times \mathbb{R}^d} h_1 h_2 v \cdot (\nabla_x u + \nabla_x \varphi_\star) \mathrm{d}\mu.$$

We next prove that

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} (v \cdot \nabla_x \psi_{h_1}) h_2 \mathrm{d}\mu = \int_{\mathbb{T}^d} \nabla_x \psi_{h_1} \cdot \nabla_x \psi_{\mathrm{T}h_2} \mathrm{d}x.$$

To this end, using (5.8) and integrating by parts in x, there holds

$$\begin{aligned}
\int_{\mathbb{T}^d \times \mathbb{R}^d} (v \cdot \nabla_x \psi_{h_1}) h_2 \mathrm{d}\mu &= \int_{\mathbb{T}^d \times \mathbb{R}^d} \psi_{h_1} \big(- v \cdot \nabla_x h_2 + (\nabla_x u + \nabla_x \varphi_\star) h_2 \big) \mathrm{d}\mu \\
&= \int_{\mathbb{T}^d} \psi_{h_1} \left(\int_{\mathbb{R}^d} \big(- v \cdot \nabla_x h_2 + (\nabla_x u + \nabla_x \varphi_\star) h_2 \big) f_\star \mathrm{d}v \right) \mathrm{d}x,
\end{aligned}$$

where the last identity uses $\mathrm{d}\mu = f_\star \mathrm{d}x\mathrm{d}v$ and Fubini's theorem. Since

$$\int_{\mathbb{R}^d} v \cdot \nabla_x \psi_{h_2} f_\star \mathrm{d}v = \rho_\star \nabla_x \psi_{h_2} \cdot \left(\int_{\mathbb{R}^d} v \mathrm{M}(v) \mathrm{d}v \right) = 0,$$

we may add the quantity $v \cdot \nabla_x \psi_{h_2} f_\star$ within the inner integral of the previous identity, and obtain

$$\begin{aligned}
\int_{\mathbb{T}^d \times \mathbb{R}^d} (v \cdot \nabla_x \psi_{h_1}) h_2 \mathrm{d}\mu &= -\int_{\mathbb{T}^d} \psi_{h_1} \left(\int_{\mathbb{R}^d} \big(v \cdot \nabla_x h_2 + v \cdot \nabla_x \psi_{h_2} - (\nabla_x u + \nabla_x \varphi_\star) h_2 \big) f_\star \mathrm{d}v \right) \mathrm{d}x \\
&= -\int_{\mathbb{T}^d} \psi_{h_1} \left(\int_{\mathbb{R}^d} (\mathrm{T}h_2) f_\star \mathrm{d}v \right) \mathrm{d}x \\
&= -\int_{\mathbb{T}^d} \psi_{h_1} \hat{\rho}_{\mathrm{T}h_2} \mathrm{d}x \\
&= \int_{\mathbb{T}^d} \nabla_x \psi_{h_1} \cdot \nabla_x \psi_{\mathrm{T}h_2} \mathrm{d}x,
\end{aligned}$$

where the last identity follows from integration by parts on the torus and the definition of $\psi_h = (-\Delta)^{-1} \hat{\rho}_h$. Similarly, integration by parts on the torus yields

$$\langle \hat{\rho}_{\mathrm{T}h_1}, \psi_{h_2} \rangle_{L^2(\mathbb{T}^d)} = -\int_{\mathbb{T}^d} \nabla_x \psi_{\mathrm{T}h_1} \cdot \nabla_x \psi_{h_2} \mathrm{d}x.$$

Collecting all the terms, we therefore conclude that

$$\begin{aligned}
(\mathrm{T}h_1, h_2)_{\mathcal{H}} &= -\int_{\mathbb{T}^d \times \mathbb{R}^d} h_1 (v \cdot \nabla_x h_2) \mathrm{d}\mu + \int_{\mathbb{T}^d \times \mathbb{R}^d} h_1 (\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v h_2 \mathrm{d}\mu \\
&\qquad + \int_{\mathbb{T}^d} \nabla_x \psi_{h_1} \cdot \nabla_x \psi_{\mathrm{T}h_2} \mathrm{d}x - \int_{\mathbb{T}^d} \nabla_x \psi_{\mathrm{T}h_1} \cdot \nabla_x \psi_{h_2} \mathrm{d}x \\
&= -(h_1, \mathrm{T}h_2)_{\mathcal{H}}.
\end{aligned}$$

Step 4. L **is symmetric.** Let $h_1, h_2 \in \mathcal{A}$ be arbitrary. Using (5.9), integration by parts in v yields

$$\int_{\mathbb{R}^d} (\Delta_v h_1) h_2 f_\star \mathrm{d}v = -\int_{\mathbb{R}^d} \nabla_v h_1 \cdot \nabla_v h_2 f_\star \mathrm{d}v + \int_{\mathbb{R}^d} \nabla_v h_1 \cdot (v h_2) f_\star \mathrm{d}v.$$

We therefore compute

$$\int_{\mathbb{R}^d} (\mathrm{L}h_1) h_2 f_\star \mathrm{d}v = \int_{\mathbb{R}^d} (\Delta_v h_1 - v \cdot \nabla_v h_1) h_2 f_\star \mathrm{d}v = -\int_{\mathbb{R}^d} \nabla_v h_1 \cdot \nabla_v h_2 f_\star \mathrm{d}v. \tag{5.10}$$

In the special case when $h_1 = h \in \mathcal{A}$ and $h_2 \equiv 1$, a standard cutoff approximation argument applied to h_2 yields

$$\hat{\rho}_{\mathrm{L}h} = -\int_{\mathbb{R}^d} \nabla_v h \cdot \nabla_v (1) f_\star \mathrm{d}v = 0.$$

Consequently, $\psi_{\mathrm{L}h} = 0$ for all $h \in \mathcal{A}$, and hence

$$\langle \hat{\rho}_{\mathrm{L}h_1}, \psi_{h_2} \rangle_{L^2(\mathbb{T}^d)} = 0 = \langle \hat{\rho}_{h_1}, \psi_{\mathrm{L}h_2} \rangle_{L^2(\mathbb{T}^d)}. \tag{5.11}$$

Finally, using (5.10), Fubini's theorem, and the factorisation $\mathrm{d}\mu = f_\star \mathrm{d}x\mathrm{d}v$, we deduce that

$$\langle \mathrm{L}h_1, h_2 \rangle_{L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)} = \int_{\mathbb{T}^d \times \mathbb{R}^d} (\mathrm{L}h_1) h_2 \mathrm{d}\mu = -\int_{\mathbb{T}^d \times \mathbb{R}^d} \nabla_v h_1 \cdot \nabla_v h_2 \mathrm{d}\mu = \langle h_1, \mathrm{L}h_2 \rangle_{L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)}. \tag{5.12}$$

By arbitrariness of h_1, h_2, we conclude that L is symmetric. □

5.3.2 Microscopic Coercivity

Let $\Pi : \mathcal{H} \to \mathcal{H}$ be given by

$$(\Pi h)(x) := \int_{\mathbb{R}^d} h(x, v) \mathrm{M}(v) \mathrm{d}v, \qquad h \in \mathcal{H}.$$

We first establish basic properties of Π, and then prove that the above operator is indeed the orthogonal projection onto ker(L). We then conclude by verifying Assumption 5.3.1 (*ii*).

Lemma 5.3.6. *Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisfy Assumption* 5.2.1. *Then,* Π *is a bounded self-adjoint operator satisfying* $\Pi \circ \Pi = \Pi$.

Proof. We first verify that Π is well-defined and bounded. Let $h \in \mathcal{H}$ be arbitrary. Since $(\mathbb{R}^d, \mathrm{M}(v)\mathrm{d}v)$ is a probability space, Jensen's inequality yields, for a.e. $x \in \mathbb{T}^d$,

$$((\Pi h)(x))^2 = \left(\int_{\mathbb{R}^d} h(x, v) \mathrm{M}(v) \mathrm{d}v \right)^2 \leq \int_{\mathbb{R}^d} h(x, v)^2 \mathrm{M}(v) \mathrm{d}v.$$

Integrating with respect to $\mathrm{d}\mu(x, w) = \rho_\star(x) \mathrm{M}(w) \mathrm{d}x \mathrm{d}w$ and applying Fubini's theorem yields

$$\begin{aligned} \|\Pi h\|^2_{L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)} &= \int_{\mathbb{T}^d \times \mathbb{R}^d} (\Pi h(x))^2 \mathrm{d}\mu(x, v) \\ &\leq \int_{\mathbb{T}^d \times \mathbb{R}^d} \left(\int_{\mathbb{R}^d} h(x, w)^2 \mathrm{M}(w) \mathrm{d}w \right) \mathrm{d}\mu(x, v) \\ &= \left(\int_{\mathbb{R}^d} \mathrm{M}(v) \mathrm{d}v \right) \left(\int_{\mathbb{T}^d \times \mathbb{R}^d} h(x, w)^2 \mathrm{d}\mu(x, w) \right) \\ &= \|h\|^2_{L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)}. \end{aligned}$$

Hence Π defines a bounded operator on $\mathcal{H}$. We next prove that Π is self-adjoint. To this end, for any $h \in \mathcal{H}$ we write

$$(\Pi h)(x) = \int_{\mathbb{R}^d} h(x, v) \mathrm{M}(v) \mathrm{d}v = \frac{1}{\rho_\star(x)} \int_{\mathbb{R}^d} h(x, v) \rho_\star(x) \mathrm{M}(v) \mathrm{d}v = \frac{\hat{\rho}_h(x)}{\rho_\star(x)} \quad \text{a.e.}, \tag{5.13}$$

where we used that $\rho_\star > 0$ a.e. (see Theorem 2.2.18) to justify the factorisation. This implies that

$$\hat{\rho}_{\Pi h} = \int_{\mathbb{R}^d} (\Pi h) f_\star \mathrm{d}v = \int_{\mathbb{R}^d} \hat{\rho}_h \mathrm{M}(v)\mathrm{d}v = \hat{\rho}_h \quad \text{a.e.}, \tag{5.14}$$

and hence

$$\psi_{\Pi h} = (-\Delta)^{-1}\hat{\rho}_{\Pi h} = (-\Delta)^{-1}\hat{\rho}_h = \psi_h.$$

Consequently, for any $h_1, h_2 \in \mathcal{H}$, we compute

$$\begin{aligned}(\Pi h_1, h_2)_{\mathcal{H}} &= \int_{\mathbb{T}^d\times\mathbb{R}^d} (\Pi h_1) h_2 \mathrm{d}\mu + \int_{\mathbb{T}^d} \hat{\rho}_{\Pi h_1}\psi_{h_2}\mathrm{d}x \\ &= \int_{\mathbb{T}^d} (\Pi h_1)(\Pi h_2)\rho_\star \mathrm{d}x + \int_{\mathbb{T}^d} \hat{\rho}_{h_1}\psi_{\Pi h_2}\mathrm{d}x \\ &= \int_{\mathbb{T}^d\times\mathbb{R}^d} h_1(\Pi h_2)\mathrm{d}\mu + \int_{\mathbb{T}^d} \hat{\rho}_{h_1}\psi_{\Pi h_2}\mathrm{d}x = (h_1, \Pi h_2)_{\mathcal{H}}.\end{aligned}$$

By arbitrariness of $h_1, h_2 \in \mathcal{H}$, it follows that Π is symmetric. Since Π is everywhere-defined on $\mathcal{H}$, and by definition of symmetric operators (see Section 1.2 in [50]), it follows that

$$\mathcal{H} = \mathcal{D}(\Pi) \subset \mathcal{D}(\Pi^*) \subset \mathcal{H}.$$

Hence, Π^* is also everywhere defined on $\mathcal{H}$. Since $\Pi \subset \Pi^*$ and both operators have domain $\mathcal{H}$, we conclude that $\Pi = \Pi^*$, that is, Π is self-adjoint. Finally, idempotency is immediate. Indeed, let $h \in \mathcal{H}$ be arbitrary. Then, since Πh is independent of v, there holds

$$\Pi(\Pi h)(x) = \int_{\mathbb{R}^d} (\Pi h)(x)\mathrm{M}(v)\mathrm{d}v = \Pi h(x),$$

and therefore $\Pi \circ \Pi = \Pi$. □

Lemma 5.3.7. *Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisfy Assumption 5.2.1. Then, Π is the orthogonal projection onto* $\ker(\mathrm{L})$.

Proof. Let $h \in \mathcal{H}$ be arbitrary. Then, since Πh is independent of v, and L is a differential operator in the velocity space, it follows that $\mathrm{L}(\Pi h) = 0$. Hence, $\mathcal{R}(\Pi) \subset \ker(\mathrm{L})$. We next prove the reverse inclusion. To this end, let first $h \in \mathcal{A}$ be arbitrary. Then, it is immediate to verify that

$$\mathrm{L}h = \frac{1}{\mathrm{M}(v)}\nabla_v \cdot (\mathrm{M}(v)\nabla_v h).$$

Therefore, integration by parts yields

$$\int_{\mathbb{R}^d} h(\mathrm{L}h)\mathrm{M}(v)\mathrm{d}v = \int_{\mathbb{R}^d} h\nabla_v \cdot (\mathrm{M}(v)\nabla_v h)\mathrm{d}v = -\int_{\mathbb{R}^d} |\nabla_v h|^2 \mathrm{M}(v)\mathrm{d}v.$$

If $h \in \ker(\mathrm{L})$, then the left-hand side of the above identity vanishes, and hence we obtain

$$\int_{\mathbb{R}^d} |\nabla_v h|^2 \mathrm{M}(v)\mathrm{d}v = 0.$$

This implies that $\nabla_v h = 0$ a.e. By arbitrariness of $h \in \ker(\mathrm{L})$, it follows that

$$\ker(\mathrm{L}) = \left\{h \in \mathcal{A} : h(x,\cdot) = \text{const. for a.e. } x \in \mathbb{T}^d\right\} = \mathrm{span}\{1\},$$

where $1 : \mathbb{R}^d \to \mathbb{R},\ v \mapsto 1$. Using that

$$\langle 1, 1\rangle_{L^2(\mathbb{R}^d;\mathrm{M}(v)\mathrm{d}v)} = \int_{\mathbb{R}^d} \mathrm{M}(v)\mathrm{d}v = 1,$$

we therefore conclude that the orthogonal projection of $h \in \mathcal{A}$ onto $\ker(\mathrm{L})$ satisfies

$$\frac{\langle h, 1\rangle_{L^2(\mathbb{R}^d;\mathrm{M}(v)\mathrm{d}v)}}{\langle 1, 1\rangle_{L^2(\mathbb{R}^d;\mathrm{M}(v)\mathrm{d}v)}} 1 = \int_{\mathbb{R}^d} h\mathrm{M}(v)\mathrm{d}v = \Pi h.$$

This concludes the proof. □

Lemma 5.3.8. *Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisfy Assumption 5.2.1. For all $h \in \mathcal{A}$, there holds*

$$-(\mathrm{L}h, h)_{\mathcal{H}} \geq \|h - \Pi h\|_{\mathcal{H}}^2.$$

Proof. Let $h \in \mathcal{A}$ be arbitrary. Using (5.11) and (5.12), we write

$$-(\mathrm{L}h, h)_{\mathcal{H}} = \int_{\mathbb{T}^d \times \mathbb{R}^d} |\nabla_v h|^2 \mathrm{d}\mu.$$

Then, Fubini's theorem and the Gaussian–Poincaré inequality (see, e.g., Theorem 1.6.4 in [9]) yield

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} |\nabla_v h|^2 \mathrm{d}\mu = \int_{\mathbb{T}^d} \left(\int_{\mathbb{R}^d} |\nabla_v h|^2 \mathrm{M}(v)\mathrm{d}v \right) \rho_\star \mathrm{d}x \geq \int_{\mathbb{T}^d} \left(\int_{\mathbb{R}^d} |h - \Pi h|^2 \mathrm{M}(v)\mathrm{d}v \right) \rho_\star \mathrm{d}x = \int_{\mathbb{T}^d \times \mathbb{R}^d} |h - \Pi h|^2 \mathrm{d}\mu.$$

From (5.14), we also have

$$\hat{\rho}_{h-\Pi h} = \int_{\mathbb{R}^d} (h - \Pi h) f_\star \mathrm{d}v = \hat{\rho}_h - \hat{\rho}_{\Pi h} = \hat{\rho}_h - \hat{\rho}_h = 0,$$

which then implies that $\psi_{h-\Pi h} = (-\Delta)^{-1} 0 = 0$. Therefore,

$$\int_{\mathbb{T}^d \times \mathbb{R}^d} |h - \Pi h|^2 \mathrm{d}\mu = \|h - \Pi h\|_{\mathcal{H}}^2.$$

The result follows by arbitrariness of $h \in \mathcal{A}$. □

5.3.3 Macroscopic Coercivity

We first establish a preliminary result together with a weighted Poincaré inequality. This allows the verification of Assumption 5.3.1 (*iii*).

Claim 5.3.9. *Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisfy Assumption 5.2.1. Then, there exist constants $c, C > 0$ such that $c \leq \rho_\star(x) \leq C$ for a.e. $x \in \mathbb{T}^d$.*

Proof. By Assumption 5.2.1, there holds, for a.e. $x \in \mathbb{T}^d$,

$$\rho_\star \leq \|\rho_\star\|_{L^\infty(\mathbb{T}^d)} =: C < +\infty.$$

Moreover, by Claim 5.2.2, there holds $\varphi_\star \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$ for all $1 < p < +\infty$. Choosing $p > \frac{d}{2}$, the continuous embedding $W^{2,p}(\mathbb{T}^d) \hookrightarrow \mathcal{C}(\mathbb{T}^d)$ yields in particular $\varphi_\star \in L^\infty_{\mathrm{zm}}(\mathbb{T}^d)$. Combining this with $u \in L^\infty(\mathbb{T}^d)$, the Euler–Lagrange representation of $\rho_\star$ yields

$$\rho_\star = \boldsymbol{m} \frac{e^{-u-\varphi_\star}}{\int_{\mathbb{T}^d} e^{-u-\varphi_\star} \mathrm{d}y} \geq \boldsymbol{m} \frac{e^{-\|u\|_{L^\infty(\mathbb{T}^d)} - \|\varphi_\star\|_{L^\infty(\mathbb{T}^d)}}}{\int_{\mathbb{T}^d} e^{-u-\varphi_\star} \mathrm{d}y} =: c > 0 \quad \text{a.e.}$$ □

Lemma 5.3.10 (Weighted Poincaré Inequality).
Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisfy Assumption 5.2.1. Then, there exists $C_0 \geq 0$ such that

$$\int_{\mathbb{T}^d} |\nabla h|^2 \rho_\star \mathrm{d}x \geq C_0 \left(\int_{\mathbb{T}^d} |h|^2 \rho_\star \mathrm{d}x - \left(\frac{1}{\boldsymbol{m}} \int_{\mathbb{T}^d} h \rho_\star \mathrm{d}x \right)^2 \right)$$

for all $h \in H^1(\mathbb{T}^d; \rho_\star \mathrm{d}x)$.

Proof. Assume by contradiction that there exists a sequence $\{h_n\}_{n\in\mathbb{N}} \subset H^1_{\mathrm{zm}}(\mathbb{T}^d;\rho_\star \mathrm{d}x)$ such that

$$\int_{\mathbb{T}^d} |\nabla h_n|^2 \rho_\star \mathrm{d}x \to 0, \qquad \text{and} \qquad \int_{\mathbb{T}^d} h_n^2 \rho_\star \mathrm{d}x = 1 \quad \text{for all } n \in \mathbb{N}. \tag{5.15}$$

Step 1. Compactness Arguments. Since $c \le \rho_\star \le C$ a.e. by Claim 5.3.9, the norms $\|\cdot\|_{L^2(\mathbb{T}^d)}$ and $\|\cdot\|_{L^2(\mathbb{T}^d;\rho_\star \mathrm{d}x)}$ are equivalent. In particular,

$$\int_{\mathbb{T}^d} |\nabla h_n|^2 \mathrm{d}x \le \frac{1}{c} \int_{\mathbb{T}^d} |\nabla h_n|^2 \rho_\star \mathrm{d}x \to 0.$$

Hence, $\nabla h_n \to 0$ strongly in $L^2(\mathbb{T}^d)^d$. This implies that the sequence ∇h_n is bounded in $L^2(\mathbb{T}^d)^d$. Define next

$$\bar{h}_n := \frac{1}{|\mathbb{T}^d|} \int_{\mathbb{T}^d} h_n \mathrm{d}x, \qquad \tilde{h}_n := h_n - \bar{h}_n.$$

The unweighted average $\bar{h}_n$ is well-defined and uniformly bounded for all $n \in \mathbb{N}$. Indeed, Hölder's inequality, together with (5.15), yields

$$|\bar{h}_n| \le \frac{1}{|\mathbb{T}^d|} \int_{\mathbb{T}^d} |h_n| \mathrm{d}x \le \frac{1}{\sqrt{|\mathbb{T}^d|}} \left(\int_{\mathbb{T}^d} |h_n|^2 \mathrm{d}x \right)^{\frac{1}{2}} \le \frac{1}{\sqrt{|\mathbb{T}^d|}} \left(\frac{1}{c} \int_{\mathbb{T}^d} |h_n|^2 \rho_\star \mathrm{d}x \right)^{\frac{1}{2}} = \sqrt{\frac{1}{c|\mathbb{T}^d|}}.$$

By construction, $\tilde{h}_n \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$ with $\nabla \tilde{h}_n = \nabla h_n$ for all $n \in \mathbb{N}$. Since $\nabla \tilde{h}_n$ is bounded in $L^2(\mathbb{T}^d)$, the standard Poincaré inequality (see Lemma A.2.6) implies that $\{\tilde{h}_n\}$ is bounded in $H^1_{\mathrm{zm}}(\mathbb{T}^d)$. Moreover, since $\{\bar{h}_n\}$ is bounded in $\mathbb{R}$, the sequence $h_n = \tilde{h}_n + \bar{h}_n$ is bounded in $H^1(\mathbb{T}^d)$. Since the the latter space is reflexive, there exist a subsequence (not relabelled) and $h \in H^1(\mathbb{T}^d)$ such that [15, Thm 3.18]

$$h_n \rightharpoonup h \quad \text{weakly in } H^1(\mathbb{T}^d).$$

By the Rellich–Kondrachov theorem A.1.5, up to a further subsequence,

$$h_n \to h \quad \text{strongly in } L^2(\mathbb{T}^d).$$

Finally, since $\nabla h_n \to 0$ strongly in $L^2(\mathbb{T}^d)^d$, uniqueness of the weak limits implies that $\nabla h = 0$, that is, h is constant a.e.

Step 2. Contradiction Argument. Using $c \le \rho_\star \le C$ again,

$$1 = \int_{\mathbb{T}^d} h_n^2 \rho_\star \mathrm{d}x \to \int_{\mathbb{T}^d} h^2 \rho_\star \mathrm{d}x,$$

where the convergence follows from strong $L^2(\mathbb{T}^d)$ convergence together with boundedness of $\rho_\star \in L^\infty(\mathbb{T}^d)$. Since h is constant and $\int_{\mathbb{T}^d} \rho_\star \mathrm{d}x = \boldsymbol{m}$, we deduce that

$$h = \frac{1}{\sqrt{\boldsymbol{m}}} \quad \text{a.e.}$$

Finally, using that $h_n \in H^1_{\mathrm{zm}}(\mathbb{T}^d;\rho_\star \mathrm{d}x)$,

$$0 = \int_{\mathbb{T}^d} h_n \rho_\star \mathrm{d}x \to \int_{\mathbb{T}^d} h \rho_\star \mathrm{d}x = \sqrt{\boldsymbol{m}},$$

which is a contradiction. Therefore, no such sequence exists, and the weighted Poincaré inequality follows. □

Remark 5.3.11. *The same argument yields a normalised version of the weighted Poincaré inequality. More precisely, there exists $\tilde{C} \geq 0$ such that*

$$\int_{\mathbb{T}^d} |\nabla h|^2 \frac{\rho_\star}{\boldsymbol{m}} \mathrm{d}x \geq \tilde{C} \left(\int_{\mathbb{T}^d} |h|^2 \frac{\rho_\star}{\boldsymbol{m}} \mathrm{d}x - \left(\int_{\mathbb{T}^d} h \frac{\rho_\star}{\boldsymbol{m}} \mathrm{d}x \right)^2 \right)$$

holds for all $h \in H^1(\mathbb{T}^d; \boldsymbol{m}^{-1}\rho_\star \mathrm{d}x)$.

Lemma 5.3.12. *Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisfy Assumption 5.2.1. Then, there exists $C_{\rho_\star} \geq 0$ such that, for any $h \in \mathcal{A}$,*

$$\int_{\mathbb{T}^d} |\nabla_x(\Pi h) + \nabla_x \psi_h|^2 \rho_\star \mathrm{d}x \geq C_{\rho_\star} \left(\int_{\mathbb{T}^d} |\Pi h + \psi_h|^2 \rho_\star \mathrm{d}x - \frac{1}{\boldsymbol{m}} \left(\int_{\mathbb{T}^d} \psi_h \rho_\star \mathrm{d}x \right)^2 \right).$$

Proof. Fix $h \in \mathcal{A}$. We first establish that $\Pi h, \psi_h \in H^1(\mathbb{T}^d)$. We conclude by applying the weighted Poincaré inequality.

Step 1. $\Pi h, \psi_h \in H^1(\mathbb{T}^d)$. Since $h \in \mathcal{A} \subset \mathcal{H}$, Claim 5.2.7 yields $\psi_h \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$. We next consider Πh. Fix $1 \leq i \leq d$. Since $h \in \mathcal{C}^1_c(\mathbb{T}^d \times \mathbb{R}^d)$, we have

$$\left| \frac{\partial h}{\partial x_i}(x,v) \mathrm{M}(v) \right| \leq \mathrm{M}(v) \left\| \frac{\partial h}{\partial x_i} \right\|_{L^\infty(\mathbb{T}^d \times \mathbb{R}^d)} \qquad \text{for a.e. } (x,v) \in \mathbb{T}^d \times \mathbb{R}^d.$$

The right-hand side is integrable in $v \in \mathbb{R}^d$ and independent of $x \in \mathbb{T}^d$. Hence, the dominated convergence theorem [10, Thm 2.8.1] yields

$$\frac{\partial(\Pi h)}{\partial x_i}(x) = \int_{\mathbb{R}^d} \frac{\partial h}{\partial x_i}(x,v) \mathrm{M}(v) \mathrm{d}v, \qquad \text{for all } x \in \mathbb{T}^d.$$

By arbitrariness of $1 \leq i \leq d$, and since $h \in \mathcal{C}^1_c(\mathbb{T}^d \times \mathbb{R}^d)$, it follows that $\Pi h \in \mathcal{C}^1(\mathbb{T}^d)$. In particular, $\Pi h \in H^1(\mathbb{T}^d)$.

Step 2. Poincaré's Inequality. For notational convenience, we let $\tilde{h} := \Pi h + \psi_h$, where $\tilde{h} \in H^1(\mathbb{T}^d)$. Using that $0 < c \leq \rho_\star \leq C$ a.e. (see Claim 5.3.9), we estimate

$$\int_{\mathbb{T}^d} (\tilde{h}^2 + |\nabla_x \tilde{h}|^2) \frac{\rho_\star}{\boldsymbol{m}} \mathrm{d}x \leq \frac{C}{\boldsymbol{m}} \int_{\mathbb{T}^d} (\tilde{h}^2 + |\nabla_x \tilde{h}|^2) \mathrm{d}x = \frac{C}{\boldsymbol{m}} \|\tilde{h}\|^2_{H^1(\mathbb{T}^d)} < +\infty.$$

Hence, $\tilde{h} \in H^1(\mathbb{T}^d; \boldsymbol{m}^{-1}\rho_\star \mathrm{d}x)$. Applying the weighted Poincaré inequality with the probability measure $\boldsymbol{m}^{-1}\rho_\star \mathrm{d}x$, there exists $C_0 \geq 0$ such that

$$\int_{\mathbb{T}^d} |\nabla_x \tilde{h}|^2 \frac{\rho_\star}{\boldsymbol{m}} \mathrm{d}x \geq C_0 \left(\int_{\mathbb{T}^d} \tilde{h}^2 \frac{\rho_\star}{\boldsymbol{m}} \mathrm{d}x - \left(\int_{\mathbb{T}^d} \tilde{h} \frac{\rho_\star}{\boldsymbol{m}} \mathrm{d}x \right)^2 \right).$$

Since $h \in L^2_{\mathrm{zm}}(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)$, Fubini's theorem yields

$$\int_{\mathbb{T}^d} \Pi h \rho_\star \mathrm{d}x = \int_{\mathbb{T}^d \times \mathbb{R}^d} h \rho_\star \mathrm{M}(v) \mathrm{d}x \mathrm{d}v = \int_{\mathbb{T}^d \times \mathbb{R}^d} h \mathrm{d}\mu = 0,$$

and hence

$$\int_{\mathbb{T}^d} \tilde{h} \rho_\star \mathrm{d}x = \int_{\mathbb{T}^d} \psi_h \rho_\star \mathrm{d}x.$$

Substituting this identity in the preceding inequality, and multiplying both sides by $\boldsymbol{m}$ yields

$$\int_{\mathbb{T}^d} |\nabla_x \tilde{h}|^2 \rho_\star \mathrm{d}x \geq C_0 \boldsymbol{m} \left(\int_{\mathbb{T}^d} \tilde{h}^2 \rho_\star \mathrm{d}x - \frac{1}{\boldsymbol{m}} \left(\int_{\mathbb{T}^d} \psi_h \rho_\star \mathrm{d}x \right)^2 \right).$$

By arbitrariness of $h \in \mathcal{A}$, the result follows with $C_{\rho_\star} := C_0 \boldsymbol{m}$. □

Lemma 5.3.13. *Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisfy Assumption 5.2.1. Then, there exists $\lambda_M \geq 0$ such that, for all $h \in \mathcal{A}$,*

$$\|\mathrm{T}\Pi h\|_{\mathcal{H}}^2 \geq \lambda_M \|\Pi h\|_{\mathcal{H}}^2.$$

Proof. Let $h \in \mathcal{A}$ be arbitrary. Since Πh is independent of v, it follows that $\nabla_v(\Pi h) = 0$. Moreover, using (5.14), there holds $\psi_{\Pi h} = \psi_h$. Hence, we have

$$\mathrm{T}\Pi h = v \cdot (\nabla_x(\Pi h) + \nabla_x \psi_h). \tag{5.16}$$

We next compute

$$\hat{\rho}_{\mathrm{T}\Pi h} = \int_{\mathbb{R}^d} \mathrm{T}\Pi h \rho_\star \mathrm{M}(v)\mathrm{d}v = \rho_\star(\nabla_x(\Pi h) + \nabla_x\psi_h) \cdot \left(\int_{\mathbb{R}^d} v\mathrm{M}(v)\mathrm{d}v \right) = 0,$$

which implies that $\psi_{\mathrm{T}\Pi h} = 0$. Therefore,

$$\|\mathrm{T}\Pi h\|_{\mathcal{H}}^2 = \int_{\mathbb{T}^d \times \mathbb{R}^d} |v \cdot (\nabla_x(\Pi h) + \nabla_x \psi_h)|^2 \mathrm{d}\mu.$$

Using the identity

$$\int_{\mathbb{R}^d} (v \cdot \mathbf{e})^2 \mathrm{M}(v)\mathrm{d}v = 1 \qquad \text{for all } \mathbf{e} \in \mathbb{S}^{d-1},$$

together with Fubini's theorem, yields

$$\|\mathrm{T}\Pi h\|_{\mathcal{H}}^2 = \int_{\mathbb{T}^d} |\nabla_x(\Pi h) + \nabla_x \psi_h|^2 \rho_\star \mathrm{d}x.$$

Combining this with Lemma 5.3.12, it follows that

$$\|\mathrm{T}\Pi h\|_{\mathcal{H}}^2 \geq C_{\rho_\star} \left(\int_{\mathbb{T}^d} (\Pi h + \psi_h)^2 \rho_\star \mathrm{d}x - \frac{1}{\boldsymbol{m}} \left(\int_{\mathbb{T}^d} \psi_h \rho_\star \mathrm{d}x \right)^2 \right). \tag{5.17}$$

Moreover, the Cauchy–Schwarz inequality yields

$$\frac{1}{\boldsymbol{m}} \left(\int_{\mathbb{T}^d} \psi_h \rho_\star \mathrm{d}x \right)^2 \leq \frac{1}{\boldsymbol{m}} \left(\int_{\mathbb{T}^d} \psi_h^2 \rho_\star \mathrm{d}x \right) \left(\int_{\mathbb{T}^d} \rho_\star \mathrm{d}x \right) = \int_{\mathbb{T}^d} \psi_h^2 \rho_\star \mathrm{d}x.$$

Consequently, expanding the square in (5.17), we obtain

$$\|\mathrm{T}\Pi h\|_{\mathcal{H}}^2 \geq C_{\rho_\star} \left(\int_{\mathbb{T}^d} (\Pi h)^2 \rho_\star \mathrm{d}x + 2 \int_{\mathbb{T}^d} \Pi h \psi_h \rho_\star \mathrm{d}x \right).$$

By Fubini's theorem, we write

$$\int_{\mathbb{T}^d} (\Pi h)^2 \rho_\star \mathrm{d}x = \int_{\mathbb{T}^d \times \mathbb{R}^d} (\Pi h)^2 \rho_\star \mathrm{M}(v)\mathrm{d}x\mathrm{d}v = \int_{\mathbb{T}^d \times \mathbb{R}^d} (\Pi h)^2 \mathrm{d}\mu = \|\Pi h\|_{L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)}^2.$$

For the second term, the identities $\rho_\star \Pi h = \hat{\rho}_h$ (see (5.13)) and $\hat{\rho}_h = \hat{\rho}_{\Pi h}$ yield

$$2 \int_{\mathbb{T}^d} \Pi h \psi_h \rho_\star \mathrm{d}x = 2 \int_{\mathbb{T}^d} \hat{\rho}_h \psi_h \mathrm{d}x = 2 \int_{\mathbb{T}^d} |\nabla_x \psi_h|^2 \mathrm{d}x \geq \|\nabla_x \psi_h\|_{L^2(\mathbb{T}^d)}^2 = \|\nabla_x \psi_{\Pi h}\|_{L^2(\mathbb{T}^d)}^2.$$

Collecting the above estimates, we conclude that

$$\|\mathrm{T}\Pi h\|_{\mathcal{H}}^2 \geq C_{\rho_\star} \left(\|\Pi h\|_{L^2(\mathbb{T}^d \times \mathbb{R}^d; \mathrm{d}\mu)}^2 + \|\nabla_x \psi_{\Pi h}\|_{L^2(\mathbb{T}^d)}^2 \right) = C_{\rho_\star} \|\Pi h\|_{\mathcal{H}}^2.$$

Therefore, the result follows with $\lambda_M = C_{\rho_\star}$. □

5.3.4 Parabolic Macroscopic Dynamics

Lemma 5.3.14. *Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisfy Assumption 5.2.1. For all $h \in \mathcal{A}$, there holds $\Pi\mathrm{T}\Pi\Pi h = 0$.*

Proof. From (5.16), we compute

$$\Pi\mathrm{T}\Pi\Pi h = (\nabla_x(\Pi h) + \nabla_x \psi_h) \cdot \int_{\mathbb{R}^d} v\mathrm{M}(v)\mathrm{d}v = 0, \qquad \text{for all } h \in \mathcal{A}. \qquad \square$$

5.3.5 Bounded Auxiliary Operators

We define the auxiliary operator $\mathrm{A} : \mathcal{D}(\mathrm{A}) \to \mathcal{H}$ by

$$\mathrm{A}h = (\mathrm{Id}_{\mathcal{H}} + (\mathrm{T}\Pi)^* \mathrm{T}\Pi)^{-1} (\mathrm{T}\Pi)^* h.$$

By Lemma B.1.5, A is densely defined and bounded on $\mathcal{H}$, with $\mathcal{D}(\mathrm{A}) \supset \mathcal{D}(\mathrm{T})$ and $\mathcal{R}(\mathrm{A}) \subset \ker(\mathrm{L})$. In what follows, we impose additional regularity assumptions on the external potential u so that Assumption 5.3.1 (v) is satisfied.

Lemma 5.3.15. *Let $u \in H^1 \cap L^\infty(\mathbb{T}^d)$ satisfy Assumption 5.2.1. For any $h \in \mathcal{A}$, there holds*

$$\|\mathrm{AL}h\|_{\mathcal{H}} \leq \frac{1}{2} \|(\mathrm{Id}_{\mathcal{H}} - \Pi)h\|_{\mathcal{H}}.$$

Proof. We first prove that $\mathrm{AL} = -\mathrm{A}$ on $\mathcal{A}$, and then establish the desired inequality.

Step 1. $\mathrm{AL} = -\mathrm{A}$**.** Let $h \in \mathcal{A}$ be arbitrary. We compute

$$\Pi\mathrm{T}h = \int_{\mathbb{R}^d} \big(v \cdot \nabla_x h - (\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v h + v \cdot \nabla_x \psi_h\big)\mathrm{M}(v)\mathrm{d}v$$

The last term vanishes, since ψ_h is independent of v and

$$\int_{\mathbb{R}^d} v\mathrm{M}(v)\mathrm{d}v = 0.$$

For the first addendum, since $h \in \mathcal{C}_c^\infty(\mathbb{T}^d \times \mathbb{R}^d)$, we can exchange the gradient with the integral sign and write

$$\int_{\mathbb{R}^d} v \cdot \nabla_x h \mathrm{M}(v)\mathrm{d}v = \nabla_x \cdot j_h,$$

where

$$j_h := \int_{\mathbb{R}^d} v h f_\star \mathrm{d}v.$$

For the second term, integration by parts yields

$$\int_{\mathbb{R}^d} (\nabla_x u + \nabla_x \varphi_\star) \cdot \nabla_v h \mathrm{M}(v)\mathrm{d}v = -(\nabla_x u + \nabla_x \varphi_\star) \cdot j_h.$$

Combining the above calculations, we obtain

$$\Pi\mathrm{T}h = \nabla_x \cdot j_h - (\nabla_x u + \nabla_x \varphi_\star) \cdot j_h. \tag{5.18}$$

Moreover, there holds $j_{\mathrm{L}h} = -j_h$. Indeed, we compute

$$j_{\mathrm{L}h} = \int_{\mathbb{R}^d} v(\mathrm{L}h)\mathrm{M}(v)\mathrm{d}v = \int_{\mathbb{R}^d} v\Delta_v h \mathrm{M}(v)\mathrm{d}v - \int_{\mathbb{R}^d} v(v \cdot \nabla_v h)\mathrm{M}(v)\mathrm{d}v.$$

For the first term, integration by parts yields, componentwise,

$$\begin{aligned}\int_{\mathbb{R}^d} v_i \Delta_v h \mathrm{M}(v)\mathrm{d}v &= \int_{\mathbb{R}^d} \Delta_v h v_i \mathrm{M}(v)\mathrm{d}v \\ &= \int_{\mathbb{R}^d} h \Delta_v (v_i \mathrm{M}(v))\mathrm{d}v \\ &= \int_{\mathbb{R}^d} h \left(\frac{\partial \mathrm{M}}{\partial v_i} - \frac{\partial \mathrm{M}}{\partial v_i} - v_i \Delta_v \mathrm{M} \right) \mathrm{d}v \\ &= -\int_{\mathbb{R}^d} h v_i \Delta_v \mathrm{M} \mathrm{d}v.\end{aligned}$$

Similarly, for the second term we obtain

$$-\int_{\mathbb{R}^d} (v \cdot \nabla_v h) v_i \mathrm{M}(v)\mathrm{d}v = \int_{\mathbb{R}^d} h \left(v_i \Delta_v \mathrm{M} + \frac{\partial \mathrm{M}}{\partial v_i} \right) \mathrm{d}v.$$

Substituting the above identities yields

$$(j_{\mathrm{L}h})_i = \int_{\mathbb{R}^d} h \frac{\partial \mathrm{M}}{\partial v_i} \mathrm{d}v = -\int_{\mathbb{R}^d} v_i h \mathrm{M}(v)\mathrm{d}v = -(j_h)_i, \qquad 1 \le i \le d.$$

This implies that $j_{\mathrm{L}h} = -j_h$ as claimed. Combining this with the identity (5.18), we therefore obtain

$$\Pi \mathrm{T} \mathrm{L} h = -\Pi \mathrm{T} h. \tag{5.19}$$

We next let $f \in \mathcal{A}$ be arbitrary. Then, by definition of A, $\mathrm{A}f = g$ if and only if

$$g + (\mathrm{T}\Pi)^*(\mathrm{T}\Pi)g = (\mathrm{T}\Pi)^* f = -\Pi \mathrm{T} f, \tag{5.20}$$

where the last identity uses that Π is self-adjoint and T is antisymmetric (see Lemmas 5.3.6 and 5.3.5, respectively). Combining this with (5.19) yields $\mathrm{AL}h = -\mathrm{A}h$ as wished.

Step 2. Conclusion. Let $h \in \mathcal{A}$ be arbitrary, and set $g := \mathrm{A}h$. Then, arguing as in (5.20) yields

$$(g, (\mathrm{T}\Pi)^* h)_{\mathcal{H}} = (g, g + (\mathrm{T}\Pi)^*(\mathrm{T}\Pi)g)_{\mathcal{H}} = \|g\|_{\mathcal{H}}^2 + \|\mathrm{T}\Pi g\|_{\mathcal{H}}^2.$$

Since $\mathcal{R}(\mathrm{A}) \subset \ker(\mathrm{L})$ and Π is the projection operator onto $\ker(\mathrm{L})$, it follows that $\Pi\mathrm{A} = \mathrm{A}$, and thus

$$\Pi g = \Pi \mathrm{A} h = \mathrm{A} h = g.$$

Hence, the previous identity yields

$$(g, (\mathrm{T}\Pi)^* h)_{\mathcal{H}} = \|g\|_{\mathcal{H}}^2 + \|\mathrm{T}g\|_{\mathcal{H}}^2 = \|\mathrm{A}h\|_{\mathcal{H}}^2 + \|\mathrm{TA}h\|_{\mathcal{H}}^2. \tag{5.21}$$

Moreover, since $\Pi\mathrm{T}\Pi = 0$ (see Lemma 5.3.14) and Π is self-adjoint, we compute

$$(\mathrm{T}\Pi \mathrm{A} h, \Pi h)_{\mathcal{H}} = (\Pi \mathrm{T} \Pi \mathrm{A} h, h)_{\mathcal{H}} = 0.$$

Therefore,

$$(g, (\mathrm{T}\Pi)^* h)_{\mathcal{H}} = (\mathrm{T}\Pi g, h)_{\mathcal{H}} = (\mathrm{T}\Pi \mathrm{A} h, h)_{\mathcal{H}} = (\mathrm{TA}h, (\mathrm{Id}_{\mathcal{H}} - \Pi)h)_{\mathcal{H}}.$$

Combining this with (5.21), the Cauchy–Schwarz inequality yields

$$\|\mathrm{A}h\|_{\mathcal{H}}^2 + \|\mathrm{TA}h\|_{\mathcal{H}}^2 = (\mathrm{TA}h, (\mathrm{Id}_{\mathcal{H}} - \Pi)h)_{\mathcal{H}} \le \|\mathrm{TA}h\|_{\mathcal{H}} \|(\mathrm{Id}_{\mathcal{H}} - \Pi)h\|_{\mathcal{H}} \le \frac{1}{2\lambda}\|\mathrm{TA}h\|_{\mathcal{H}}^2 + \frac{\lambda}{2}\|(\mathrm{Id}_{\mathcal{H}} - \Pi)h\|_{\mathcal{H}}^2,$$

for all $\lambda > 0$. Letting $\lambda = \frac{1}{2}$ yields

$$\|\mathrm{A}h\|_{\mathcal{H}}^2 \le \frac{1}{4}\|(\mathrm{Id}_{\mathcal{H}} - \Pi)h\|_{\mathcal{H}}^2.$$

Since $\mathrm{AL}h = -\mathrm{A}h$, the desired inequality follows. □

We conclude this subsection by establishing the boundedness of the operator $\mathrm{AT}(\mathrm{Id}_{\mathcal{H}} - \Pi)$. More precisely, we aim to prove that

$$\|\mathrm{AT}(\mathrm{Id}_{\mathcal{H}} - \Pi)h\|_{\mathcal{H}} \leq \lambda \|(\mathrm{Id}_{\mathcal{H}} - \Pi)h\|_{\mathcal{H}} \qquad \text{for all } h \in \mathcal{A}.$$

This property follows from additional regularity assumptions on the external potential u. Although these regularity requirements can be weakened, doing so introduces further technical complications; we refer the reader to [2] for a detailed discussion.

The proof of this property relies on several intermediate results, which we establish first.

Claim 5.3.16. *Let $u \in W^{2,\infty}(\mathbb{T}^d)$ satisfy Assumption 5.2.1. Then, the corresponding free energy stationary spatial density and self-consistent potential satisfy*

$$\rho_\star \in W^{2,\infty} \cap \mathcal{C}^1(\mathbb{T}^d) \qquad \text{and} \qquad \varphi_\star \in W^{4,p}_{\mathrm{zm}} \cap \mathcal{C}^3(\mathbb{T}^d)$$

for all $1 \leq p < +\infty$.

Proof. Since u satisfies Assumption 5.2.1, Claim 5.2.2 yields $\varphi_\star \in W^{2,p}_{\mathrm{zm}}(\mathbb{T}^d)$ for all $1 \leq p < +\infty$. Therefore, and since $u \in W^{2,\infty}(\mathbb{T}^d)$, Proposition 3.1.5 yields

$$\rho_\star \in W^{2,p} \cap \mathcal{C}^1(\mathbb{T}^d) \qquad \text{and} \qquad \varphi_\star \in W^{4,p}_{\mathrm{zm}} \cap \mathcal{C}^3(\mathbb{T}^d)$$

for all $1 \leq p < +\infty$. Next, by Claim 5.2.2 there holds

$$\frac{\partial \rho_\star}{\partial x_i} = -\left(\frac{\partial u}{\partial x_i} + \frac{\partial \varphi_\star}{\partial x_i}\right)\rho_\star, \quad \text{for all } 1 \leq i \leq d.$$

Since

$$\rho_\star\,,\ \frac{\partial u}{\partial x_i}\,,\ \frac{\partial \varphi_\star}{\partial x_i} \in W^{1,\infty}(\mathbb{T}^d) \quad \text{for all } 1 \leq i \leq d,$$

the Sobolev product rule (see, e.g., Proposition 9.4 in [15]) yields

$$\frac{\partial \rho_\star}{\partial x_i} \in W^{1,\infty}(\mathbb{T}^d) \quad \text{for all } 1 \leq i \leq d.$$

We therefore conclude that $\rho_\star \in W^{2,\infty}(\mathbb{T}^d)$. □

In the following, we set $w_\star := u + \varphi_\star$. Whenever $u \in W^{2,\infty}(\mathbb{T}^d)$ satisfies Assumption 5.2.1, Claim 5.3.16 yields $\varphi_\star \in W^{2,\infty}(\mathbb{T}^d)$, and therefore we obtain $w_\star \in W^{2,\infty}(\mathbb{T}^d)$. This observation will be used repeatedly throughout the sequel. We next establish an additional weighted Poincaré inequality.

Lemma 5.3.17 (Poincaré Inequality with Total Field Weight).
Let $u \in W^{2,\infty}(\mathbb{T}^d)$ satisfy Assumption 5.2.1. Then, there exist a constant $C_1 > 0$ such that

$$\int_{\mathbb{T}^d} |\nabla_x h|^2 \rho_\star \mathrm{d}x \geq C_1 \int_{\mathbb{T}^d} |h|^2 |\nabla_x w_\star|^2 \rho_\star \mathrm{d}x$$

for all $h \in H^1_{\mathrm{zm}}(\mathbb{T}^d; \rho_\star \mathrm{d}x)$.

Proof. Let $h \in H^1_{\mathrm{zm}}(\mathbb{T}^d; \rho_\star \mathrm{d}x)$ be arbitrary. Using that $w_\star \in W^{2,\infty}(\mathbb{T}^d)$, together with the weighted Poincaré inequality (see Lemma 5.3.10), yields

$$\begin{aligned}\int_{\mathbb{T}^d} |h|^2 |\nabla_x w_\star|^2 \rho_\star \mathrm{d}x &\leq \|\nabla_x w_\star\|_{L^\infty(\mathbb{T}^d)^d} \int_{\mathbb{T}^d} |h|^2 \rho_\star \mathrm{d}x \\ &\leq C_0 \|\nabla_x w_\star\|_{L^\infty(\mathbb{T}^d)^d} \int_{\mathbb{T}^d} |\nabla h|^2 \rho_\star \mathrm{d}x.\end{aligned}$$

By arbitrariness of h, the inequality follows with $C_1 := C_0 \|\nabla_x w_\star\|_{L^\infty(\mathbb{T}^d)^d}$. □

In the sequel, we let $\mathrm{Hess}(\cdot)$ denote the Hessian matrix and Id_d denote the $d \times d$ identity matrix.

Lemma 5.3.18. *Let $u \in W^{2,\infty}(\mathbb{T}^d)$ satisfy Assumption 5.2.1. Then, there exist a constant $C_2 > 0$ such that*

$$\int_{\mathbb{T}^d} \nabla h^\top \left(\mathrm{Hess}(\rho_\star) - \frac{1}{2}\Delta\rho_\star \mathrm{Id}_d \right) \nabla h \mathrm{d}x \leq C_2 \int_{\mathbb{T}^d} |\nabla h|^2 \rho_\star \mathrm{d}x$$

for all $h \in H^1(\mathbb{T}^d)$.

Proof. By Claim 5.3.16, there holds $\rho_\star \in W^{2,\infty}(\mathbb{T}^d)$. Hence, $\mathrm{Hess}(\rho_\star)$ and $\Delta\rho_\star$ are well-defined. In particular, Claim 5.2.2 yields

$$\nabla\rho_\star = -\nabla w_\star \rho_\star \quad \text{a.e.}$$

Since $w_\star \in W^{2,\infty}(\mathbb{T}^d)$ and $\rho_\star \in W^{1,\infty}(\mathbb{T}^d)$, the Sobolev product rule yields

$$\Delta\rho_\star = \nabla \cdot \nabla\rho_\star = (|\nabla w_\star|^2 - \Delta w_\star)\rho_\star \in L^\infty(\mathbb{T}^d).$$

An analogous argument shows that

$$\mathrm{Hess}(\rho_\star) = \left(\nabla w_\star \nabla w_\star^\top - \mathrm{Hess}(w_\star) \right) \rho_\star \in L^\infty(\mathbb{T}^d)^{d\times d}.$$

Set, for $\xi \in \mathbb{R}^d$,

$$P(x)\xi := \left(\mathrm{Hess}(\rho_\star) - \frac{1}{2}\Delta\rho_\star \mathrm{Id}_d \right) \xi.$$

Then, for a.e. $x \in \mathbb{T}^d$,

$$\xi^\top P(x)\xi = \rho_\star\, \xi^\top Q(x)\xi,$$

where

$$Q = \nabla w_\star \nabla w_\star^\top - \mathrm{Hess}(w_\star) - \frac{1}{2}|\nabla w_\star|^2 \mathrm{Id}_d + \frac{1}{2}(\Delta w_\star)\mathrm{Id}_d.$$

We estimate each term using $\|w_\star\|_{W^{2,\infty}(\mathbb{T}^d)} < \infty$:

$$|(\nabla w_\star \cdot \xi)^2| \leq |\nabla w_\star|^2 |\xi|^2,$$

$$|\xi^\top \mathrm{Hess}(w_\star)\xi| \leq \|\mathrm{Hess}(w_\star)\|_{L^\infty}|\xi|^2, \qquad |\Delta w_\star| \leq \sqrt{d}\,\|\mathrm{Hess}(w_\star)\|_{L^\infty}.$$

Therefore,

$$|\xi^\top Q\xi| \leq C_d \|w_\star\|^2_{W^{2,\infty}(\mathbb{T}^d)} |\xi|^2.$$

Multiplying by $\rho_\star$, and letting $\xi = \nabla h$ for an arbitrary $h \in H^1(\mathbb{T}^d)$, yields

$$\left| \nabla h^\top \left(\mathrm{Hess}(\rho_\star) - \frac{1}{2}\Delta\rho_\star \mathrm{Id}_d \right) \nabla h \right| \leq C_d \|w_\star\|^2_{W^{2,\infty}(\mathbb{T}^d)} |\nabla h|^2 \rho_\star.$$

Integrating over $\mathbb{T}^d$ concludes the proof. □

The following result establishes the desired boundedness property of the operator $\mathrm{AT}(\mathrm{Id}_{\mathcal{H}} - \Pi)$ and concludes the previous discussion.

Lemma 5.3.19. *Let $u \in W^{2,\infty}(\mathbb{T}^d)$ satisfy Assumption 5.2.1. Then, there exists $\bar{C} \geq 0$ such that*

$$\|\mathrm{AT}(\mathrm{Id}_{\mathcal{H}} - \Pi)h\|_{\mathcal{H}} \leq \bar{C}\|(\mathrm{Id}_{\mathcal{H}} - \Pi)h\|_{\mathcal{H}}$$

for all $h \in \mathcal{A}$.

The proof of Lemma 5.3.19 follows a sequence of intermediate estimates and computations that are analogous to those in Lemma 21 of [2]. It also relies on Lemmas 5.3.10, 5.3.17, and 5.3.18, which correspond to Lemma 6, Corollary, and Lemma 9 in [2]. Moreover, under the stated assumptions, Lemmas 11 and 12 in [2] remain valid and are used in the argument, while Lemmas 9 and 10 are unnecessary in the present setting due to the boundedness of $\mathbb{T}^d$. Since the derivations closely mirror those already established in the cited work, we omit the details here.

5.3.6 Linear Exponential Stability of Variationally Optimal Stationary States

The preceding analysis provides sufficient conditions ensuring the linear exponential stability of stationary states of the VPFP system. In particular, for any $u \in W^{2,\infty}(\mathbb{T}^d)$ such that $\rho_\star(u) \in L^\infty(\mathbb{T}^d)$, Assumption 5.3.1 is satisfied. Consequently, Theorem 5.3.2 implies the desired stability result.

We conclude this discussion by establishing sufficient conditions under which the external potentials obtained from the inverse confinement problem studied in Chapter 4 induce stationary states that are linearly exponentially stable.

Theorem 5.3.20. *Let Assumption* 4.2.2 *hold with $m \geq 3$ and $r > d$. Then, any minimiser $u_\star$ of J_{reg} on $\mathcal{W}_{\mathrm{reg}}(m, r, s, E_S, \{(S_i, z_i)\}_{i=1}^{n_S})$ yields a linearly exponentially stable stationary state $f_\star(u_\star)(x, v) = \rho_\star(u_\star)(x)\mathrm{M}(v)$ in the sense of Theorem* 5.3.2.

Proof. Since $m \geq 3$ and $r > d$, the Sobolev embedding theorem yields the continuous embeddings

$$W^{m,r}(\mathbb{T}^d) \hookrightarrow \mathcal{C}^2(\mathbb{T}^d) \hookrightarrow W^{2,\infty}(\mathbb{T}^d) \hookrightarrow H^1(\mathbb{T}^d) \cap L^\infty(\mathbb{T}^d).$$

Consequently, Proposition 4.3.2 implies that J_{reg} attains its minimum on $\mathcal{W}_{\mathrm{reg}}(m, r, s, E_S, \{(S_i, z_i)\}_{i=1}^{n_S})$. Moreover, by Corollary 4.3.4, every minimiser $u_\star$ of J_{reg} satisfies

$$\rho_\star(u_\star) \in \mathcal{C}^{m-1}(\mathbb{T}^d).$$

In particular, $\rho_\star(u_\star) \in L^\infty(\mathbb{T}^d)$, and the preceding analysis therefore ensures that Assumption 5.3.1 is satisfied. Theorem 5.3.2 then yields the linear exponential stability of the stationary state associated with any such minimiser $u_\star$. □

5.4 Numerical Experiments

To complement the theoretical hypocoercivity analysis developed in the previous sections, we investigate numerically the decay properties of the linearised VPFP system and the effect of an additional feedback control mechanism. The objective of these experiments is twofold: first, to illustrate the convergence behaviour predicted by the hypocoercivity analysis; and second, to assess whether a suitable control strategy can enhance convergence towards equilibrium.

Throughout this section, we restrict our attention to the one-dimensional setting. The numerical scheme combines spectral discretisation in the spatial variable with finite-difference approximations in velocity, while the self-consistent Poisson field is computed using a Fourier-based solver. We first study the uncontrolled dynamics, both in the deterministic setting and under stochastic perturbations, before introducing the feedback control mechanism and comparing the resulting controlled and uncontrolled evolutions.

5.4.1 Discretisation of the Linearised System and Computation of the Poisson Field

In one spatial dimension, we identify $\mathbb{T}^1 = \mathbb{R}/(2\pi\mathbb{Z})$ and truncate the velocity domain to $[-V_{\max}, V_{\max}]$, where $V_{\max} \gg 1$. The spatial domain is discretised by the uniform grid

$$\{\mathbf{x}_{j_x}\}_{j_x=0}^{N_x-1}, \qquad \mathbf{x}_{j_x} = j_x w_x, \qquad w_x = \frac{2\pi}{N_x},$$

where $N_x \in \mathbb{N}$ denotes the number of spatial grid points, and the indexing is understood modulo N_x to enforce periodic boundary conditions. Similarly, the truncated velocity domain is discretised by the uniform grid

$$\{\mathbf{v}_{j_v}\}_{j_v=0}^{N_v}, \qquad \mathbf{v}_{j_v} = -V_{\max} + j_v w_v, \qquad w_v = \frac{2V_{\max}}{N_v},$$

where $N_v \in \mathbb{N}$ and $N_v + 1$ denotes the number of velocity grid points. Finally, time is discretised as $t_n = nw_t$, where $n \in \mathbb{N}$, and $0 < w_t \ll 1$ denotes the discretisation step. We denote the fully discrete solution by

$$h^n_{j_x\,,\,j_v} \approx h(t_n, \mathbf{x}_{j_x}, \mathbf{v}_{j_v}).$$

Free Energy Solutions. The simulations are based on free energy stationary states of the form

$$f^\ell_\star(x,v) = \rho_\star(u^{\alpha,\ell}_\star)(x)\,\mathrm{M}(v), \qquad 1 \le \ell \le 4,$$

where $u^{\alpha,\ell}_\star$ denotes the solution of the optimal control problem (4.18) with regularisation parameter $\alpha = 0.01$ and reference density ρ_{ref_ℓ}. This is computed using the numerical scheme introduced in Section 4.4. The function $\rho_\star(u^{\alpha,\ell}_\star)(x)$ denotes the associated stationary spatial density. As discussed therein, both $\rho_\star(u^{\alpha,\ell}_\star)$ and the corresponding self-consistent potential $\varphi_\star(u^{\alpha,\ell}_\star)$ are smooth and therefore satisfy the assumptions required for the hypocoercivity analysis in the previous section. Moreover, since the stationary state is normalised to have unit mass, it follows that

$$\int_{\mathbb{T}^1\times\mathbb{R}} \rho_\star(u^{\alpha,\ell}_\star)(x)\mathrm{M}(v)\mathrm{d}x\mathrm{d}v = 1, \qquad 1 \le \ell \le 4.$$

Hence, μ_ℓ is a probability measure for each $1 \le \ell \le 4$.

Weighted Density and Poisson field. The weighted spatial density is computed at each time step $n \in \mathbb{N}$ as

$$\hat\rho_h(t_n, \mathbf{x}_{j_x}) \approx \hat\rho^n_{j_x} := \sum_{j_v=0}^{N_v} h^n_{j_x\,,\,j_v}\, f^\ell_\star(\mathbf{x}_{j_x}, \mathbf{v}_{j_v})\, w_v, \qquad \text{for all } 0 \le j_x \le N_x - 1. \tag{5.22}$$

The Poisson equation $-\Delta_x \psi_h = \hat\rho_h$ is then solved spectrally. Denoting by $\mathcal{F}_x$ the discrete Fourier transform in x, we compute

$$\mathcal{F}_x\left(\psi^n\right)(k) := \frac{\mathcal{F}_x\left(\hat\rho^n\right)(k)}{k^2}, \qquad k \in \mathbb{Z}\setminus\{0\}, \tag{5.23}$$

with $\mathcal{F}_x(\psi^n)(0) = 0$ to enforce zero mean. The self-consistent field is then obtained by spectral differentiation:

$$\frac{\partial \psi_h(t_n, \mathbf{x}_{j_x})}{\partial x} \approx \mathcal{F}_x^{-1}\left(ik\,\mathcal{F}_x\left(\psi^n\right)(k)\right).$$

Discretisation of Transport and Collision operators. The transport and collision operators are evaluated explicitly at time t_n using a mixed spectral and finite difference discretisation. In the spatial variable x, derivatives are computed spectrally. For each fixed time index $n \in \mathbb{N}$ and velocity index $0 \le j_v \le N_v$, we apply a discrete Fourier transform in x and compute

$$\frac{\partial h(t_n, \mathbf{x}_{j_x}, \mathbf{v}_{j_v})}{\partial x} \approx \mathcal{F}_x^{-1}\Big(ik\,\mathcal{F}_x\left(h^n_{j_x\,,\,j_v}\right)(k)\Big), \qquad \text{for all } 0 \le j_x \le N_x - 1.$$

In the velocity variable v, all derivatives are approximated by centred finite differences on the uniform grid, that is,

$$\frac{\partial h(t_n, \mathbf{x}_{j_x}, \mathbf{v}_{j_v})}{\partial v} \approx \frac{h^n_{j_x, j_v+1} - h^n_{j_x, j_v-1}}{2w_v}, \qquad \frac{\partial^2 h(t_n, \mathbf{x}_{j_x}, \mathbf{v}_{j_v})}{\partial v^2} \approx \frac{h^n_{j_x\,,\,j_v+1} - 2h^n_{j_x\,,\,j_v} + h^n_{j_x\,,\,j_v-1}}{w_v^2},$$

with one-sided finite differences at $v = \pm V_{\max}$. The transport and collision operators are then assembled using the above spatial spectral and velocity finite-difference approximations. We denote by $\mathcal{R}$ the discretised transport–collision operator.

Stochastic forcing. To investigate the robustness of the hypocoercive decay properties under external perturbations, we introduce an additive external forcing term driven by a temporally correlated stochastic process. More precisely, the dynamics of the perturbation h are augmented by a forcing term $\eta(t,x,v)$ evolving according to an Ornstein–Uhlenbeck process in time:

$$\begin{aligned}&\frac{\partial h}{\partial t} + \mathrm{T}h = \mathrm{L}h + \eta,\\ &\mathrm{d}\eta(t,x,v) = -\frac{1}{\tau_{\mathrm{noise}}}\eta(t,x,v)\,\mathrm{d}t + \sqrt{\frac{2\sigma^2}{\tau_{\mathrm{noise}}}}\mathrm{d}W_t(x,v),\end{aligned}$$

where $W_t(x,v)$ denotes a cylindrical Wiener process and $\tau_{\mathrm{noise}} > 0$ controls the temporal correlation of the forcing. The parameter $\sigma > 0$ determines the amplitude of the stationary fluctuations of the stochastic process. In particular, the Ornstein–Uhlenbeck dynamics introduce temporal correlations into the external forcing while preserving the zero μ_ℓ-mean constraint imposed on the noise realisations. The stochastic process is constructed such that each realisation remains in the μ_ℓ-zero-mean subspace, thereby preserving compatibility with the functional-analytic framework of the hypocoercivity analysis.

For the numerical experiments, the stochastic forcing term is constructed as follows. Let

$$\{\zeta^n\}_{n\in\mathbb{N}}$$

be a sequence of independent Gaussian random fields on the phase-space grid, with entries distributed according to

$$\zeta^n_{j_x\,,\,j_v} \sim \mathcal{N}(0,1), \qquad 0 \le j_x \le N_x - 1, \quad 0 \le j_v \le N_v, \tag{5.24}$$

independently for each time step $n \in \mathbb{N}$. To enforce the zero μ_ℓ-mean condition, each Gaussian realisation is projected onto the admissible perturbation space by defining

$$\xi^n_{j_x\,,\,j_v} := \zeta^n_{j_x\,,\,j_v} - \sum_{i_x\,,\,i_v} \zeta^n_{i_x\,,\,i_v} f^\ell_\star(\mathbf{x}_{i_x}, \mathbf{v}_{i_v}) w_x w_v, \tag{5.25}$$

which ensures that

$$\sum_{j_x\,,\,j_v} \xi^n_{j_x\,,\,j_v} f^\ell_\star(\mathbf{x}_{j_x}, \mathbf{v}_{j_v}) w_x w_v = 0.$$

The projected Gaussian fields $\xi^{\,n}$ are then used to generate the external coloured forcing through a discrete Ornstein–Uhlenbeck process. Denoting by η^n the value of the stochastic forcing at time t_n, we discretise the evolution of the stochastic process through an Euler approximation scheme with initial condition $\eta^0 = 0$:

$$\eta^{n+1} = \eta^n - \frac{w_t}{\tau_{\mathrm{noise}}}\eta^n + \sigma\sqrt{\frac{2w_t}{\tau_{\mathrm{noise}}}}\xi^{\,n}. \tag{5.26}$$

Time Integration. We employ an explicit Euler method to perform time integration. The update is given by

$$h^{n+1}_{j_x\,,\,j_v} = h^n_{j_x\,,\,j_v} + w_t\left(\mathcal{R}(h^n_{j_x\,,\,j_v}) + \eta^n_{j_x\,,\,j_v}\right). \tag{5.27}$$

The explicit Euler discretisation is conditionally stable for sufficiently small time steps. In the present simulations, stability is assessed through a refinement study in which the time step is reduced while keeping all other parameters unchanged. Since no visible changes are observed in the resulting decay curves, the numerical results are insensitive to further temporal refinement, indicating that the chosen time step operates within the stable regime.

Initial condition. The initial condition is constructed as a small random perturbation around equilibrium. More precisely, a Gaussian random field is generated on the phase space grid, and then smoothed using a Gaussian kernel to remove high-frequency components. Finally, the perturbation is projected to be mean-free with respect to the invariant measure, that is,

$$\int_{\mathbb{T}^1\times\mathbb{R}} h_0(x,v)\mathrm{d}\mu_\ell = 0.$$

The pseudocode of the numerical procedure is provided in Algorithm 5.4.1.

Algorithm 5.4.1 Time evolution of the linearised VPFP system with stochastic forcing.

Require: Spatial grid N_x, velocity grid N_v, time step w_t, final time T, equilibrium $f_\star^\ell$, correlation parameter τ_{noise}, noise amplitude σ, μ_ℓ-mean-free initial condition h_0.

Initialisation: Set $h^0 \leftarrow h_0$, $\eta^0 \leftarrow 0$.
for $n = 0, \dots, T/w_t - 1$ **do**
    Compute weighted density as in (5.22).
    Solve Poisson Equation with FFT as in (5.23).
    Compute Derivatives using spectral differentiation in x and centred finite differences in v.
    Assemble the discretised transport–collision operator $\mathcal{R}(h^n)$.
    Time update as in (5.27).
    Generate stochastic forcing according to (5.24), (5.25) and (5.26).
end for
return $\{h^n\}_{n=0}^{T/w_t}$

Simulation protocol. Unless stated otherwise, all numerical experiments presented in this section use the same discretisation parameters, namely $N_x = N_v = 128$, $w_t = 8.5 \cdot 10^{-5}$, and final time $T = 8$, together with the same initial perturbation h_0. Whenever stochastic forcing is considered, identical stochastic forcing realisations η^n are used across all corresponding experiments. The truncation parameter $V_{\max} = 10$ is chosen sufficiently large such that the Maxwellian tails are negligible at the velocity boundaries. A grid refinement study demonstrated that the decay characteristics reported below are robust with respect to spatial and velocity discretisation refinement. Consequently, differences in the observed decay behaviour primarily reflect the influence of the underlying equilibrium state or the introduced control mechanism, rather than variations in the initial perturbation or external forcing.

Numerical Results. Figure 5.4.1 illustrates the evolution of the squared weighted $L^2(\mathbb{T}^1 \times \mathbb{R}; \mathrm{d}\mu_\ell)$-norm of the perturbation $h(t)$ for the linearised VPFP system across the four equilibrium configurations $1 \le \ell \le 4$, considering both deterministic dynamics and dynamics under stochastic forcing. In the deterministic setting, the perturbation decays asymptotically towards equilibrium, in agreement with the exponential convergence predicted by the hypocoercivity analysis developed in the preceding sections. When stochastic forcing is introduced, this decay is interrupted by persistent fluctuations, and the system converges towards a statistically stationary regime with non-zero perturbation amplitude. The observed convergence behaviour varies significantly between equilibrium configurations, highlighting the dependence of the hypocoercive decay properties on the underlying invariant measure.

5.4.2 Linear Spectral Feedback Control

The hypocoercivity analysis establishes exponential convergence of the linearised system towards equilibrium. However, the associated decay rate depends strongly on the underlying stationary state $f_\star$ and may become slow for configurations with weakly dissipative modes. To investigate whether this convergence can be accelerated, we introduce an additional dissipative feedback mechanism acting on the perturbation

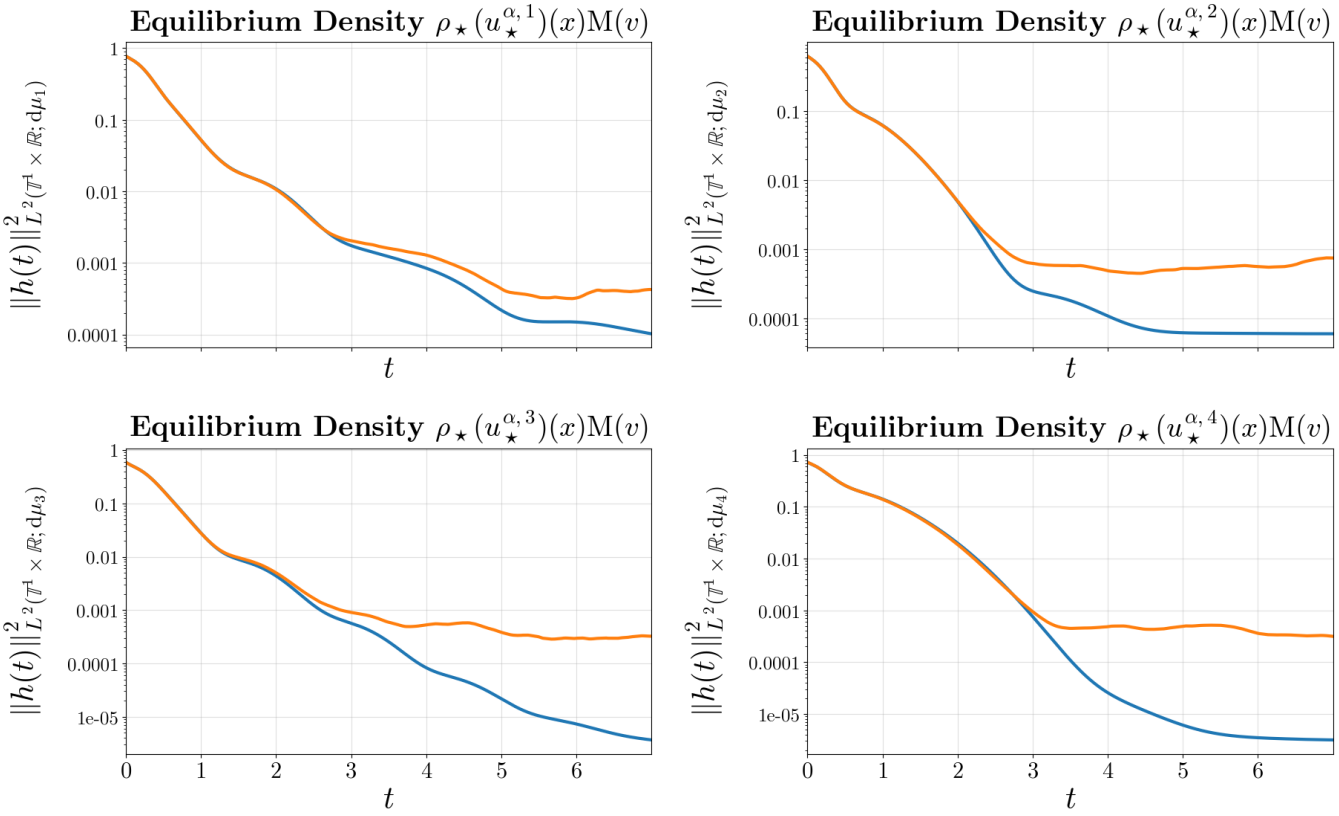


Figure 5.4.1: Evolution of the squared weighted $L^2(\mathbb{T}^1 \times \mathbb{R}; \mathrm{d}\mu_\ell)$-norm of the perturbation $h(t)$ for the linearised VPFP system in the deterministic case and the stochastically forced case across the four equilibrium configurations $1 \leq \ell \leq 4$ (Log-y Scale).

h. The controlled linearised dynamics are obtained by augmenting the transport–collision operator with a linear feedback contribution,

$$\frac{\partial h}{\partial t} + \mathrm{T}h = \mathrm{L}h + \mathcal{C}h,$$

where the control operator $\mathcal{C}$ is designed to introduce additional damping into the perturbation dynamics. We first consider a static linear feedback controller of the form

$$\mathcal{C}_{\mathrm{LFb}}h := -K_{\mathrm{FG}}h, \tag{5.28}$$

where $K_{\mathrm{FG}} > 0$ denotes the feedback gain. This controller corresponds to uniform damping across the whole frequency spectrum and provides a reference case for assessing the effect of feedback on the linearised perturbation dynamics.

Although the full-frequency controller is mathematically natural, it is however not representative of a physically realisable control approach. In practice, a feedback device cannot independently control arbitrarily small phase-space scales. High-frequency Fourier modes correspond to increasingly fine phase-space structures and would require an actuation mechanism with unrealistically high resolution. To account for this limitation, we introduce a spectrally truncated feedback controller by restricting the control action

to the low-frequency spatial modes. More precisely, we consider a spectral feedback controller

$$\mathcal{C}_{\mathrm{SFb}}(h) := -K_{\mathrm{FG}}\mathcal{P}_{\mathrm{low}}(h), \tag{5.29}$$

where $\mathcal{P}_{\mathrm{low}}$ denotes the projection onto the controlled Fourier subspace. The projection is implemented in Fourier space according to

$$\mathcal{F}_x(\mathcal{P}_{\mathrm{low}}(h))(k) = \begin{cases} \mathcal{F}_x(h)(k), & |k| \leq K_{\mathrm{cut}}, \\ 0, & |k| > K_{\mathrm{cut}}, \end{cases}$$

where K_{cut} is the prescribed cut-off frequency. The motivation for restricting the feedback to low frequencies is twofold. First, the slow convergence of hypocoercive systems is typically governed by large-scale structures, which are associated with low Fourier modes. Consequently, damping these modes directly targets the components responsible for the dominant convergence timescale. Second, excluding high-frequency modes avoids introducing artificial actuation at phase-space scales that are already controlled by the intrinsic transport and collision mechanisms. The high-frequency components of the perturbation decay primarily through kinetic dissipation, and additional damping of these modes is neither physically necessary nor practically achievable.

The numerical implementation therefore considers both the idealised full-frequency linear feedback controller (5.28) and the low-frequency spectral feedback controller (5.29). For the latter, the cut-off frequency is selected as

$$K_{\mathrm{cut}} = k_{\mathrm{cutoff}} \cdot k_{\max},$$

where $k_{\max}$ denotes the maximum Fourier frequency, and $k_{\mathrm{cutoff}} \in (0, 1)$ specifies the fraction of spatial modes controlled by the feedback mechanism. The controlled simulations follow the same numerical procedure described in Algorithm 5.4.1, with the only modification being the inclusion of the feedback contribution in the time integration step. The explicit Euler update is replaced by

$$h^{n+1}_{j_x\,,\,j_v} = h^n_{j_x\,,\,j_v} + w_t \left(\mathcal{R}(h^n)_{j_x\,,\,j_v} + \mathcal{C}(h^n)_{j_x\,,\,j_v} + \eta^n_{j_x\,,\,j_v} \right),$$

where $\mathcal{C}$ denotes either the full-frequency feedback operator $\mathcal{C}_{\mathrm{LFb}}$ or the spectrally truncated feedback operator $\mathcal{C}_{\mathrm{SFb}}$.

The influence of the feedback mechanism on the decay properties of the linearised VPFP system is assessed by comparing the evolution of the squared weighted $L^2(\mathbb{T}^1 \times \mathbb{R}; \mathrm{d}\mu_\ell)$-norm for the uncontrolled and controlled dynamics, as shown in Figure 5.4.2. The same feedback gain $K_{\mathrm{FG}} = 0.25$ is used for both the full-frequency and low-frequency spectral feedback controllers. This choice isolates the effect of the spectral truncation, ensuring that differences in the observed convergence behaviour arise primarily from the restriction of the control action to the low-frequency restriction rather than from variations in the feedback intensity. In the deterministic setting, both feedback strategies enhance convergence towards equilibrium by introducing additional dissipation into the perturbation dynamics. The full-frequency feedback controller produces the strongest damping effect, as expected, since it acts across the entire frequency spectrum. However, the spectrally truncated feedback controller achieves a comparable reduction of the transient decay time despite acting only on the low-frequency component of the solution. This indicates that the dominant contribution to the slow hypocoercive convergence is associated with large-scale spatial structures, which are effectively targeted by the low-frequency feedback mechanism. In the presence of stochastic forcing, the uncontrolled system reaches a statistically stationary regime characterised by persistent fluctuations around equilibrium. The introduction of feedback reduces the amplitude of these fluctuations by increasing the effective dissipation of the perturbation modes. This effect is particularly noticeable for equilibrium configurations exhibiting slower intrinsic hypocoercive convergence, such as the case $\ell = 3$, corresponding to the equilibrium configuration with bimodal spatial density.

Overall, the numerical results demonstrate that additional linear feedback can effectively accelerate convergence towards equilibrium and improve robustness against external perturbations. Moreover, the comparable performance of the low-frequency spectral controller highlights that efficient control can be achieved by targeting only the slowly dissipating modes, avoiding unnecessary actuation of high-frequency components that are already strongly damped by the intrinsic transport–collision dynamics.

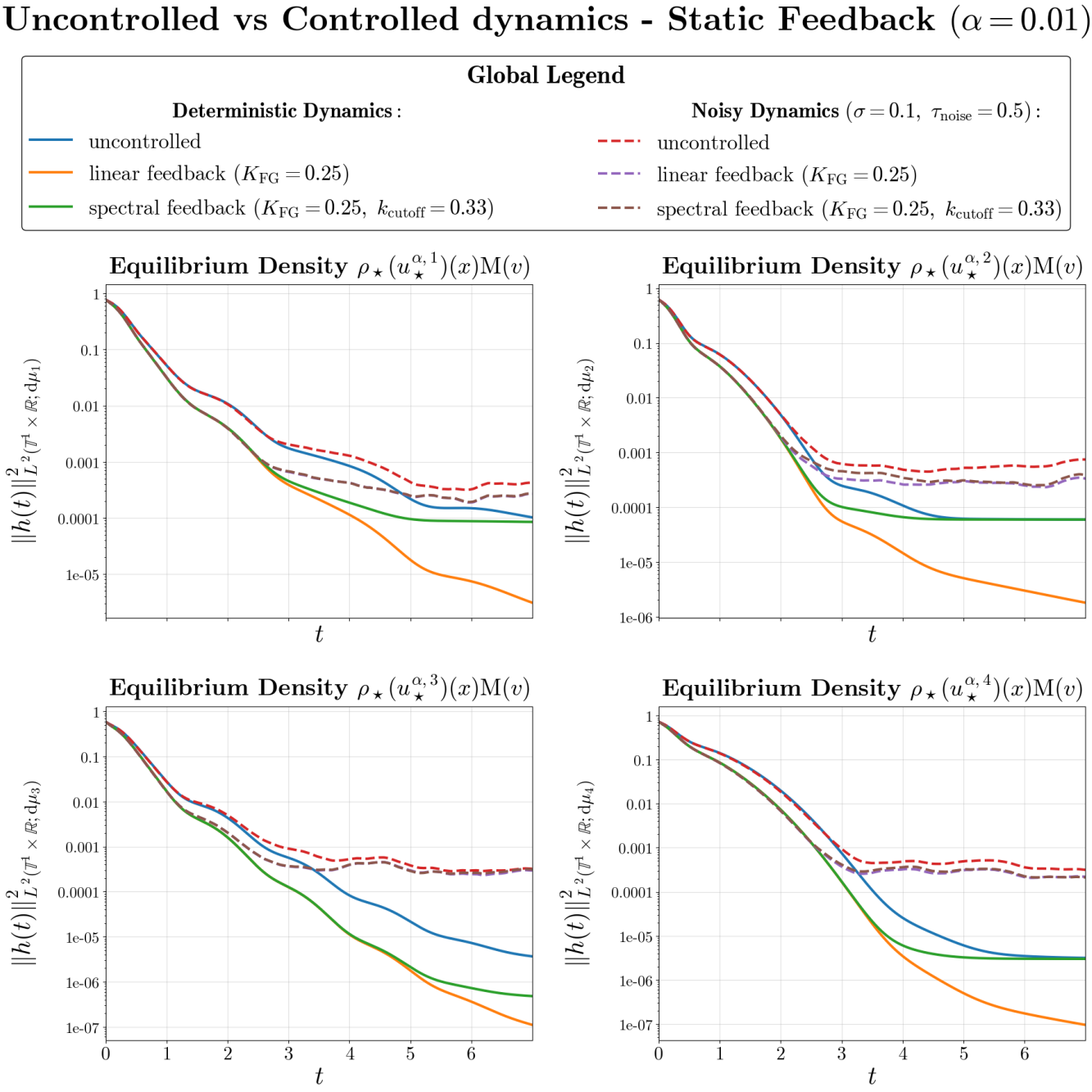


Figure 5.4.2: Evolution of the squared weighted $L^2(\mathbb{T}^1 \times \mathbb{R}; \mathrm{d}\mu_\ell)$-norm of the perturbation $h(t)$ for the linearised VPFP system under uncontrolled dynamics, full-frequency linear feedback, and low-frequency spectral feedback, in both deterministic and stochastically forced settings, across the four equilibrium configurations $1 \leq \ell \leq 4$ (Log-y Scale).

5.4.3 Hybrid LQR Controller

The previous results demonstrate that restricting the feedback action to low-frequency spatial modes is sufficient to achieve a substantial improvement in the hypocoercive convergence rate. A natural next step is to investigate whether an optimal state-feedback controller can further accelerate convergence. Ideally, one would design a linear-quadratic regulator (LQR) for the full semi-discrete system, thereby optimally controlling the entire phase-space dynamics. However, for the spatial and velocity resolutions considered in the numerical experiments, the resulting discretised system is extremely high-dimensional. Computing the full LQR gain requires solving a large-scale algebraic Riccati equation, which is computationally prohibitive in this setting. To overcome this limitation, we introduce a hybrid LQR controller. Rather than applying optimal control to the full state, the LQR feedback is constructed only on a low-dimensional subspace spanned by the least stable modes of the discretised transport–collision operator, while the remaining components are controlled using the low-frequency spectral feedback introduced in the previous subsection. This hybrid strategy preserves the computational tractability of the spectral controller while concentrating the optimal feedback on the modes that are expected to dominate the long-time convergence behaviour.

Let $\tilde{h}(t) \in \mathbb{R}^{N_x(N_v+1)}$ denote the vector obtained by stacking the discrete values

$$h(t, \mathbf{x}_{j_x}, \mathbf{v}_{j_v}) =: h_{j_x,j_v}(t), \qquad 0 \le j_x \le N_x - 1, \quad 0 \le j_v \le N_v,$$

of the perturbation into a single column vector. The discretisation described previously induces a matrix representation

$$A \in \mathbb{R}^{N_x(N_v+1)\times N_x(N_v+1)}$$

of the discretised linearised transport–collision operator $\mathcal{R}$, so that the semi-discrete uncontrolled dynamics can be written as

$$\frac{\mathrm{d}\tilde{h}}{\mathrm{d}t} = A\tilde{h}.$$

Since the modes with real parts closest to zero correspond to the slowest decaying components of the semigroup generated by A, we define the reduced unstable/slow subspace

$$\mathcal{V}_u := \mathrm{span}\{z_1, \dots, z_{n_{\text{LQR modes}}}\},$$

where $0 < n_{\text{LQR modes}} \le N_x(N_v + 1)$ denotes the number of modes controlled by the LQR feedback. The computed eigenvectors are subsequently orthonormalised with respect to the Euclidean inner product, and we denote by

$$V_u = \begin{bmatrix} z_1 & \cdots & z_{n_{\text{LQR modes}}} \end{bmatrix}$$

the resulting matrix whose columns form an orthonormal basis of $\mathcal{V}_u$, satisfying

$$V_u^\top V_u = \mathrm{Id}.$$

The semi-discrete perturbation $\tilde{h}$ is then decomposed as its projection onto the slow subspace and onto its orthogonal complement,

$$\tilde{h} = V_u z + \tilde{h}_\perp, \qquad z = V_u^\top \tilde{h},$$

where $z \in \mathbb{R}^{n_{\text{LQR modes}}}$ contains the modal amplitudes associated with the least stable eigenvectors. Projecting the dynamics onto $\mathcal{V}_u$ yields the reduced-order system

$$\dot{z} = A_u z + B_u u,$$

where

$$A_u = V_u^\top A V_u, \qquad B_u = V_u^\top B V_u.$$

Here, B denotes the low-frequency actuation operator associated with the spectral feedback mechanism,

$$B\tilde{h} = \mathcal{P}_{\text{low}}(\tilde{h}),$$

where $\mathcal{P}_{\mathrm{low}}$ is the Fourier projection introduced in the previous subsection. The optimal feedback law on the reduced-order system is obtained by solving the continuous algebraic Riccati equation

$$A_u^\top P + P A_u - P B_u R_u^{-1} B_u^\top P + Q_u = 0,$$

where P is the positive definite solution of the Riccati equation, and Q_u and R_u are the state and control weighting matrices, respectively. The reduced-order LQR gain is then given by

$$K_u = R_u^{-1} B_u^\top P.$$

The resulting feedback action on the slow subspace is therefore

$$\mathcal{C}_{\mathrm{LQR}}(\tilde{h}) = -V_u K_u V_u^\top \tilde{h}.$$

To avoid neglecting the remaining components of the perturbation, the complementary subspace is controlled using the low-frequency spectral feedback introduced previously. More precisely, the orthogonal component

$$\tilde{h}_\perp = \tilde{h} - V_u V_u^\top \tilde{h}$$

is stabilised through the feedback contribution

$$\mathcal{C}_{\mathrm{SFb}}(\tilde{h}_\perp) = -K_{\mathrm{FG}} \mathcal{P}_{\mathrm{low}}(\tilde{h}_\perp).$$

Combining both contributions gives the hybrid LQR controller

$$\mathcal{C}_{\mathrm{HLQR}}(\tilde{h}) = -V_u K_u V_u^\top \tilde{h} - K_{\mathrm{FG}} \mathcal{P}_{\mathrm{low}} \left(\tilde{h} - V_u V_u^\top \tilde{h} \right).$$

The resulting controlled dynamics are therefore given by

$$\frac{\mathrm{d}\tilde{h}}{\mathrm{d}t} = A\tilde{h} + \mathcal{C}_{\mathrm{HLQR}}(\tilde{h}),$$

with the numerical time integration performed using the same explicit Euler discretisation as in the previous sections.

Figure 5.4.3 compares the evolution of the squared weighted $L^2(\mathbb{T}^1 \times \mathbb{R}; \mathrm{d}\mu_\ell)$-norm of the perturbation $h(t)$ for the uncontrolled linearised VPFP system and for the system controlled by the hybrid LQR feedback mechanism. In the deterministic setting, the hybrid controller consistently accelerates the convergence towards equilibrium across all four equilibrium configurations. The improvement is particularly pronounced for configurations exhibiting slower intrinsic hypocoercive decay, where the dominant weakly dissipative modes limit the convergence rate of the uncontrolled dynamics. By applying the LQR feedback directly to the least stable modes of the discretised transport–collision operator, the controller introduces targeted damping precisely in the components responsible for the slow asymptotic behaviour. In the presence of stochastic forcing, the hybrid LQR controller reduces the amplitude of the fluctuations by enhancing the dissipation of the slow dynamical modes onto which the external forcing continuously projects.

For the computation of the solution to the algebraic Riccati equation, we use the same cost parameters for all equilibrium configurations, namely

$$Q_u = 5 \cdot 10^5 \operatorname{diag}\left(f_\star^\ell(\mathbf{x}_{j_x}, \mathbf{v}_{j_v}) \right), \qquad R_u = \mathrm{Id},$$

where $\operatorname{diag}(\cdot)$ denotes the diagonal matrix whose entries are given by the stacked values of the equilibrium distribution $f_\star^\ell(\mathbf{x}_{j_x}, \mathbf{v}_{j_v})$. The dimension of the reduced LQR subspace is chosen as $n_{\mathrm{LQR\ modes}} = 50$ for all equilibrium configurations, except for the case $\ell = 3$, where $n_{\mathrm{LQR\ modes}} = 70$ is used to account for the slower convergence behaviour observed for this configuration. Furthermore, the spectral feedback parameters are fixed across all experiments, with feedback gain $K_{\mathrm{FG}} = 0.25$ and cut-off parameter $k_{\mathrm{cutoff}} = 0.33$.

These results demonstrate that combining reduced-order optimal control with low-frequency spectral feedback provides an effective strategy for accelerating hypocoercive convergence while maintaining computational tractability. The hybrid approach retains the advantages of the spectral controller on the full phase-space dynamics while concentrating the more expensive LQR optimisation on the small subspace that governs the long-time convergence behaviour.

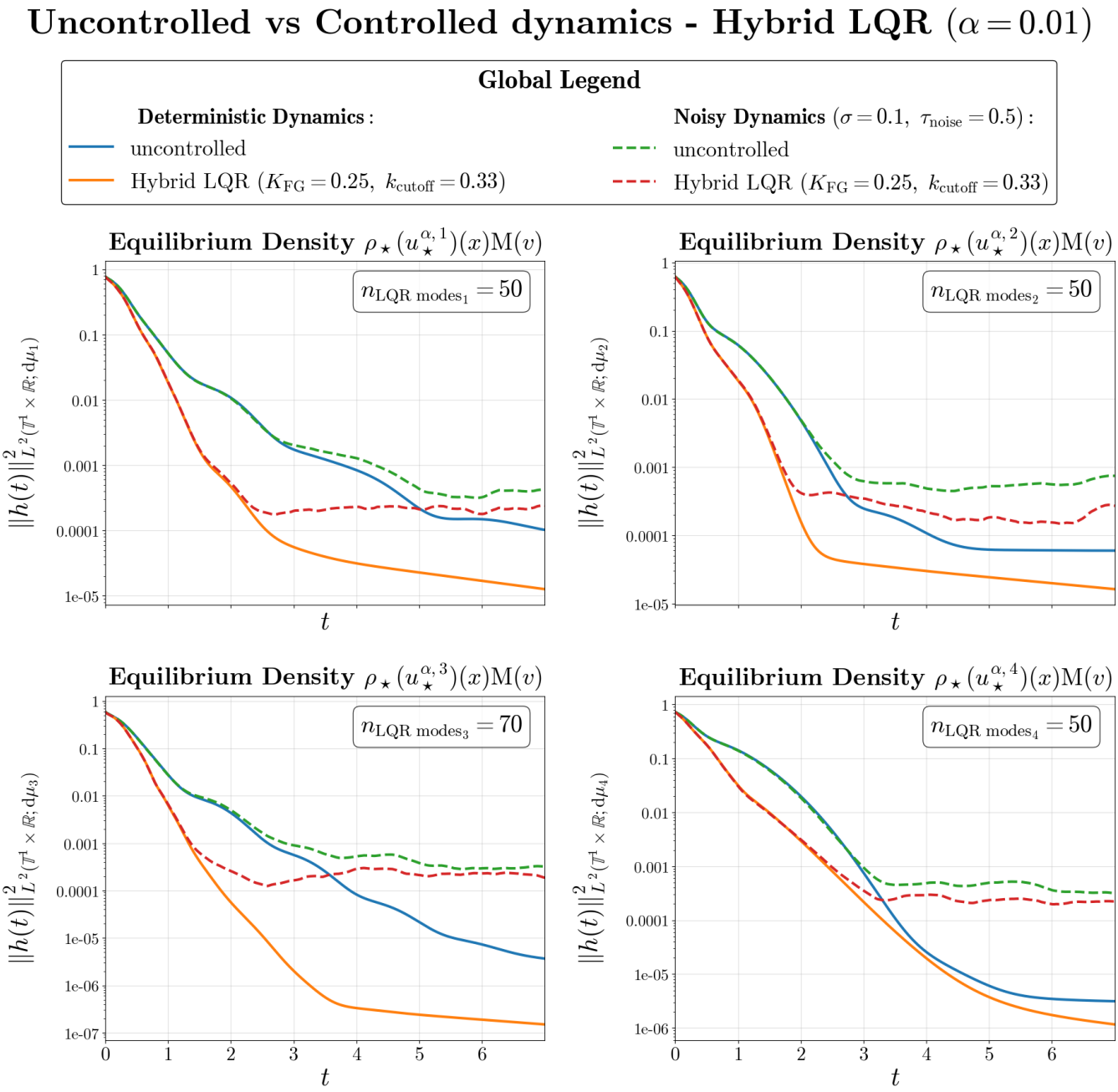


Figure 5.4.3: Evolution of the squared weighted $L^2(\mathbb{T}^1 \times \mathbb{R}; \mathrm{d}\mu_\ell)$-norm of the perturbation $h(t)$ for the linearised VPFP system under uncontrolled dynamics and hybrid LQR feedback, in both deterministic and stochastically forced settings, across the four equilibrium configurations $1 \leq \ell \leq 4$ (Log-y Scale).

5.4.4 Comparison: Spectral Feedback vs Hybrid LQR

As a final assessment of the effectiveness of the hybrid LQR strategy, we compare its performance with the low-frequency spectral feedback controller introduced previously. Figure 5.4.4 presents the evolution of the squared weighted $L^2(\mathbb{T}^1 \times \mathbb{R}; \mathrm{d}\mu_\ell)$-norm of the perturbation $h(t)$ for the controlled linearised VPFP system under both control strategies, considering deterministic dynamics as well as dynamics subject to stochastic forcing. All simulations are performed using the same initial perturbation and, in the stochastic setting, the same realisation of the external forcing, ensuring that differences in the observed behaviour are solely attributable to the control mechanisms.

To provide a consistent comparison, the spectral feedback parameters are kept identical for both controllers, with feedback gain $K_{\mathrm{FG}} = 0.25$ and cut-off parameter $k_{\mathrm{cutoff}} = 0.33$. The hybrid LQR controller is constructed using the same cost matrices Q_u and R_u, as well as the same number of controlled LQR modes $n_{\mathrm{LQR\ modes}}$, as specified in the previous subsection. This allows the impact of the additional optimal feedback component to be isolated from the influence of parameter selection.

In both deterministic and stochastically forced settings, the hybrid LQR controller consistently outperforms the low-frequency spectral feedback mechanism, leading to a faster reduction of the perturbation norm across all equilibrium configurations. This improvement is particularly evident in the long-time regime, where the convergence rate is governed by the slowest dissipating modes of the transport–collision operator. While the spectral controller provides uniform damping of the low-frequency components, the hybrid approach exploits the reduced-order LQR optimisation to apply targeted feedback to the least stable dynamical modes. In the presence of stochastic forcing, the hybrid LQR controller also achieves a stronger reduction of the stationary fluctuation amplitude. By directly controlling the dominant slowly decaying modes, it limits the accumulation of energy introduced by the external perturbations and improves the overall robustness of the system against persistent stochastic excitation.
To quantify the decay behaviour beyond the direct comparison of the norm evolution, we introduce the decay rate

$$\lambda(t) = -\log\left(\frac{\|h(t)\|_{L^2(\mathbb{T}^1\times\mathbb{R};\mathrm{d}\mu_\ell)}}{\|h(0)\|_{L^2(\mathbb{T}^1\times\mathbb{R};\mathrm{d}\mu_\ell)}}\right).$$

This quantity measures the cumulative exponential decay of the perturbation norm relative to its initial value. For an ideal exponential convergence with rate λ, $\lambda(t)$ grows approximately linearly as $\lambda(t) \approx \lambda t$, allowing the effective convergence rates of the low-frequency spectral feedback and hybrid LQR controllers to be compared independently of the initial perturbation amplitude.

Figure 5.4.5 presents the evolution of $\lambda(t)$ for the deterministic simulations considered in Figure 5.4.4, comparing the uncontrolled system, the low-frequency spectral feedback controller, and the hybrid LQR controller across the four equilibrium configurations. For the uncontrolled dynamics, the decay rate increases slowly, reflecting the intrinsic hypocoercive convergence associated with the underlying equilibrium state. The spectral feedback controller produces a larger growth of $\lambda(t)$, indicating an enhanced effective decay rate through additional damping of the low-frequency modes. The hybrid LQR controller yields the fastest increase of the decay rate for all equilibrium configurations, demonstrating that the targeted optimisation of the least stable modes provides additional acceleration compared with uniform low-frequency damping. Moreover, the smoother evolution of $\lambda(t)$ under the hybrid controller suggests a more uniform dissipation of the dominant slow modes, reducing the transient effects associated with non-normal hypocoercive dynamics. In all cases, the decay rate $\lambda(t)$ grows approximately linearly during the transient, while it plateaus towards the end of the simulated trajectories, due to numerical errors and artificial energy conservation due to the discretisation.

Overall, the comparison demonstrates that the hybrid LQR strategy provides a more efficient control mechanism than purely spectral feedback while preserving computational feasibility. The results confirm that combining optimal control on a reduced set of critical modes with physically motivated low-frequency damping on the remaining components constitutes an effective approach for accelerating hypocoercive convergence in high-dimensional kinetic systems.

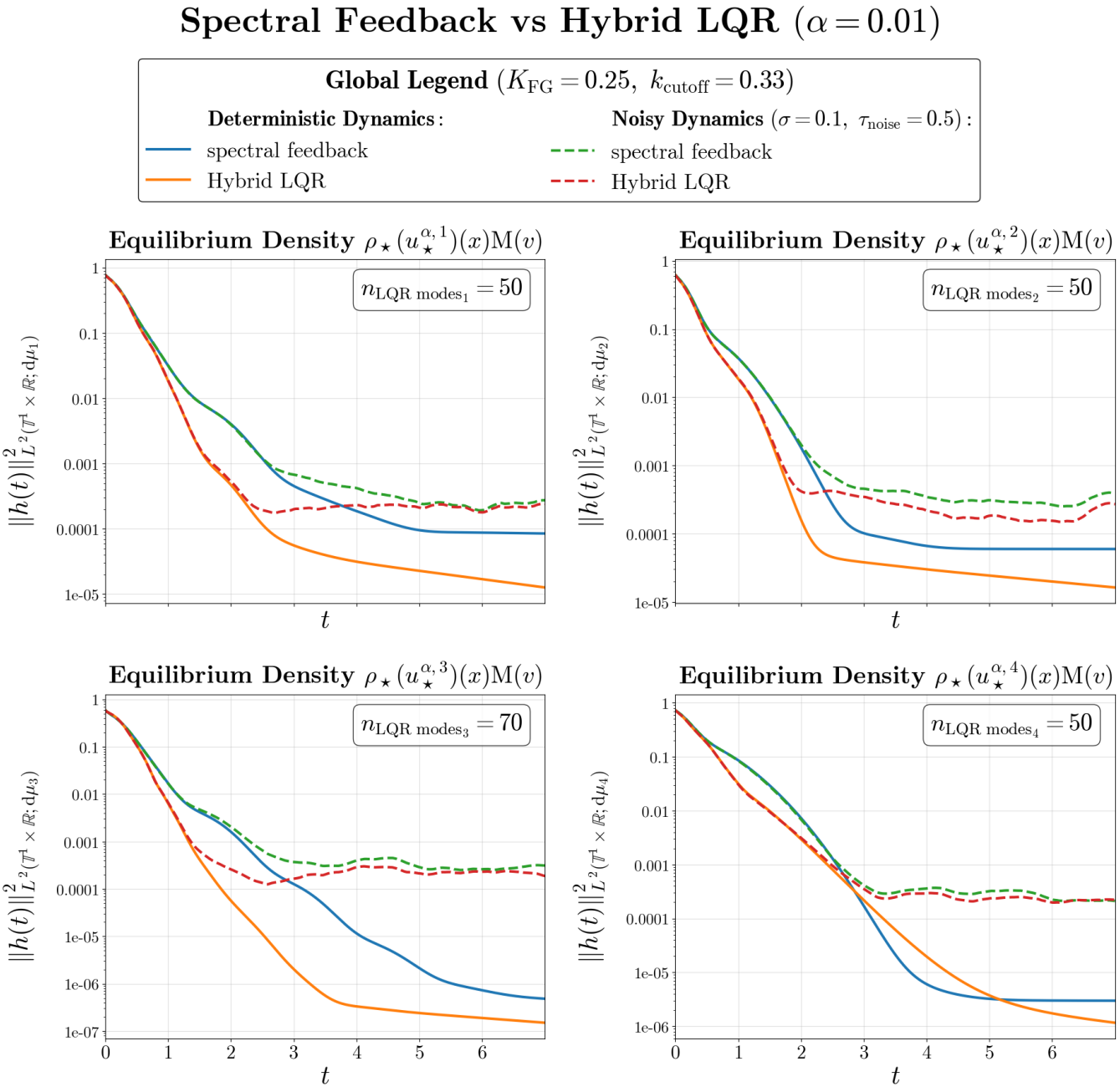


Figure 5.4.4: Evolution of the squared weighted $L^2(\mathbb{T}^1 \times \mathbb{R}; \mathrm{d}\mu_\ell)$-norm of the perturbation $h(t)$ for the controlled linearised VPFP system under spectral feedback and hybrid LQR feedback, in both deterministic and stochastically forced settings, across the four equilibrium configurations $1 \leq \ell \leq 4$ (Log-y Scale).

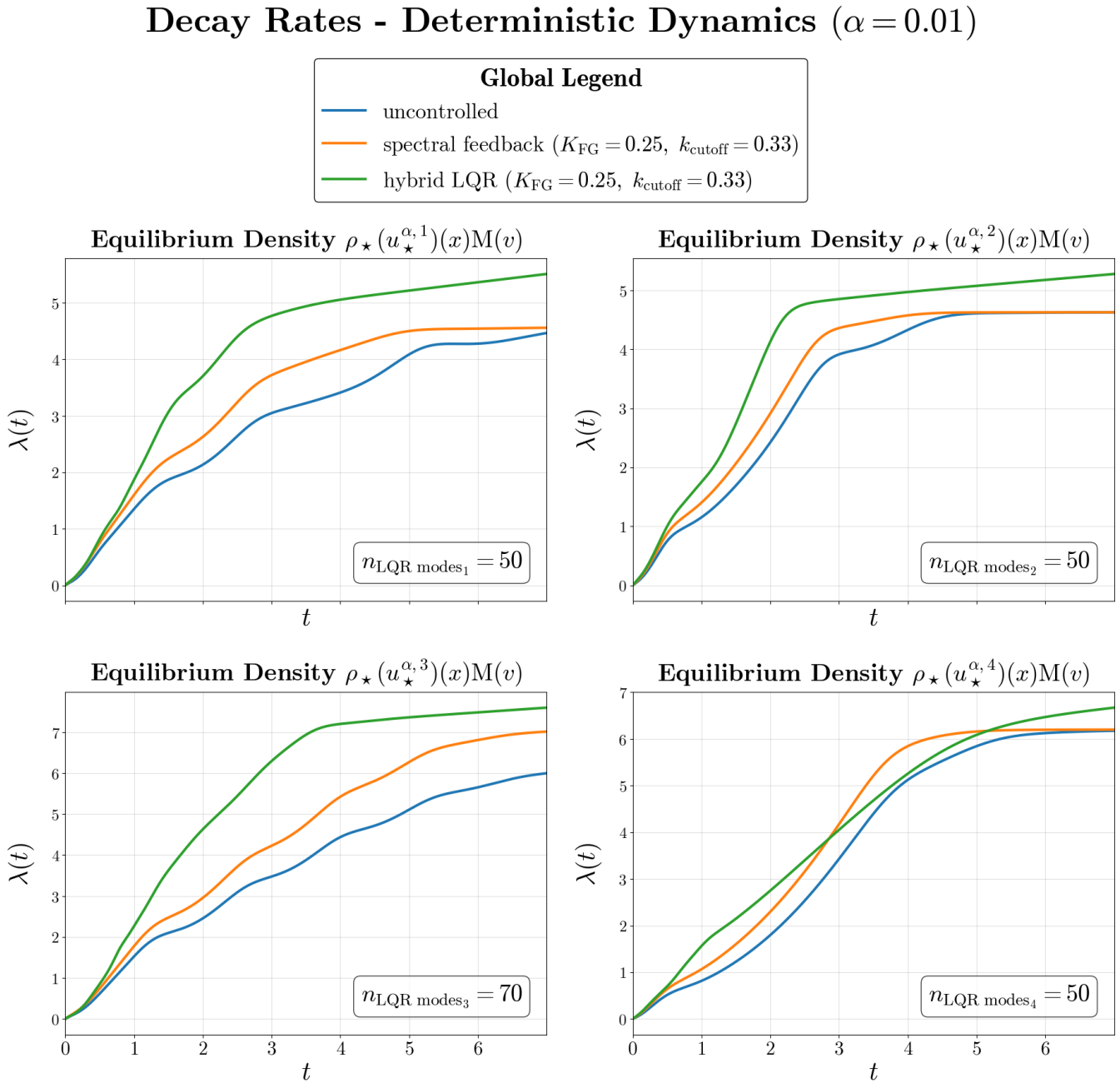


Figure 5.4.5: Cumulative decay rate $\lambda(t) = -\log(\|h(t)\|_{L^2(\mathbb{T}^1\times\mathbb{R};\mathrm{d}\mu_\ell)}/\|h(0)\|_{L^2(\mathbb{T}^1\times\mathbb{R};\mathrm{d}\mu_\ell)})$ for the deterministic linearised VPFP system under uncontrolled dynamics, low-frequency spectral feedback, and hybrid LQR feedback across the four equilibrium configurations $1 \leq \ell \leq 4$.

Conclusion

The numerical experiments presented in this section provide a computational validation of the hypocoercive convergence properties established in the previous sections and demonstrate the potential of feedback control for accelerating convergence towards equilibrium in the linearised VPFP system. For the uncontrolled dynamics, the simulations confirm the predicted exponential convergence towards equilibrium, while also illustrating the strong dependence of the observed convergence rate on the underlying stationary state. In particular, equilibria associated with weaker dissipative mechanisms exhibit significantly slower convergence, highlighting the importance of identifying and controlling the dominant slowly decaying modes.

The introduction of additional linear feedback substantially improves the convergence behaviour. The full-frequency feedback controller provides a strong reference case by uniformly damping the perturbation, while the low-frequency spectral feedback achieves a comparable improvement by acting only on the large-scale spatial structures responsible for the slow asymptotic dynamics. This confirms that the dominant bottleneck in the hypocoercive convergence is associated with low-frequency modes, and that efficient control can be achieved without requiring actuation of the full phase-space spectrum.

Building on this observation, the hybrid LQR controller combines reduced-order optimal control of the least stable modes with spectral damping of the remaining components. The numerical results show that this approach consistently outperforms purely spectral feedback, both in deterministic simulations and in the presence of persistent stochastic perturbations. By concentrating the optimisation effort on the dynamically relevant modes, the hybrid strategy achieves enhanced convergence rates while avoiding the computational cost of a full-dimensional LQR design.

Overall, these experiments demonstrate that hypocoercive kinetic systems can be effectively stabilised and accelerated through appropriately designed feedback mechanisms. The results suggest that combining analytical information on the slow convergence structure with reduced-order optimal control provides a promising framework for designing computationally feasible control strategies for high-dimensional kinetic equations.

Chapter 6

Conclusion

The topics developed in this thesis form a logical chain, with each chapter providing the necessary foundations for the results that follow. Viewed collectively, these results establish a framework for studying confinement through the dependence of stationary VPFP equilibria on the external potential. The central object throughout the thesis is the solution operator $u \mapsto (\rho_\star(u), \varphi_\star(u))$, whose construction, regularity, approximate inversion, and dynamical consequences provide the common thread linking the individual chapters. The dependence between these developments is briefly summarised below.

Chapter 2 established, for each admissible external potential u, the existence and uniqueness of a free energy minimiser $f_\star$, together with its associated spatial density $\rho_\star$ and self-consistent potential $\varphi_\star$ (Proposition 2.2.16), and identified this minimiser with a weak solution of the stationary VPFP system (Theorem 2.2.30). These results determine, for every admissible u, a unique stationary state, but they do not describe the dependence of this state on variations of the external potential. Once u is allowed to vary, the associated solution operators $u \mapsto \rho_\star(u)$ and $u \mapsto \varphi_\star(u)$ become the central objects of study.

This is the subject of Chapter 3. The elliptic bootstrapping argument shows that the regularity of u propagates pointwise to $\rho_\star$ and $\varphi_\star$ (Proposition 3.1.5), while the weak sequential continuity result of Theorem 3.3.1 extends this to continuity of the solution operators between the relevant Banach spaces. This continuity property makes optimisation over the space of external potentials mathematically meaningful in Chapter 4: minimising sequences of potentials can be passed to the limit while preserving the correspondence between potentials and their associated stationary states.

Chapter 4 reverses the perspective of the preceding chapters. Rather than determining the stationary state generated by a prescribed potential u, it seeks the potential u that approximates a prescribed reference density ρ_{ref}. The non-attainability discussion shows that this objective cannot, in general, be achieved exactly, motivating the variational relaxation through a Kullback–Leibler tracking functional. The well-posedness of the resulting regularised optimisation problem (Proposition 4.3.2) is established by applying the continuity theory developed in Chapter 3 directly along a minimising sequence of admissible potentials.

The confinement problem, as formulated, produces a static object, namely an optimal potential $u_\star$ and its associated equilibrium pair $(\rho_\star(u_\star), \varphi_\star(u_\star))$. It does not, however, guarantee that the time-dependent VPFP dynamics converge towards this equilibrium, nor does it provide information on the rate of convergence. Chapter 5 addresses this question at the linearised level by establishing exponential decay of the linearised VPFP dynamics around any free energy stationary state satisfying the standing assumptions on u. Subsequent numerical experiments show that the decay rate depends sensitively on the choice of equilibrium and motivate the control strategies developed in Section 5.4, aimed at accelerating convergence towards a given confinement-problem equilibrium.

The results established in this thesis rely on a number of structural assumptions that define the framework in which the analysis is carried out. The restriction to the torus $\mathbb{T}^d$ as spatial domain, together with a uniform neutralising background charge ρ_0, is the principal such choice. As discussed in Section 1.1, this setting avoids the existence difficulties that dominate the corresponding theory on $\mathbb{R}^d$, where confine-

ment depends on the behaviour of the confining potential at infinity. We conjecture that extending the analysis to bounded convex domains with sufficiently regular boundaries and suitable boundary conditions requires primarily technical adaptations of the arguments developed here, rather than fundamentally new ideas. In contrast, extending the results to unbounded spatial domains, such as $\mathbb{R}^d$, introduces substantial analytical difficulties that are absent on the torus and that were deliberately avoided in this thesis. In particular, both the construction of stationary states and the analysis of their dependence on the external potential would require different existence and regularity theories, respectively, beyond those developed in Chapters 2 and 3. Such extensions therefore remain outside the scope of the present work.

Two further simplifications, each introduced explicitly in the relevant chapter, are worth recalling. First, the stability analysis of Chapter 5 is performed at the linearised level, while the nonlinear stability of the equilibria produced by the confinement problem remains an open question. Second, the numerical investigation of Chapter 4 was restricted to the one-dimensional setting, leaving the qualitative behaviour of optimal potentials in higher dimensions unexplored.

6.1 Open Problems

Several questions arising from the developments of this thesis remain open, either because they lie beyond its intended scope or because their resolution requires analytical tools not developed here. They are summarised below according to the chapter in which they originate.

A first open problem concerns Chapter 2. The stationary states considered throughout this thesis are, by construction, free energy minimisers, and their identification with weak solutions of the stationary VPFP system was established under additional regularity assumptions on $\varphi_\star$ rather than directly at the level of the weak formulation. It remains open under which assumptions the renormalisation techniques in the spirit of the Ambrosio–DiPerna–Lions theory [19, 22], as employed by [29] for the VPFP system, or approximation arguments of the type developed by [17], can be used in the toroidal setting to characterise weak solutions of the stationary system directly, without passing through the variational formulation. Such a characterisation would extend the subsequent theory of Chapters 3–5 to a broader class of stationary states than free energy minimisers alone.

Closely related to the first, a second open problem concerns a dimensional obstruction arising from the variational framework in Chapter 2 and affecting the analysis in Chapter 3. The variational argument only guarantees that $\varphi_\star$ solves the Poisson–Boltzmann–Emden equation with H^1 regularity. For dimensions $d \geq 3$, this regularity is insufficient to identify free energy minimisers with weak stationary solutions, and equally insufficient to initiate the elliptic bootstrapping argument of Section 3.1 without a further integrability assumption on $\varphi_\star$. The same lack of regularity prevents the application of the implicit function theorem required to extend the Fréchet differentiability results of Section 3.4 to higher dimensions. This dimensional obstruction is well documented in the broader literature on Gelfand-type equations (see, e.g., [39]), of which the Poisson–Boltzmann–Emden equation is an instance. In the recent work of [12], H^2 regularity of the solutions to the Poisson–Boltzmann–Emden equation was established in dimension $d = 3$ via a Schauder fixed-point argument, thereby resolving the obstruction in the physically relevant dimensions (see Section 2.2.6 for a detailed discussion). However, it remains open whether the arguments developed in this literature can be adapted to the Poisson–Boltzmann–Emden equation to yield the additional integrability required in dimensions $d \geq 4$.

A third open problem concerns the continuous Fréchet differentiability of the solution operators $u \mapsto \rho_\star(u)$ and $u \mapsto \varphi_\star(u)$. In particular, [11] exploit this differentiability property to solve their confinement problem in an optimise-then-discretise manner, deriving optimality conditions directly at the continuous level, in contrast to the discretise-then-optimise strategy adopted in Chapter 4 of this thesis. An interesting direction for future research is therefore to exploit the continuous Fréchet differentiability of the solution operators established in dimensions $d = 1, 2$ to derive a continuous optimality system for the Kullback–Leibler tracking functional of Chapter 4. Such a result would enable a direct comparison between the two optimisation paradigms and provide the analytical foundation for an optimise-then-discretise

treatment of the confinement problem in the present setting.

In Chapter 5, the numerical experiments show that the exponential decay rate of the linearised VPFP dynamics depends sensitively on the equilibrium about which the system is linearised. However, no analytical description of this dependence was obtained. An interesting direction for future research is therefore to study the mapping $u \mapsto \lambda(u)$ from the external potential to the associated decay rate, and in particular its continuity properties, along the lines of the analysis of $u \mapsto \rho_\star(u)$ and $u \mapsto \varphi_\star(u)$ developed in Chapter 3. Such a study would provide a basis for identifying structural properties of the external potentials u that guarantee faster convergence to equilibrium and would place the acceleration strategies explored numerically in Section 5.4 on a rigorous analytical foundation.

A fifth open problem is directly tied to the preceding one. The understanding of the dependence of the decay rate λ on the external potential u would open the way to a dynamically informed formulation of the inverse problem. Indeed, the confinement problem of Chapter 4 already allows additional confinement requirements to be imposed through constraint functionals, and a constraint of the form $\lambda(u) \geq \bar{\lambda}$, for a prescribed target rate $\bar{\lambda}$, could be incorporated within this formulation. The resulting optimal potentials would then generate equilibria with a prescribed rate of convergence by construction, thereby bridging the static nature of the confinement problem addressed in Chapter 4 and the dynamical nature of the underlying VPFP system.

The dynamically informed inverse problem discussed above is one direction in which the confinement problem of Chapter 4 could be enriched by additional structure. A further such direction concerns the physical realisability of the external potential itself. A typical engineering formulation of the confinement problem replaces the generic electrostatic potential u considered in this thesis with an externally imposed electromagnetic field, coupled to the particle dynamics through the Lorentz force and closed self-consistently via Maxwell's equations, as in [3]. It is left for future work to formulate and study an inverse problem in this setting, in which the Euler–Lagrange and Poisson–Boltzmann–Emden equations that uniquely characterise the stationary VPFP system considered here are coupled with Maxwell's equations, so that the resulting confining field is not only optimal with respect to a prescribed reference density but also physically realisable as an electromagnetic field.

Finally, the control strategies developed in Section 5.4 were assessed solely through numerical experiments, and a rigorous controllability analysis of the linearised VPFP dynamics remains an open problem. Such an analysis would require, first, a systematic characterisation of the admissible control mechanisms and their influence on the linearised dynamics, and, second, an investigation of the controllability properties of the resulting system, including the determination of the fastest achievable convergence rates. Establishing these results would provide a rigorous theoretical framework for the acceleration strategies explored numerically in this thesis and would further connect the static optimisation perspective of the confinement problem with the dynamical objective of steering the VPFP system towards equilibrium.

6.2 Closing Remarks

The confinement problem studied in this thesis was motivated, as discussed in Chapter 1, by analogous problems in kinetic plasma physics and its engineering applications. The setting adopted here, namely an electrostatic potential on the torus, with confinement interpreted in the entropic sense of approximating a prescribed reference density, is a substantial simplification of these physically motivated problems, and the results obtained should be understood within this context. Nevertheless, the four-stage structure developed in this thesis – existence and variational characterisation of equilibria, regularity of the associated solution operators, a well-posed inverse problem exploiting this regularity, and dynamical stability of the resulting equilibria – is not intrinsically tied to the periodic electrostatic setting. It may therefore provide a framework for analogous confinement problems on non-compact domains or for models in which confinement is achieved through a magnetic rather than an electrostatic mechanism.

Bibliography

[1] R. A. Adams and J. J. F. Fournier. *Sobolev Spaces*. 2nd. Vol. 140. Pure and Applied Mathematics. Amsterdam: Academic Press, 2003, p. 305. ISBN: 9780120441433.

[2] L. Addala et al. "L^2-Hypocoercivity and large time asymptotics of the linearized Vlasov–Poisson–Fokker–Planck system". In: *Journal of Statistical Physics* 184.1 (2021), p. 4. DOI: 10.1007/s10955-021-02784-4.

[3] G. Albi et al. "Instantaneous control strategies for magnetically confined Fusion plasma". In: *Journal of Computational Physics* 527 (2025), p. 113804. DOI: 10.1016/j.jcp.2025.113804.

[4] H. Amann. *Ordinary differential equations: an introduction to nonlinear analysis*. Vol. 13. De Gruyter Studies in Mathematics. Berlin and New York: De Gruyter, 1990. ISBN: 9783110115154. DOI: 10.1515/9783110853698.

[5] L. Armijo. "Minimization of functions having Lipschitz continuous first partial derivatives". In: *Pacific Journal of Mathematics* 16.1 (1966), pp. 1–3. DOI: 10.2140/pjm.1966.16.1.

[6] A. Arnold et al. "On convex Sobolev inequalities and the rate of convergence to equilibrium for Fokker–Planck type equations". In: *Communications in Partial Differential Equations* 26.1–2 (2001), pp. 43–100. DOI: 10.1081/PDE-100002246.

[7] H. Bahouri, J. Y. Chemin, and R. Danchin. *Fourier analysis and nonlinear partial differential equations*. Vol. 343. Grundlehren der mathematischen Wissenschaften. Berlin, Heidelberg: Springer Science & Business Media, 2011. ISBN: 978-3-642-16830-7.

[8] J. A. Bittencourt. *Fundamentals of Plasma Physics*. 3rd. New York, NY: Springer, 2004, p. 679. ISBN: 978-0-387-20975-3. DOI: 10.1007/978-1-4757-4030-1.

[9] V. I. Bogachev. *Gaussian measures*. Vol. 62. Mathematical Surveys and Monographs. American Mathematical Society, 1998. ISBN: 978-0-8218-1054-5.

[10] V. I. Bogachev. *Measure Theory*. 1st. Berlin, Heidelberg: Springer Berlin Heidelberg, 2007, pp. XVII, 1075. ISBN: 978-3-540-34513-8. DOI: 10.1007/978-3-540-34514-5.

[11] A. Borzì, G. Infante, and G. Mascali. "Optimal design of equilibrium solutions of the Vlasov–Poisson system by an external electric field". In: *SIAM Journal on Applied Mathematics* 85.1 (2025), pp. 143–163. DOI: 10.1137/24M1643177.

[12] A. Borzì et al. *Analysis and optimal design of equilibria in the Vlasov–Poisson system*. 2026. arXiv: 2607.25698 [math.OC].

[13] F. Bouchut and J. Dolbeault. "On long time asymptotics of the Vlasov–Fokker–Planck equation and of the Vlasov–Poisson–Fokker–Planck system with Coulombic and Newtonian potentials". In: *Differential and Integral Equations* 8.3 (1995), pp. 487–514. DOI: 10.57262/die/1369316501.

[14] T. Breiten and K. K. Kunisch. "Improving the convergence rates for the kinetic Fokker–Planck equation by optimal control". In: *SIAM Journal on Control and Optimization* 61.3 (2023), pp. 1557–1581. DOI: 10.1137/22M1499698.

[15] H. Brezis. *Functional analysis, Sobolev spaces and partial differential equations*. 1st. Universitext. New York, NY: Springer, 2010, p. 600. ISBN: 978-0-387-70913-0. DOI: 10.1007/978-0-387-70914-7.

[16] M. P. do Carmo. *Riemannian Geometry*. 1st. Mathematics: Theory & Applications. Boston, MA: Birkhäuser Boston, 1992, pp. xv+300. ISBN: 978-0-8176-3490-2.

[17] J. A. Carrillo. "Global weak solutions for the initial-boundary-value problems Vlasov–Poisson–Fokker–Planck system". In: *Mathematical Methods in the Applied Sciences* 21.10 (1998), pp. 907–938. DOI: 10.1002/(SICI)1099-1476(19980710)21:10<907::AID-MMA977>3.0.CO;2-W.

[18] H. Cartan. *Differential Calculus*. Translated from the French edition, 1967. London and Boston: Kershaw Publishing Co. Ltd. and Houghton Mifflin Co., 1971. ISBN: 0395120330.

[19] C. De Lellis. "Ordinary differential equations with rough coefficients and the renormalization theorem of Ambrosio [after Ambrosio, DiPerna, Lions]". In: *Astérisque* 317.972 (2008), pp. 175–203.

[20] L. Desvillettes and J. Dolbeault. "On long time asymptotics of the Vlasov–Poisson–Boltzmann equation". In: *Communications in Partial Differential Equations* 16.2–3 (1991), pp. 451–489. DOI: 10.1080/03605309108820765.

[21] J. Dieudonné. *Foundations of Modern Analysis*. Vol. 10. Pure and Applied Mathematics. New York: Academic Press, 1960.

[22] R. J. DiPerna and P. L. Lions. "On the Cauchy Problem for Boltzmann equations: global existence and weak stability". In: *Annals of Mathematics* 130.2 (1989), pp. 321–366. DOI: 10.2307/1971423.

[23] J. Dolbeault. "Free energy and solutions of the Vlasov–Poisson–Fokker–Planck system: external potential and confinement (large time behavior and steady states)". In: *Journal de Mathématiques Pures et Appliquées* 78.2 (1999), pp. 121–157. DOI: 10.1016/S0021-7824(01)80006-4.

[24] J. Dolbeault and X. Li. "Generalized logarithmic Hardy–Littlewood–Sobolev inequality". In: *International Mathematics Research Notices* 2021.23 (2021), pp. 17862–17874. DOI: 10.1093/imrn/rnz324.

[25] J. Dolbeault, C. Mouhot, and C. Schmeiser. "Hypocoercivity for linear kinetic equations conserving mass". In: *Transactions of the American Mathematical Society* 367.6 (2015), pp. 3807–3828. DOI: 10.1090/S0002-9947-2015-06012-7.

[26] K. Dressler. "Steady states in plasma physics—the Vlasov–Fokker–Planck equation". In: *Mathematical Methods in the Applied Sciences* 12.6 (1990), pp. 471–487. DOI: 10.1002/mma.1670120603.

[27] N. Dunford and J. T. Schwartz. *Linear operators. Part I: general theory*. Vol. 7. Pure and Applied Mathematics. New York: Interscience Publishers, 1958. ISBN: 0470226056.

[28] L. Einkemmer et al. "Suppressing instability in a Vlasov–Poisson system by an external electric field through constrained optimization". In: *Journal of Computational Physics* 498 (2024), p. 112662. ISSN: 0021-9991. DOI: 10.1016/j.jcp.2023.112662.

[29] N. El Ghani and N. Masmoudi. "Diffusion limit of the Vlasov–Poisson–Fokker–Planck system". In: *Communications in Mathematical Sciences* 8.2 (2010), pp. 463–479.

[30] L. C. Evans. *Partial differential equations*. 2nd. Vol. 19. Graduate Studies in Mathematics. Providence, RI: American Mathematical Society, 2010. ISBN: 9780821849743.

[31] M. R. Feix and P. Bertrand. "A universal model: the Vlasov equation". In: *Transport Theory and Statistical Physics* 34.1–2 (2005), pp. 7–62. DOI: 10.1080/00411450500253703.

[32] I. M. Gelfand and S. V. Fomin. *Calculus of Variations*. Reprint of the 1963 Prentice-Hall edition, revised and edited by Richard A. Silverman. Mineola, NY: Dover Publications, 2000. ISBN: 978-0-486-41448-5.

[33] D. Gilbarg and N. S. Trudinger. *Elliptic partial differential equations of second order*. 2nd. Classics in Mathematics. Berlin, Heidelberg: Springer, 2001. ISBN: 978-3-540-41160-4. DOI: 10.1007/978-3-642-61798-0.

[34] R. Glassey, J. Schaeffer, and Y. Zheng. “Steady states of the Vlasov–Poisson–Fokker–Planck system”. In: *Journal of Mathematical Analysis and Applications* 202.3 (1996), pp. 1058–1075. ISSN: 0022-247X. DOI: 10.1006/jmaa.1996.0360.

[35] S. N. Gomes and G. A. Pavliotis. “Mean field limits for interacting diffusions in a two-scale potential”. In: *Journal of Nonlinear Science* 28.3 (2018), pp. 905–941. ISSN: 1432-1467. DOI: 10.1007/s00332-017-9433-y.

[36] D. Han-Kwan. “On the confinement of a Tokamak plasma”. In: *SIAM Journal on Mathematical Analysis* 42.6 (2010), pp. 2337–2367. DOI: 10.1137/090774574.

[37] E. Hebey. *Sobolev spaces on Riemannian manifolds*. Vol. 1635. Lecture Notes in Mathematics. Berlin, Heidelberg: Springer-Verlag, 1996. ISBN: 978-3-540-61722-8. DOI: 10.1007/BFb0092907.

[38] K. H. Hoffmann, G. Leugering, and F. Tröltzsch, eds. *Optimal control of partial differential equations. International Conference in Chemnitz, Germany, April 20–25, 1998*. Vol. 133. International Series of Numerical Mathematics. Basel: Birkhäuser, 1999. ISBN: 978-3-7643-6151-8. DOI: 10.1007/978-3-0348-8691-8.

[39] D. D. Joseph and T. S. Lundgren. “Quasilinear Dirichlet problems driven by positive sources”. In: *Archive for Rational Mechanics and Analysis* 49.4 (1973), pp. 241–269. ISSN: 1432-0673. DOI: 10.1007/BF00250508.

[40] D. Kalise, L. M. Moschen, and G. A. Pavliotis. *Linearization-based feedback stabilization of McKean–Vlasov PDEs*. arXiv:2507.12411v3. 2025. DOI: 10.48550/arXiv.2507.12411. arXiv: 2507.12411.

[41] A. N. Kaufman and B. I. Cohen. “Theoretical Plasma Physics”. In: *Journal of Plasma Physics* 85.6 (2019), p. 205850601. DOI: 10.1017/S0022377819000667.

[42] A. Manzoni, A. Quarteroni, and S. Salsa. *Optimal control of partial differential equations: analysis, approximation, and applications*. Vol. 207. Applied Mathematical Sciences. Cham, Switzerland: Springer, 2021. ISBN: 978-3-030-77225-3. DOI: 10.1007/978-3-030-77226-0.

[43] L. M. Milne-Thomson. *The calculus of finite differences*. English. 1st. London: Macmillan and Company, Limited, 1933.

[44] J. W. Neuberger. *Sobolev gradients and differential equations*. 2nd. Vol. 1670. Lecture Notes in Mathematics. Springer Berlin Heidelberg, 2010, pp. XIII, 289. ISBN: 978-3-642-04040-5. DOI: 10.1007/978-3-642-04041-2.

[45] J. Nocedal and S. J. Wright. *Numerical optimization*. 2nd. Springer Series in Operations Research and Financial Engineering. New York, NY: Springer, 2006. ISBN: 978-0-387-30303-1. DOI: 10.1007/978-0-387-40065-5.

[46] G. A. Pavliotis. *Stochastic processes and applications*. Vol. 60. Texts in Applied Mathematics. New York: Springer, 2014. ISBN: 978-1-4939-1322-0. DOI: 10.1007/978-1-4939-1323-7.

[47] A. G. Peeters and D. Strintzi. “The Fokker–Planck equation and its Application in plasma physics”. In: *Annalen der Physik* 520.2–3 (2008), pp. 142–157. ISSN: 0003-3804. DOI: 10.1002/andp.200852002-310.

[48] F. Rezakhanlou and C. Villani. *Entropy methods for the Boltzmann equation. Lectures from a Special Semester at the Centre Émile Borel, Institut H. Poincaré, Paris, 2001*. Ed. by François Golse and Stefano Olla. 1st. Vol. 1911. Lecture Notes in Mathematics. Springer Berlin Heidelberg, 2008, pp. XII, 113. ISBN: 978-3-540-73705-6. DOI: 10.1007/978-3-540-73705-6.

[49] Y. Saad. “Acceleration methods for fixed-point iterations”. In: *Acta Numerica* 34 (2025), pp. 805–890.

[50] K. Schmüdgen. *Unbounded self-adjoint operators on Hilbert space*. Vol. 265. Graduate Texts in Mathematics. Springer Dordrecht, 2012, pp. xx, 432. ISBN: 978-94-007-4752-4. DOI: 10.1007/978-94-007-4753-1.

[51] M. E. Taylor. *Partial differential equations I: basic theory*. 2nd. Vol. 115. Applied Mathematical Sciences. Springer New York, 2011. ISBN: 978-1-4419-7054-1. DOI: 10.1007/978-1-4419-7055-8.

[52] A. G. R. Thomas et al. "A review of Vlasov–Fokker–Planck numerical modeling of inertial confinement fusion plasma". In: *Journal of Computational Physics* 231.3 (2012). Special Issue: Computational Plasma Physics, pp. 1051–1079. ISSN: 0021-9991. DOI: 10.1016/j.jcp.2011.09.028.

[53] C. Villani. *Hypocoercivity*. Vol. 202. Memoirs of the American Mathematical Society 950. Providence, RI: American Mathematical Society, 2009. ISBN: 978-0-8218-4498-4. DOI: 10.1090/memo/0950.

[54] W. P. Ziemer. *Weakly differentiable functions: Sobolev spaces and functions of bounded variation*. Vol. 120. Graduate Texts in Mathematics. New York: Springer-Verlag, 1989. ISBN: 9780387970172.

Appendix

Appendix A

Sobolev Spaces and Sobolev Embeddings on the Flat Torus

We denote by $\mathbb{T}^d$ the d-dimensional flat torus with 2π-periodicity (for brevity, the torus). The torus is a boundaryless, compact d-dimensional Riemannian manifold [16]. There exists a bijective correspondence between functions $f : \mathbb{T}^d \to \mathbb{R}$ defined on the torus and 2π-periodic functions $\tilde{f} : \mathbb{R}^d \to \mathbb{R}$, where periodicity is understood as

$$\tilde{f}(x) = \tilde{f}(x + 2\pi k) \quad \text{for all } k \in \mathbb{Z}^d,$$

where the periodicity condition is understood in the appropriate sense (e.g. pointwise for smooth functions and almost everywhere for equivalence classes of Lebesgue functions). In particular, the spaces $L^p(\mathbb{T}^d)$ and $L^p_{\mathrm{per}}([0, 2\pi]^d)$ are isometrically isomorphic for any $1 \le p \le \infty$. Consequently, we will not distinguish notationally between f and its periodic representative $\tilde{f}$, and we will identify functions on $\mathbb{T}^d$ with their 2π-periodic extensions to $\mathbb{R}^d$.

A.1 Sobolev Embeddings on the Torus

A body of literature develops extensions of Sobolev inequalities to the setting of compact Riemannian manifolds, including results such as the Rellich–Kondrachov compactness theorem and Morrey's embedding theorem; see, for instance, [37]. Since these results are scattered across the literature, we briefly collect and present the versions relevant to our discussion.

Theorem A.1.1 (**Sobolev–Gagliardo–Nirenberg, Subcritical Embedding**).
Let $1 \le p < d$. Then, the injection

$$W^{1,p}(\mathbb{T}^d) \hookrightarrow L^q(\mathbb{T}^d)\,, \qquad q \in [p, p^\star]$$

is continuous, where $p^\star$ is the critical Sobolev exponent, given by

$$\frac{1}{p^\star} = \frac{1}{p} - \frac{1}{d}\,.$$

Proof. Let $f \in W^{1,p}(\mathbb{T}^d)$. Then $f \in W^{1,p}_{\mathrm{per}}([0, 2\pi]^d)$, and in particular $f \in W^{1,p}([0, 2\pi]^d)$. By the Sobolev–Gagliardo–Nirenberg inequality on bounded $\mathcal{C}^1$ domains (see, e.g., Corollary 9.14 in [15]), there exists $C = C(p, d) > 0$ such that

$$\|f\|_{L^{p^\star}([0,2\pi]^d)} \le C\|f\|_{W^{1,p}([0,2\pi]^d)}.$$

By the isometric isomorphism $W^{1,p}(\mathbb{T}^d) \cong W^{1,p}_{\mathrm{per}}([0, 2\pi]^d)$, this inequality coincides with the corresponding estimate on the torus:

$$\|f\|_{L^{p^\star}(\mathbb{T}^d)} \le C\|f\|_{W^{1,p}(\mathbb{T}^d)}.$$

This proves the continuity of the embedding

$$W^{1,p}(\mathbb{T}^d) \hookrightarrow L^{p^\star}(\mathbb{T}^d).$$

The general result follows by standard interpolation arguments – see, e.g., Remark 2 in [15, Chapter 4]. □

Theorem A.1.2 (**Critical Embedding**).
Let $p = d$. Then, the injection

$$W^{1,p}(\mathbb{T}^d) \hookrightarrow L^q(\mathbb{T}^d)\,, \qquad q \in [p, +\infty)$$

is continuous.

Theorem A.1.3 (**Morrey, Supercritical Embedding**).
Let $p > d$. Then, the injection

$$W^{1,p}(\mathbb{T}^d) \hookrightarrow \mathcal{C}(\mathbb{T}^d)$$

is continuous. In particular, $W^{1,p}(\mathbb{T}^d) \hookrightarrow L^\infty(\mathbb{T}^d)$ is a continuous injection.

The proof of Theorem A.1.2 (resp. Theorem A.1.3) is identical in spirit to that of Theorem A.1.1 and is therefore omitted. Repeated applications of the previous theorems yield the following result (cf. Corollary 9.13 in [15]).

Corollary A.1.4. *Let $m \geq 1$ be an integer and $p \in [1, +\infty)$. Then, the following are continuous injections:*

$$\begin{aligned}
&i)\quad W^{m,p}(\mathbb{T}^d) \hookrightarrow L^q(\mathbb{T}^d) \qquad q \in [p, p^\star_m] \qquad \text{where } \frac{1}{p^\star_m} = \frac{1}{p} - \frac{m}{d} \qquad \text{if } p < \frac{d}{m}\,;\\
&ii)\quad W^{m,p}(\mathbb{T}^d) \hookrightarrow L^q(\mathbb{T}^d) \qquad q \in [p, \infty) \qquad\qquad\qquad\qquad \text{if } p = \frac{d}{m}\,;\\
&iii)\quad W^{m,p}(\mathbb{T}^d) \hookrightarrow \mathcal{C}(\mathbb{T}^d) \qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{if } p > \frac{d}{m}\,.
\end{aligned}$$

We conclude this section with the Rellich–Kondrachov theorem on the torus (cf. Theorem 9.16 in [15] and Theorem 3.6 in [37]).

Theorem A.1.5 (**Rellich–Kondrachov**).
Let $p \geq 1$. The following inejctions are compact:

$$\begin{aligned}
&i)\quad W^{1,p}(\mathbb{T}^d) \hookrightarrow L^q(\mathbb{T}^d) \qquad q \in [p, p^\star) \qquad \text{where } \frac{1}{p^\star} = \frac{1}{p} - \frac{1}{d} \qquad \text{if } p < d\,;\\
&ii)\quad W^{1,p}(\mathbb{T}^d) \hookrightarrow L^q(\mathbb{T}^d) \qquad q \in [p, \infty) \qquad\qquad\qquad\qquad \text{if } p = d\,;\\
&iii)\quad W^{1,p}(\mathbb{T}^d) \hookrightarrow \mathcal{C}(\mathbb{T}^d) \qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{if } p > d\,.
\end{aligned}$$

In particular, $W^{1,p}(\mathbb{T}^d) \hookrightarrow L^p(\mathbb{T}^d)$ is a compact injection for all p and d.

A.2 Fourier Characterisation of Sobolev Spaces on the Torus

This section collects classical results from Fourier Analysis on Sobolev spaces $H^s(\mathbb{T}^d)$ – see, e.g., Chapter 1 in [7] for an introduction. Throughout this presentation, and unless specified otherwise, we let $s \in \mathbb{R}$ be

arbitrary.

Let $f \in \mathcal{D}'(\mathbb{T}^d)$ be a distribution. For all $k \in \mathbb{Z}^d$, we define

$$\hat{f}_k := \frac{1}{\sqrt{|\mathbb{T}^d|}} \langle f, e^{-ik\cdot x} \rangle \,.$$

The space $H^s(\mathbb{T}^d)$ consists of all distributions $f \in \mathcal{D}'(\mathbb{T}^d)$ such that

$$\|f\|^2_{H^s(\mathbb{T}^d)} = \sum_{k \in \mathbb{Z}^d} (1 + |k|^2)^s |\hat{f}_k|^2 < +\infty \,. \tag{A.1}$$

The map $f \mapsto \|f\|_{H^s(\mathbb{T}^d)}$ defines a norm. In particular, the space $H^s(\mathbb{T}^d)$ equipped with the above norm is a Hilbert space, where the norm is induced by the inner product

$$(f, g)_{H^s(\mathbb{T}^d)} := \sum_{k \in \mathbb{Z}^d} (1 + |k|^2)^s \hat{f}_k \overline{\hat{g}_k} \,.$$

For all $f \in H^s(\mathbb{T}^d)$ and $N \in \mathbb{N}$, we denote the N-th Fourier partial sum by

$$S_N(f)(x) := \sum_{|k| \leq N} \hat{f}_k e^{ik\cdot x} \,.$$

Lemma A.2.1. *Let $f \in H^s(\mathbb{T}^d)$ be arbitrary. Then, $S_N(f) \to f$ in $H^s(\mathbb{T}^d)$ as $N \to \infty$. In particular,*

$$f = \sum_{k \in \mathbb{Z}^d} \hat{f}_k e^{ik\cdot x} \quad \text{in } \mathcal{D}'(\mathbb{T}^d) \,.$$

Proof. For all $N \in \mathbb{N}$, we compute

$$\|f - S_N(f)\|^2_{H^s(\mathbb{T}^d)} = \sum_{|k| > N} (1 + |k|^2)^s |\hat{f}_k|^2 \,,$$

where the right-hand side is the tail of a convergent series – see (A.1). We thus deduce that for all $\varepsilon > 0$ there exists $\bar{N} \in \mathbb{N}$ such that

$$\|f - S_N(f)\|^2_{H^s(\mathbb{T}^d)} < \varepsilon \quad \text{for all } N \geq \bar{N}.$$

Therefore $S_N(f) \to f$ in $H^s(\mathbb{T}^d)$, and hence also in $\mathcal{D}'(\mathbb{T}^d)$:

$$f = \lim_{N \to \infty} S_N(f) = \sum_{k \in \mathbb{Z}^d} \hat{f}_k e^{ik\cdot x} \,. \qquad \square$$

Corollary A.2.2. ***(Friedrichs' Lemma on the Torus)***
$\mathcal{C}^\infty(\mathbb{T}^d)$ is dense in $H^s(\mathbb{T}^d)$.

Proof. Let $f \in H^s(\mathbb{T}^d)$ be arbitrary. By Lemma A.2.1, for all $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$\|f - S_N(f)\|_{H^s(\mathbb{T}^d)} < \varepsilon.$$

It remains to show that $S_N(f) \in \mathcal{C}^\infty(\mathbb{T}^d)$. To this end, we write

$$S_N(f)(x) = \sum_{|k| \leq N} \hat{f}_k e^{ik\cdot x} = \frac{1}{2} \sum_{|k| \leq N} \left(\hat{f}_k e^{ik\cdot x} + \hat{f}_{-k} e^{-ik\cdot x} \right).$$

By definition of Fourier coefficients and Euler's formula, for all $k \in \mathbb{N}^d$ we have

$$\begin{aligned}
\hat{f}_k e^{ik\cdot x} + \hat{f}_{-k} e^{-ik\cdot x} &= \frac{1}{\sqrt{|\mathbb{T}^d|}} \Big(\langle f, e^{-ik\cdot x}\rangle e^{ik\cdot x} + \langle f, e^{ik\cdot x}\rangle e^{-ik\cdot x} \Big) \\
&= \frac{1}{\sqrt{|\mathbb{T}^d|}} \Big(\big(\langle f, \cos(k\cdot x)\rangle - i\langle f, \sin(k\cdot x)\rangle\big)\big(\cos(k\cdot x) + i\sin(k\cdot x)\big) \\
&\qquad\qquad + \big(\langle f, \cos(k\cdot x)\rangle + i\langle f, \sin(k\cdot x)\rangle\big)\big(\cos(k\cdot x) - i\sin(k\cdot x)\big) \Big) \\
&= \frac{2}{\sqrt{|\mathbb{T}^d|}} \big(\langle f, \cos(k\cdot x)\rangle \cos(k\cdot x) + \langle f, \sin(k\cdot x)\rangle \sin(k\cdot x)\big).
\end{aligned}$$

Substituting this identity in the expression of $S_N(f)$ yields

$$S_N(f)(x) = \frac{1}{\sqrt{|\mathbb{T}^d|}} \sum_{|k|\le N} \big(\langle f, \cos(k\cdot x)\rangle \cos(k\cdot x) + \langle f, \sin(k\cdot x)\rangle \sin(k\cdot x)\big) \in \mathcal{C}^\infty(\mathbb{T}^d),$$

as wished. □

Let $H^s_{\mathrm{zm}}(\mathbb{T}^d)$ denote the zero-mean subspace of $H^s(\mathbb{T}^d)$, that is,

$$H^s_{\mathrm{zm}}(\mathbb{T}^d) := \left\{ f \in H^s(\mathbb{T}^d) : \frac{1}{|\mathbb{T}^d|}\langle f, 1\rangle = 0 \right\},$$

where 1 denotes the function $\varphi \equiv 1 \in \mathcal{D}(\mathbb{T}^d)$.

Lemma A.2.3. *The space $H^s_{\mathrm{zm}}(\mathbb{T}^d)$ equipped with the norm* (A.1) *is a reflexive Banach space. Furthermore, on $H^s_{\mathrm{zm}}(\mathbb{T}^d)$ the norm* (A.1) *is equivalent to the norm*

$$\|f\|^2_{H^s_{\mathrm{zm}}(\mathbb{T}^d)} := \sum_{k\in\mathbb{Z}^d\setminus\{0\}} |k|^{2s} |\hat{f}_k|^2 . \tag{A.2}$$

Proof. The linear functional

$$h : H^s(\mathbb{T}^d) \to \mathbb{R},\ f \mapsto \frac{1}{|\mathbb{T}^d|}\langle f, 1\rangle$$

is bounded, since

$$\frac{1}{|\mathbb{T}^d|}|\langle f, 1\rangle| = \frac{1}{\sqrt{|\mathbb{T}^d|}}|\hat{f}_0| \le \frac{1}{\sqrt{|\mathbb{T}^d|}}\|f\|_{H^s(\mathbb{T}^d)} \qquad \text{for all } f \in H^s(\mathbb{T}^d).$$

Therefore, h is continuous. Since $H^s_{\mathrm{zm}}(\mathbb{T}^d) = \ker(h)$, it follows that $H^s_{\mathrm{zm}}(\mathbb{T}^d)$ is a closed linear subspace of the Banach space $H^s(\mathbb{T}^d)$, hence itself a Banach space. Furthermore, since $H^s(\mathbb{T}^d)$ is a Hilbert space when equipped with the norm (A.1), it is uniformly convex, and therefore reflexive. Consequently, $H^s_{\mathrm{zm}}(\mathbb{T}^d)$ is itself reflexive [15, Prop 3.20].

It is straightforward to verify that $\|\cdot\|_{H^s_{\mathrm{zm}}(\mathbb{T}^d)}$ defines indeed a norm. Indeed, nonnegativity, homogeneity and the triangle inequality follow by the analogous properties of the absolute value. For the definiteness, it suffices to use Lemma A.2.1.

We conclude with the equivalence of the two norms. By definition of $H^s_{\mathrm{zm}}(\mathbb{T}^d)$, it holds

$$\frac{1}{\sqrt{|\mathbb{T}^d|}}\hat{f}_0 = \frac{1}{|\mathbb{T}^d|}\langle f, 1\rangle = 0 \quad \text{for all } f \in H^s_{\mathrm{zm}}(\mathbb{T}^d).$$

Furthermore, for all $k \in \mathbb{Z}^d \setminus \{0\}$, it holds

$$|k|^2 \leq 1 + |k|^2 \leq 2|k|^2 .$$

Hence, if $s \geq 0$, we have

$$|k|^{2s} \leq (1 + |k|^2)^s \leq 2^s |k|^{2s} .$$

If $s < 0$, the inequalities are reversed. Therefore, we conclude that there exists $C_s > 0$ such that

$$\frac{1}{C_s} \|f\|^2_{H^s(\mathbb{T}^d)} \leq \|f\|^2_{H^s_{\mathrm{zm}}(\mathbb{T}^d)} \leq C_s \|f\|^2_{H^s(\mathbb{T}^d)}. \qquad \square$$

Remark A.2.4. *The norm* (A.2) *is precisely the standard seminorm on the homogeneous space* $\dot{H}^s(\mathbb{T}^d)$ *(see [7]).*

Lemma A.2.5 (**Fourier Differentiation Rule**)**.**
Let $f \in H^s(\mathbb{T}^d)$*. For every multi-index* $\alpha \in \mathbb{N}^d$*, it holds*

$$\widehat{\partial^\alpha f}_k = (ik)^\alpha \hat{f}_k \quad \text{for all } k \in \mathbb{Z}^d,$$

and $\partial^\alpha f \in H^{s-|\alpha|}(\mathbb{T}^d)$*.*

Proof. Let $f \in H^s(\mathbb{T}^d) \subset \mathcal{D}'(\mathbb{T}^d)$. For all $k \in \mathbb{Z}^d$, we compute

$$\widehat{\partial^\alpha f}_k = \frac{1}{\sqrt{|\mathbb{T}^d|}} \langle \partial^\alpha f, e^{-ik\cdot x} \rangle = \frac{(-1)^{|\alpha|}}{\sqrt{|\mathbb{T}^d|}} \langle f, \partial^\alpha (e^{-ik\cdot x}) \rangle.$$

Since $\partial^\alpha (e^{-ik\cdot x}) = (-ik)^\alpha e^{-ik\cdot x}$, we obtain

$$\widehat{\partial^\alpha f}_k = (ik)^\alpha \hat{f}_k.$$

It follows that

$$\|\partial^\alpha f\|^2_{H^{s-|\alpha|}(\mathbb{T}^d)} = \sum_{k \in \mathbb{Z}^d} (1 + |k|^2)^{s-|\alpha|} |k|^{2|\alpha|} |\hat{f}_k|^2.$$

Using $|k|^{2|\alpha|} \leq (1 + |k|^2)^{|\alpha|}$, we deduce

$$\|\partial^\alpha f\|^2_{H^{s-|\alpha|}(\mathbb{T}^d)} \leq \sum_{k \in \mathbb{Z}^d} (1 + |k|^2)^s |\hat{f}_k|^2 = \|f\|^2_{H^s(\mathbb{T}^d)} < \infty,$$

which proves $\partial^\alpha f \in H^{s-|\alpha|}(\mathbb{T}^d)$. $\square$

Lemma A.2.6 (**Poincaré Inequality in** $H^1_{\mathrm{zm}}(\mathbb{T}^d)$)**.**
Let $f \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$*. Then,*

$$\|f\|^2_{H^1(\mathbb{T}^d)} \leq 2 \|\nabla f\|^2_{L^2(\mathbb{T}^d)^d} .$$

Proof. Let $f \in H^1_{\mathrm{zm}}(\mathbb{T}^d)$. Then $\hat{f}_0 = 0$, and by the definition of the Sobolev norm on the torus,

$$\|f\|^2_{H^1(\mathbb{T}^d)} = \sum_{k \in \mathbb{Z}^d} (1 + |k|^2) |\hat{f}_k|^2 = \sum_{k \in \mathbb{Z}^d \setminus \{0\}} (1 + |k|^2) |\hat{f}_k|^2.$$

For all $k \in \mathbb{Z}^d \setminus \{0\}$, we use the trivial estimate

$$1 + |k|^2 \leq 2|k|^2$$

to deduce that

$$\|f\|_{H^1(\mathbb{T}^d)}^2 \leq 2 \sum_{k \in \mathbb{Z}^d \setminus \{0\}} |k|^2 |\hat{f}_k|^2.$$

On the other hand, the Fourier differentiation rule yields

$$\nabla f(x) = \sum_{k \in \mathbb{Z}^d} (ik)\, \hat{f}_k e^{ik \cdot x}.$$

Hence, we compute

$$\|\nabla f\|_{L^2(\mathbb{T}^d)^d}^2 = \sum_{i=1}^{d} \left\| \frac{\partial f}{\partial x_i} \right\|_{L^2(\mathbb{T}^d)}^2 = \sum_{k \in \mathbb{Z}^d} |k|^2 |\hat{f}_k|^2 = \sum_{k \in \mathbb{Z}^d \setminus \{0\}} |k|^2 |\hat{f}_k|^2,$$

where we used that the $k = 0$ mode vanishes since $\hat{f}_0 = 0$. Combining the previous estimates gives

$$\|f\|_{H^1(\mathbb{T}^d)}^2 \leq 2 \|\nabla f\|_{L^2(\mathbb{T}^d)^d}^2,$$

which concludes the proof. □

Appendix B

Hypocoercivity Method

In this section, we present the abstract hypocoercivity method for transport-collision linear evolution equations on a Hilbert space developed in [25].

Let $(\mathcal{H}, (\,\cdot\,,\,\cdot\,)_{\mathcal{H}})$ be a separable Hilbert space over $\mathbb{R}$ and $\|\cdot\|_{\mathcal{H}}$ denote the norm induced by the inner product. Consider the following linear evolution equation on $\mathcal{H}$:

$$\frac{\mathrm{d}F}{\mathrm{d}t} + \mathrm{T}F = \mathrm{L}F, \tag{B.1}$$

where the (possibly unbounded) densely defined operators $\mathrm{T} : \mathcal{D}(\mathrm{T}) \subset \mathcal{H} \to \mathcal{H}$ and $\mathrm{L} : \mathcal{D}(\mathrm{L}) \subset \mathcal{H} \to \mathcal{H}$ are the *transport* and *collision* operators, respectively, and F belongs to a suitably defined closed subspace of time-dependent $\mathcal{H}$-valued functions. Let $\Pi : \mathcal{H} \to \ker(\mathrm{L})$ denote the orthogonal projection onto the null space of L, and define the auxiliary operator

$$\mathrm{A} := \left(\mathrm{Id}_{\mathcal{H}} + (\mathrm{T}\Pi)^*\mathrm{T}\Pi\right)^{-1}(\mathrm{T}\Pi)^*.$$

In the remainder of this chapter, we first state the main assumptions required for the hypocoercivity analysis and derive preliminary results on the operators involved. Building on this, we then derive a Lyapunov functional that captures the combined effect of transport and collision mechanisms and prove exponential decay in the Hilbert space norm.

B.1 Hypocoercivity Assumptions and Preliminary Results

We first introduce the structural assumptions required throughout this chapter, including the symmetry properties of the transport and collision operators, the coercivity assumptions on microscopic and macroscopic dynamics, and the boundedness properties of the auxiliary operator. Under these assumptions, we derive fundamental properties of the auxiliary operator A and establish several estimates that will be used in the construction of the Lyapunov functional in the following section.

As in [53], the following assumption is imposed to ensure that finitely many compositions of operators involving T and L are well-defined, without domain issues.

Assumption B.1.1. *There exists a dense topological vector space $\mathcal{S} \subset \mathcal{H}$ such that*

$$\mathcal{S} \subset \mathcal{D}(\mathrm{T}) \cap \mathcal{D}(\mathrm{L}) \cap \mathcal{D}((\mathrm{T}\Pi)^*)$$

The next assumption collects the main structural and coercivity properties required for the abstract hypocoercivity method introduced in [25]. It combines the algebraic properties of the transport and collision operators with the microscopic and macroscopic coercivity conditions and the boundedness estimates involving the auxiliary operator A.

Assumption B.1.2. *Let $\lambda_m, \lambda_M, C_M$ be positive constants such that, for any $f \in \mathcal{D}(\mathrm{T}) \cap \mathcal{D}(\mathrm{L})$, the following hold:*

(*i*) T *and* L *are densely defined, with* T *antisymmetric and* L *symmetric, and* TΠ *is densely defined;*

(*ii*) *microscopic coercivity:* $-(\mathrm{L}f, f)_{\mathcal{H}} \geq \lambda_m \|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}}^2$;

(*iii*) *macroscopic coercivity:* $\|\mathrm{T}\Pi f\|_{\mathcal{H}}^2 \geq \lambda_M \|\Pi f\|_{\mathcal{H}}^2$;

(*iv*) *parabolic macroscopic dynamics:* $\Pi\mathrm{T}\Pi f = 0$;

(*v*) *bounded auxiliary operators:* $\|\mathrm{AT}(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}} + \|\mathrm{AL}f\|_{\mathcal{H}} \leq C_M \|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}}$.

The first condition guarantees the density and symmetry properties required for the operator-theoretic framework used below. The next three conditions quantify the dissipative effects at the microscopic and macroscopic levels, while the final condition controls the coupling between the transport and collision mechanisms through the auxiliary operator.

Remark B.1.3. *By antisymmetry (respectively symmetry) of* T *(respectively* L*), it follows that* T *(respectively* L*) is closable. In particular, this implies that* $\ker(\mathrm{L})$ *is closed (see, e.g., Proposition 1.5 in [50]).*

Remark B.1.4. *Since* Π *is the orthogonal projection onto the closed subspace* $\ker(\mathrm{L})$*, it is a bounded self-adjoint – hence continuous – operator satisfying* $\Pi \circ \Pi = \Pi$*. Moreover, since* T *is closable and* Π *is bounded, the operator* TΠ *is closable. Together with the assumption that* $\mathcal{D}(\mathrm{T}\Pi)$ *is dense, this implies that* $(\mathrm{T}\Pi)^*$ *is densely defined (see [50]).*

We next turn to establishing fundamental properties of the auxiliary operator.

Lemma B.1.5. *Let Assumption* B.1.2 *hold. Then,* $\mathrm{A} : \mathcal{D}(\mathrm{A}) \to \mathcal{H}$ *is a densely-defined linear operator which extends to a bounded operator on* $\mathcal{H}$ *satisfying* $\Pi\mathrm{A} = \mathrm{A}$ *and* $\|\mathrm{A}\| \leq \frac{1}{2}$.

Proof. We first establish that $\mathcal{R}((\mathrm{T}\Pi)^*) \subset \ker(\mathrm{L})$ in Step 1. This property is then used in Step 2 to prove that A is densely defined and linear, with $\mathcal{R}(\mathrm{A}) \subset \ker(\mathrm{L})$. We conclude with the identity $\Pi\mathrm{A} = \mathrm{A}$ in Step 3 and the boundedness of A in Step 4.

Step 1. We establish that $\mathcal{R}((\mathrm{T}\Pi)^*) \subset \ker(\mathrm{L})$. Let $g \in \ker(\mathrm{L})^\perp$ and $f \in \mathcal{D}((\mathrm{T}\Pi)^*)$ be arbitrary. Then,

$$(f, \mathrm{T}\Pi g)_{\mathcal{H}} = (f, \mathrm{T}(\Pi g))_{\mathcal{H}} = (f, \mathrm{T}0)_{\mathcal{H}} = 0,$$

where the two identities follow from the definition of Π and the linearity of T, respectively. Combining this with the definition of Hilbert adjoint yields

$$((\mathrm{T}\Pi)^* f, g)_{\mathcal{H}} = 0.$$

By arbitrariness of $g \in \ker(\mathrm{L})^\perp$, it follows that $(\mathrm{T}\Pi)^* f \in (\ker(\mathrm{L})^\perp)^\perp = \ker(\mathrm{L})$ for any $f \in \mathcal{D}((\mathrm{T}\Pi)^*)$, that is, $\mathcal{R}((\mathrm{T}\Pi)^*) \subset \ker(\mathrm{L})$ as wished.

Step 2. We first show that the restriction

$$\mathrm{B} := \left(\mathrm{Id}_{\mathcal{H}} + (\mathrm{T}\Pi)^*\mathrm{T}\Pi\right)\big|_{\ker(\mathrm{L})}$$

is a bijective map onto $\ker(\mathrm{L})$, so that $(\mathrm{Id}_{\mathcal{H}} + (\mathrm{T}\Pi)^*\mathrm{T}\Pi)^{-1}$ exists on $\ker(\mathrm{L})$. To this end, let $f \in \ker(\mathrm{L})$ be such that $\mathrm{B}f = 0$. Then,

$$0 = (f, (\mathrm{Id}_{\mathcal{H}} + (\mathrm{T}\Pi)^*\mathrm{T}\Pi)f)_{\mathcal{H}} = \|f\|_{\mathcal{H}}^2 + (f, (\mathrm{T}\Pi)^*\mathrm{T}\Pi f)_{\mathcal{H}} = \|f\|_{\mathcal{H}}^2 + (\mathrm{T}\Pi f, \mathrm{T}\Pi f)_{\mathcal{H}} = \|f\|_{\mathcal{H}}^2 + \|\mathrm{T}\Pi f\|_{\mathcal{H}}^2.$$

It follows that $f = 0$, and thus B is injective. To show surjectivity, let $f \in \ker(\mathrm{L})$ be arbitrary. Then, by Cauchy-Schwarz and the above calculation, we have

$$\|f\|_{\mathcal{H}}^2 \leq \|f\|_{\mathcal{H}}^2 + \|\mathrm{T}\Pi f\|_{\mathcal{H}}^2 = (f, \mathrm{B}f)_{\mathcal{H}} \leq \|f\|_{\mathcal{H}} \cdot \|\mathrm{B}f\|_{\mathcal{H}} \quad \Longrightarrow \quad \|Bf\|_{\mathcal{H}} \geq \|f\|_{\mathcal{H}}. \tag{B.2}$$

In other words, B is coercive, and thus its range is closed. Furthermore, since B is self-adjoint, we have

$$\mathcal{R}(\mathrm{B}) = \overline{\mathcal{R}(\mathrm{B})} = \ker(\mathrm{B})^{\perp} = \{0\}^{\perp} = \ker(\mathrm{L}).$$

Therefore, B is bijective. This implies that B^{-1} is well-defined on $\ker(\mathrm{L})$. Using that $\mathcal{R}((\mathrm{T}\Pi)^*) \subset \ker(\mathrm{L})$ from Step 1, the operator $\mathrm{A} = \mathrm{B}^{-1}(\mathrm{T}\Pi)^*$ is well-defined on $\mathcal{D}((\mathrm{T}\Pi)^*)$. Moreover, $\mathcal{D}(\mathrm{A}) = \mathcal{D}((\mathrm{T}\Pi)^*)$. Since $(\mathrm{T}\Pi)^*$ is densely defined (see Remark B.1.4), it follows that A is densely defined.

Step 3. We establish that $\Pi\mathrm{A} = \mathrm{A}$. Let $x \in \mathcal{D}(\mathrm{A})$ be arbitrary, and set $\mathrm{A}x =: y \in \mathcal{R}(\mathrm{A})$. Since $\mathcal{R}(\mathrm{A}) \subset \ker(\mathrm{L})$, there exists a sequence $\{y_n\}_{n\in\mathbb{N}} \subset \ker(\mathrm{L})$ such that $y_n \to y$ in $\mathcal{H}$. But then, by continuity of Π it follows that

$$(\Pi\mathrm{A})(x) = \Pi(\mathrm{A}x) = \Pi y = \Pi(\lim_{n\to\infty} y_n) = \lim_{n\in\mathbb{N}} \Pi y_n = \lim_{n\to\infty} y_n = y = \mathrm{A}x.$$

By arbitrariness of $x \in \mathcal{D}(\mathrm{A})$, it follows that $\Pi\mathrm{A} = \mathrm{A}$.

Step 4. Finally, we prove boundedness of A. Let $x \in \mathcal{D}(\mathrm{A})$ be arbitrary and set $y := \mathrm{A}x = \mathrm{B}^{-1}(\mathrm{T}\Pi)^*x$, so that $\mathrm{B}y = (\mathrm{T}\Pi)^*x$. Therefore, by definition of Hilbert adjoint and the Cauchy-Schwarz inequality, we obtain

$$(y, \mathrm{B}y)_{\mathcal{H}} = (y, (\mathrm{T}\Pi)^*x)_{\mathcal{H}} = (\mathrm{T}\Pi y, x)_{\mathcal{H}} \leq \|\mathrm{T}\Pi y\|_{\mathcal{H}} \cdot \|x\|_{\mathcal{H}}.$$

On the other hand, proceeding as in (B.2) yields

$$(y, \mathrm{B}y)_{\mathcal{H}} = \|y\|_{\mathcal{H}}^2 + \|\mathrm{T}\Pi y\|_{\mathcal{H}}^2.$$

Combining the two expressions, we obtain

$$\|y\|_{\mathcal{H}}^2 + \|\mathrm{T}\Pi y\|_{\mathcal{H}}^2 \leq \|\mathrm{T}\Pi y\|_{\mathcal{H}} \cdot \|x\|_{\mathcal{H}}.$$

Letting $a := \|y\|_{\mathcal{H}}$ and $b := \|\mathrm{T}\Pi y\|_{\mathcal{H}}$, the above inequality becomes

$$a^2 \leq b\|x\|_{\mathcal{H}} - b^2 \leq \frac{\|x\|_{\mathcal{H}}^2}{4},$$

where the last upper bound follows from the fact that $b \mapsto b\|x\|_{\mathcal{H}} - b^2$ achieves a maximum at $b = \frac{\|x\|_{\mathcal{H}}}{2}$ with value $\frac{\|x\|_{\mathcal{H}}^2}{4}$. Substituting $a = \|y\|_{\mathcal{H}}$, and taking the square root on both sides yields

$$\|\mathrm{A}x\|_{\mathcal{H}} = \|y\|_{\mathcal{H}} \leq \frac{\|x\|_{\mathcal{H}}}{2}.$$

By arbitrariness of $x \in \mathcal{D}(\mathrm{A})$, A is bounded with $\|\mathrm{A}\| \leq \frac{1}{2}$. □

We next recall Lemma 1 from [25] and present a modified proof for completeness.

Lemma B.1.6. *Let Assumption* B.1.2 *hold. Then,* A *and* TA *are bounded; in particular, for all* $f \in \mathcal{D}(\mathrm{T})$*, there holds*

$$\|\mathrm{A}f\|_{\mathcal{H}} \leq \frac{1}{2}\|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}} \qquad \text{and} \qquad \|\mathrm{TA}f\|_{\mathcal{H}} \leq \|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}}.$$

Proof. Let $f \in \mathcal{D}(\mathrm{A})$ and $g \in \mathcal{R}(\mathrm{A})$ be arbitrary. Since Π is self-adjoint and $\Pi\mathrm{A} = \mathrm{A}$ by Lemma B.1.5, we obtain

$$\mathrm{A}f = g \qquad \Longleftrightarrow \qquad \mathrm{TA}f = \mathrm{T}\Pi\mathrm{A}f = \mathrm{T}\Pi g.$$

Therefore,

$$\|g\|_{\mathcal{H}}^2 + \|\mathrm{T}\Pi g\|_{\mathcal{H}}^2 = ((\mathrm{Id}_{\mathcal{H}} + (\mathrm{T}\Pi)^*\mathrm{T}\Pi)g, g)_{\mathcal{H}} = ((\mathrm{T}\Pi)^* f, g)_{\mathcal{H}} = (f, \mathrm{T}\Pi g)_{\mathcal{H}},$$

where we used that $(\mathrm{Id}_{\mathcal{H}} + (\mathrm{T}\Pi)^*\mathrm{T}\Pi)\mathrm{A} = (\mathrm{T}\Pi)^*$. Moreover, since Π is self-adjoint and $\Pi\mathrm{T}\Pi = 0$, it follows that

$$(f, \mathrm{T}\Pi g)_{\mathcal{H}} = (f, \mathrm{T}\Pi g)_{\mathcal{H}} - (f, \Pi\mathrm{T}\Pi g)_{\mathcal{H}} = (f, \mathrm{T}\Pi g)_{\mathcal{H}} - (\Pi f, \mathrm{T}\Pi g)_{\mathcal{H}} = ((\mathrm{Id}_{\mathcal{H}} - \Pi)f, \mathrm{T}\Pi g)_{\mathcal{H}}.$$

Combining the previous identities yields

$$\|g\|_{\mathcal{H}}^2 + \|\mathrm{T}\Pi g\|_{\mathcal{H}}^2 = ((\mathrm{Id}_{\mathcal{H}} - \Pi)f, \mathrm{T}\Pi g)_{\mathcal{H}}.$$

Hence, by Cauchy-Schwarz and Young's inequality,

$$\|g\|_{\mathcal{H}}^2 + \|\mathrm{T}\Pi g\|_{\mathcal{H}}^2 \leq \|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}} \|\mathrm{T}\Pi g\|_{\mathcal{H}} \leq \tfrac{1}{4}\|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}}^2 + \|\mathrm{T}\Pi g\|_{\mathcal{H}}^2, \tag{B.3}$$

that is,

$$\|\mathrm{A}f\|_{\mathcal{H}} = \|g\|_{\mathcal{H}} \leq \tfrac{1}{2}\|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}}.$$

Next, define

$$a := \|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}}, \qquad b := \|g\|_{\mathcal{H}}^2, \qquad x := \|\mathrm{T}\Pi g\|_{\mathcal{H}}.$$

Then, the first inequality in (B.3) yields $x^2 - ax + b \leq 0$. This implies that

$$0 \leq x \leq \frac{1}{2}(a + \sqrt{a^2 - 4b}) \leq a.$$

Substituting a and x using $g = \mathrm{A}f$ as well as $\Pi\mathrm{A} = \mathrm{A}$, we conclude that

$$\|\mathrm{TA}f\|_{\mathcal{H}} = \|\mathrm{T}\Pi\mathrm{A}f\|_{\mathcal{H}} = \|\mathrm{T}\Pi g\|_{\mathcal{H}} = x \leq a = \|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}}. \qquad \square$$

Lemma B.1.7. *Let Assumption* B.1.2 *hold. Then, the operator* $A\mathrm{T}\Pi$ *satisfies the following coercivity-type estimate:*

$$(\mathrm{AT}\Pi f, f)_{\mathcal{H}} \geq \frac{\lambda_M}{1 + \lambda_M}\|\Pi f\|_{\mathcal{H}}^2$$

for all $f \in \mathcal{D}(\mathrm{T})$.

Proof. Let $\mathrm{B} := (\mathrm{T}\Pi)^*\mathrm{T}\Pi$ and note that $\mathrm{AT}\Pi = (\mathrm{Id}_{\mathcal{H}} + \mathrm{B})^{-1}\mathrm{B}$. By macroscopic coercivity, for all $f \in \mathcal{D}(\mathrm{B})$, there holds

$$(\mathrm{B}f, f)_{\mathcal{H}} = ((\mathrm{T}\Pi)^*\mathrm{T}\Pi f, f)_{\mathcal{H}} = (\mathrm{T}\Pi f, \mathrm{T}\Pi f)_{\mathcal{H}} = \|\mathrm{T}\Pi f\|_{\mathcal{H}}^2 \geq \lambda_M \|\Pi f\|_{\mathcal{H}}^2. \tag{B.4}$$

Next, as in Step 2 of Lemma B.1.5, $(\mathrm{Id}_{\mathcal{H}} + \mathrm{B})^{-1}\mathrm{B}$ maps into $\ker(\mathrm{L})$. Moreover, since $\ker(\mathrm{L})$ is closed, $\mathcal{H}$ is isometrically isomorphic to $\ker(\mathrm{L}) \oplus \ker(\mathrm{L})^{\perp}$. Therefore, using that $\Pi^2 = \Pi$, for all $f \in \mathcal{D}(\mathrm{B})$ we obtain

$$(\mathrm{AT}\Pi f, f)_{\mathcal{H}} = (\mathrm{AT}\Pi^2 f, \Pi f)_{\mathcal{H}} = ((\mathrm{Id}_{\mathcal{H}} + \mathrm{B})^{-1}\mathrm{B}\Pi f, \Pi f)_{\mathcal{H}}.$$

Furthermore, we compute

$$\mathrm{B} = (\mathrm{Id}_{\mathcal{H}} + \mathrm{B}) - \mathrm{Id}_{\mathcal{H}} \qquad \Longrightarrow \qquad (\mathrm{Id}_{\mathcal{H}} + \mathrm{B})^{-1}\mathrm{B} = \mathrm{Id}_{\mathcal{H}} - (\mathrm{Id}_{\mathcal{H}} + \mathrm{B})^{-1}.$$

In other words, for all $f \in \mathcal{D}(\mathrm{B})$, there holds

$$((\mathrm{Id}_{\mathcal{H}} + \mathrm{B})^{-1}\mathrm{B}\Pi f, \Pi f)_{\mathcal{H}} = \Big((\mathrm{Id}_{\mathcal{H}} - (\mathrm{Id}_{\mathcal{H}} + \mathrm{B})^{-1})\Pi f, \Pi f\Big)_{\mathcal{H}} = \|\Pi f\|_{\mathcal{H}}^2 - \Big((\mathrm{Id}_{\mathcal{H}} + \mathrm{B})^{-1})\Pi f, \Pi f\Big)_{\mathcal{H}}. \quad \text{(B.5)}$$

By Cauchy-Scwharz and the coercivity of B in (B.4), for all $g \in \mathcal{D}(\mathrm{B})$ we compute

$$\|(\mathrm{Id}_{\mathcal{H}} + \mathrm{B})g\|_{\mathcal{H}}\|g\|_{\mathcal{H}} \geq |((\mathrm{Id}_{\mathcal{H}} + \mathrm{B})g, g)_{\mathcal{H}}| \geq (1 + \lambda_M)\|g\|_{\mathcal{H}}^2.$$

Letting $g = (\mathrm{Id}_{\mathcal{H}} + \mathrm{B})^{-1}\Pi f$ and noting that $g \in \mathcal{D}(\mathrm{B})$ for all $f \in \mathcal{D}(\mathrm{B})$, the previous inequality yields

$$\|(\mathrm{Id}_{\mathcal{H}} + \mathrm{B})^{-1}\Pi f\|_{\mathcal{H}} \leq \frac{1}{1 + \lambda_M}\|\Pi f\|_{\mathcal{H}}.$$

Using the Cauchy-Schwarz inequality in (B.5) yields the desired bound:

$$((\mathrm{Id}_{\mathcal{H}} + \mathrm{B})^{-1}\mathrm{B}\Pi f, \Pi f)_{\mathcal{H}} \geq \left(1 - \frac{1}{1 + \lambda_M}\right)\|\Pi f\|_{\mathcal{H}}^2 = \frac{\lambda_M}{1 + \lambda_M}\|\Pi f\|_{\mathcal{H}}^2. \qquad \square$$

B.2 Hypocoercivity and Decay Rates

Building on the operator framework developed in the preceding discussion, this section constructs a Lyapunov functional and derives explicit exponential decay estimates for the solutions to the linear evolution equation (B.1). The analysis quantifies the combined effect of transport and collision dynamics and establishes convergence to equilibrium in the Hilbert space norm.

Under the hypotheses in Assumption B.1.2, any (sufficiently smooth) nontrivial solution F to the linear evolution equation (B.1) satisfies

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|F(t)\|_{\mathcal{H}}^2 = \frac{1}{2}((\mathrm{L}-\mathrm{T})F(t), F(t))_{\mathcal{H}} + \frac{1}{2}(F(t), (\mathrm{L}-\mathrm{T})F(t))_{\mathcal{H}} = (\mathrm{L}F(t), F(t))_{\mathcal{H}} \leq -\lambda_m\|(\mathrm{Id}_{\mathcal{H}} - \Pi)F(t)\|_{\mathcal{H}}^2.$$

Nevertheless, this estimate alone is not enough to guarantee exponential decay of $\|F(t)\|_{\mathcal{H}}$. To establish exponential stability of the solution in the $\mathcal{H}$-norm sense, let us introduce the Lyapunov functional $\mathrm{H}_\delta : \mathcal{H} \to \mathbb{F}$ defined by

$$\mathrm{H}_\delta(f) = \frac{1}{2}\|f\|_{\mathcal{H}}^2 + \delta(\mathrm{A}f, f)_{\mathcal{H}}$$

for some $\delta > 0$ to be chosen appropriately. By continuity of the inner product and A and the (anti)symmetry property of T and L, we obtain

$$\begin{aligned}
\mathrm{D}_\delta(F)(t) := -\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{H}_\delta(F)(t) &= -(\mathrm{L}F(t), F(t))_{\mathcal{H}} - \delta(\mathrm{A}(\mathrm{L}-\mathrm{T})F(t), F(t))_{\mathcal{H}} - \delta(\mathrm{A}F(t), (\mathrm{L}-\mathrm{T})F(t))_{\mathcal{H}} \\
&= -(\mathrm{L}F(t), F(t))_{\mathcal{H}} - \delta(\mathrm{AL}F(t), F(t))_{\mathcal{H}} + \delta(\mathrm{AT}(\mathrm{Id}_{\mathcal{H}} - \Pi)F(t), F(t))_{\mathcal{H}} \\
&\quad + \delta(\mathrm{AT}\Pi F(t), F(t))_{\mathcal{H}} - \delta(\mathrm{TA}F(t), F(t))_{\mathcal{H}} - \delta(\mathrm{LA}F(t), F(t))_{\mathcal{H}}.
\end{aligned}$$

Since $\mathcal{R}(\mathrm{A}) \subset \ker(\mathrm{L})$ by Lemma B.1.5, there holds $\mathrm{LA}f = 0$ for all $f \in \mathcal{D}(\mathrm{A})$. Hence, the last summand in the expression of $\mathrm{D}_\delta(F)$ vanishes, yielding

$$\mathrm{D}_\delta(F) = -(\mathrm{L}F, F)_{\mathcal{H}} - \delta(\mathrm{AL}F, F)_{\mathcal{H}} + \delta(\mathrm{AT}(\mathrm{Id}_{\mathcal{H}} - \Pi)F, F)_{\mathcal{H}} + \delta(\mathrm{AT}\Pi F, F)_{\mathcal{H}} - \delta(\mathrm{TA}F, F)_{\mathcal{H}}.$$

We next state Theorem 2 in [25], which describes the two main properties of the hypocoercivity method for real-valued operators through a Lyapunov decay estimate involving H_δ and its derivative D_δ. We also include its proof for completeness.

Theorem B.2.1. *Let Assumptions* B.1.1 *and* B.1.2 *hold. Then, the linear functionals* H_δ *and* D_δ *are densely defined. Furthermore, there exists* $\delta_\star > 0$ *such that for all* $\delta \in (0, \delta_\star)$:

(*i*) H_δ *and* $\|\cdot\|_{\mathcal{H}}^2$ *are equivalent, in the sense that*

$$\frac{1}{2}(1-\delta)\|f\|_{\mathcal{H}}^2 \leq \mathrm{H}_\delta(f) \leq \frac{1}{2}(1+\delta)\|f\|_{\mathcal{H}}^2 \qquad \forall\, f \in \mathcal{S};$$

(*ii*) *for some* $\lambda > 0$ *depending on* δ*, there holds*

$$\lambda \mathrm{H}_\delta(f) \leq \mathrm{D}_\delta(f) \qquad \forall\, f \in \mathcal{S}.$$

Proof. First, note that $\mathcal{S} \subset \mathcal{D}(\mathrm{H}_\delta), \mathcal{D}(\mathrm{D}_\delta)$, hence the two functionals are densely defined. Next, by Cauchy-Schwarz and the first inequality in Lemma B.1.6, for any $f \in \mathcal{S}$, there holds

$$|(\mathrm{A}f, f)_{\mathcal{H}}| \leq \|\mathrm{A}f\|_{\mathcal{H}}\|f\|_{\mathcal{H}} \leq \frac{1}{2}\|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}}\|f\|_{\mathcal{H}} \leq \frac{1}{2}\|f\|_{\mathcal{H}}^2,$$

where the last inequality follows from the fact that $\mathrm{Id}_{\mathcal{H}} - \Pi$ is the projection operator onto $\ker(\mathrm{L})^\perp$, hence it satisfies $\|\mathrm{Id}_{\mathcal{H}} - \Pi\| \leq 1$. Therefore, (*i*) follows for all $\delta \in (0, 1)$.

The Cauchy-Schwarz inequality, Lemma B.1.7 and conditions $(ii) - (iv)$ in Assumption B.1.2 imply

$$\begin{aligned} -(\mathrm{L}f, f)_{\mathcal{H}} &\geq \lambda_m \|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}}^2 \\ \delta(\mathrm{AT}(\mathrm{Id}_{\mathcal{H}} - \Pi)f, f)_{\mathcal{H}} - \delta(\mathrm{AL}f, f)_{\mathcal{H}} &\geq -\delta C_M \|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}}\|f\|_{\mathcal{H}} \\ \delta(\mathrm{AT}\Pi f, f)_{\mathcal{H}} &\geq \frac{\delta\lambda_M}{1+\lambda_M}\|\Pi f\|_{\mathcal{H}}^2. \end{aligned}$$

Next, since $\Pi\mathrm{T}\Pi = 0$, it follows that $\mathrm{T}\Pi$ maps into $\ker(\mathrm{L})^\perp$. Moreover, since A maps into $\ker(\mathrm{L})$, we have $\mathrm{TA} = \mathrm{T}\Pi\mathrm{A}$, and thus TA maps into $\ker(\mathrm{L})^\perp$. Therefore, the isometric isomorphism $\mathcal{H} \cong \ker(\mathrm{L}) \oplus \ker(\mathrm{L})^\perp$, the Cauchy-Schwarz inequality and the second inequality in Lemma B.1.6 imply that

$$\delta(\mathrm{TA}f, f)_{\mathcal{H}} = \delta(\mathrm{TA}f, (\mathrm{Id}_{\mathcal{H}} - \Pi)f)_{\mathcal{H}} \geq -\delta\|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}}^2.$$

Combining the above bounds, for all $f \in \mathcal{S}$, we have

$$\mathrm{D}_\delta(f) \geq (\lambda_m - \delta)a^2 + \frac{\delta\lambda_M}{1+\lambda_M}b^2 - \delta C_M ab,$$

where $a := \|(\mathrm{Id}_{\mathcal{H}} - \Pi)f\|_{\mathcal{H}}$ and $b := \|\Pi f\|_{\mathcal{H}}$. Furthermore, the Cauchy-Schwarz inequality and the first inequality in Lemma B.1.6 yield

$$\mathrm{H}_\delta(f) \leq \frac{1}{2}\|f\|_{\mathcal{H}}^2 + \frac{\delta}{2}ab = \frac{1}{2}(a^2 + b^2) + \frac{\delta}{2}ab,$$

where the last equality uses the isomorphism $\mathcal{H} \cong \ker(\mathrm{L}) \oplus \ker(\mathrm{L})^\perp$. Therefore, the largest value of $\lambda > 0$ for which $\lambda\mathrm{H}_\delta(f) \leq \mathrm{D}_\delta(f)$ holds for all $f \in \mathcal{S}$ can be estimated by the largest value of λ for which the map

$$(a, b) \mapsto (\lambda_m - \delta)a^2 + \frac{\delta\lambda_M}{1+\lambda_M}b^2 - \delta C_M ab - \frac{\lambda}{2}(a^2 + b^2) - \frac{\lambda}{2}\delta ab = (a, b)M(a, b)^\top$$

is a nonnegative quadratic form of $(a, b) \in \mathbb{R}^2$, where

$$M := \begin{pmatrix} \lambda_m - \delta - \frac{\lambda}{2} & -\frac{\delta}{2}(C_M + \frac{\lambda}{2}) \\ -\frac{\delta}{2}(C_M + \frac{\lambda}{2}) & \frac{\delta\lambda_M}{1+\lambda_M} - \frac{\lambda}{2} \end{pmatrix}.$$

Define next

$$\delta_\star := \min\left\{1,\, \lambda_m,\, \frac{4\lambda_m\lambda_M}{4\lambda_M + C_M^2(1+\lambda_M)}\right\},$$

and let $\delta \in (0, \delta_\star)$ be arbitrary. Letting $h(\delta, \lambda)$ denote $\det(M)$ as a function of $\delta > 0$ and $\lambda \geq 0$, the above map is a negative semidefinite quadratic form if and only if

$$h(\delta, \lambda) \geq 0 \qquad \text{and} \qquad \lambda_m - \delta - \frac{\lambda}{2} < 0$$

for a suitable $\lambda > 0$. Since $h(\delta, 0) < 0$ and $\lim_{\lambda \to +\infty} = +\infty$ for any $\delta \in (0, \delta_\star)$, by continuity of the determinant there exists $\lambda > 0$ such that $h(\delta, \lambda) = 0$, which automatically satisfies $\lambda_m - \delta - \frac{\lambda}{2} < 0$, ensuring that M is negative semidefinite. □

Combining the preceding Theorem with Grönwall's inequality (see, e.g., Lemma (6.1) in [4]) yields the following corollary, which illustrates the utility of the Lyapunov approach for evolution equations of the form (B.1).

Corollary B.2.2. *Let Assumptions* B.1.1 *and* B.1.2 *hold. Then, any sufficiently smooth solution* $F : [0, +\infty) \to \mathcal{H}$ *to* (B.1) *with initial datum* $F_0 \in \mathcal{H}$ *satisfies*

$$\mathrm{H}_\delta(F)(t) \leq \mathrm{H}_\delta(F_0) e^{-\lambda t} \quad \forall\, t \geq 0$$

as well as

$$\|F(t)\|_{\mathcal{H}}^2 \leq \frac{1+\delta}{1-\delta} \|F_0\|_{\mathcal{H}}^2 e^{-\lambda t} \quad \forall\, t \geq 0$$

for any $\delta \in (0, \delta_\star)$*, where* $\delta_\star$ *and* $\lambda > 0$ *are as in Theorem* B.2.1.

In other words, under the given assumptions, sufficiently smooth solutions to the linear evolution equation decay exponentially over time in both a tailored Lyapunov functional and the standard Hilbert space norm. The main advantage of this framework is that it provides a quantitative, robust estimate of convergence to equilibrium that accounts for the interplay between transport and collision effects.